You said:

NPTEL Week1 De...
PDF

lecture pdf attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too as i am preparing for introduction to virtual reality for nptel exam

ChatGPT said:

NPTEL Introduction to Virtual Reality — Week 1 Notes

Topic: Definition of VR and History | Based on your 40-slide lecture PDF

I'll explain your lecture in **very simple, 7th-class-level English**, with easy examples, memory tricks, important exam points, and NPTEL-style multiple-choice questions (MCQs) with answers.

The main topics in your PDF are the definition of Virtual Reality (VR), examples of VR, awareness and perception, VR developer bias, the history of VR, realism versus simplicity, FPS (frames per second), and consumer VR. NPTEL Week1 Definition of VR, history.pdf

1. What is Virtual Reality (VR)?

Imagine you wear a special headset and see yourself inside a virtual world. You may feel like you are standing in a forest, flying in the sky, or playing inside a game—even though your real body is still in your room.

That is the basic idea of Virtual Reality.

Easy definition: Virtual Reality is a computer-created experience that makes a person feel as if they are inside an artificial world.

The definition given in your lecture

Your PDF gives a more scientific definition:

> Inducing targeted behavior in an organism by using artificial sensory stimulation, while the organism has little or no awareness of the interference. NPTEL Week1 Definition of VR, history.pdf

Let's break it into small pieces.

Difficult words	Easy meaning
Inducing	Causing something to happen
Targeted behavior	A particular action or response
Organism	A living thing
Artificial	Made by humans, not natural
Sensory stimulation	Giving information to the senses, such as sight and sound

Awareness	Knowing or realizing what is happening
Interference	Something influencing your normal experience

Example: You wear a VR headset and see a virtual cliff. You may feel afraid of falling, even though you are actually standing safely in a room.

Your eyes receive artificial images, and your brain may react as if the situation is real.

Memory trick

Remember these three steps:

S → B → A

- **S — Stimulation:** Give artificial sights or sounds.
- **B — Behavior:** Cause a particular reaction.
- **A — Awareness:** The person may not fully realize how the artificial experience is influencing them.

This is a memory aid for the scientific definition in your lecture.

2. Examples and uses of VR

Your lecture shows many different examples of VR, not just gaming. NPTEL Week1 Definition of VR, history.pdf

1. Socializing in virtual spaces

People meet and communicate through virtual characters called avatars.

2. Architecture and real estate

You can explore a virtual house before it is built or before you buy it.

3. Training and simulation

People can practise activities in a simulated environment.

4. Birdly VR

A person uses a special machine and VR to experience flying like a bird.

5. VR with robots

A person can use a virtual interface to help control or interact with a robot.

6. First-person shooter games

The player experiences the game from the viewpoint of the character they control.

Other examples named in your slides include movies, panoramas, Earth VR, body swapping, and virtual experiences involving animals.
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Remember for the exam: VR is not limited to games. It can be used for social interaction, architecture, entertainment, experiments, and robot interaction.

3. VR in science versus mainstream VR

Your lecture makes an important distinction.

Scientific use of VR	Mainstream use of VR
Studying the brain and nervous system	Entertainment and experiences
Understanding how organisms perceive the world	Playing games and exploring virtual places
Studying neural processes	Helping users feel present in a virtual environment

The scientific fields mentioned in the PDF are neuroscience, neurobiology, and perceptual psychology. NPTEL Week1 Definition of VR, history.pdf

What are place cells and grid cells?

- **Place cells:** Brain cells associated with recognizing or representing particular locations.
- **Grid cells:** Brain cells associated with representing locations using a grid-like pattern.

Think of place cells as helping you remember *where a place is*, and grid cells as helping your brain organize positions, like points on a map.

Your lecture mentions these while discussing awareness and navigation. Do not assume that the PDF defines awareness simply as the existence of these cells; it raises the relationship as a scientific question.

Memory trick: Place = Place on a map; Grid = Grid on a map.

4. Awareness and the danger of developer bias

This is one of the most interesting concepts in your lecture.

What is awareness?

Awareness means knowing what is happening around you or realizing that something is influencing your experience.

For example, when you wear a VR headset, you may know that the world is artificial but still feel that you are inside it.

What is developer bias?

Imagine that you create a VR game. You know exactly how it works, and you really want your teacher to like it.

You might ignore small problems because you already know what the game is supposed to look like. A new user, however, might immediately notice those problems.

This is called **developer bias** in your lecture.

The PDF explains two reasons for this bias:

1. The developer wants the system to succeed.
2. The developer already knows what the system is supposed to do. NPTEL Week1 Definition of VR, history.pdf

How can developers avoid this bias?

The lecture recommends testing VR experiences on:

- People who have never tried that experience before.
- People who have never tried VR before.
- People who experience motion sickness.

Why? Because the creator may have become used to the system's flaws and may no longer notice them.

Memory trick: B-T-T

- **B** — Bias can hide problems.
- **T** — Test with new users.
- **T** — Test with users who may experience discomfort.

NPTEL exam point: A VR developer can be both the scientist and the laboratory rat. This means the developer is both creating and experiencing their own system, which can cause bias.

5. Is everything VR?

Your lecture asks whether several activities should be considered VR. These are useful for conceptual MCQs.

Activity	What to remember
Using an Oculus Quest 2 VR headset	A clear example of VR
Playing a first-person video game	Can provide a virtual experience, but a normal screen game is not automatically immersive VR
Playing a third-person game	The viewpoint is usually behind or outside the controlled character
Augmented reality (AR)	Adds digital content to the real-world view; related to VR, but not the same thing
Watching a movie	Usually not VR by itself
Video conferencing	Usually not VR by itself
Listening to music	Sensory stimulation, but not automatically VR
Reading a book	Usually not VR by itself

These activities appear as discussion questions in your PDF; the table gives you a simple way to understand their differences.

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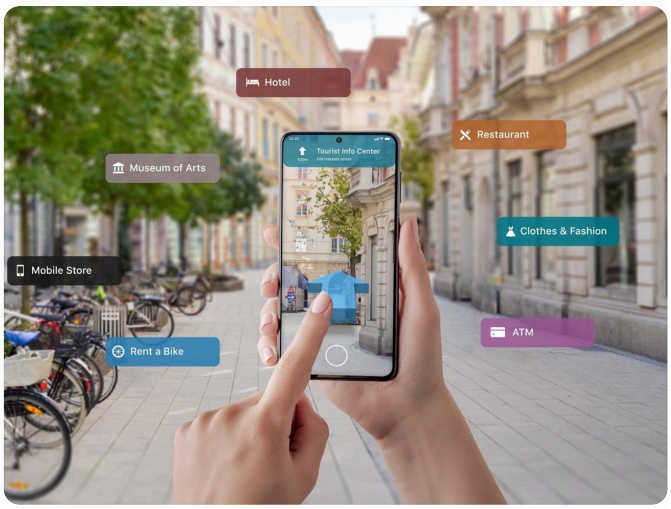
VR versus AR

Virtual Reality (VR)
Replaces or largely blocks your normal view with a virtual environment.



Augmented Reality (AR)
Adds digital objects or information to a view of the real world.

Memory trick: VR = Virtual world; AR = Adds to reality.

6. When did VR start? History of VR

This part of the lecture asks a deeper question: did VR begin with computer headsets, or did humans start trying to create realistic experiences much earlier?

The slides discuss paintings, motion pictures, animation, computer games, and consumer VR. NPTEL Week1 Definition of VR, history.pdf

Important timeline to remember

1. Ancient paintings
Around 30,000 years ago — Bhimbetka example in the slides
People used images to represent the world. The lecture asks whether such attempts to represent reality could be considered an early step toward VR.

2. 1878 — Motion studies

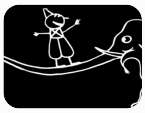
Eadweard Muybridge

Photographic sequences showed movement, helping people study how motion could be represented visually.

3. 1895 — Early cinema

Lumière brothers

Motion pictures made moving visual experiences possible. Early films had limitations such as blur, shakiness, low resolution, and black-and-white images.



4. 1908 — Animation

Fantasmagorie

Animation showed that movement and imagined worlds could be created using drawings.

5. 1960–1990 — First wave of VR

Early VR development

VR attracted significant interest, but cost, bulky hardware, resolution, latency, and limited content created difficulties.

6. 1995 — Nintendo Virtual Boy

The lecture describes the Virtual Boy as marking the death of that wave of consumer VR.

7. 2012 — Palmer Luckey

The lecture identifies Palmer Luckey's work in 2012 as the point when consumer VR interest returned.

Historical details follow the lecture's examples and framing. The painting examples are presented as a question about the origins of VR, not as proof that prehistoric paintings were modern VR systems.

The easiest way to remember the timeline

Paintings → Photographs → Movies → Animation → Early VR → Virtual Boy → Palmer Luckey

If an MCQ asks about the lecture's consumer VR history, remember these exact points:

- **1960–1990:** First wave of VR.
- **1995:** Nintendo Virtual Boy.
- **2012:** Palmer Luckey and renewed consumer VR interest.

7. Realism versus simplicity

Your lecture also discusses realism versus simplicity in cartoons and computer games.

Imagine two games:

- **Game A:** Looks extremely realistic but requires expensive computers.
- **Game B:** Uses simple graphics but runs on a cheap, portable device.

Which is better? It depends on the goal.

Realism	Simplicity
More detailed appearance	Fewer details
Can require more computing power	Can require fewer resources
May improve visual realism	May reduce cost and improve portability

The lecture also points out that cheaper and more portable systems can be useful, even if they are less realistic.

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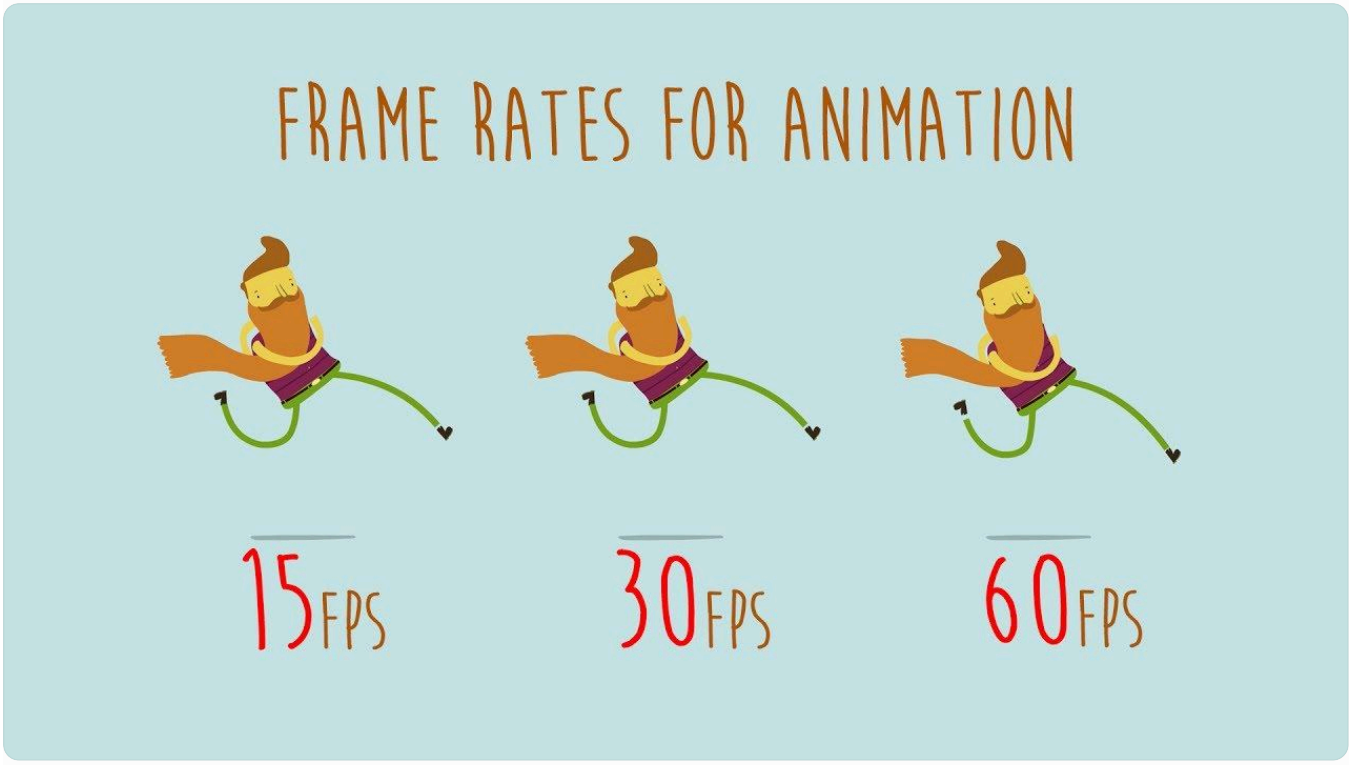
Remember: The most realistic system is not always the most practical system.

8. FPS — Frames Per Second



FPS means **Frames Per Second**.

A frame is one still picture. When many pictures appear quickly, your eyes and brain can perceive movement.

Think of a flipbook: if you turn the pages slowly, you see separate pictures. If you turn them quickly, the pictures seem to move.

Numbers you must memorize

Question from the lecture	Answer
FPS mentioned as sufficient to perceive motion	2 FPS
FPS mentioned as sufficient to perceive continuous motion	12 FPS
Traditional movie-theatre frame rate in the lecture	24 FPS

These are the specific figures stated in your slides. NPTEL Week1 Definition of VR, history.pdf

What happens when FPS increases?

Generally, increasing FPS can make motion look smoother, but it can also increase the computing and hardware requirements.

The lecture says that the suitable FPS for VR depends on:

- Hardware specifications.
- Distance from the screen.
- Types of interaction with the content.

It also mentions possible causes of discomfort, including judder, flicker, tracking problems, and perception of stationarity.
NPTEL Week1 Definition of VR, history.pdf

Memory trick: 2 → 12 → 24

- 2: Motion can be perceived.
- 12: Continuous motion, according to the lecture.
- 24: Traditional cinema.

Important: These are lecture-specific teaching figures, not universal minimum requirements for every VR system. Modern VR often needs higher refresh rates for comfort.

9. Why did early consumer VR struggle?

According to the final slides, early consumer VR had several problems. NPTEL Week1 Definition of VR, history.pdf

Remember this list:

C-R-L-C-H-B-E

- **C** — Cost was high.
- **R** — Resolution was limited.
- **L** — Latency was a problem.
- **C** — Content was limited.
- **H** — Hardware was heavy.
- **B** — Devices were bulky.
- **E** — Devices were expensive.

Latency means the delay between an action and the system responding to it. For example, if you turn your head but the VR image updates late, you may feel uncomfortable.

Exam tip: If an MCQ asks why early consumer VR struggled, look for an option mentioning high cost, low resolution, latency, limited content, or bulky hardware.

10. NPTEL-style MCQs with answers

Try answering these questions before revealing the correct answers. These are practice questions based on your lecture, not official NPTEL exam questions.

Practice test

0 / 24 answered

Choose one answer for each question. Your score and explanations appear when you submit.

1. According to the lecture, VR involves inducing targeted behavior using what?

- A. Natural sunlight only
- B. Artificial sensory stimulation
- C. Printed books only
- D. Physical exercise only

2. What does 'organism' mean in the VR definition?

- A. Only a computer
- B. Only a robot
- C. A living thing
- D. A computer program

3. Which is a mainstream use of VR highlighted in the lecture?

- A. Entertainment
- B. Only studying cells
- C. Only measuring temperature
- D. Only printing documents

4. Which brain cells are mentioned in the lecture's discussion of awareness?

- A. Red cells and white cells
- B. Place cells and grid cells
- C. Skin cells and bone cells
- D. Muscle cells and blood cells

5. What is developer bias?

- A. A VR headset battery problem
- B. A developer overlooking flaws in their own creation
- C. A type of computer virus
- D. A high frame rate

6. Who should test a VR experience to reduce developer bias?

- A. Only its creator
- B. Only experienced developers
- C. New users, including people who have never tried VR
- D. Nobody

7. What does AR stand for?

- A. Artificial Resolution
- B. Augmented Reality
- C. Automatic Rendering
- D. Advanced Recording

8. Which is the best example of VR rather than ordinary movie watching?

- A. Reading a newspaper
- B. Watching a normal film on a TV
- C. Exploring a virtual world through a VR headset
- D. Listening to a radio

9. Which historical figure is linked to motion studies in 1878 in the lecture?

- A. Palmer Luckey
- B. Eadweard Muybridge
- C. Claude Monet
- D. Hans Vredeman de Vries

10. The film L'arrivée d'un train à La Ciotat is associated with which year in the lecture?

- A. 1878
- B. 1895
- C. 1908
- D. 2012

11. Which animated film is mentioned with the year 1908?

- A. Gravity
- B. Fantasmagorie
- C. South Park
- D. Star Wars: The Clone Wars

12. What did the lecture associate with the year 1995?

- A. The first cave paintings
- B. Nintendo Virtual Boy
- C. The Lumière brothers' film
- D. Palmer Luckey's 2012 work

13. Which year is linked to renewed consumer VR interest through Palmer Luckey?

- A. 1878
- B. 1908
- C. 1995
- D. 2012

14. What does FPS stand for?

- A. Frames Per Second
- B. Files Per System
- C. Frequency Processing Speed
- D. First Person Simulation

15. According to the lecture, how many FPS are enough to perceive motion?

- A. 1 FPS
- B. 2 FPS
- C. 12 FPS
- D. 24 FPS

16. According to the lecture, how many FPS are enough to perceive continuous motion?

- A. 2 FPS
- B. 5 FPS
- C. 12 FPS
- D. 60 FPS

17. What frame rate does the lecture give for a traditional movie theatre?

- A. 12 FPS
- B. 15 FPS
- C. 24 FPS
- D. 120 FPS

18. Which is a factor affecting the appropriate FPS for VR?

- A. Hardware specifications
- B. Distance to the screen
- C. Type of interaction
- D. All of the above

19. Which is a possible cause of discomfort in VR listed in the lecture?

- A. Judder
- B. Flicker
- C. Tracking problems
- D. All of the above

20. Which was a problem for early consumer VR?

- A. Very low cost in all systems
- B. Unlimited content
- C. Heavy and bulky hardware
- D. No latency ever

21. What is the main trade-off in realism versus simplicity?

- A. Realism and simplicity are identical
- B. More realism may require greater resources
- C. Simple graphics always need more power
- D. Realism never matters

22. Which application is shown in the lecture as an example of VR?

- A. Architecture and real estate
- B. Only ordinary telephone calls
- C. Only paper examination
- D. Only spreadsheet calculations

23. What does latency mean in VR?

- A. A delay in system response
- B. The weight of a headset
- C. The number of virtual characters
- D. The colour of a screen

24. What is the best lesson for a VR developer?

- A. Test only on yourself
- B. Ignore motion sickness
- C. Test with different users to identify flaws
- D. Always use the most expensive hardware

Submit answers and see score

11. Last-minute revision sheet

Memorize these points before your NPTEL exam.

Topic	One-line answer
VR definition	Artificial sensory stimulation that can influence an organism's behavior with little or no awareness of the interference
Mainstream VR	Primarily entertainment and virtual experiences in this lecture

Scientific VR	Used to study neural processes and perception
Place cells	Associated with representing locations
Grid cells	Associated with grid-like spatial representation
Developer bias	The creator may overlook flaws in their own system
AR	Adds digital information to the real world
1878	Muybridge's motion studies
1895	Lumière brothers' motion picture
1908	*Fantasmagorie*
1960–1990	First wave of VR
1995	Nintendo Virtual Boy
2012	Palmer Luckey; renewed consumer VR interest
FPS	Frames Per Second
2 FPS	Motion perception, according to the lecture
12 FPS	Continuous motion, according to the lecture
24 FPS	Traditional movie-theatre figure in the lecture
Early VR problems	Cost, resolution, latency, content, weight, and bulk

NPTEL Week1 Definition of VR, history.pdf

Final memory challenge

Before finishing, try saying these answers without looking at the notes:

1. What is VR?
2. What is developer bias?
3. What is the difference between VR and AR?
4. Which years are associated with Muybridge, the Lumière brothers, Fantasmagorie, Nintendo Virtual Boy, and Palmer Luckey?
5. What are 2, 12, and 24 FPS associated with?
6. Why did early consumer VR struggle?
7. How can VR developers avoid bias?

If you can answer these seven questions and score well on the practice test, you will have a strong foundation in the main concepts covered in this Week 1 PDF.

You said:

Quiz 1 - 2026.pdf
PDF

explain them easily

ChatGPT said:

NPTEL VR Quiz 1 — Easy Explanation of All 10 Questions

Your new PDF contains 10 quiz questions. Let's learn each one in very simple English, with examples and memory tricks so you can remember the answers in the exam. Quiz 1 - 2026.pdf

Question 1. What is NOT required in the definition of VR?

Correct answer: C

Complete absence of awareness of interference

Understand it like this: Imagine you wear a VR headset and see a virtual mountain. Your brain reacts to it, even though you know that you are wearing a headset.

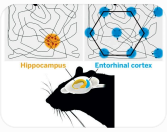
The definition says you have **little or no awareness of the interference**. It does not say that you must be completely unaware.

Remember these four important elements:

- An organism (a living thing).
- Artificial sensory stimulation.
- Targeted behaviour.
- Little or no awareness of the interference.

Memory trick: VR needs *little or no awareness*, not necessarily *zero awareness*.

Question 2. Which brain cells are mentioned in the lecture?



Correct answer: C — Place cells

Easy explanation

Imagine your brain has a map of your school. When you enter your classroom, particular brain cells help represent your location.

These are called **place cells**. The lecture also mentions grid cells, which help represent space in a grid-like pattern.

Memory trick: Place cells = remembering a place.

Exam point: The quiz asks specifically for *place cells*, not grid cells.

Question 3. Why does developer bias happen?

Correct answer: C

The developer acts as both the experiment designer and the experimental subject.

Imagine you create a game. You know where every object is, and you know how the game should work.

Because you created it, you may fail to notice its problems. A new player might notice them immediately.

This is called **developer bias**.

How to avoid it? Test your VR application with people who have never used it before, including people who have never tried VR.

Memory trick: Creator tests their own creation → may miss flaws → developer bias.

Question 4. What is a panoramic VR image?

Correct answer: C

A 2D spherical projection that creates an illusion of immersion.

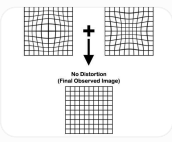
Imagine standing in the middle of a room and looking around at a 360-degree photograph.

You can look left, right, up and down. But the photograph itself is still a **2D image** projected onto the inside of a sphere.

It does not automatically contain true 3D depth or let you move freely through the scene.

Memory trick: Panorama = 2D picture wrapped around you.

Question 5. Why are images pre-distorted before displaying them in a VR headset?



Correct answer: B

To compensate for lens-induced optical distortion.

A VR headset has lenses between your eyes and the screen. These lenses can make straight lines appear curved or distorted.

To solve this problem, the computer deliberately distorts the image before displaying it. After the image passes through the lenses, it appears more correct to your eyes.

Think of it like this:

Computer distorts image → Lens corrects the distortion → You see the proper image.

Memory trick: Pre-distortion is used to correct lens distortion.

Question 6. How many FPS are needed for continuous apparent motion?

Correct answer: D — 12 FPS

Remember 2 → 12 → 24

2	12	24
Motion is visible	Continuous apparent motion	Cinema frame rate

FPS means **Frames Per Second**.

Imagine a flipbook. Each page has a slightly different drawing. When you flip the pages, the drawings seem to move.

- **2 FPS:** You can perceive movement, but individual frames remain visible.
- **12 FPS:** The lecture says continuous apparent motion begins.
- **24 FPS:** The lecture identifies this as the traditional cinema frame rate, mainly chosen to reduce flicker.

Memory trick: 2 = motion, 12 = continuous motion, 24 = cinema.

For this particular quiz, choose **12 FPS**.

Question 7. Which was the first functional VR headset?

Correct answer: D

Ivan Sutherland's Sword of Damocles (1968)

Imagine wearing a device on your head that changes the displayed view when you turn your head.

This is important because the virtual view responds to your movement. The Sword of Damocles introduced head tracking and is widely regarded in the quiz as the first functional VR headset.

Memory trick: Sword of Damocles = 1968 = head tracking.

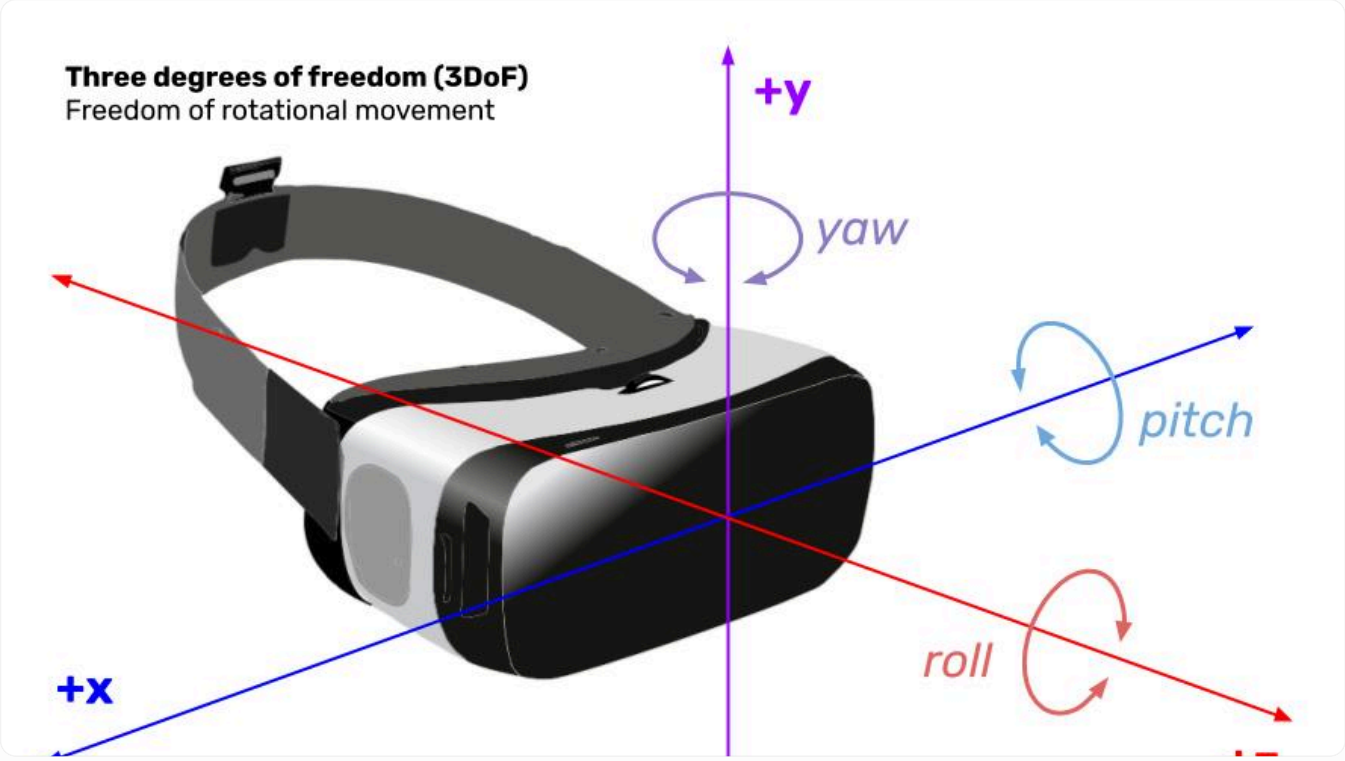
Do not confuse it with the Nintendo Virtual Boy (1995) or Oculus Quest.

Question 8. How many degrees of freedom does smartphone-based VR Cardboard track?

Correct answer: B — 3 DOF

DOF means Degrees of Freedom.

Imagine your phone is inside a simple VR viewer. You can turn your head in three rotational directions.



The three rotations are:

- **Pitch:** Look up and down.
- **Yaw:** Turn left and right.
- **Roll:** Tilt your head sideways.

A standard Cardboard setup tracks these rotations using the phone's motion sensors. It does not track your actual position as you walk around.

Memory trick: Cardboard = 3 rotations = 3 DOF.

For comparison, a 6-DOF system tracks three rotations plus three positional movements: moving left/right, up/down and forward/backward.

Question 9. Are realistic graphics essential for immersion?

Correct answer: False

Realistic graphics are not essential for immersion.

Imagine playing a cartoon game. The characters look simple, but you become so involved that you feel like you are inside the game.

That is immersion: feeling involved or present in the experience.

VR can use realistic graphics, cartoons or abstract environments. Realism alone does not determine whether an experience feels immersive.

Memory trick: Immersion does not mean photorealism.

Question 10. Are the traditional five human senses enough to understand VR?

Correct answer: False

VR must consider more than the traditional five senses.

We commonly learn about five senses:

1. Sight
2. Hearing
3. Smell
4. Taste

5. Touch

But VR also needs to consider other ways the body senses the world.

Sensory system	Easy example
Vestibular system	Knowing that your head is moving or your body is accelerating
Proprioception	Knowing where your arms and legs are without looking
Pressure sensing	Feeling something press against your skin
Temperature sensing	Feeling hot or cold
Pain receptors	Detecting potentially harmful stimuli

For example, in a VR game, your eyes may show that you are flying, but your inner ear may tell you that you are standing still. This mismatch can contribute to discomfort.

Memory trick: Human perception is much richer than five senses.

Quick revision test

Answer all 10, then check your score.

1. Which is NOT required by the VR definition?

- A. Artificial stimulation
- B. Targeted behaviour
- C. Complete absence of awareness
- D. An organism

2. Which cells are the answer to Question 2?

- A. Place cells
- B. Rod cells
- C. Mirror neurons
- D. Photoreceptors

3. Developer bias happens because the developer…

- A. Never uses a computer
- B. Tests their own creation and may miss flaws
- C. Always has low resolution
- D. Uses too many sensors

4. A panorama is fundamentally…

- A. A complete 3D model
- B. A 2D image projected onto a sphere
- C. A depth camera
- D. A moving robot

5. Why pre-distort VR images?

- A. To correct lens distortion
- B. To increase sound volume

C. To track walking

D. To remove audio

6. Continuous apparent motion begins at how many FPS according to the quiz?

A. 2

B. 5

C. 12

D. 60

7. Who created the Sword of Damocles?

A. Palmer Luckey

B. Ivan Sutherland

C. Muybridge

D. Claude Monet

8. Standard Cardboard VR tracks how many DOF?

A. 2

B. 3

C. 6

D. 10

9. Realistic graphics are essential for immersion. True or False?

A. True

B. False

10. The five traditional senses are sufficient for understanding VR. True or False?

A. True

B. False

Check answers

My advice for your NPTEL exam: First memorize the definitions and numbers—**place cells, developer bias, panorama = 2D spherical projection, pre-distortion, 12 FPS, Sword of Damocles = 1968, Cardboard = 3 DOF**. Then practise the questions until you can answer them without looking at the notes.

You said:

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[Music] So um so in this video right you saw that all everything was almost realistic but there was still a lot of mismatch. There are other mismatches that we'll learn about uh in VR but also the mismatch between your propriception and what you see your body is where it is in space. So some researchers actually think that after using VR we might change the structure of our brains especially related to this propriception. So some say we'll improve propriception and they're quite optimistic like it's going to be a positive change. we can increase reaction rates or improve memory or something. Um, but it's not 100% right. So, so let's be optimistic. But please watch this video if you want to learn more about propreception related um research. So that's a big question. Will VR improve propriception or change

it? I want to say damage it but you know. Okay. So let's move on. Um now what let's talk about other sensory conflicts. So modern virtual reality devices such as Oculus Quest and even more so the cardboards that we have in class they interfere with some senses but not the others. Right? We completely ignore other senses. Uh so and this causes mismatch between multiple senses and this contradictory information needs to be analyzed by the brain and the brain may react in a variety of ways. So it could be the case that you will not be consciously aware of the conflict but then you may feel fatigue right or develop a headache or even worse have dizziness and c and nausea right or like really or feel tired for for days after the experience. So that's not good. In other cases the brain might react by making us so consciously aware of the conflict that we immediately understand that the experience is not real artificial. So this would result in loss of immers immersiveness in uh immersion in VR experiences. So uh to therefore to make a comfortable and effective VR experience, how would we do that? You need to do trials with human subjects and it's essential to understanding how the brains will react across population. So particularly if you develop your VR experiences for the class projects. So always ask other people to try it before releasing it, especially people who didn't do VR at all. So you really don't want the last thing you want to do is to make people sick with your uh final projects, right? So one of the most important examples of bad sensory conflict in in VR is vection. So vection is the illusion of self motion. And this conflict arises when your vision sense reports that uh you're moving, you're accelerating, but your balance sense say no I'm motionless. So that's happens when you move your avatar in VR with a controller. So your body remains still, but visually you feel like you're accelerating. So how to eliminate the nausea? That is often related or vaction often leads to nausea. So how to eliminate this uh nausea is an ongoing research in VR. So here's an example of vection not in VR but in real world. Uh so vection is an illusion of motion. These people feel like they're moving but they're not. So this bench is stationary. It's the walls that are moving. So this is a very unusual scenario, right? There are some amusement parts like that. So we're not used to seeing room spinning around us, right? So this is uh then the people these people feel illusion of self motion and many of them get sick because of that. So because of this large scale motion happens. Okay. So calibration. Yeah, I saw request calibration. Let's see if they Oh, yeah, they replaced it. This would be cool. How is paral error corrected in Cardboard VR? Oh, there is no parallax in Cardboard VR uh for panoramas, but there is parallax with rotation only. tracking in your cardboard experiences. So, it's very important to uh differentiate between panoramas and unity based or 3D model experiences in in cardboards. They're quite different. There's always three degrees of freedom tracking, but there is parallels in one and no parallels in another. I think we'll have a lecture to just look into it in more details. But now in homework 2, you can you can experience them differently. So, if there's panoramas, then you're right. There is no parallax. But um like shadows of truth see if when you rotate there is there's some at least there's binocular disparities right so there is some kind of differences between due to the distance between the eyes view but yeah there's uh only paral due to that we'll we'll talk maybe in more details the math related to that so they they're different experiences in VR but um that's an open question So consciousness I don't know conscious nobody knows the answer unity is free for students okay so maybe need ideas to implement locomotions in cardboard very good question so if you want to do that jump into homework 4 we give you a good idea I think okay so let me go back to our slides another conflict that is not between senses which is not uncomfortable But really, really interesting. Have a look at this. What do you hear? Ba ba ba ba ba? But look what happens when we change the picture. Ba ba ba ba ba ba. And yet the sound hasn't changed. In every clip, you are only ever hearing bar with a B. It's an illusion known as the Mgherk effect. Take another look. Concentrate first on the right of the screen. Now to the left of the screen. The illusion occurs because what you are seeing clashes with what you are hearing in the illusion um what we see overrides the mgherk effect that's the name of this illusion it's not uncomfortable so the brain is totally fine with this contradiction between audio and visual information but there's a conflict and it will have to pick vision because we are largely visual animals we have more neurons devoted to that so we trust vision more than hearing. Okay. So, another very bad problem is sensory adaptation. It's it's not a problem. It's great for us. We adapt to all of these uh unnecessary sensory stimulus that the brain will just ignore after a while. So, this rel this comes to this is true about different senses like loudness. We adapt to that. after a while we live in the loud city we stop noticing the noise right or in the airplane you don't notice after a couple of minutes uh in case of vision I don't show vision but we adapt to darkness we adapt to lightness it takes up to like sometimes seconds to minutes smells temperature we adapt to those right uh the touch sense of touch so this this clothes and uh jewelry that we put on in the morning we don't notice them right the brain after a while it just ignores the sensory stimulation. So maybe in the end of the day uh you get tired, you feel tired and you want to take them off because the brain was doing a lot of work ignoring it. Nevertheless, that's what's called adaptation. So I want to ask you if we have this adaptation related to many senses and many stimuli um what would be one example that is relevant to v virtual reality? Adaptation to what? Can you guess? Yeah, roller coasters, right? Adaptation to low resolution. Yeah, after a while probably like, "Oh, this is the greatest thing." A lot of things, right? So, very good. The weight of the headset. Yeah, I guess you It's really heavy though. I cannot adapt to that to be honest, but maybe if you're really engaged in some environments, online classes. Yeah. So what so um what does it mean for VR? Yeah, those are good examples. But I think I was asking I was kind of hoping to get vection. Vection is an adaptation. So let's let's talk about it. So after long periods of time, the sensory mismatches that we have in virtual reality, including what leads to vections, illusion of motion. So the sensory mismatches

may lead to adaptation as well. Uh so think about it like astronauts they train themselves to through this regular exposure to get adapted to high G forces. Similarly you guys those who spend hours or days playing video games or those people who do car racing. So they develop adaptation to vetin and vetin is the biggest what theory is the biggest cause of uh nausea and discomfort. So then people who are adapted through video games they experience less fction in VR too. So this becomes really critical for VR developers because you're constantly exposed to this VR while developing the environments. But and if you even test your um your experiences on your game of friends, they mo may also report that your experiences are comfortable. But you really really need to test your experiences on people who are newcomers to virtual reality, but also maybe newcomers to video games and or racing cars. So um however, it's not so bad. Adaptation is not so bad. Even though you're adapted uh you can through series of exercises you may also in VR develop perceptual training. So then you can detect all these things like latency can be detected through specific exercises also. Um curvature or like distortion parameters they can you can improve your perception of distortion and improve your ability to set up parameters by specific exercises and exposure. So that's all good. I have a question for you and I have to I think I have to finish because we have still 15 slides. I think that would be too long to take take um but we'll we'll we'll see this. How long would it take to adapt to the upside world? Uh so we'll we'll start next time with this. Imagine you have upside down goggles and everything seems upside down. Would you be able to adapt to that? Would it be comfortable? Maybe not at first, but if if you can adapt, how long? So, the correct answer is at least on one experiment that I was going to show is 7 days. And we'll see that experiment starting next time. [Music] lecture pdf and lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

Worked for 30s

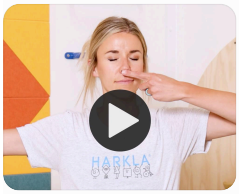
ChatGPT said:

NPTEL Introduction to Virtual Reality — Week 2

Easy notes, memory tricks and MCQs

Your lecture is mainly about **how our brain understands the world and why VR sometimes tricks us or makes us feel uncomfortable**. I'll explain it like a 7th-class lesson, using everyday examples and simple words. These notes follow your uploaded Week 2 slides and the lecturer's explanation. NPTEL WEEK 2 Bird's Eye View, Software, Perception.pdf NPTEL WEEK 2 Bird's Eye View, Software, Perception.pdf

Part 1: Proprioception — How do you know where your body is?



Imagine closing your eyes and touching your nose with your finger.

How did you know where your hand and nose were without looking?

Your body has a sense called **proprioception**.

Simple definition: Proprioception is the sense that tells you where your body parts are and how they are moving, even when you cannot see them.

Examples:

- You can touch your nose with your eyes closed.
- You can walk without watching every movement of your feet.
- You know whether your arm is raised or lowered.

Your muscles, tendons and joints send information to your brain about body position and movement.

Why is proprioception important in VR?

Suppose you wear a VR headset and see a virtual hand moving. But your real hand does not move in the same way.

Your eyes tell you one thing, while your body's position sense tells you something else. This mismatch can confuse your brain.

The lecturer raises an interesting question: can repeated VR use improve or change proprioception? Research is ongoing, so we should not assume that VR always improves it or damages it.

Memory trick: Proprioception = *position of your own body.*

Part 2: Sensory conflict — When your senses disagree

Imagine you are sitting on a chair wearing a VR headset.

- Your eyes see a car racing forward.
- Your inner ear says your body is sitting still.
- Your body does not feel the acceleration shown in the virtual world.

Your senses are giving your brain different messages. This is called **sensory conflict**.

What can sensory conflict cause?

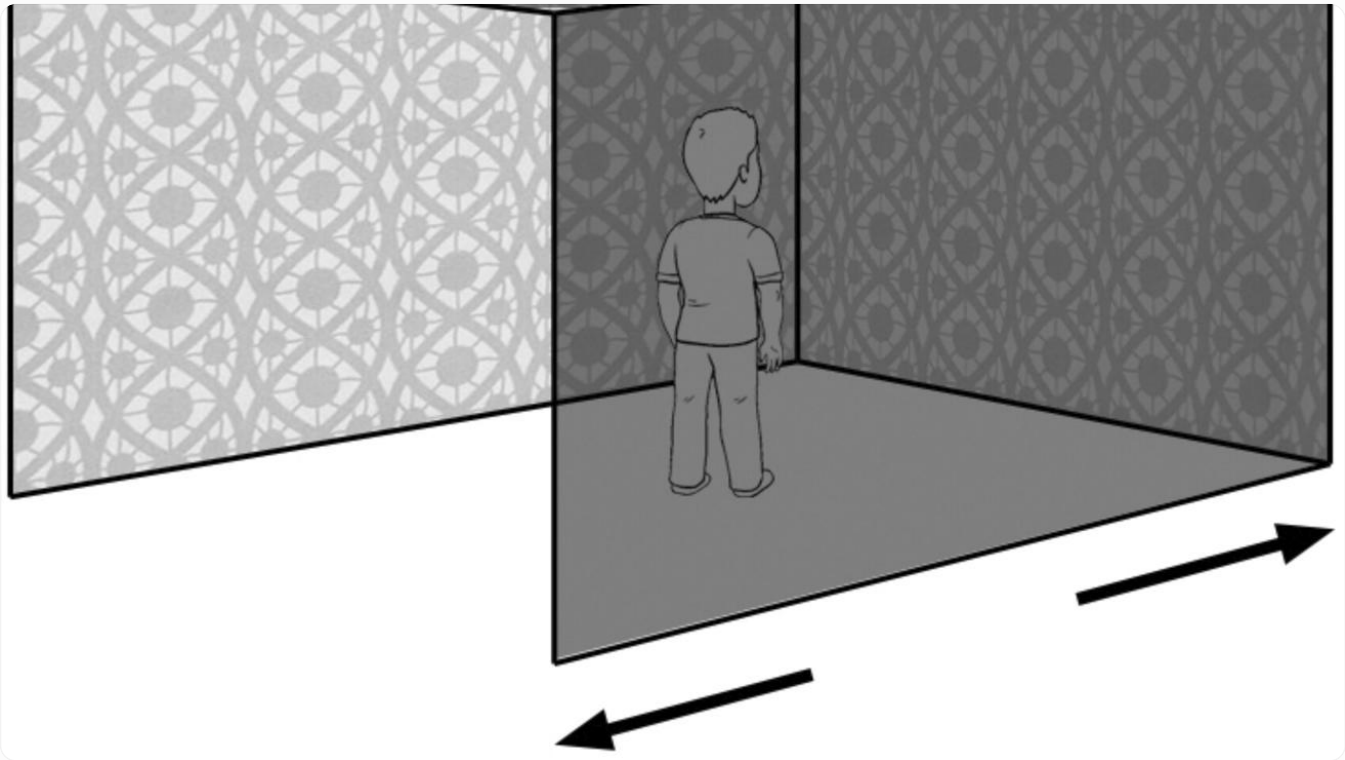
- Fatigue — feeling tired
- Headache
- Dizziness and nausea — feeling sick
- Loss of immersion — realizing the virtual experience does not feel real

The brain tries to understand the conflicting information. Sometimes you feel physically uncomfortable; sometimes you simply notice that the virtual world is artificial.

Memory trick: Sensory conflict = senses disagree.

NPTEL point: VR developers must test their applications on different people, especially beginners, because a developer who is used to the experience may not notice its problems.

Part 3: Vection — Feeling movement when you are not moving



This is one of the most important terms in your lecture.

Vection is the illusion of self-motion: you feel that you are moving even though your body is stationary.

Imagine sitting in a stationary train. The train beside you begins moving, and for a moment you feel as if your own train is moving in the opposite direction.

Your eyes see movement, but your body is not actually moving.

In VR, the same thing can happen when you move your virtual character using a controller while your real body remains still.

Why can vection make you feel sick?

1. Your eyes report movement.
2. Your balance system reports little or no corresponding movement.
3. Your brain receives conflicting signals.
4. You may feel dizzy or nauseated.

Researchers continue to investigate ways to reduce VR-related nausea.

Memory trick: Vection = *virtual movement felt as real movement*.

Do not confuse these terms:

Term	Simple meaning
Proprioception	Knowing where your body parts are
Sensory conflict	Two or more senses disagree
Vection	Feeling that you are moving when you are not

Part 4: Parallax — How do we understand depth?

Depth perception means understanding **how far away an object is** and seeing the world as three-dimensional.

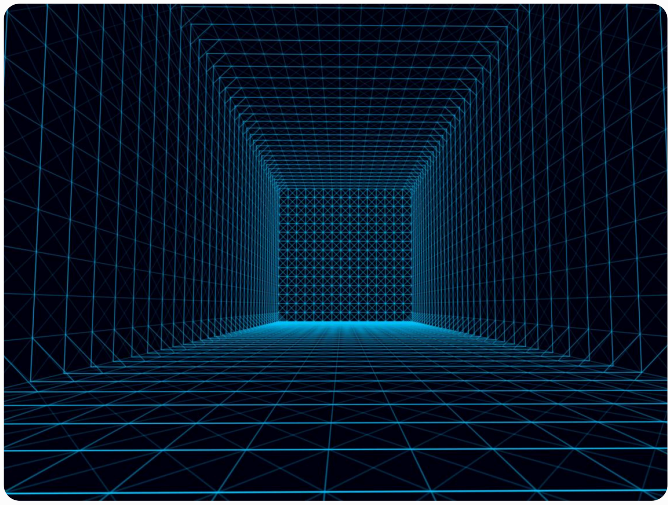
Imagine looking out of a moving bus:

- Nearby trees seem to move quickly across your view.
- Faraway mountains seem to move slowly.

This difference in apparent movement helps your brain estimate distance. It is called **motion parallax**.

Panoramas versus 3D VR

360° panorama
A 2D picture wrapped around a virtual sphere. You can look around, but you cannot move freely through the original scene with correct depth changes.



3D virtual environment
A computer-generated 3D scene can update objects' positions and sizes as you move around.

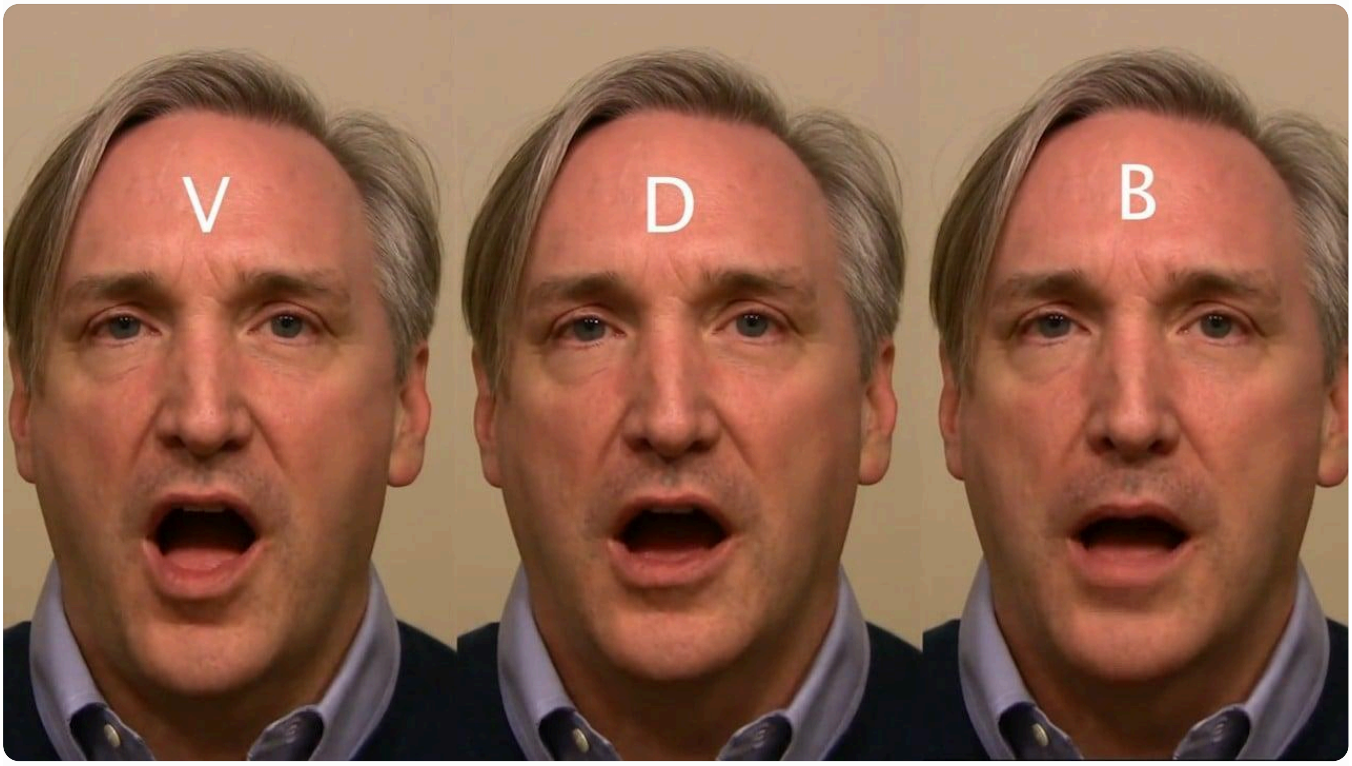
Your lecturer explains that Google Cardboard can provide different kinds of experiences.

- **Panorama:** Usually provides rotational viewing, but no genuine motion parallax from moving your head sideways.
- **3D scene:** Can provide motion parallax when the software updates the view based on your position.

Standard smartphone Cardboard typically tracks three rotational movements: pitch, yaw and roll. That is called **3 degrees of freedom (3-DOF)**. It does not normally track your actual position in space.

Memory trick: Parallax = different apparent movement because of distance.

Part 5: The McGurk effect — When your eyes change what you hear



Imagine hearing someone say “ba” while watching their mouth move as if they are saying a different sound.

You may perceive a third sound, even though the audio has not changed.

This is called the **McGurk effect**.

It happens because your brain combines information from your eyes and ears when understanding speech. What you see can influence what you think you hear.

Remember the difference

- **Vection:** Your senses disagree about movement.
- **McGurk effect:** What you see changes what you perceive hearing.

The McGurk effect is a useful example of how the brain combines different senses. It is not necessarily uncomfortable.

Memory trick: McGurk = mouth movement affects hearing.

Part 6: Sensory adaptation — Your brain gets used to things

Have you ever entered a room that smelled strongly of perfume? After some time, you may barely notice the smell.

That is **sensory adaptation**.

Your brain becomes less responsive to a constant or repeated stimulus, so you notice it less.

Other examples:

- You stop noticing the sound of a fan.
- You become used to the feeling of wearing a watch.
- You get used to a room's temperature.
- You adjust when entering a dark room.

How does adaptation affect VR?

A developer may spend hours testing their own VR game. Over time, they may get used to the headset's weight, visual distortions or motion mismatch.

Their friends who play games regularly may also be used to similar experiences.

But a beginner might feel dizzy or sick immediately.

Therefore, **VR developers should test their applications on people who are new to VR**, rather than relying only on themselves or experienced gamers.

The lecture also mentions that perceptual training can help people learn to detect certain problems, such as latency or image distortion.

Memory trick: Adaptation = repeated experience becomes less noticeable.

Important: Adaptation does not mean that every discomfort disappears or that VR becomes safe for everyone.

Part 7: The upside-down goggles experiment

Imagine wearing special glasses that turn everything upside down.

At first, the world looks strange. Walking and reaching for objects may become difficult because what you see does not match what your brain expects.

With repeated experience, the brain can adapt to the changed visual information.

Your lecturer mentions an experiment in which adaptation took **at least seven days**.

Remember that this is the result described for that particular experiment, not a universal rule that everybody adapts in exactly seven days.

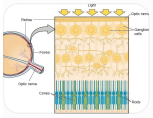
Memory trick: Upside-down goggles → mismatch first → adaptation with practice.

Part 8: Human senses and their receptors

Your brain does not directly receive light, sound or touch as they exist outside your body. Your sense organs convert these forms of energy into electrical signals that travel through nerves to the brain.

This conversion is called **sensory transduction**.

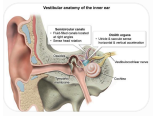
Important receptors



1. Photoreceptors

Sense organ: Eyes

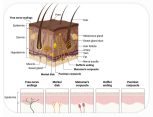
Detect light. The two main types are rods and cones.



2. Mechanoreceptors

Examples: Hearing, touch and balance

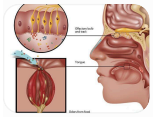
Respond to mechanical forces, movement or vibration. The inner ear also contains structures involved in balance.



3. Thermal receptors

Sense organ: Skin

Help detect changes in temperature, such as hot and cold.



4. Chemoreceptors

Examples: Smell and taste

Respond to chemical substances.

The balance system

The vestibular system is inside your inner ear. It helps your brain understand head movement, orientation and balance.

It works somewhat like sensors used in smartphones and VR headsets.

The five traditional senses are not enough

VR also needs to consider:

- Proprioception — body position.
- Vestibular sense — balance and head movement.

- Temperature.
- Pressure and other touch sensations.
- Pain.

Memory trick for receptors:

- Photo = light.
- Mechano = force or movement.
- Thermal = temperature.
- Chemo = chemicals.

Part 9: How does your brain understand what you see?

Imagine looking at a dog. Your brain does not understand the entire scene in one simple step. Visual information is processed through different stages.

Step 1: Light enters the eye
The retina receives light.

↓

Step 2: Basic visual features
Lines, edges, corners and simple patterns.

↓

Step 3: Shapes and objects
The brain combines simpler features.

↓

Step 4: Recognizing things
Faces, people, places and familiar objects.

↓

Step 5: Meaningful experience
Visual information combines with memory, attention and other senses.

This is called **hierarchical visual processing**. In simple words, the brain builds a bigger understanding from smaller pieces of information.

Your lecture gives these approximate figures:

Brain fact from the lecture	Number
Neurons in the human body	86 billion
Neurons in the cerebral cortex	21 billion
Cerebral cortex devoted to vision	30%
Cerebral cortex devoted to touch	8%
Cerebral cortex devoted to hearing	3%

The cerebral cortex is the outer layer of the brain involved in many complex functions, including perception, memory and language. These figures are the lecture's estimates, rather than exact measurements for every individual. NPTEL WEEK 2 Bird's Eye View, Software, Perception.pdf

Memory trick: 86 → 21 → 30, 8, 3.

Do not conclude that an animal with more neurons must be more intelligent. The lecturer explains that intelligence cannot be determined simply by counting neurons.

Part 10: Optical illusions — Why can the brain make mistakes?

An optical illusion happens when what you perceive differs from what is physically present.

Your brain uses clues from past experience to interpret images. Sometimes these clues lead to an incorrect conclusion.

A. Interposition (overlap)

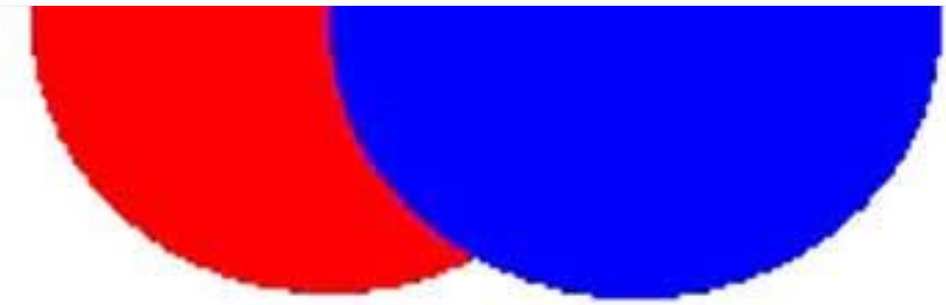


Figure 2. Interposition. The blue circle is reported to be closer since it overlaps the red circle.

Imagine one circle partly covering another circle.

Your brain usually concludes that the circle covering the other one is **in front**.

This depth cue is called **interposition**, also known as overlap.

The lecturer explains that T-shaped boundaries between objects can help the brain determine which object is in front.

Memory trick: If A covers B, the brain usually sees A in front of B.

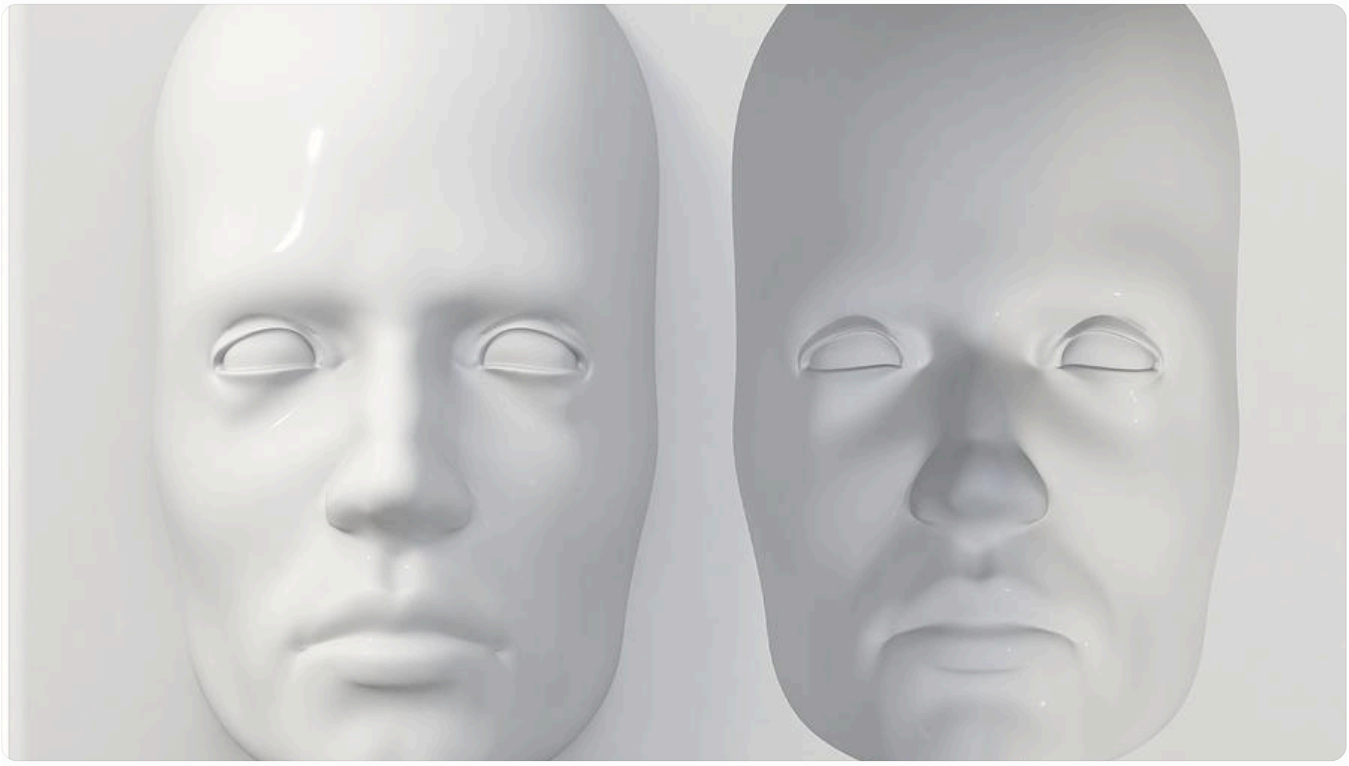
B. The spinning dancer illusion

This illusion can make the same silhouette appear to rotate clockwise or anticlockwise.

Why? The image does not provide enough clear depth information to determine the direction uniquely, so your brain can interpret it in different ways.

Remember: Ambiguous visual information can produce different perceptions.

C. The hollow-face illusion and convexity bias



A hollow mask curves inward, but your brain may perceive it as a normal face that curves outward.

Why does this happen?

We usually see real faces from the outside, where the nose and other features project outward. The brain has learned to expect faces to be convex.

When depth clues are unclear, this expectation can dominate what you see. This is called a **convexity bias** in the lecture.

Memory trick: Convex = bulges outward; concave = curves inward.

D. The invisible gorilla — Attention

Imagine you are told to count how many times players in white shirts pass a basketball. You concentrate so hard on counting that you may fail to notice a person in a gorilla costume walking through the scene.

This is an example of **inattentional blindness**: failing to notice something visible because your attention is focused elsewhere.

It teaches us that seeing something and consciously noticing it are not always the same.

Memory trick: Attention on one task → something else may go unnoticed.

Four illusion concepts to memorize

Concept	What it teaches
Interposition	Overlap gives a clue about depth
Spinning dancer	Ambiguous depth can change perceived rotation
Convexity bias	The brain expects familiar shapes, such as faces, to bulge outward
Inattentional blindness	Focused attention can cause us to miss visible events

These concepts matter in VR because developers must understand how the brain interprets incomplete or misleading visual information.

Part 11: NPTEL-style MCQs

These 20 practice questions cover the main concepts from your Week 2 lecture. Try answering them without looking at the notes first.

Week 2 practice quiz

0/20 answered

Select one option per question, then submit to see your score and explanations.

1. What is proprioception?

- A. Sense of smell
- B. Sense of body position and movement
- C. Sense of colour
- D. Sense of taste

2. What is sensory conflict?

- A. When senses give contradictory information
- B. When all senses stop working
- C. When a computer shuts down
- D. When the room becomes dark

3. What is vection?

- A. Feeling hot
- B. Feeling self-motion when stationary
- C. Recognizing a face
- D. Hearing a sound

4. Which is a possible effect of sensory conflict in VR?

- A. Nausea
- B. Headache
- C. Fatigue
- D. All of the above

5. What is motion parallax?

- A. Nearby and distant objects appear to move differently when you move
- B. A type of sound
- C. A smell illusion
- D. A headset battery feature

6. A panorama in VR is fundamentally what?

- A. A complete 3D model
- B. A 2D image projected onto a sphere
- C. A physical room
- D. A motion sensor

7. How many rotational degrees of freedom does standard Cardboard VR track?

- A. 1
- B. 2
- C. 3
- D. 6

8. What is the McGurk effect?

- A. Visual speech influences what sound you perceive
- B. A headset tracks walking
- C. A person adapts to darkness
- D. A type of depth cue

9. What is sensory adaptation?

- A. Becoming less responsive to a constant stimulus
- B. Seeing a new colour
- C. Losing all senses
- D. Increasing headset resolution

10. Why should developers test VR with beginners?

- A. Beginners always have better computers
- B. Experienced users may have adapted to flaws
- C. Beginners cannot see images
- D. Testing is unnecessary

11. Which experiment is mentioned in the lecture as taking at least seven days for adaptation?

- A. Upside-down goggles
- B. A smell test
- C. A sound test
- D. A colour-blindness test

12. Which receptors detect light?

- A. Chemoreceptors
- B. Photoreceptors
- C. Thermal receptors
- D. Mechanoreceptors

13. Which receptors respond to mechanical forces or vibrations?

- A. Photoreceptors
- B. Chemoreceptors
- C. Mechanoreceptors
- D. Only thermal receptors

14. Which receptors are associated with chemical substances in taste and smell?

- A. Chemoreceptors
- B. Photoreceptors
- C. Mechanoreceptors
- D. Thermal receptors

15. Which system helps detect head movement and balance?

- A. Vestibular system
- B. Taste system
- C. Only the visual system
- D. Digestive system

16. What is hierarchical visual processing?

- A. Building complex recognition from simpler visual features
- B. Hearing a sound without seeing anything
- C. Turning a picture into a smell
- D. Measuring body temperature

17. What does interposition tell the brain?

- A. Which object is likely in front
- B. How loud a sound is
- C. How warm an object is
- D. How fast a computer runs

18. What is convexity bias in the hollow-face illusion?

- A. The brain tends to interpret a face as bulging outward
- B. The brain cannot see faces
- C. The brain detects sound incorrectly
- D. The eyes stop moving

19. What is inattentional blindness?

- A. Missing a visible event because attention is elsewhere
- B. Being unable to hear
- C. A loss of body position sense
- D. An inability to recognize temperature

20. Which statement about animal intelligence is supported by the lecture?

- A. More neurons always mean more intelligence
- B. Neuron counts alone do not determine intelligence
- C. Animals have no neurons
- D. Only humans can perceive the world

Submit and check answers

Final revision: Learn these 12 facts first

1. **Proprioception** — knowing your body's position.
2. **Sensory conflict** — senses disagree.
3. **Vection** — feeling movement while stationary.
4. **Motion parallax** — apparent movement differs with distance.
5. **McGurk effect** — visual speech changes perceived sound.
6. **Sensory adaptation** — constant stimuli become less noticeable.

7. **Photoreceptors** — detect light.
8. **Mechanoreceptors** — detect mechanical forces.
9. **Chemoreceptors** — detect chemicals.
10. **Interposition** — overlap helps identify what is in front.
11. **Convexity bias** — the brain tends to interpret faces as bulging outward.
12. **Inattentional blindness** — missing something visible because attention is elsewhere.

My study suggestion: Read these 12 facts twice, then attempt the quiz. After that, revise the examples of vection, the McGurk effect and the hollow-face illusion, because these are especially useful for conceptual NPTEL questions.

You said:

NPTEL Quiz 2 - 20...
PDF

explain easily

ChatGPT said:

NPTEL Introduction to Virtual Reality – Quiz 2

I'll explain all **10 questions in very simple English**, like you're learning them for the first time. I'll also give you easy memory tricks to remember the correct answers for your exam.

Question 1: How does a visual VR system work?

Correct answer: C

Think of wearing a VR headset and moving your head. The headset detects your movement, the computer updates the virtual scene, and your eyes see the result.

1. Tracking hardware
Detects head movement using sensors and cameras

↓

2. Virtual World Generator
Computer creates and updates the virtual scene

↓

3. Display
Shows the virtual scene

↓

4. Sense organ → Brain
Eyes receive the image; the brain interprets it

Memory trick: Track → Create → Display → See → Brain.

Question 2: Why does an HMD need more tracking than a CAVE?

Correct answer: C

- **HMD** (Head-Mounted Display): a headset worn on your head. It moves whenever your head moves.
- **CAVE:** a room-like VR environment with displays around a user. The displays stay in place.

An HMD must track where your head is, which direction it faces, and how your body moves when needed. A CAVE already surrounds you with stationary displays.

Memory trick: HMD moves with your head; CAVE stays in the room.

Question 3: What does a gyroscope do?

Correct answer: B

A gyroscope is a sensor inside an IMU (Inertial Measurement Unit).

Imagine turning your head left or right. The gyroscope measures **how quickly your head rotates**.

- Angular velocity = speed of rotation.
- Orientation = the direction your head is facing.
- The system estimates orientation by combining rotation measurements over time.
- Small measurement errors can accumulate, causing **drift**. Other sensors can help correct it.

Memory trick: Gyroscope = **rotation**, not position.

Question 4: What happens in a 3-DOF headset when you lean forward?

Correct answer: C

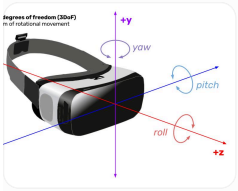
First, understand DOF.

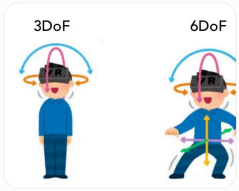
DOF = Degrees of Freedom, meaning the ways a person or object can move.

A 3-DOF headset tracks only three kinds of rotation:

- **Pitch:** looking up and down.
- **Yaw:** looking left and right.
- **Roll:** tilting your head sideways.

It does **not** track your position moving forward, backward, left, right, up, or down.





For example, if you physically lean forward to look under a virtual table, a 3-DOF headset cannot correctly update your viewpoint based on that movement. The virtual world may appear to move along with your head.

Memory trick:

- 3-DOF = rotation only.
- 6-DOF = rotation + position.

Question 5: Which is NOT a binocular depth cue?

Correct answer: C — Interposition

Depth cues help your brain understand which objects are near and which are far.

There are two important types:

Type	Meaning	Example
Binocular cue	Uses information from both eyes	Convergence, binocular disparity
Monocular cue	Can work using one eye	Interposition (overlap)

What is interposition?

Imagine a tree covering part of a house. Your brain understands that the tree is in front of the house.

This is called **interposition or occlusion**. It is a monocular depth cue because you can recognize it even with one eye closed.

Memory trick: Interposition = one object hides another.

Question 6: How many DOFs determine the position and orientation of both ears?

Correct answer: C — 6

Your ears are attached to your head. If a VR system knows your head's position and orientation, it can determine where your ears are and how they are oriented.

Six DOFs consist of:

3 positional movements		
Left / right	Up / down	Forward / backward
3 rotational movements		
Pitch	Yaw	Roll

Memory trick: 3 movements + 3 rotations = **6 DOF**.

Question 7: Does the McGurk Effect show that hearing always dominates vision?

Correct answer: False

The McGurk Effect shows that **what you see can change what you think you hear**.

Imagine:

- A person says one sound, such as “ba.”
- But their mouth appears to make a different sound.
- Your brain combines the visual and auditory information.
- You may perceive a different sound from the one actually played.

This demonstrates that vision does not always lose to hearing.

Memory trick: McGurk = eyes can influence ears.

Question 8: What percentage of the cerebral cortex is devoted to vision?

Correct answer: D — 30%

According to your lecture:

Sense	Approximate percentage stated
Vision	**30%**
Touch	8%
Hearing	3%

These are the figures to remember for this quiz.

Why is this important in VR? Because vision plays a major role in creating the feeling that we are inside a virtual world.

Memory trick: Remember **30–8–3**: Vision, Touch, Hearing.

Question 9: What is vection?

Correct answer: B

Vection is the illusion that you are moving when your body is actually stationary.

Imagine sitting on a bench while the walls around you appear to move. Your brain may feel as if you are moving even though you are sitting still.

This happens because:

- Your eyes suggest that you are moving.
- Your vestibular system (the balance system in your inner ear) says your body is stationary.
- The conflicting signals can cause dizziness, nausea, or VR sickness.

Memory trick: Vection = visual motion creates a feeling of self-motion.

Do not confuse it with actual physical acceleration.

Question 10: How many DOFs describe gaze and viewpoint in an eye-tracked HMD?

Correct answer: B — More than 9 DOFs

Your lecture counts these components:

Component	DOFs
Head translation (position)	3
Head rotation	3
Eye rotation	3
Accommodation	1
Total	**10 DOFs**

What is accommodation?

Accommodation is the eye changing the shape of its lens to focus on objects at different distances.

Calculation:

$$3 + 3 + 3 + 1 = \boxed{10}$$

Ten is more than nine, so the answer is **B**.

Memory trick: Head position 3 + head rotation 3 + eye rotation 3 + focus 1 = **10**.

Quick revision sheet

Q.	Topic	Answer to remember
1	VR information flow	Tracking → Virtual World Generator → Display → Sense organ → Brain
2	HMD vs CAVE	HMD moves with the head
3	Gyroscope	Measures angular velocity
4	3-DOF headset	Tracks rotation, not position
5	Monocular depth cue	Interposition
6	Head pose / ears	6 DOFs
7	McGurk Effect	False — vision can influence hearing
8	Cortex devoted to vision	30% according to lecture
9	Vection	Illusion of self-motion
10	Eye-tracked HMD	10 DOFs, so more than 9

Test yourself

10-question practice quiz

0/10 answered

1. Which component detects your head movement in VR?

- A. Display
- B. Tracking hardware
- C. Brain
- D. Virtual table

2. Which movement can a 3-DOF headset track?

A. Leaning forward

B. Walking left

C. Turning your head

D. Moving upward

3. What does a gyroscope measure?

A. Absolute position

B. Angular velocity

C. Temperature

D. Sound

4. A tree hides part of a house. Which depth cue is this?

A. Convergence

B. Accommodation

C. Interposition

D. Vection

5. How many DOFs make up full head pose?

A. 2

B. 3

C. 6

D. 10

6. What does the McGurk Effect demonstrate?

A. Vision can influence hearing

B. Hearing always dominates vision

C. VR cannot affect perception

D. Balance controls speech

7. According to the lecture, what percentage is devoted to visual processing?

A. 3%

B. 8%

C. 30%

D. 50%

8. You feel you are moving while sitting still. What is this?

A. Drift

B. Accommodation

C. Interposition

D. Vection

9. Which system helps your body sense balance?

A. Vestibular system

B. Display system

C. Rendering engine

D. Gyroscope alone

10. What is the total in the lecture's eye-tracked HMD example?

A. 6 DOFs

B. 8 DOFs

C. 9 DOFs

D. 10 DOFs

Check answers

My suggestion: First memorize the **quick revision sheet**, then attempt the quiz without looking at the answers. That is a good way to prepare for the NPTEL exam.

You said:

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[Music] I'm unable to answer any of the matrix questions. What should I do? What about after this lecture? And we'll have um we'll have lots of quizzes uh for this material if you keep H if it keeps happening that you can't then it's a problem but if after these lectures listening to them again online maybe if you can do things then you'll be fine and you'll be able to do 3D model in VR so that this is essential you'll be dealing with 3D matrices and quitterians um in Unity it will ask you to set up these yaw patrol uh angles quturnians or matrices is so there's no way around it. That's the language that Unity will talk with you. Okay. So let's continue. We still have 5 minutes and maybe I can talk a little bit in general then. So there is more to learn about matrices than just themselves. So the inverses and compositions become extremely important for rendering and visualization. So let's start with compositions. compositions are directly related to this animations like this. So if you were to describe mathematically this animation then each transition from frame to frame would be a matrix and then or a composition of matrices. So let's set it up. So if you have your current local frame described by say some matrix M1. So now you can see that we could say okay this cube from the global coordinate frame where we have a global coordinate frame somewhere it's placed here in this local coordinate frame and that's m1 the matrix that determines where it is now in the first frame. So that's good. And then if the second frame local corner frame is this. So again you prescribe the matrix that will place the cube in the second frame animation or how it should look in second frame. Then it is often convenient to keep the following matrix M12 instead. So this matrix M12 describes how to transform M1 into M22 into M2. So it could be some very simple canonical say rotation only or um let's if it's not rigid like some kind of shear or or scale up but if it's rigid transformation then you could say oh it's just rotation about this axis. So it's very simple step that describes how to get from this local query frame to this one. So how to get from M1 to M2 frame. So then it becomes convenient because to render at frame time at time frame t2 all you have to do is to know where it started m1 and what is the transition matrix is and then you just have to multiply these two matrices or compose them m1 * m12 to get to m_sub_2. So again compositions have this very intuitive geometric meaning is that if you start with some matrix and then that matrix or that frame has to transform in certain way prescribed by this transition matrix then multiplying them will give you where the point in this second frame. So what is really cool is that then you can do it iteratively. So all you have to do then is to have this iterative update from each previous frame to get to the next frame through this transition matrix and this iterative update is like crucial to many methods. So like in Oculus SDK we had this method iterative update how take previous state multiply by transition from that state to the current state to get to this state and it generally applies not only to matrices but quturnians as well. So anything that describes the current state of the system. So amazingly the inverses will allow us to represent the arrows in opposite directions. So inverses of the matrices also have this amazing visualization or visual interpretation. So let me draw inverse arrows here. So it turns out that if you have this frame and to get to the next one it's described by this transformation then the inverse would represent the inverse um so the inverse um of m23 would represent the inverse um direction how to get from this frame to this frame. Similarly for M12 the inverse would represent the arrow in the opposite direction. So then you can play movie in inverse or in reverse play your animations in reverse just by having an reverse

iterative update. So let me write it down again. So if you have any frame MI to get to the previous frame, you just need to multiply that frame or compose it with the inverse of the transition that would describe how to get there. So um so then you can get from any matrix or from any frame to the previous frame. So say from frame m3 you m compose it with m23 inverse to get to m_sub_2. So this is m_sub_2 then. And then m_sub_2 * m12 inverse would get you to m1. Uh so it's very intuitive. Just do it in reverse order. [Music] lecture pdf and lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

NPTEL Virtual Reality: Matrices, Rotations and Quaternions — Easy Notes

Don't worry if you cannot answer the matrix questions yet! These topics look difficult because they use mathematical symbols, but the basic ideas are simple. Let's learn them as if you're in 7th class, with examples, memory tricks, and MCQs.

The most important idea is this:

Matrices and quaternions are ways to tell a computer how an object should move or rotate in a 3D world.

For example, when you turn your head while wearing a VR headset, the computer must understand which direction you are facing so it can show the correct view.

1. What is a matrix?

A matrix is simply a rectangular arrangement of numbers.

For example:

$$M = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

This is called a 2×2 matrix because it has 2 rows and 2 columns.

In VR, a matrix can describe what happens to an object when it is transformed.

Imagine a small square in a game. A matrix can tell the computer to:

- Make the square bigger or smaller.
- Rotate it.
- Slant it.
- Reflect it like a mirror.

The most important matrix trick

Look at the **columns**, not the rows!

Consider:

$$M = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

The first column is $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$. It tells you where the original horizontal unit vector $(1, 0)$ goes.

The second column is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. It tells you where the original vertical unit vector $(0, 1)$ goes.

Column 1	Column 2
$(2, -1)$	$(1, 1)$
Where X goes	Where Y goes

Exam shortcut: In a 2D matrix, the first column describes the transformed X direction and the second describes the transformed Y direction.

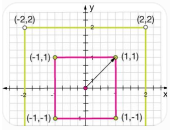
For a 3×3 matrix, the same rule applies:

- First column → X direction.

- Second column → Y direction.
- Third column → Z direction.

Memory trick: X, Y, Z = first, second, third columns.

2. Types of transformations

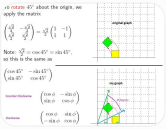


A. Scaling

Makes an object bigger or smaller.

Example: A small cube becomes a large cube.

Memory trick: **Scale = size changes.**

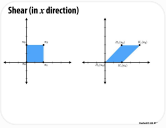


B. Rotation

Turns an object without changing its size or shape.

Example: Turning a square by 45°.

Memory trick: **Rotation = turning.**

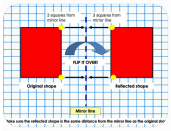


C. Shearing

Slants an object sideways.

Example: The top of a square shifts to one side.

Memory trick: **Shear = slant.**



D. Reflection

Creates a mirror image.

Example: A shape flips across an axis.

Memory trick: **Reflection = mirror.**

3. What is a rigid transformation?

Imagine a wooden square.

You can move it or turn it, but you cannot stretch, shrink, bend, or squash it while keeping it the same rigid object.

A rigid transformation preserves distances and shape.

Transformation	Rigid?
Translation (moving)	Yes
Rotation (turning)	Yes
Scaling (changing size)	No
Shearing (slanting)	No
Reflection (mirror flip)	No, if only rotations and translations are allowed

Exam memory trick: A rigid object can **move and rotate**, but cannot change shape or become its mirror image using rotations and translations alone.

4. Can a matrix turn a square into any shape?

No! This is an important quiz concept.

A normal 2×2 linear transformation changes the square into a parallelogram, which can include a tilted square or a flattened parallelogram.

It cannot turn a square into:

- A circle.
- A triangle.

- A pentagon.
- An irregular, non-parallelogram shape.

Why? A linear transformation maps straight lines to straight lines and preserves the relationships that make the square's opposite sides parallel.

Exam trick: Square → parallelogram is possible. Square → circle or triangle is not possible with a 2×2 linear transformation.

5. What is matrix composition?

Composition means **combining transformations**.

Imagine moving a character in a VR game:

1. First, the character is at position A.
2. Next, it rotates.
3. Then, it moves to another position.

Instead of calculating everything from the beginning each time, we can combine the transformations using matrix multiplication.

$$M_2 = M_1 M_{12}$$

Here:

- M_1 = the starting transformation.
- M_{12} = the transformation that changes the starting state to the next state.
- M_2 = the new transformation.

Think of it like following directions: first walk forward, then turn right. Both actions together give you the final position and direction.

Memory trick: Composition = **combine steps**.

Important: Matrix multiplication is generally **not commutative**. This means the order matters:

$$AB \neq BA$$

Rotating an object and then moving it may give a different result from moving it and then rotating it.

6. What is an inverse matrix?

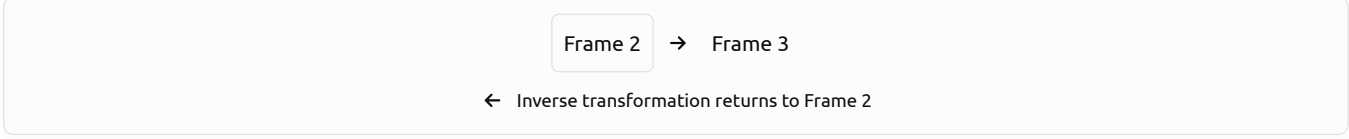
An inverse transformation undoes another transformation.

Imagine rotating a toy by 30° clockwise. To return it to its original orientation, rotate it 30° counterclockwise.

That opposite operation is the inverse rotation.

$$M_2 = M_3 M_{23}^{-1}$$

This means that if M_{23} takes you from state 2 to state 3, its inverse takes you back from state 3 to state 2.



Memory trick: Inverse = **undo**. Composition = **combine**.

7. How do we rotate objects in 3D?

In a 3D world, we have three axes: X, Y, and Z.

Your lecturer teaches three important rotations called **yaw, pitch, and roll**.

Yaw — around Y-axis

Turn left or right, like looking left and right.

Memory: Yaw → Y.

Pitch — around X-axis

Tilt the nose up or down, like an airplane climbing or diving.

Memory: Pitch → airplane nose.

Roll — around Z-axis

Tilt sideways, like an airplane lowering one wing.

Memory: Roll → airplane rolls onto a wing.

For your lecture's convention, remember:

Rotation	Axis
Yaw	Y
Pitch	X
Roll	Z

A computer can combine yaw, pitch, and roll to describe a 3D orientation. However, the order in which rotations are applied matters.

8. What are Euler angles?

Euler angles are a way of describing an object's orientation using three angles.

These angles are typically yaw, pitch, and roll.

For example, a VR headset can tell Unity:

- Turn left by a certain angle.
- Tilt up by a certain angle.
- Roll sideways by a certain angle.

Together, these describe the direction in which the headset faces.

What is gimbal lock?

Euler angles have a problem called **gimbal lock**. At certain orientations, two rotation axes effectively line up, and one independent rotational direction is lost.

Imagine three rings that allow an object to rotate. In a special position, two rings line up, making it difficult to control all three directions independently.

Memory trick: Euler angles are easy to understand, but they can suffer from **gimbal lock**.

9. What is a quaternion?

This is another way to describe rotation in 3D.

A quaternion uses **four numbers**, usually written as:

$$q = (w, x, y, z)$$

Don't let the four numbers scare you. For now, remember that they work together to represent a rotation.

For a rotation by angle θ around a unit axis (a_x, a_y, a_z), a quaternion is:

$$q = \left(\cos\frac{\theta}{2}, a_x \sin\frac{\theta}{2}, a_y \sin\frac{\theta}{2}, a_z \sin\frac{\theta}{2}\right)$$

You do not need to memorize the whole formula immediately. Remember these key ideas:

- A quaternion has four components.
- It can represent a 3D rotation.
- It avoids the gimbal-lock problem associated with Euler-angle representations.
- Unity uses quaternions for rotations.

Example 1: No rotation

$$q = (1, 0, 0, 0)$$

This represents **no rotation**.

Example 2: Rotation around X-axis

For a pitch rotation, the quaternion's X component carries the axis-related rotation information. The Y and Z components are zero when the rotation axis is exactly X.

Memory trick: Quaternion = **4 numbers, 3D rotation, no gimbal lock**.

10. What is a unit quaternion?

A unit quaternion has a length of exactly 1.

For $q = (w, x, y, z)$:

$$w^2 + x^2 + y^2 + z^2 = 1$$

Example:

$$q = (1, 0, 0, 0)$$

Its squared length is:

$$1^2 + 0^2 + 0^2 + 0^2 = 1$$

So it is a unit quaternion.

But consider:

$$q = (1, 1, 0, 0)$$

Its squared length is $1 + 1 = 2$, so it is **not** a unit quaternion.

The lecturer warns that using a non-unit quaternion where a rotation quaternion is expected can lead to incorrect transformations or scaling-like effects in some implementations.

Memory trick: Unit quaternion = **squared components add up to 1**.

11. What are quaternion inverse and duplicate?

Your lecturer gives two useful shortcuts.

Suppose:

$$q = (w, x, y, z)$$

Inverse

For a unit quaternion, the inverse is:

$$q^{-1} = (w, -x, -y, -z)$$

It represents the opposite rotation and undoes the original rotation.

Memory: Inverse → change the signs of the last three components.

Duplicate

The quaternion

$$-q = (-w, -x, -y, -z)$$

represents the **same physical rotation** as q.

For example, q and $-q$ are two mathematical representations of the same orientation.

Memory: Duplicate → change the signs of all four components.

Operation	What changes?	Meaning
Inverse	Last 3 signs	Undo the rotation
Duplicate	All 4 signs	Same rotation

12. The 2D rotation matrix — one formula to recognize

For a rotation by angle θ, the standard 2D rotation matrix is:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

You may see this in the exam. You do not need to panic if you don't remember all the calculations.

For now, remember:

- It rotates a 2D object.
- It preserves its size and shape.
- Its columns describe where the original X and Y directions go.

Memory trick: Rotation matrix → sine and cosine, with the negative sign in the top-right position.

MCQs for your NPTEL exam

Try answering these questions first. Then press **Check answers** to see your score and explanations.

20-question practice 0/20 answered

1. What is the main purpose of a transformation matrix in VR?

- A. To play music
- B. To describe changes in an object's position, orientation or shape
- C. To store passwords
- D. To increase internet speed

2. In a 2×2 matrix, which column tells you where the X unit vector goes?

- A. First column
- B. Second column
- C. First row

D. Last row

3. In a 3×3 matrix, which column describes the transformed Z unit vector?

A. First

B. Second

C. Third

D. None

4. Which transformation changes the size of an object?

A. Rotation

B. Translation

C. Scaling

D. None

5. Which transformation slants a square into a parallelogram?

A. Shearing

B. Translation

C. Identity

D. Rotation only

6. Which pair consists of rigid transformations?

A. Scaling and shearing

B. Translation and rotation

C. Scaling and reflection

D. Shearing and scaling

7. Can a linear transformation turn a square into a circle?

A. Yes, always

B. Yes, if rotated

C. No

D. Only in VR

8. What does composition mean?

A. Combining transformations

B. Deleting a matrix

C. Making a circle

D. Changing a computer

9. What does an inverse transformation do?

A. Repeats the same action forever

B. Undoes a transformation

C. Always doubles an object

D. Creates a mirror automatically

10. Yaw is rotation around which axis in this lecture?

- A. X
- B. Y
- C. Z
- D. No axis

11. Pitch is rotation around which axis?

- A. X
- B. Y
- C. Z
- D. All axes equally

12. Roll is rotation around which axis?

- A. X
- B. Y
- C. Z
- D. None

13. How many angles are used in the usual yaw-pitch-roll representation?

- A. One
- B. Two
- C. Three
- D. Four

14. What is a major problem associated with Euler angles?

- A. Gimbal lock
- B. They always scale objects
- C. They cannot describe orientation
- D. They use no numbers

15. How many components does a quaternion have?

- A. 2
- B. 3
- C. 4
- D. 9

16. What does (1, 0, 0, 0) represent as a quaternion?

- A. A 90° rotation
- B. No rotation
- C. A scale of 2
- D. A mirror image

17. What must the squared components of a unit quaternion add up to?

- A. 0
- B. 1
- C. 3
- D. 4

18. For a unit quaternion q=(w,x,y,z), what is its inverse?

- A. (-w,-x,-y,-z)
- B. (w,-x,-y,-z)
- C. (w,x,y,z+1)
- D. (0,0,0,0)

19. Which quaternion represents the same rotation as q?

- A. q with all four signs reversed
- B. Only the first component reversed
- C. Only the second component reversed
- D. The zero quaternion

20. Why are quaternions useful in VR?

- A. They eliminate the need for displays
- B. They represent 3D rotations without Euler-angle gimbal lock
- C. They only work in 2D
- D. They automatically create games

Check answers

Final revision: memorize these 10 facts

1. **Matrix** = a group of numbers that can describe a transformation.
2. **Columns** = where the X, Y and Z unit vectors go.
3. **Scaling** = size changes.
4. **Shearing** = slanting.
5. **Composition** = combine transformations.
6. **Inverse** = undo a transformation.
7. **Yaw–Pitch–Roll** = Y–X–Z axes in this lecture.
8. **Euler angles** = three angles; can suffer from gimbal lock.
9. **Quaternion** = four numbers representing a 3D rotation.
10. **Unit quaternion** = squared components add up to 1.

My advice: Don't try to master every formula at once. First learn what each mathematical term *does*. Then practise identifying transformations from matrices and solving the MCQs. Once these basics become familiar, the formulas will feel much easier.

You said:

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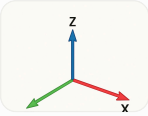
explain easily

ChatGPT said:

I'll explain your **Quiz 3 – Matrices, Rotations, and Quaternions** in very easy English, like a 7th-class student, with simple examples and memory tricks.

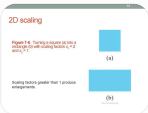
First, understand these 5 basic ideas



1. Rotation

Turning an object without changing its size or shape.

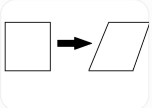
Example: Turning a book on a table.



2. Scaling

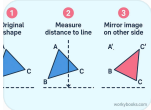
Making an object bigger or smaller.

Example: Stretching a square into a rectangle.



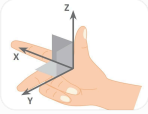
3. Shearing

Slanting an object sideways.



4. Reflection

Showing the mirror image of an object.



5. Matrix

A table of numbers that tells a computer how to transform an object.

The most important rule to remember

A **rotation matrix** only turns an object. It must not stretch, shrink, or mirror it.

- Every column has length **1**.
- Columns are perpendicular to each other.
- Determinant is **+1**.

Memory trick: **1 – 90° – +1**

- Length 1 for each column.
- Columns at 90° to each other.
- Determinant +1.

Explanation of all 10 questions

Q1. Which matrix is NOT a valid 3D rotation matrix?

Answer: C FINAL_QUIZ3.docx.pdf

Matrix C makes one axis twice as long. That means it stretches the object instead of only rotating it.

Remember:

- Rotation = turn only.
- Scaling = change size.
- If a column length is 2 instead of 1, it is not a valid rotation matrix.

Exam trick: If any column has a length other than 1, reject the rotation matrix.

Q2. Which vector becomes the new X-axis?

Answer: B FINAL_QUIZ3.docx.pdf

Look at the **first column of the transformation matrix**.

For a matrix

$$M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

the new X-axis is

$$\begin{bmatrix} a \\ d \\ g \end{bmatrix}$$

Why? The original X-axis is the vector $(1, 0, 0)$. Multiplying the matrix by this vector selects its first column.

Memory trick: X = First column, Y = Second column, Z = Third column.

Q3. What operation does the matrix perform?

Answer: C — Rotation about the Z-axis by 90° FINAL_QUIZ3.docx.pdf

The standard 2D rotation matrix for 90° is

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

It turns objects in the XY-plane by 90°. The Z-axis stays unchanged.

Memory trick: Rotation changes direction, not size.

Q4. What is TRUE about the given matrix?

Answer: C — It preserves distances between every pair of points. FINAL_QUIZ3.docx.pdf

Imagine a square. You rotate it, and it is still the same square.

- Its sides remain the same length.
- Its angles remain the same.
- Its shape and size remain unchanged.

Only its direction changes.

Memory trick: Rotation = same shape + same size.

Q5. What if the determinant is –1?

Answer: B — Reflection (mirror transformation). FINAL_QUIZ3.docx.pdf

For a matrix whose columns are unit length and perpendicular:

Determinant	Meaning
+1	Valid rotation, if the other conditions are satisfied
–1	Reflection or mirror transformation

Imagine looking at your reflection in a mirror. The orientation is reversed.

Memory trick: Plus one = proper rotation; minus one = mirror.

Q6. Quaternion for a 120° rotation about the positive X-axis

Answer: A FINAL_QUIZ3.docx.pdf

A quaternion is a way of representing rotation using four numbers.

The formula is

$$q = \left(\cos\frac{\theta}{2}, a_x \sin\frac{\theta}{2}, a_y \sin\frac{\theta}{2}, a_z \sin\frac{\theta}{2}\right)$$

Here:

- Angle $\theta = 120^\circ$.
- X-axis direction is $(1, 0, 0)$.
- Half of 120° is 60°.

Therefore,

$$q = (\cos 60^\circ, \sin 60^\circ, 0, 0)$$

So the quaternion is

$$\boxed{q = (0.5,\ 0.866,\ 0,\ 0)}$$

Memory trick: Divide the rotation angle by 2 first. For X-axis rotation, only the X component is nonzero among the three axis components.

Q7. Which quaternion is the inverse of q?

Answer: A FINAL_QUIZ3.docx.pdf

Suppose a unit quaternion is

$$q = (w, x, y, z)$$

Its inverse is

$$\boxed{q^{-1} = (w, -x, -y, -z)}$$

Notice what happens:

- Keep the first number the same.
- Change the signs of the remaining three numbers.

Example:

$$q = (0.5, 0.866, 0, 0)$$

Then

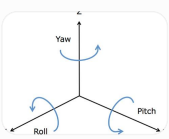
$$q^{-1} = (0.5, -0.866, 0, 0)$$

The inverse represents the rotation that undoes the original rotation.

Memory trick: First number stays; other signs flip.

Q8. A camera rotates around its vertical axis. Which rotation is it?

Answer: C — Yaw FINAL_QUIZ3.docx.pdf



Yaw — Y-axis

Turn your head left and right, as if saying “no.”

Pitch — X-axis

Move your head up and down, as if saying “yes.”

Roll — Z-axis

Tilt your head toward your shoulder.

Memory trick: Yaw = left/right, Pitch = up/down, Roll = tilt.

These are the axis conventions used in your quiz. FINAL_QUIZ3.docx.pdf

Q9. Which condition proves a matrix cannot be a rotation matrix?

Answer: C — One column has length 1.4. FINAL_QUIZ3.docx.pdf

Every column of a valid rotation matrix must have length exactly 1.

If one column has length 1.4, that axis is stretched.

Imagine a unit stick that should be 1 metre long. If the transformation makes it 1.4 metres long, it is changing the size instead of performing only a rotation.

Memory trick: Column length must be exactly 1, not 1.4 or 2.

Q10. Which statement is FALSE about rotation matrices?

Answer: D — They may include scaling along one axis. FINAL_QUIZ3.docx.pdf

A rotation matrix must preserve:

- Distances
- Angles
- Shape
- Size

Scaling changes the size, so a matrix that includes scaling along an axis is not a pure rotation matrix.

Memory trick: Rotation never stretches.

Final revision table

Question	Answer	Remember this
1	C	Length 2 means scaling
2	B	First column = new X-axis
3	C	90° rotation about Z
4	C	Rotation preserves distances
5	B	Determinant −1 means reflection here
6	A	Quaternion uses half the angle
7	A	Keep w; flip x, y, z signs
8	C	Vertical-axis rotation = yaw
9	C	Every column length must be 1
10	D	Rotation cannot include scaling

Test yourself

1. Which column represents the new X-axis?

- First column
- Second column
- Third column

2. What is the determinant of a proper rotation matrix?

- −1
- 0
- +1

3. Which rotation is around the vertical Y-axis?

- Pitch
- Yaw
- Roll

4. A column has length 2. What does this indicate?

- Pure rotation
- Scaling
- No transformation

5. For a unit quaternion inverse, what happens to the first component?

- Its sign changes
- It stays the same
- It becomes zero

Try again

My advice: first memorize the **three rotation-matrix rules**, **X/Y/Z columns**, **yaw-pitch-roll**, and **quaternion inverse**. These ideas cover most of the concepts tested in this quiz.

You said:

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[Music] Okay. What is what are the coordinates of 01 in the sign coordinate frame? It's 1 Z, right? So because that that's how we compose transformation matrices. We just look where these transformation matrices take the unit vectors and that's how we build our matrices. The first column is where X goes to and the second column is where Y goes to. Okay. So that's the conversion matrix for which takes you from right-handed coordinate frame to left-handed coordinate frame in um in 2D. Right? And let for the sake of sanity let's see what would be the coordinates of min -21. So this point -21 in y coordinate frame or original coordinate frame. What would be this coordinates in the sign coordinate frame? Well its coordinates in s prime is 1 and y prime is minus2. But if you then multiply this point by the matrix that we just combined you get 1 - 2. Right? So essentially this matrix takes X and switches it with Y, right? Just switch the S with Y. That's all that's what it says. X becomes Y, Y becomes S. So I have a question for you. And of course it's relevant, right? cuz I showed you how we

have completely different um handedness in many of the tools that we might use in and in our uh textbooks and u a lot of math textbooks and that are out there in our textbook for the course. Uh so how would you switch from one to another? So that's that's the motivation. And here I have a question for you. So what if I have my lecture coordinate frame which is S prime Y prime Z prime right in lectures we keep Z as straight ahead X is to the side and Y up and then your patrol are all like consistent with textbooks roll is the winds uh like that s pitch is the uh nose pitch so it's all consistent in our lectures but unity is like this so how to convert from unity to our lectures which is how to switch S into Z, right? S has to become Z and Z has to become S. So what would be the matrix that accomplishes it? And that's all that's all to it, right? That's that's how to switch from right-handed to left-handed. So if you have some coordinates x uh 1 y1 z1 in uh lecture slides to switch to unity you just have to switch x and z. So you get z1 y1 s1. But what is the matrix that would accomplish it? Because then you can apply it not only to vectors but to other matrices transformation matrices that we talked about in class. You could just apply that matrix to them to get the transformation matrix in unity which essentially switches the rows and correspondingly. So, so what is the matrix that transforms X into Z and Z into X while Y stays unchanged? Yeah. Right. So that's it. B. Hey, so many of you answered. That means you followed all until here. 20 people at least followed quite confidently and knows now that when we looking at the matrix we just know where cores get transformed with them right so that's all to it there's very a very good and intuitive geometric representation of these matrices and that's what this whole series of lectures was [Music] lecture pdf and lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

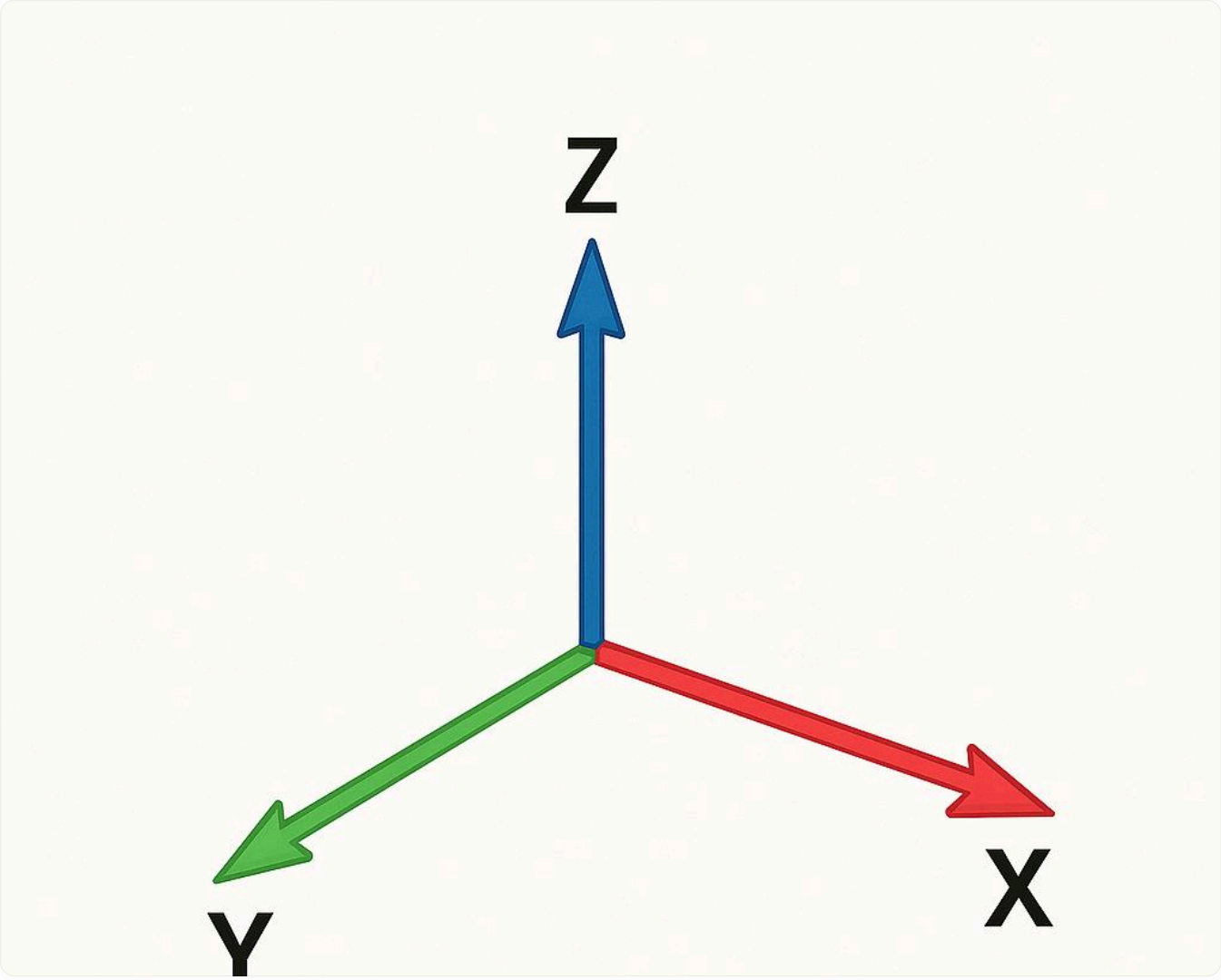
Worked for 17s

ChatGPT said:

NPTEL Virtual Reality — Easy Notes for Week 4

Your lecture is mainly about **coordinate systems, transformation matrices, inverses, quaternions, and 4×4 homogeneous transformation matrices**. Let’s learn them with simple examples and memory tricks.

1. What is a coordinate system?



A coordinate system is like a map that tells us where something is.

- **X-axis:** one direction sideways.
- **Y-axis:** up and down.
- **Z-axis:** forward and backward.

For example, a point $(2, 3, 4)$ means:

- Move 2 units along X.
- Move 3 units along Y.
- Move 4 units along Z.

Global frame vs local frame

Imagine a person standing in a room.

- **Global coordinate frame:** the room's coordinate system.
- **Local coordinate frame:** the coordinate system attached to the person's head, hand, or leg.

If the person turns their head, the head's local frame turns too, but the room's global frame stays fixed.

Memory trick: Global = whole world; Local = one object.

2. How does a matrix move the axes?

A matrix is a table of numbers that tells us where the coordinate axes go after a transformation.

The most important rule from your lecture is:

- First column = where the original X-axis goes.

- Second column = where the original Y-axis goes.
- Third column = where the original Z-axis goes.

For example, suppose the X-axis becomes the Z-axis, the Y-axis stays the same, and the Z-axis becomes the X-axis.

Then:

$$X \to Z, \qquad Y \to Y, \qquad Z \to X$$

The transformation matrix is

$$\boxed{M = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}}$$

Why? Its columns are the new X, Y, and Z directions:

$$X \to \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad Y \to \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad Z \to \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Memory trick: Don't memorize the matrix first. Ask, “Where does X go? Where does Y go? Where does Z go?”

3. Right-handed and left-handed coordinate systems

Different software and textbooks may use different coordinate-system conventions.

Your lecture uses a coordinate system where:

- X points sideways.
- Y points up.
- Z points forward.

To convert coordinates to a system that swaps X and Z, use:

$$(x, y, z) \to (z, y, x)$$

Example:

$$(2, 3, 4) \to (4, 3, 2)$$

The matrix is the same one from Section 2.

Important exam point: Swapping two axes changes the handedness of the coordinate system. This conversion is not a pure rotation because its determinant is -1.

Memory trick: To swap X and Z, exchange the first and third coordinates. Keep Y unchanged.

4. What is a matrix inverse?

An inverse is a transformation that **undoes what the original transformation did**.

Imagine this:

Original position

↓

Rotate and move

↓

New position

Apply the inverse to go back to the original position.

For a matrix M, its inverse is written as M^{-1}.

$$M^{-1}M = I$$

Here, I is the identity matrix. It means “do nothing.”

Example

- Move 5 steps forward.
- To undo it, move 5 steps backward.

Likewise, to undo a rotation, rotate by the opposite angle.

Memory trick: Inverse = Undo.

5. Why does the camera use an inverse matrix?

Suppose a camera moves, but a chair in the room stays still.

From the room's point of view, the chair has not moved. But from the camera's point of view, the chair's coordinates change.

To calculate the chair's coordinates relative to the camera, use the **inverse of the camera's transformation matrix**, assuming the chair's coordinates are given in the global frame.

$$\boxed{\text{Camera-relative coordinates} = M_{\text{camera}}^{-1}(\text{global coordinates})}$$

This is important in VR because the headset moves while objects in the virtual world may remain stationary.

Memory trick: To see the world from a moving camera, undo the camera's movement.

6. Why does the order of transformations matter?

Imagine you first rotate a book and then move it.

That is not always the same as moving the book first and then rotating it.

In general,

$$AB \neq BA$$

For example:

- Rotate, then translate.
- Translate, then rotate.

These can produce different final positions.

How do we undo multiple steps?

Suppose you first rotate and then translate. To undo both:

1. Undo the translation.
2. Undo the rotation.

The inverse of a product follows this rule:

$$\boxed{(AB)^{-1} = B^{-1}A^{-1}}$$

Memory trick: Undo in reverse order.

This rule is especially important in exam questions involving transformation matrices. Pasted text (3)(2).txt

7. What is a quaternion?

A quaternion is another way to represent a 3D rotation. Instead of using a 3×3 rotation matrix with 9 entries, a quaternion uses 4 numbers.

$$q = (w, x, y, z)$$

For a rotation by angle θ around a unit axis (a_x, a_y, a_z):

$$q = \left(\cos\frac{\theta}{2}, a_x \sin\frac{\theta}{2}, a_y \sin\frac{\theta}{2}, a_z \sin\frac{\theta}{2}\right)$$

You usually do not need to fear the formula. Remember these facts:

- A quaternion has **4 components**.
- It represents rotation.
- The rotation angle is divided by 2 in the formula.
- The axis vector must be normalized first.

What does normalization mean?

Suppose the rotation axis is

$$v = (1, 1, 1)$$

Its length is $\sqrt{3}$. To make it a unit vector, divide each component by $\sqrt{3}$:

$$\boxed{v_{\text{unit}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)}$$

A unit vector has length exactly 1.

Memory trick: Normalize first, then create the quaternion.

8. What is a 4×4 homogeneous transformation matrix?

This is one of the most important topics in your lecture.

A 3×3 rotation matrix can represent rotation, but translation needs an additional operation. A **4×4 homogeneous transformation matrix combines rotation and translation into one matrix**. Pasted text (2)(2).txt

Its general form is

$$\boxed{T = \begin{bmatrix} & & & t_x \\ & R & & t_y \\ & & & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

Here, R is the 3×3 rotation matrix, and t_x, t_y, t_z represent translation.

More clearly, the structure is:

$$T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Why is this useful? Because we can rotate and translate an object using a single matrix multiplication.

Memory trick

- Top-left 3×3 = Rotation.
- Last column's first three numbers = Translation.
- Last row = $0, 0, 0, 1$.

In 3D, there are 6 degrees of freedom: 3 translations and 3 rotations.

9. What does “rotate, then translate” mean?

Your lecturer explains the 4×4 matrix using the **global coordinate frame**.

For a point represented by a homogeneous vector p:

$$p' = Tp$$

The transformation matrix has the form

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$$

Therefore,

$$p' = Rp + t$$

In simple words:

1. Rp: rotate the point.
2. $+t$: move the rotated point.

Exam trick: With this convention, the point experiences rotation first, followed by translation in the global frame. Pasted text (3)(2).txt

10. How do we find the inverse of a 4×4 transformation?

You do not need to memorize a complicated 4×4 inverse calculation for every question.

If

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$$

then

$$\boxed{T^{-1} = \begin{bmatrix} R^{-1} & -R^{-1}t \\ 0 & 1 \end{bmatrix}}$$

For a rotation matrix, $R^{-1} = R^T$, where R^T is the transpose.

The important idea is that the inverse must undo **both the rotation and the translation**. Simply changing the signs of the translation values is not always enough when rotation is also present.

Memory trick: Undo the translation and rotation in reverse order.

Practice MCQs

Try these questions before checking your answers.

1. What does the first column of a transformation matrix represent?

- Where the original X-axis goes
- Where the original Y-axis goes
- The translation only

2. If (2, 3, 4) is converted by swapping X and Z, what is the result?

- (2, 4, 3)
- (4, 3, 2)
- (3, 2, 4)

3. What does a matrix inverse do?

- Doubles a transformation
- Undoes a transformation
- Always rotates 90°

4. In general, is AB equal to BA?

- Always yes
- Always zero
- Not necessarily

5. How many components does a quaternion have?

- 2
- 4
- 9

6. What must you do to the rotation axis before constructing a quaternion?

- Normalize it
- Double it
- Convert it into a matrix first

7. What does a 4×4 homogeneous transformation matrix combine?

- Only scaling and reflection
- Rotation and translation
- Only translation

8. In the standard homogeneous matrix, what is the bottom row?

- 1, 1, 1, 1
- 0, 0, 0, 1
- 0, 1, 0, 0

9. A 3D rigid object has how many degrees of freedom?

- 3
- 4
- 6

10. A camera moves, and you want a stationary object's coordinates relative to the camera. Which matrix is useful?

- The inverse camera transformation
- Always the identity matrix
- Only the camera's translation

Restart quiz

Score: 0 / 10 answered correctly

Last-minute revision sheet

Topic	Remember this
Matrix columns	X, Y, Z in order
Coordinate conversion	Swap the required axes
Inverse	Undo
Matrix multiplication	Order matters
Quaternion	Four numbers
Normalization	Make vector length 1
Homogeneous matrix	Rotation + translation
Inverse of product	Reverse the order
3D rigid body	6 degrees of freedom
Camera coordinates	Use inverse camera transform

My suggestion: memorize the **matrix columns, inverse rule, quaternion normalization, and 4×4 matrix structure** first. Then practise the MCQs, because these concepts are the foundation for understanding how Unity moves objects and tracks the VR camera.

You said:

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PDF

explain easily

Worked for 4s

ChatGPT said:

NPTEL VR Quiz 4 — Easy Explanation

I'll explain all 10 questions from your PDF in **very simple English**, with the correct answer, an easy example, and a memory trick for the exam.

Q1. Why does the order of transformations matter?

Correct answer: C — Matrix multiplication is not commutative.

Imagine a toy car:

- **Rotate, then move:** Turn the car first, then move it forward.
- **Move, then rotate:** Move the car first, then turn it.

The final position can be different!

Remember:

$$AB \neq BA$$

This means changing the order can change the result.

Memory trick: *Order matters in transformations.*

Q2. Why should a VR object appear optically at infinity?

Correct answer: B — It allows the eyes to remain in a relaxed accommodation state.

A VR headset has a screen very close to your eyes. But its lenses make the image appear much farther away, ideally at optical infinity.

This helps your eyes stay relaxed, like when looking at a distant object.

Memory trick: *Optical infinity = relaxed focusing.*

Q3. How do we undo "rotate, then translate"?

Correct answer: True.

To undo actions, reverse their order and undo each action.

Suppose you:

1. Rotate a toy.
2. Move it to another place.

To return it to the original state:

1. Undo the movement.
2. Undo the rotation.

Mathematically:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Memory trick: *Undo in reverse order.*

Q4. Can a 3×3 matrix represent translation?

Correct answer: A — It cannot move the origin, so it cannot represent translation.

The origin is the point $(0, 0, 0)$.

When we multiply it by any ordinary 3×3 matrix, it stays at the origin:

$$M \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

So, a 3×3 matrix alone cannot translate a point.

Memory trick: *3×3 = rotation and scaling; 4×4 = rotation plus translation.*

Q5. Why add an extra coordinate to a point?

Correct answer: D — To perform rotation and translation using one matrix multiplication.

Normally, a 3D point has three coordinates:

$$(x, y, z)$$

We add one extra coordinate, $w = 1$:

$$(x, y, z, 1)$$

Now we can use a 4×4 homogeneous transformation matrix to perform rotation and translation together.

For example, a point $(2, 3, 4)$ becomes $(2, 3, 4, 1)$.

Memory trick: *Add 1 to combine rotation and translation.*

Q6. Which axes are exchanged when converting the coordinate systems?

Correct answer: C — X and Z.

According to your quiz, when converting from the right-handed coordinate system to Unity's left-handed coordinate system, exchange the X-axis and Z-axis. The Y-axis remains unchanged.

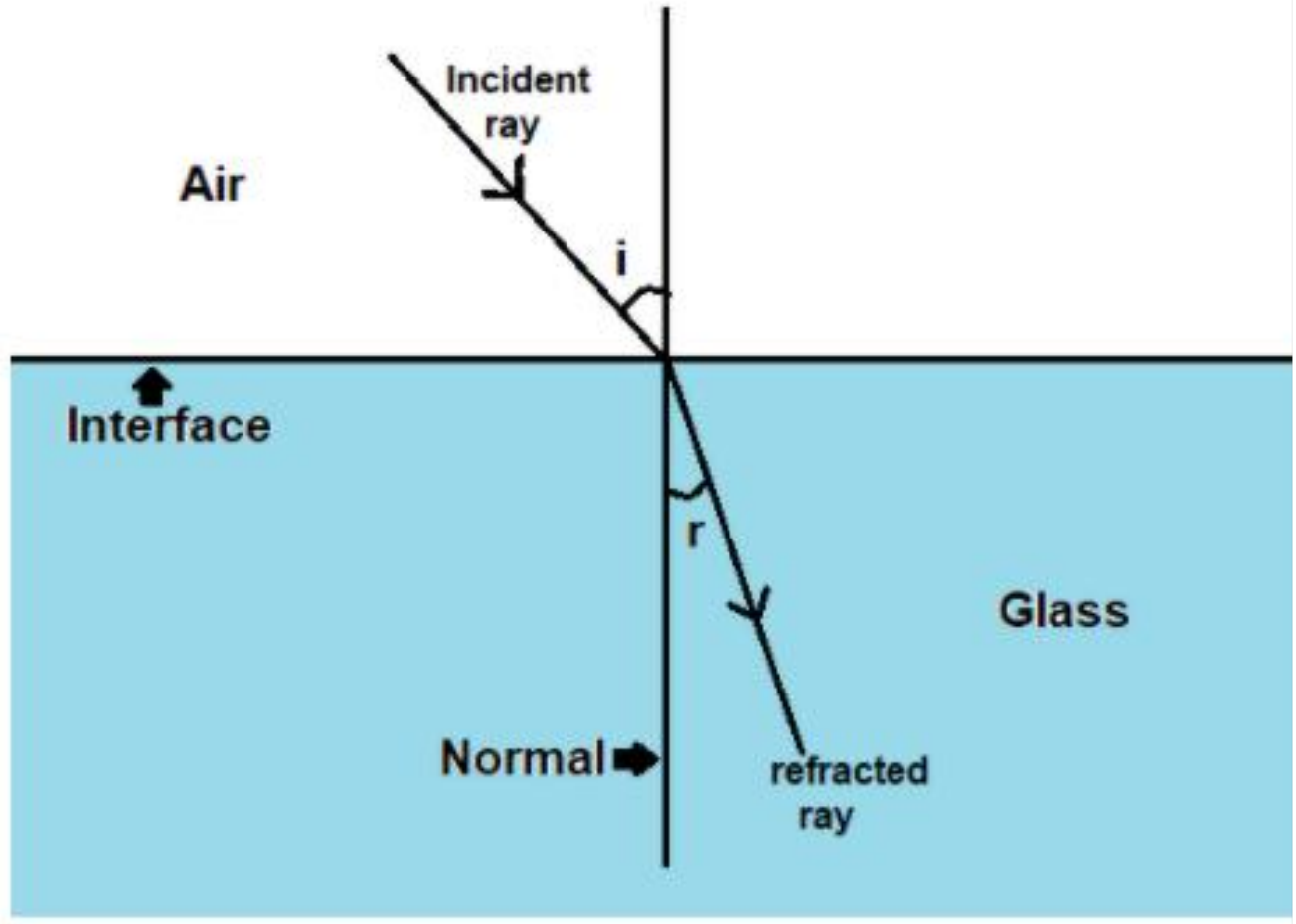
$$(x, y, z) \rightarrow (z, y, x)$$

Example: $(2, 5, 7)$ becomes $(7, 5, 2)$.

Memory trick: *Swap X and Z; keep Y.*

Q7. What happens when light travels from air into glass?

Correct answer: B — It bends toward the surface normal because light slows down.



Think of the normal as an imaginary line perpendicular to the surface.

- Air has a lower refractive index.

- Glass has a higher refractive index.
- Light slows down when it enters glass at an angle.
- The ray bends **toward the normal**.

Memory trick: *Lower index → higher index = toward the normal.*

Q8. When does total internal reflection occur?

Correct answer: E — Light travels from a higher refractive index medium to a lower one at a sufficiently large angle.

Imagine light travelling inside glass and reaching the glass–air boundary.

Total internal reflection occurs when both conditions are satisfied:

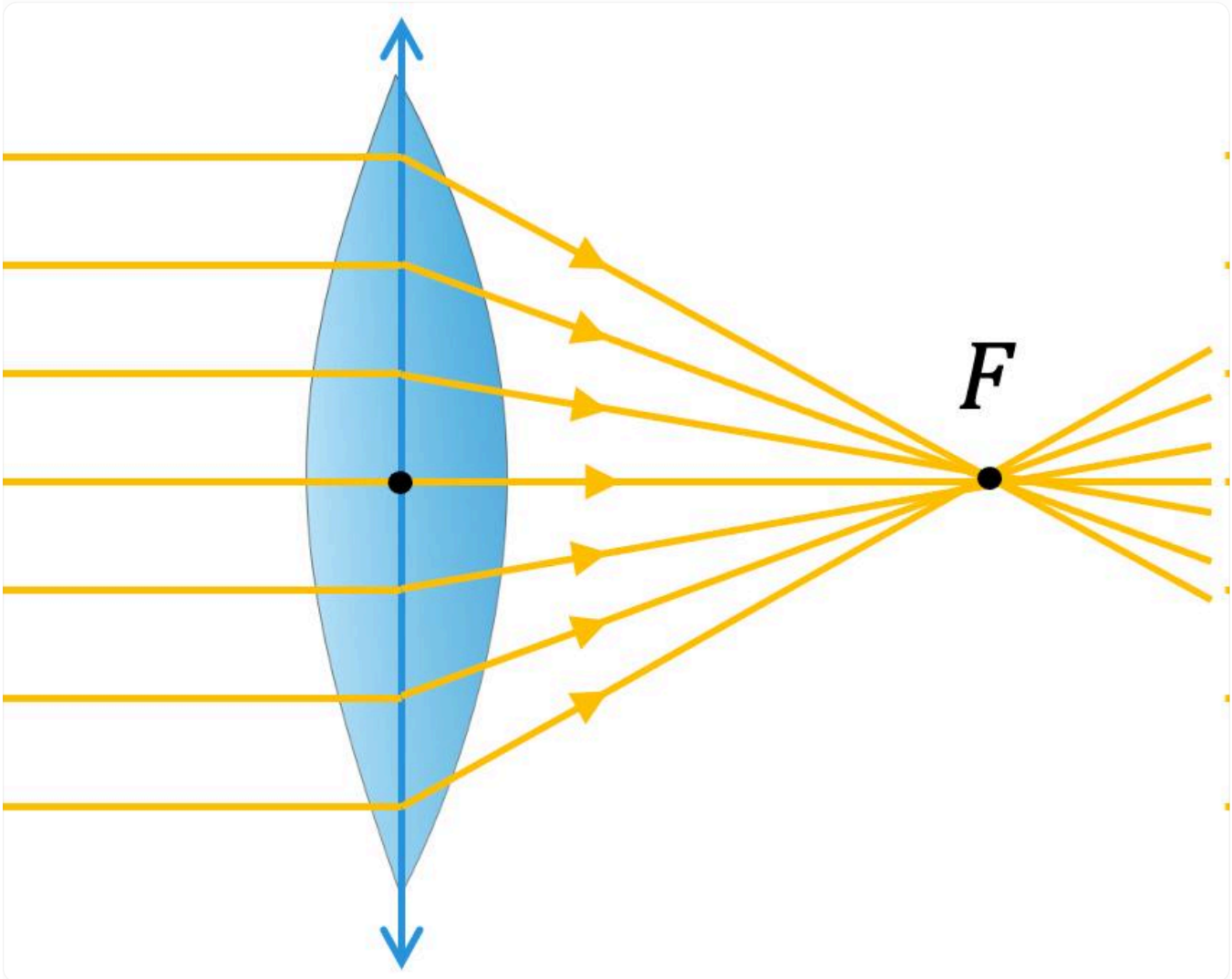
1. Light travels from glass (higher refractive index) toward air (lower refractive index).
2. The angle of incidence is greater than the critical angle.

Instead of passing out into the air, the light reflects back inside the glass.

Memory trick: *High to low + angle greater than critical = total internal reflection.*

Q9. Why does a convex lens have positive diopter power?

Correct answer: A — It converges parallel rays to a real focal point.



A convex lens is thicker in the middle and brings parallel light rays together at a focal point.

Its optical power is:

$$P = \frac{1}{f}$$

Here, P is power in diopters and f is focal length in metres.

For a convex lens, the focal length is positive, so its power is positive.

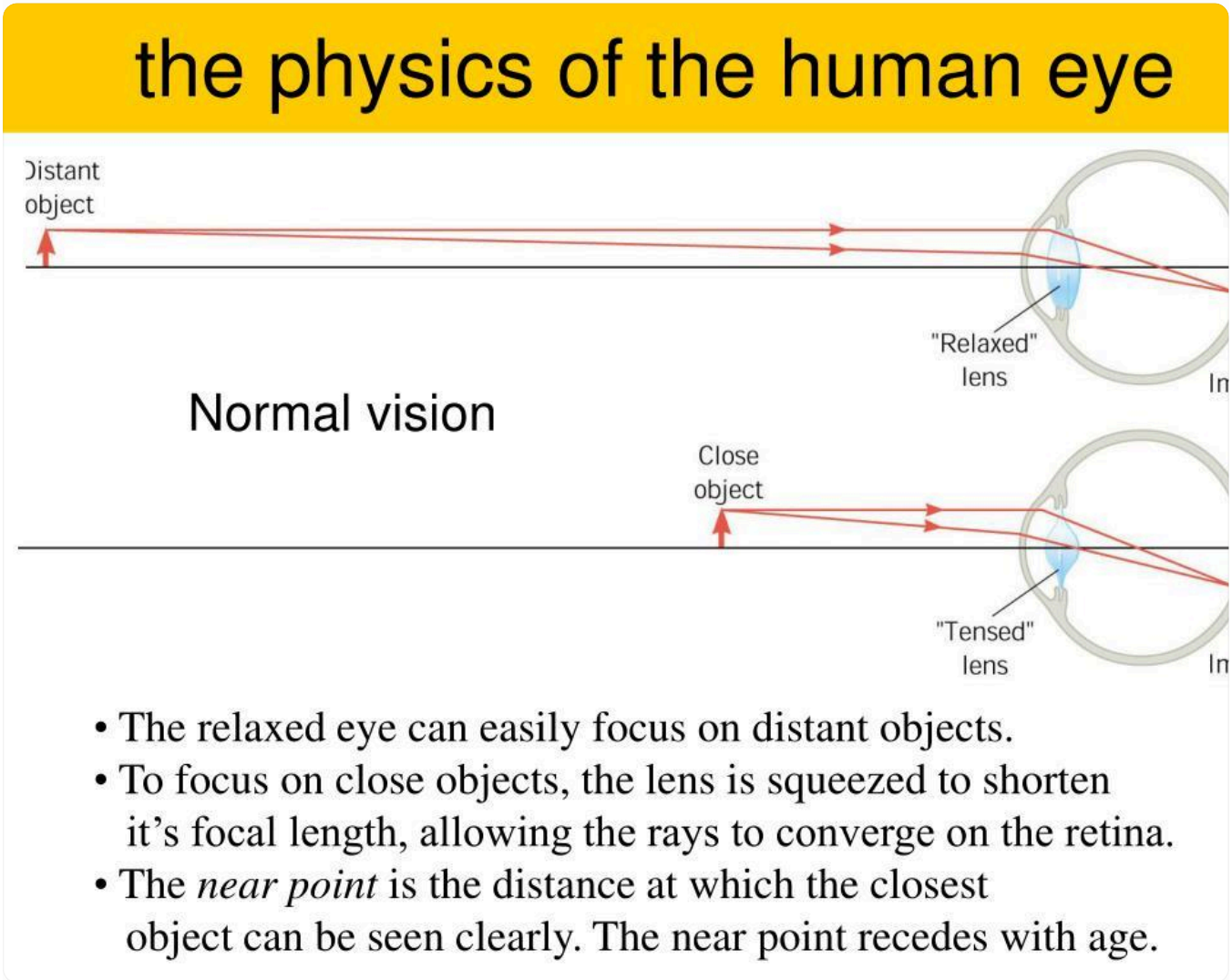
Example: If $f = 0.5$ m,

$$P = \frac{1}{0.5} = +2 \text{ D}$$

Memory trick: *Convex converges → positive power.*

Q10. What happens when a relaxed eye looks at a nearby object?

Correct answer: C — The image forms behind the retina, making the object appear blurred.



A relaxed eye is adjusted to focus on distant objects.

When you look at something nearby:

- Light rays entering your eye are more divergent.
- A relaxed eye does not bend them enough.
- The image would form **behind the retina**.
- Therefore, the object appears blurred.

Your eye normally fixes this using **accommodation**: the lens changes shape to increase its optical power and focus the image on the retina.

Memory trick: *Nearby object + relaxed eye = image behind retina.*

Quick revision table

Question	Correct answer	Remember this
1	C	Order of transformations matters
2	B	Optical infinity relaxes focusing
3	True	Undo in reverse order
4	A	3×3 cannot translate
5	D	4×4 combines rotation and translation
6	C	Swap X and Z
7	B	Air to glass: toward normal
8	E	High index to low + large angle
9	A	Convex lens converges
10	C	Nearby object + relaxed eye: behind retina

Test yourself

1. Which matrix is used to combine rotation and translation in 3D?

- A. 2×2
- B. 3×3
- C. 4×4
- D. 5×5

2. Light travels from glass into air at an angle greater than the critical angle. What occurs?

- A. Refraction toward normal
- B. Total internal reflection
- C. No change ever
- D. Diffraction always

3. What does a convex lens do to parallel light rays?

- A. Spreads them apart
- B. Absorbs them
- C. Brings them together
- D. Stops them

4. How should you undo two transformations?

- A. Use the same order
- B. Reverse the order and invert each
- C. Ignore the first
- D. Always swap X and Y

5. What happens when a normal eye looks at a nearby object without accommodation?

- A. Image forms behind the retina

B. Image always forms on retina

C. Image forms before retina

D. The pupil removes all blur

Try again

You said:

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Pasted text (2)(3)... Document

Pasted text (3)(3)... Document

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[Music] Okay. So, another interesting optical solution to aberrations is to use a completely different opt. And this is what's used in hollow lens, Google glass, and magic leaf one. So these are called wave guides. So the idea behind waveguides, it's extremely uh advanced objects. Uh I've seen demos of it. I've seen the uh prototypes. I've tried them and I think a lot of it is still in R&D and a lot of it is just speculation on my side how it's happening. Um because it's uh it's it's it's advanced and it's an R&D and it's proprietary for these companies. A lot of it cannot did not see was not released to public. So um however what we know about the solutions is that waveguides use advanced optics to bounce the rays around using um um mirrors and and optical structures. And one of the um essential parts of these devices are lasers right lasers or very little displays. So in the case of pure waveguides we just have a display which send uh the light through these waveguides and eventually it hits the retina. So there is some kind of display that's waveguides. Another solution is virtual retinal display. So it's just the laser uh either directly sending a beam directly in the retina. So kind of scanning your retina or doing it using some kind of MEMS's mirrors and some kind of reflector. So very simple wave guide to still scan the retina. Uh but combined into one you can combine virtual retinal display with more advanced waveguides and you get um even more advanced optics then. So, here is Microsoft Hollow Lens 2 R&D video. Uh, again, this is just a speculation on on the part of this enthusiast of all of the components of Microsoft Hollow Lens 2. Really good video. So, let's uh watch it. >> Hi, I'm Jay. Welcome to my channel and my first video. >> I think let's see. They are a small company based out of Redmond, Washington that design and manufacture laser beam scanning displays among other things. It's been speculated that their technology is what's inside the Hollow Lens 2. There's been no concrete evidence of this anywhere until my video now. Okay. So, as you can see, there's a laser, there is a wave guide, there's a MEMS's mirror that will um scatter this uh laser across uh the wave guide to to to scan the retina. And uh he was really happy to find that mirror. I think it was in the beginning of the video to to find this mirror that he uh found inside of the hole lens. [Music] Okay, let's watch this video first. >> So, just to explain very briefly why this is so important, we have to understand IPD, the interpopularity distance. That is the distance between the pupils of your eyes. Of course, to measure it super easy. There are many apps on the app store or a play store or you can just go in the front of a mirror, take a scale like this one with the millimeters. This is from IKEA by the way. uh you put it in front and you measure the distance from one to the other. At the point you know the number for me is 66.4 and of course the Oculus Quest 2 has a 63 or 68. So I'm right there in the middle. Bummer. And you want to be right in the center of the lenses to be able to see this screen in the right way possible with the right distortion and stuff like that. So if you sit on the right IPD, lucky you. If you don't, well luckily there's something. >> Okay. So what is IPD? IPD is the distance between your pupils. Let me draw on my screen. So it's this distance right when you look straight. So when you look at infinity, of course you verge your pupils when you look at something closer. It's very important to relax your eyes, look straight and measure your APD. So what happens if your pupils or the distance between your pupils is smaller or bigger than the distance between optical centers of your lens in HMD. So if your pupils end up looking through periphery, you see a lot more of aberrations that we just talked about than you need to. and high-end HMDs, they have the switches for hardware IPD so that the lenses would be right the optical uh centers of the lenses would be right right in front of your pupils and if you don't set it up you have an inferior performance of this devices for our cardboards we don't have an option of switching uh the IPD hardware we cannot move these lenses further off or uh closer. Uh therefore we may or may not have the best performance of this cardboards. So what should be the solution for us if you use uh cardboards? Well, you could try to tinker with your cardboard and cut new cardboard for lenses that are exactly for AIPD and see if you get a better performance and you will. uh at least you should compare your IPD to the distance between lenses and see how far off you are. So that's interesting. Nevertheless, it's very important to set it up and know what it is. And it's important to set it up not only in hardware like we discussed, but also in software. So in software, how far will be the images uh that are projected on your screen from each other? Will they really go through the optical services of the lens? So software correction is very important but that we have covered in homework one. What we didn't cover in homework one is the hardware IPD. Um so I have a question for you for those of you who have high-end HMDs. Have you corrected your IPD? And did you correct it in both software and hardware or only in software or only in hardware or no? So that's good that you're aware of this problem now and you know about all of

the things that can go wrong with spherical lenses and why is it important to look straight through the center and not through the periphery of the lens. Uh so now you can improve the performance but still it's very important for us to remember about this and um that's the end of today's lectures. [Music] [Music] Let's go with the chapter five. So we just finished with light and optics and we talked about basics of propagation of the light and how we see that light in HMD. So we talked about virgin accommodation mismatch uh but today we'll start talk looking deep in the human eye deeper in the human eye I guess and understanding the structures there and how perception happens ultimately how do we see things. So what is this eye defect? Just looking at where the straight ray if if you have parallel rays coming into the uh eye ball or through the through the pupil and through the lens and they converge before retina. What can you say about the eye defect or vision defect in this picture just looking at the picture? So these are parallel rays and they do they do not converge in retina. They converge before retina. So what is this? Yeah. So it's myopia concave lens would would correct for it. Not shortsided. The they the uh sorry sorry this is wrong you guys. You guys this is wrong. If the lens is too strong it's a shortsided eye not longsided. So you have it. Oh which is what my Okay. You're correct. Okay. Sorry. Yeah, myopia and short-sightedness are um synonyms. So, these are correct answers. Very good. Okay, I was thinking about something else. Uh distortion is when straight lines do not appear straight after looking through the lens. Is that true or false? And that's straight definition of distortion. Right? There is nothing tricky about this question. Just definition of distortion is that when straight lines do not appear straight after looking through the lens. So lens distorts straight lines. Good. Now think about laser emitting red light uh in three dimensions. So it's just red light. And if the rays have one focal point in one plane and a different focal point in another plane, what type of optical aberration is it? And it's just the definition of that aberration. Nothing else, nothing more. So you have one plane having one focal point, another plane having another focal point. Maybe they're not necessarily orthal orthogonal. What is it? It's a stigmatism, right? It's not chromatic aberration. Chromatic aberration is just different planes for the whole uh spectrum. But a stigmatism is when you have just one light and uh one wavelength and it has different focal distances in different planes. Okay. So it's a stigmatism. Okay. Not chromatic aberration. Okay. Very good. So what is the main cause of chromatic aberration? Light is combined of different wavelengths. Right? So that's what causes chromatic aberration. So for chromatic aberration, so looking back, this is just definition of a stigmatism. But chromatic aberration is when you have the full spectrum light all of the wavelengths but then they refract to different um angles and then you have different focal lengths for different uh wavelengths. So because light is combined of different wavelengths that's why chromatic aberration happens. Very good. At what distance you can no longer focus on the lines on your hand in real life. So this is something not to remember but just do an experiment. I want you to bring your hand very close in real life to your eyes and see at which you may leave the glasses on if you still have if you wear glasses which distance can you focus on the lines and see them sharply more than 5 cm what's the smallest one that you can get right okay so for me without glasses would be 50 cm mters. So, you're a very young crowd. That's what I can say. If your um distance is more than 5 cm or less than 5 cm, less than 5 cm. I would imagine you guys to be more in the range from 5 to 10, maybe 10 to 20. Yeah, these are like maybe in the late 20s, but this is this is very young. Okay, so we have a young crowd here. That's really cool. Or you have very good glasses. Okay, now let's imagine that you have the same experiment but in virtual reality. That is maybe you have a virtual hand in virtual reality or maybe it's just an object. How close can you bring it to your eyes and still let's say one eye even so just one eye bring close another eye just bring it to one eye and what would be the distance the smallest distance at which you can focus on it and that's when I'll see if you understand virgin accommodation mismatch in virtual reality and I see that we have to go through this topics again accommodation in virtual reality What is it? Does it happen in virtual reality? What is accommodation? And if you understand that definition, you'll imply what is how close you can see things in virtual reality. So in VR, there is no accommodation. All of the light rays that come out from the lens are parallel, which means that you don't have to accommodate if you don't have vision defense, of course. Um, but assuming that you're wearing your glasses and assuming that all of you could see distances in real life, objects closer than 5 cm, then you have good vision or you have good glasses, which means that you can converge parallel rays to a point on your retina. Which means that objects can be at any distance. Less than 5 cm is the correct answer. I hope you see that. Any distance you can you'll be able to focus on any object in virtual reality on any more than 20 cm, less than 20 cm, less than 5 cm. You can put an object on your eyeball and you'll be able to see it without any accommodation. There's no accommodation in virtual reality. I hope this is clear. Let's go through the next three questions because they're all related to virgin's accommodation mismatch and we'll hope we'll understand it. So let's make some assumptions because otherwise it's a little messy. It will depend on your particular vision defect or in particular uh VR headset. Let's talk about a simplified VR headset model. That is we know for sure that the lens is at the focal distance. So that the light that comes out from the display goes through the lens is parallel. We have only parallel rays always coming out from this headset independently on the content. So you can put objects virtually on your eyeball or a kilometer away. The light coming from those objects will be parallel. It's independent on the virtual depth. Right? And it's not the case for objects in real life. the because the rays coming out from closer objects are diverging more diverging than from further away objects and that's an accommodation squeezes the lens to accommodate for that. So accommodation is the amount of squeezing that you need to do or just the process of squeezing of the lens that you need to do to focus on close by objects. In VR you

don't ever have to focus all of the rays are parallel. you never need to use accommodation if it's a simplified uh headset model and we also assume that you have no vision defense otherwise it's messy. So okay that's our assumption. Simplified VR headset model. No vision defects in a person. Let's try to answer this question then. Accommodation occurs in real life but not in VR. Is that true or false? Yeah, it's true. Very good. It's true. In real life, you need to squeeze less for further away objects because rays are less divergent and more to accommodate for very diverging rays to bring it onto the retina. That's accommodation. In VR, the rays are parallel. You don't have to accommodate. Okay, fine. Next question. Virgin's accommodation mismatch occurs in real life but not in VR. So there are two questions here but I mean it's obviously false because the last statement is false like virgin accommodations does occurs in occur in VR. There is no accommodation on objects but you have to verge in them right objects can be very close in VR and you have to verge very closely but you'll still not need to accommodate your islands because the rays are parallel. So that's virgin accommodation in VR. it it does occur but in real life it occurs too if you use some other kinds of optics even when you're reading a book it's virgin's accommodation mismatch so um so this statement is false I hope uh we went through this uh definition of virgin's accommodation of both of those uh now enough but if it's still not clear go back to last lecture [Music] [Music] Okay. So, um yeah. So, I think the last question will be exactly related to this slide and you'll see how limited RGBs on our phones are in comparison to what our eyes can perceive. So basic the range at which our eyes can see things is just fascinating and uh this is because the ways the rods and cones are divided up on our periphery. So they have on on our eyes uh they have different functions and we end up with different vision of modes. So we'll talk about vision modes. So what are those vision modes? There's phototopic for day, misopic and scotopic for night. And misopic is kind of a transition mode. So it is nice to look at this and understand these modes from the point of view of light intensity. So when we talk about perception of light in terms of human vision, so how we perceive it, we we use for um luminance, sorry, for intensity of light, we use luminance. So this is what we have on the left side of this slide. This is the scale for luminance. So this takes into account how sensitive human eye is to the light. And the unit is then candelas. So roughly how much light is roughly emitted from one candle light per meter squared. So you take one candle light. I don't know what's the sizees but how much light there is in one meter squared um from from that handle light. So that's luminance but we can start talking about a pure physical notion and corresponded to this uh perception notion and um then we can start talking about the number of photons that are eliminated per meter squared per second. So how many photons were released on on in this environment per meter squared per second. So that is a very concrete physical notion and that's what we will put or correspond to luminance on the right side except that here it's a log of number of photons illuminating uh some environment per micrometer squared per second. Okay. So then we can do this correspondence right we can classify now this uh human vision modes right. So it turns out that the photo receptors our photo receptors have the really great range of light conditions under which we can see. So the range is roughly from 10 billion candles per square meter to 1 millionth of a a fraction of a candle per square meter. So notice that it's log. So that means this is 10 to the 0 or one candle light is right here. And this is 10 the minus 6. So 1 millionth fraction of a candle per square meter and we'll see still see something. That's the visual threshold. So let's start with phototopic vision first. So we can read a book in full sunlight, right? And that corresponds roughly to 100 million photons per micrometer squared per second or 10 to the 4th of candles. So 10,000 candles per meter squared squared. That's daylight, right? So then at about 100,000 photons per square micrometer/s, we can't read comfortably anymore. And that's when we enter misopic uh mode. Right? So this corresponds roughly to 10 candles per square meter and it's more difficult to read. Okay. So the phototopic uh region ends we start misopic uh region uh and we are staying in uh misopic region until this number which is 100 photons per micrometer squared per second or 10 to the -2. So 100th or one one h 100th of a candle per meter squared. So I think so what what does it mean? We um in this range we read with difficulty we can see white surface in the moonlight and then that's the threshold for cone vision. So the cones will shut down after this threshold. Okay. So then we enter the skopic um region right. So this scoptopic region is right here and this is the night vision. So about 100 photons per square micrometer per second. So it's 100 candle light per square meter as I said. So we can see white surface even on a moonless night. we can see it and that's the rods that then will be working in for us to see it and then we we hit the visual threshold. So what about virtual reality? Can we produce this range of elimination? No. Right? We can't possibly. So uh so that's a problem. That means we cannot replicate this kind of vision conditions and the vision modes with high fidelity in virtual reality. But with OLEDs, yes, of course, you just make the pixels all black and you will enter scootopic mode. So let's kind of um summarize everything that we just learned about scotopic and phototopic vision modes. So what are the photo receptors that are working in heat and what are the light intensities under which they work? So the dominant photo receptors for skic vision are rods. So at night it's mostly rods that do the vision and for phototopic vision it's cones and these are the light levels under which we enter these modes. And this is just from the last slide. And as far as the color perception well in scootopic vision due to rods it's monochromatic. that means black and white and for phototopic vision it's trichromatic which means RGB colors right so we have three cones RGB and as far as the adaptation goes it takes 35 minutes to go fully into the scoptopic mode so for example if you've been in the bright sun how long does it take for you to see in the dark well after a few minutes you'll start improving and you'll see things but you really need to wait for the full 35 seconds to fully adapt it so that this so that the rods get fully

sensitive to very low light intensities. Uh so if you really want to observe the bright night sky, you cannot just leave your car and turn off the lights and just do it. You will not get the full um experience from the show from the night show or the bright sky uh night sky. You really need to wait for 35 minutes to really enjoy it and appreciate it. So going in the opposite direction is slow is faster. So it takes only 10 minutes to adapt from dark conditions to the uh phototopic mode. So uh it's about 90% of our neurons in our vision system in our brains is devoted to phototopic vision. Although there are much fewer cones, it's still what we analyze mostly comes from cones which means that we are uh daylight animals, right? So that information is much more important than the uh night nighttime information. Okay. So let's see what kind of wavelengths will trigger rods and cones. So that's very important. You can already make a lot of conclusions just by looking at these graphs. So what particular wavelength will trigger which cones? So the rods are responsible for light across these wavelengths as you can see from 400 to 600 with a peak at 498 um nanometers wavelength. So of course that means that they respond to some others but just with less probability right so with lower and lower probability but they'll respond to this light with higher probability. And notice it's green blue light here. And then the for the cones, there are three different sensitivity curves for three different types of cones. And it's really amazing that it's exactly RGB just like what we make in our monitors or other um screens. So RGB cones, that's what we have on our retinas. So let's see the red cones the green cones the red cones are sensitive for this particular uh wavelengths and they are distributed irregularly on mostly around phobia as you remember. So let's look at particular examples then of how we can perceive colors and light under different light conditions. So let's look at this and we will derive it from this sensitivity curse. So, first of all, we don't see colors at night, right? That that's clear. If it's low, low intensity light, the cones will just shut down. And the rods, they don't see color. We don't perceive color from them. Although they u they're sensitive to various colors that is still in the environment. We just don't perceive it. We see black and white. So, that's cool. Then um it's also important to notice that on periphery where we have only rods the same situation applies as if it was in the dark. We don't see color on periphery. You could demonstrate it in a VR experience. Actually I'll show you later if we have time today. Now let's think of red rose and green leaf at dark at dusk. So we have a red rose and a green leaf. So during bright daylight you can see both red and green right because our cones will be sensitive they will be awake and they'll see red and green they'll respond to this uh wavelengths red and green the green cones and red cones will respond to that. So that's great. Um now I want to do a thought experiment. What if the light intensity so the sun goes down it becomes dusk right? What happens? the cones will shut down and they'll say, "Okay, nigh night night. I'm going nigh night night." So, the rods will pick up the light and the rods are very sensitive to this green. So, they'll still see light, just not the color. The rods are not sensitive to this wavelength. They will not perceive the light coming out of the rose. which means that at dusk the red will be black and the green will be gray lighter gray. So this is the um simulation of how it looks like. The red goes darker much faster than the green. The green remains to be in light gray color colors. So uh this is fascinating right? It's really interesting uh to understand that to understand the limits of our vision. So we can work out a bunch of examples and what is interesting is that can you simulate these conditions in virtual reality and some of them yes some of them no again depending on the screens and technology that we deal with. [Music] [Music] So let's look at what we see at phobia. So this is phobia. It's phobia. So what we see at phobia in daylight and at dusk. So in daylight clearly when we look at things straight we see them in high resolution in color. But as it gets darker and our cones go do not work anymore, they're not sensitive anymore at all to to low light levels. You become blind blind at phobia. So have you ever noticed that? Have you ever experienced that when you for example at night you sleep it's very low light conditions you can see things in gray um scale and you look at some reflection maybe a lighter reflection than the others and you cannot see it if you look straight at it but if you see slightly away if you look slightly away and it hits your periphery then you see it right. So the faint lines can be invisible if you look straight at them but if you look slightly off you you see them. So this is because straight at fobia you're blind at night. Now for uh let's look at for periphery for the rods on periphery. So what do we see on periphery? In daylight you see black white and low resolution. So when you don't look at things, you see them, you think they're in color, but they're actually not. They're actually black and white and low resolution, but the brain creates this illusion that you perceive them in colors because you remember what color they were, right? But you could actually create an experiment and put an object of unexpected color in your periphery. Will you be able to detect that color? The answer is no. It's it's a cool experiment that you can do in VR as well. Now, what happens at dusk and night? the same thing. It's also black and white and low resolution. So, so the rods work um the same kind of in that respect that they always see black and white and they always have low resolution for you. Now, uh there's some people with damaged rods. That's a condition where rods malformed or damaged in some way. So, there are no rods. All you have is just cones. And what does it mean? That means that in daylight you can only see what's called tunnel vision and it's color high resolution. But at night you're effectively blind, right? At night your cones will shut down. You don't have rods. You see nothing. Um during the day it's still quite quite difficult condition to live with because crossing a road can be difficult and you cannot drive without periphery. So periphery is very important for any kind of motion detection that might be important or crucial while driving uh vehicles say or crossing roads. Now something interesting will happen with respect to um red charts or warning lights or stuff like that. So what if you are a captain on a ship or an airplane, right? And you have red instrument lights. So have you noticed that a lot of instrument lights are red?

So what is the reason for that? So what why why do we have that? Well, because if you have low light conditions, you can enter scutotopic modes and your rods will be highly sensitive to low light conditions. However, if you just shine red light around, it will not desynthesize your uh your rods because rods do not respond to red light. So if you have red charts with this wavelength, the rods are not sensitive to that color, which means that the rods can stay in the scootopic mode while you see very bright red lights and you will not leave scootopic mode. That's really really interesting and that's the reason behind red um red warning lights all over. So um so like um but like lighouses right and towers and airplanes they have red warning lights. So that's the reason. Now if you have faint red lights, it might be difficult to to detect them because if raw rods will not respond and cones um do not respond to lowle red lights then um then it's very difficult to detect them overall. So so faint red charts are not really useful unless you look straight at them. But bright red warning lights are pretty pretty good. Okay. So, uh I want to show you an experiment that we did in high-end HMDs where there is enough field of view to uh create this experiment. So, the experiment is the following. You ask the person to look here while you have these objects that will change the color on periphery and you flip the color randomly from one to another. Right? So from uh it's uh red, yellow, blue. So this is your environment. Well, you ask the person just to look at the sphere straight. So this object hits the periphery of the eye in VR. So what happens is that the person sees this. They cannot tell the colors because these are random colors. So there's no way the person uh the the people that we experimented on uh including me could tell what those colors are. You just see brighter gray or less bright gray depending on u what the original color was and that's it. So in highend HMDs you can demonstrating it. We haven't tried it for um cardboards. So a very interesting experiment. Here's an example of some demo that you could implement in uh Cardboards VR as part of your projects would be interesting would be cool in the sense that if you create this kind of APKs then we can pass them along for future courses and anybody could uh see them. So it would be very useful from educational point of view. But I think I'm done for today because we we are done. So hopefully today was more lightweight kind of question uh session because the rods can be very sensitive to very lowle lights. But if you even shine very high level high intensity red light the cones will see it because they'll wake up for that. The red cones wake up for that. But if red cones wake up for that it doesn't mean that these rods will be sensitive like they will not detect that bright light. They will not have to change its sensitivity. So they stay in scootopic mo mode. So it's kind of cool. Very cool effects. [Music]lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

NPTEL Introduction to Virtual Reality — Easy Notes

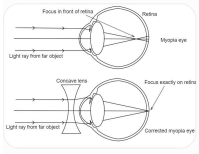
Your lecture covers two big topics:

1. **Light and VR headset optics** — how lenses help us see virtual objects.
2. **Human vision** — how our eyes see light, colours, and objects.

I'll explain them like you're in 7th class, using simple examples, memory tricks, and MCQs.

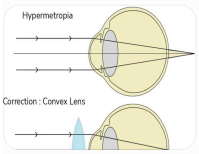
Part 1: Vision defects and corrective lenses

1. Myopia and hyperopia



Myopia (short-sightedness / near-sightedness)

- The eye lens is too strong, or the eyeball is too long.
- Light focuses **before the retina**.
- Distant objects look blurry.
- Corrected using a **concave lens** (negative power).



Hyperopia (long-sightedness / far-sightedness)

- The eye's focusing power is too weak, or the eyeball is too short.
- Light tends to focus **behind the retina**.
- Near objects can be difficult to see clearly.
- Corrected using a **convex lens** (positive power).

Easy memory trick:

- **Myopia → Minus → Concave lens.**
- **Hyperopia → Plus → Convex lens.**

Remember: accommodation means the eye changes its lens shape to focus. Young people with hyperopia may compensate for it using accommodation, but myopia cannot be corrected by squeezing the eye lens more.

2. What are reading glasses?

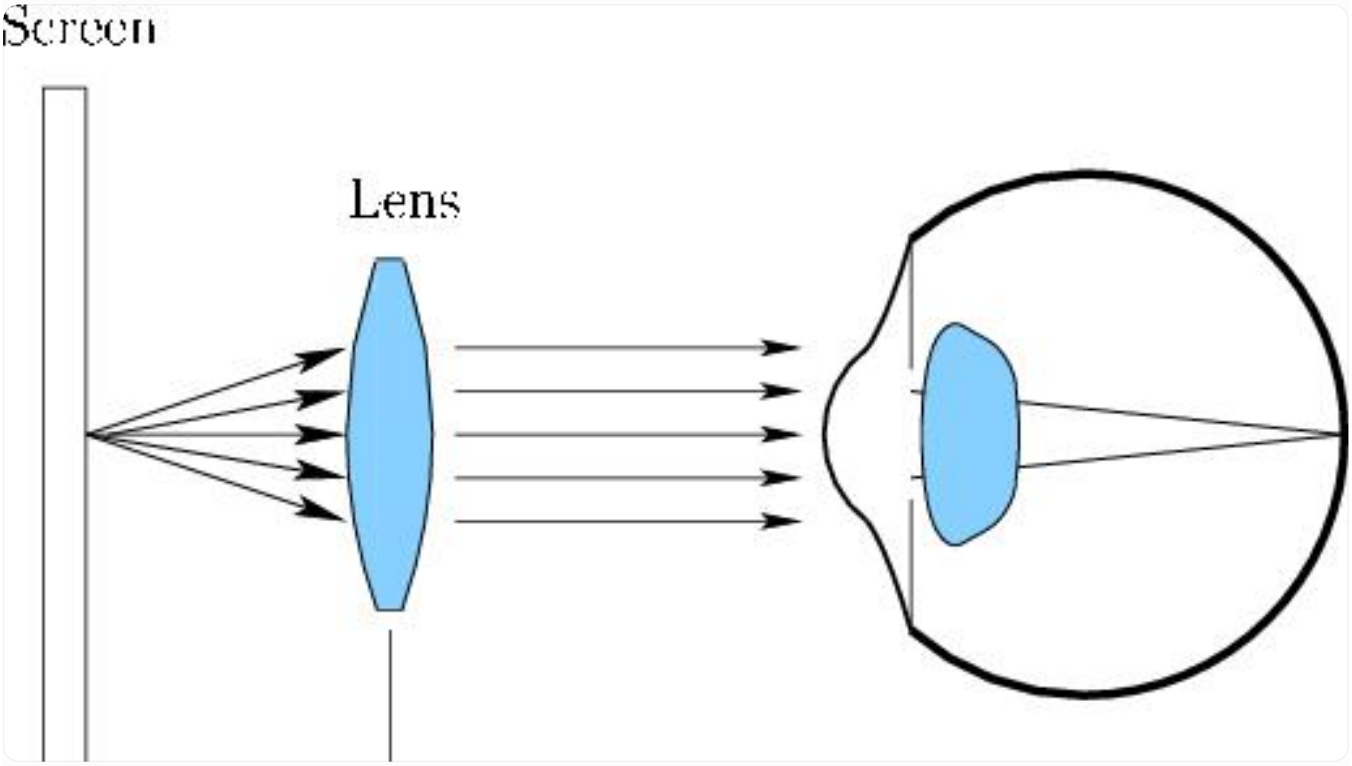
Reading glasses usually use a **convex lens** to help focus on nearby text, especially when the eye can no longer accommodate enough.

For example, imagine reading a book. The convex lens helps the light reach the retina correctly, so the letters appear sharp.

Your lecture also explains an interesting point: when a book is placed at the focal distance of a reading lens, the outgoing rays can become parallel. The eye can then see a sharp, slightly magnified virtual image that appears optically at infinity.

Part 2: How do VR headset lenses work?

3. Why does a VR headset need lenses?



A phone screen placed just a few centimetres from your eyes is normally too close for comfortable focusing. A VR headset uses lenses to magnify the screen and make the virtual image appear farther away.

In the simplified headset model from your lecture:

1. The display is placed at the lens's focal distance.
2. The lens makes the outgoing light rays parallel.
3. The eye receives parallel rays and, assuming normal vision, focuses them on the retina without needing accommodation.
4. The user sees a magnified virtual image that appears optically at infinity.

Important exam point: The virtual object can look near or far inside VR, but in this simplified model, the light reaching the eye is parallel for every virtual depth.

4. What is vergence–accommodation mismatch?

This is one of the most important concepts in your lecture.

There are two different actions:

- **Vergence:** Both eyes turn inward or outward to look at an object at a particular distance.
- **Accommodation:** The eye lens changes shape to focus the object clearly.

In real life, these actions usually work together. In the simplified VR headset, the light rays remain parallel, so accommodation stays relaxed even when your eyes turn inward to look at a nearby virtual object.

That difference is called **vergence–accommodation mismatch**.

Memory trick

Vergence = eyes turn.

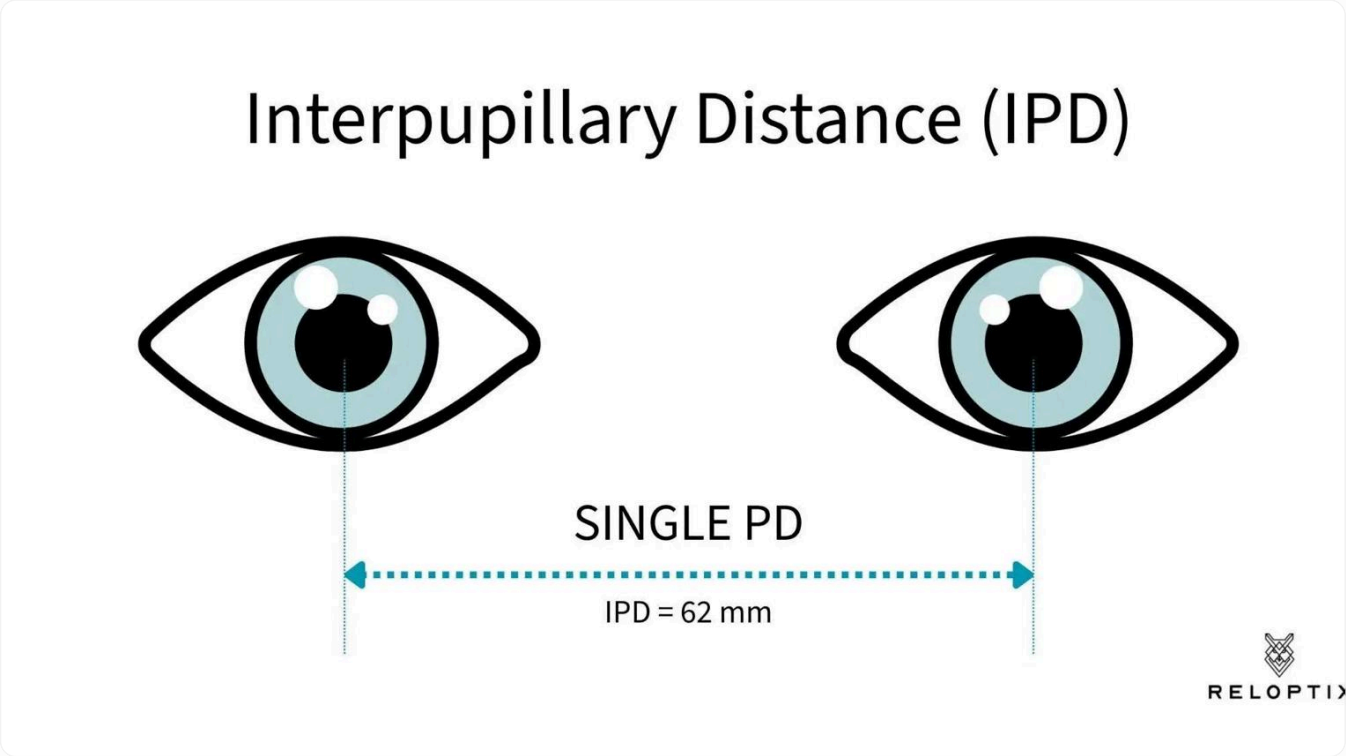
Accommodation = lens changes shape.

VR mismatch = eyes turn for virtual depth, but focusing stays nearly constant.

This mismatch can contribute to eye strain or discomfort in VR.

5. What is IPD?

IPD stands for **Interpupillary Distance**. It is the distance between the centres of your two pupils.



Why is it important?

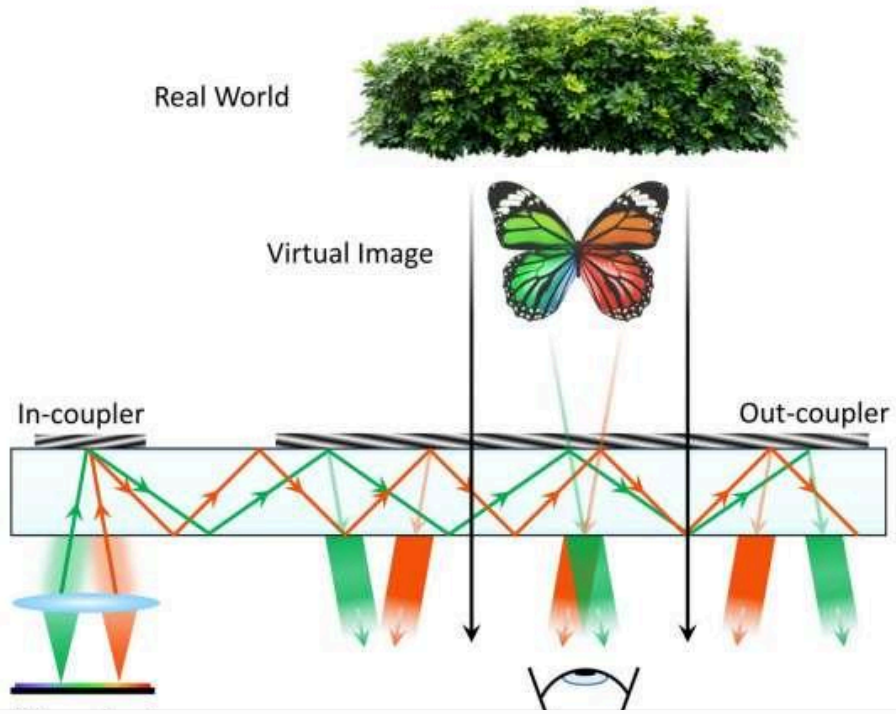
- Your pupils should line up with the optical centres of the headset lenses.
- If they do not line up, you may experience more distortion or a less clear image.
- Some headsets let you adjust lens spacing using a hardware control.
- Software can also adjust the displayed images, but software adjustment is not always a substitute for correct physical alignment.

Memory trick: IPD = *In-between Pupils Distance*.

Cardboard-style headsets often have fixed lens spacing, so they may not fit everyone's IPD equally well.

Part 3: Different optics used in VR

6. What are waveguides?



A **waveguide** is a special optical structure that guides light through a thin lens or transparent surface.

Think of it like a road for light: light enters, travels through the optical structure, and is directed towards your eye.

Waveguides are used in some augmented-reality glasses, including devices in the HoloLens family.

Your lecture also mentions **virtual retinal displays**, which use scanned light beams to create an image on the retina. Some systems use lasers and tiny moving mirrors called MEMS mirrors.

For the exam, remember:

- **Waveguide:** Guides light through an optical structure towards the eye.
- **Virtual retinal display:** Uses scanned light to form an image on the retina.
- Exact proprietary designs may vary between manufacturers.

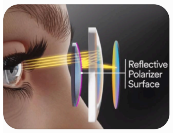
7. Types of lenses

Spherical lens

A conventional curved lens used to bend and magnify light. It can be relatively simple to manufacture, but it may produce optical aberrations.

Fresnel lens

Has many concentric steps or grooves. It can be thinner and lighter than a conventional lens with similar optical power, but the grooves can scatter light and cause glare or reduced contrast.



Pancake optics

Uses folded light paths and multiple optical elements to make a headset thinner. The trade-offs can include light loss, complexity, and cost.

What is a trade-off?

A trade-off means improving one feature may make another feature worse.

For example:

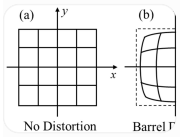
Design choice	Benefit	Possible drawback
Thicker, stronger lens	Can help make a headset compact	More weight or distortion
Thinner lens	Less weight	May need a larger optical arrangement
Fresnel lens	Lighter lens	Glare and reduced contrast
More complex optics	Can improve some image qualities	Higher cost and manufacturing difficulty

Exam tip: There is no single lens design that automatically gives the best weight, size, field of view, clarity, comfort, and cost all at once.

Part 4: Optical aberrations

An **aberration** is an imperfection in the image produced by an optical system.

8. The four important aberrations

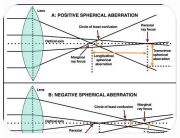


1. Optical distortion

Straight lines appear curved or a grid looks warped.

- Barrel distortion: lines bow outward.
- Pincushion distortion: lines bow inward.

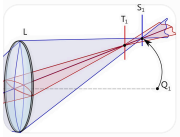
Memory: Distortion = *shape changes*.



2. Spherical aberration

Rays passing through the centre and edges of a spherical lens may focus at different distances, making parts of the image blurry.

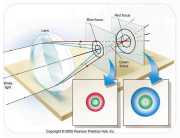
Memory: Spherical = *different focus for centre and edge rays*.



3. Astigmatism

Light in different orientations, such as horizontal and vertical, focuses differently.

Memory: Astigmatism = *different directions, different focus*.



4. Chromatic aberration

Different colours bend by different amounts, so they focus at different locations. Coloured fringes may appear near image edges.

Memory: Chromatic = *colours separate*.

Your lecture also mentions **coma**. It is an aberration that can make off-centre points appear blurred or comet-shaped, especially towards the periphery.

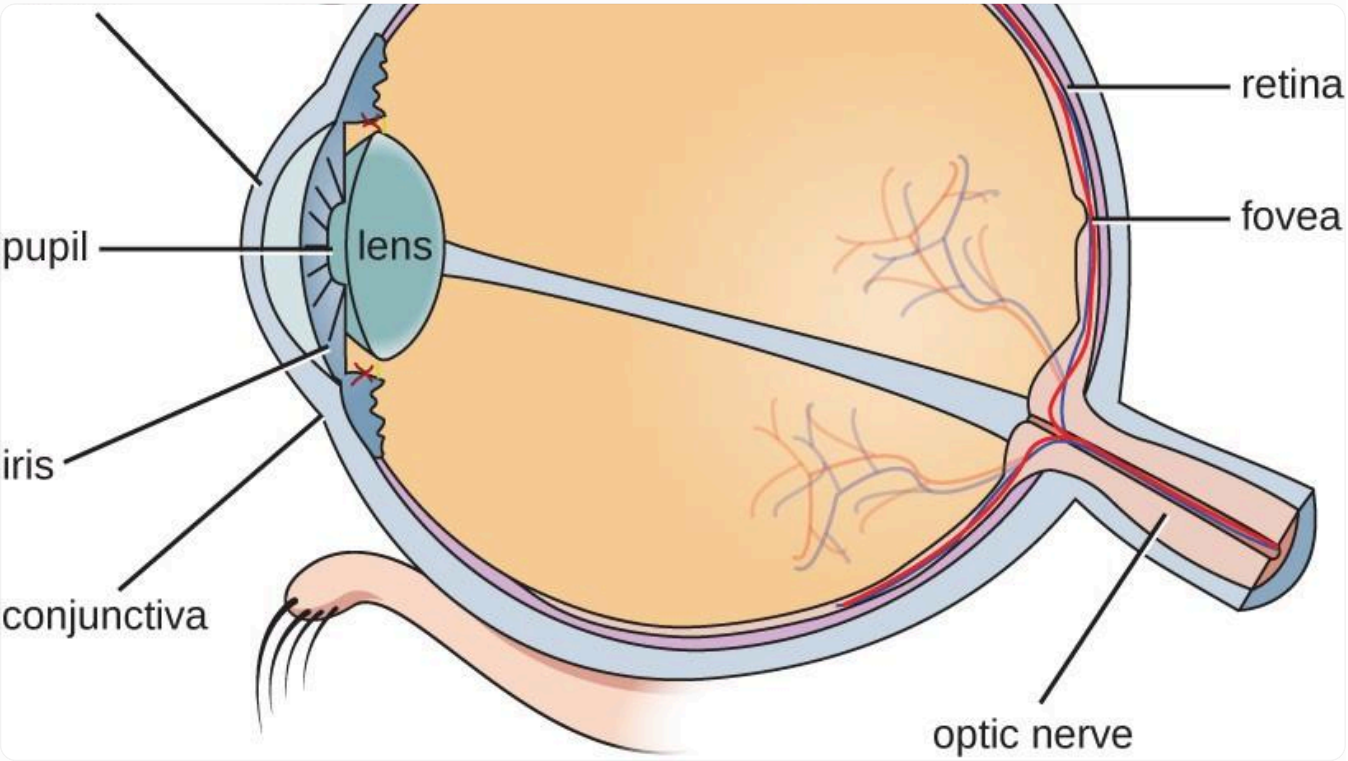
How can aberrations be corrected?

- **Software correction:** Predistort the image so that the lens makes it look correct.
- **Hardware correction:** Use specially designed optical elements or lens combinations.
- **Astigmatism in a person's eyes:** Often corrected with a suitable prescription lens.

One important detail: software correction can reduce certain aberrations, but it cannot necessarily eliminate every optical problem.

Part 5: How does the human eye see?

9. Follow the path of light through the eye



Light travels through these main parts:

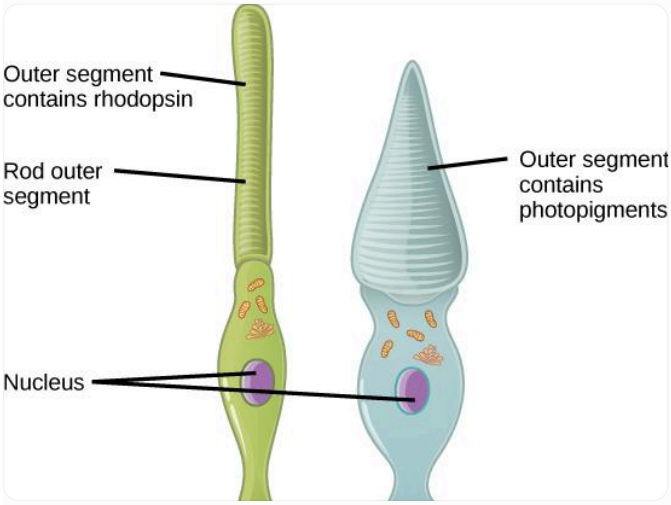
1. **Cornea:** The clear front surface of the eye. It bends incoming light.
2. **Aqueous humour:** Clear fluid in the front part of the eye.
3. **Pupil:** The opening through which light enters.
4. **Lens:** Helps focus light on the retina.
5. **Vitreous humour:** Gel-like material inside the eyeball.
6. **Retina:** Contains light-sensitive cells that detect light.
7. **Optic nerve:** Carries visual information towards the brain.

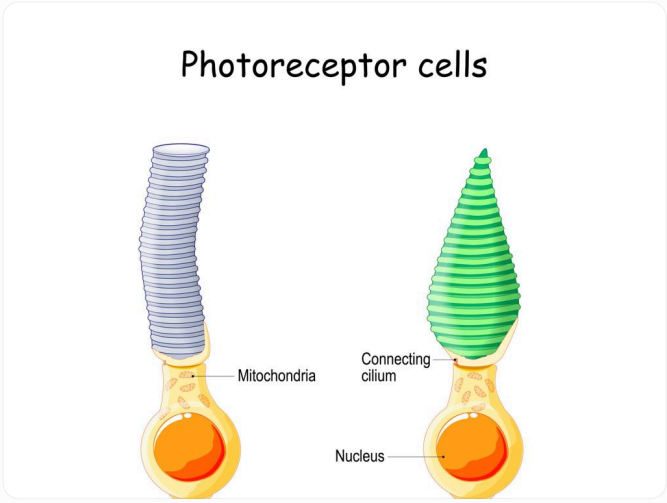
The brain processes the information so that we perceive the world correctly.

Memory trick: Cornea → Pupil → Lens → Retina → Optic nerve.

10. What are rods and cones?

The retina contains two main types of photoreceptors. Photoreceptors are cells that respond to light.





Rods **Cones**

- Work very well in dim light.
- Do not provide colour vision.
- Mainly support black-and-white vision.
- Numerous in the peripheral retina.

- Work best in brighter light.
- Provide colour vision.
- Support high visual detail.
- Concentrated heavily in the fovea.

The lecture gives approximate numbers per eye:

- Rods: **120 million**
- Cones: **6 million**

That means there are many more rods than cones.

Memory trick:

- **Rods = darkness and grayscale.**
- **Cones = colour and clarity.**

11. What are photopic, mesopic and scotopic vision?

These are the three vision modes based on the amount of light available.

Photopic vision — Daylight

Bright light

- Cones dominate.
- We see colours clearly.
- Visual detail is high.

Example: Reading a book in daylight.

Mesopic vision — Twilight

Medium light

- Both rods and cones contribute.
- Colour and detail become less reliable as light decreases.

Example: Seeing the surroundings at dusk.

Scotopic vision — Night

Very dim light

- Rods dominate.
- Colour vision is essentially absent.
- We can detect objects in very dim conditions, but fine detail is poor.

Example: Seeing a faint object on a dark night.

Memory trick:

P-M-S = Plenty, Medium, Scarcely any light.

- Photopic = bright.
- Mesopic = in between.
- Scotopic = dark.

How long does dark adaptation take?

According to your lecture:

- Moving from bright conditions into darkness can take about **35 minutes** for full scotopic adaptation.
- Adapting from darkness to bright conditions is faster, around **10 minutes** in the lecture's simplified comparison.

These are approximate figures, not exact timings for every person and situation.

12. Why does a red warning light help at night?

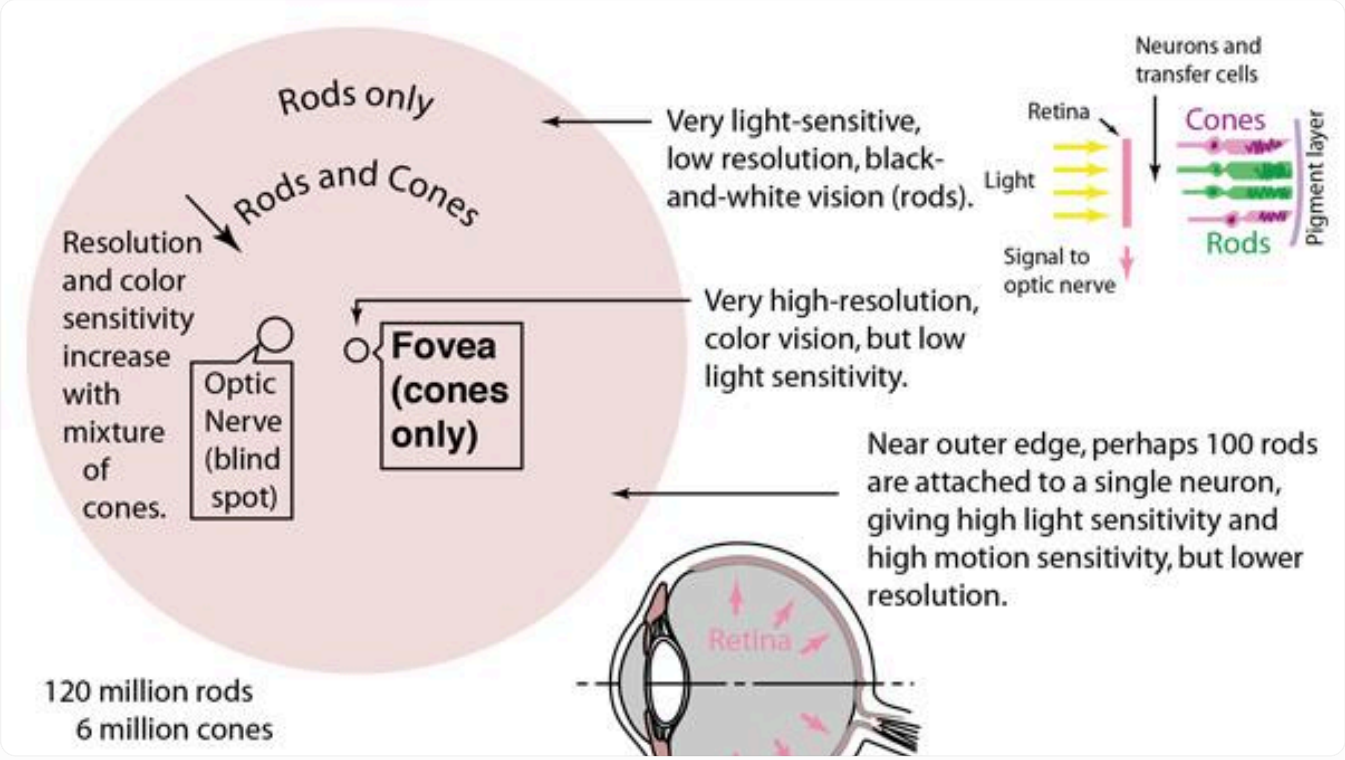
Rods are much less sensitive to red light than to blue-green light.

That is why dim red lighting is often used on aircraft or ship instrument panels at night: it can help preserve dark adaptation while allowing instruments to be seen.

One qualification: bright enough red light can still stimulate cones and affect vision. Red light does not guarantee that dark adaptation is completely unaffected.

Memory trick: *Dim red light helps preserve night vision.*

13. What is the fovea and what is the periphery?



- **Fovea:** The central retinal region used when you look directly at something. It provides the sharpest, most detailed colour vision in bright light.
- **Periphery:** The retinal areas away from the centre. They are particularly useful for noticing movement and detecting things around you, but provide less detail and less reliable colour information.

Imagine looking straight at a ball while another object moves beside you. You can often notice the movement without turning your eyes towards it. That is the usefulness of peripheral vision.

The retina also has a **blind spot**, where the optic nerve exits the eye. This area has no photoreceptors. Your brain usually fills in the missing information, so you do not normally notice it.

Memory trick: *Fovea = fine detail; periphery = movement and surroundings.*

14. Why does a red rose look dark at dusk?

In bright daylight, cones detect colours, so a red rose and green leaf appear red and green.

As the light becomes dim:

- Cone activity decreases.
- Rods become more important.
- Rods are more sensitive to blue-green light than red light.
- The red rose can appear much darker than the green leaf.

At very low light levels, colour perception largely disappears.

This is an example of how the eye's sensitivity changes with lighting conditions.

15. Why can't VR perfectly reproduce every real-world lighting condition?

A VR headset has a display with limited brightness, contrast, colours and other physical capabilities. The human eye can adapt to a very wide range of light levels.

Consequently, a headset cannot reproduce every real-world lighting condition and every vision mode perfectly. For example, making the display black does not automatically reproduce all the effects of true night vision.

Also, the eye and brain process detail, motion, colour and brightness in ways that a display cannot perfectly imitate.

Final exam revision sheet

Concept	One-line answer to remember
Myopia	Image focuses before retina; concave lens corrects it
Hyperopia	Image tends to focus behind retina; convex lens corrects it
Accommodation	Eye lens changes shape to focus
Vergence	Eyes turn to look at an object
Vergence–accommodation mismatch	Eye turning and focusing cues disagree
IPD	Distance between the centres of the pupils
Waveguide	Guides light through an optical structure
Fresnel lens	Grooved lens that can reduce weight
Optical distortion	Straight lines appear curved
Spherical aberration	Centre and edge rays focus differently
Astigmatism	Different orientations focus differently
Chromatic aberration	Different colours focus differently
Rods	Dim-light vision, largely without colour
Cones	Colour vision and fine detail in bright light
Photopic	Daylight vision
Mesopic	Twilight vision
Scotopic	Very dim-light vision
Fovea	Sharp central vision
Periphery	Wider awareness and movement detection
Blind spot	Retinal area without photoreceptors
Dark adaptation	Eyes gradually become more sensitive in darkness

MCQs for practice

Try these without looking at the notes first. Select an option to see whether you are correct and why.

1. A person cannot see distant objects clearly. The image focuses before the retina. What is the defect?

- A. Hyperopia
- B. Myopia
- C. Astigmatism
- D. Colour blindness

2. Which lens usually corrects hyperopia?

A. Concave

B. Convex

C. Flat glass only

D. Fresnel only

3. What does IPD mean?

A. Image Processing Display

B. Inter-pupil Distance

C. Internal Power Difference

D. Image Position Direction

4. In the simplified VR headset model, why does the eye need little or no accommodation?

A. The display is black

B. The outgoing rays are parallel

C. The eyes stop working

D. The pupil disappears

5. What is vergence–accommodation mismatch?

A. Both eyes are closed

B. The screen has no pixels

C. The eyes turn for depth but focusing does not change accordingly

D. The lens is always concave

6. Which optical effect makes colours focus at different locations?

A. Barrel distortion

B. Chromatic aberration

C. IPD

D. Vergence

7. Which photoreceptors are most useful in very dim light?

A. Cones

B. Rods

C. Pupils

D. Optic nerves

8. Which vision mode dominates in bright daylight?

A. Scotopic

B. Mesopic

C. Photopic

D. Blind-spot vision

9. Which part of the retina provides the sharpest central vision?

- A. Blind spot
- B. Fovea
- C. Optic nerve
- D. Outer edge only

10. Why can a red rose appear darker than a green leaf at dusk?

- A. Rods are more sensitive to red
- B. Rods are more sensitive to blue-green light than red
- C. Cones become more active in darkness
- D. Red light travels backwards

11. What is a main advantage of a Fresnel lens in a VR headset?

- A. It removes every aberration
- B. It can reduce lens thickness and weight
- C. It creates rods and cones
- D. It eliminates IPD

12. What is the blind spot?

- A. The sharpest part of the retina
- B. A place where there are no photoreceptors
- C. The pupil in darkness
- D. The centre of the lens

Score: 0 / 12 Retry quiz

Best way to remember this chapter: First memorise the three pairs — *myopia/concave, hyperopia/convex, rods/night and cones/day* — then learn the VR mismatch and the four aberrations. Those connections make many exam questions much easier.

You said:

Quiz - 5 - 2026.do…
PDF

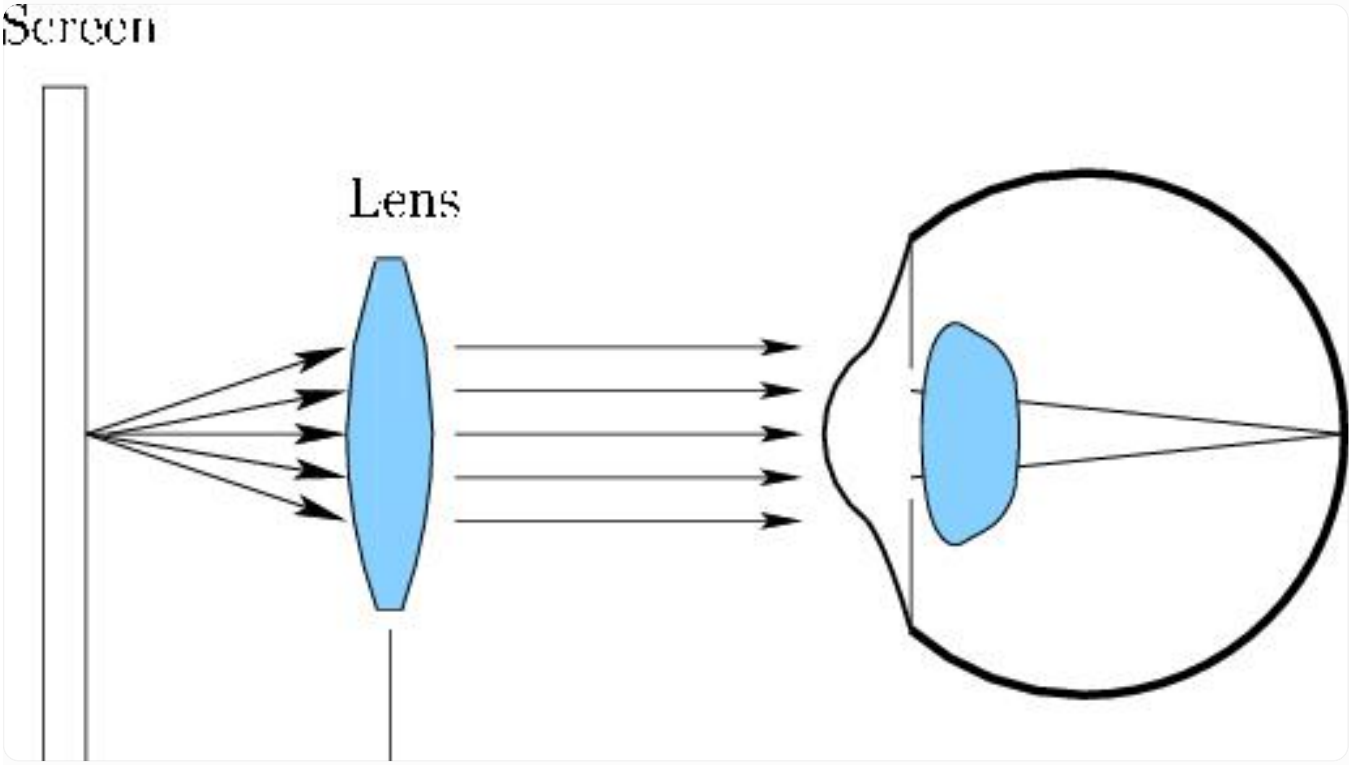
explain easily

ChatGPT said:

NPTEL VR Quiz 5 — Easy Explanation

I'll explain all 10 questions from your PDF in **very simple English**, with the correct answer, the reason, and a small memory trick to help you remember for the exam.

Question 1: Why does nearby VR viewing cause discomfort?



Correct answer: A. Vergence–accommodation mismatch

Imagine you are looking at a nearby object in VR.

- **Vergence:** Your two eyes turn inward or outward to look at an object.
- **Accommodation:** Your eye lens changes shape to focus on an object.

In a normal VR headset, the eyes may turn as if an object is near, but the eye lens keeps focusing at almost the same distance.

This difference can cause eye strain or discomfort.

Remember: Eyes turn for one distance, but focus stays at another = mismatch.

Question 2: How can a headset be made thinner?

Correct answer: B. Use a stronger convex lens with a shorter focal length.

A convex lens helps magnify the display so you can see it through the headset.

- Shorter focal length → stronger lens.
- A stronger lens can allow the display to be closer to the lens.
- This can make the headset thinner, but the lens may become thicker and heavier and may introduce more distortion.

Remember: Shorter focal length = stronger lens.

Question 3: Does perfect tracking remove eye discomfort?

Correct answer: B. Prototype B can still exhibit vergence–accommodation mismatch.

An IMU is a sensor that measures movement, such as the movement of your head or controller.

Even if the IMU tracks perfectly, the headset's optical system can still cause discomfort.

Remember: Tracking controls movement; optics affect vision.

Question 4: Lens power and IMU samples

Correct answer: D. Lens power = +25 D; 1000 samples collected.

This question has two small calculations.

Part A: Calculate lens power

Formula:

$$P = \frac{1}{f}$$

Here, f must be in **metres**.

Given focal distance:

$$f = 4\ \text{cm} = 0.04\ \text{m}$$

Therefore,

$$P = \frac{1}{0.04} = 25\ \text{D}$$

A convex lens has positive power, so the answer is **+25 D**.

Part B: Calculate IMU samples

Formula:

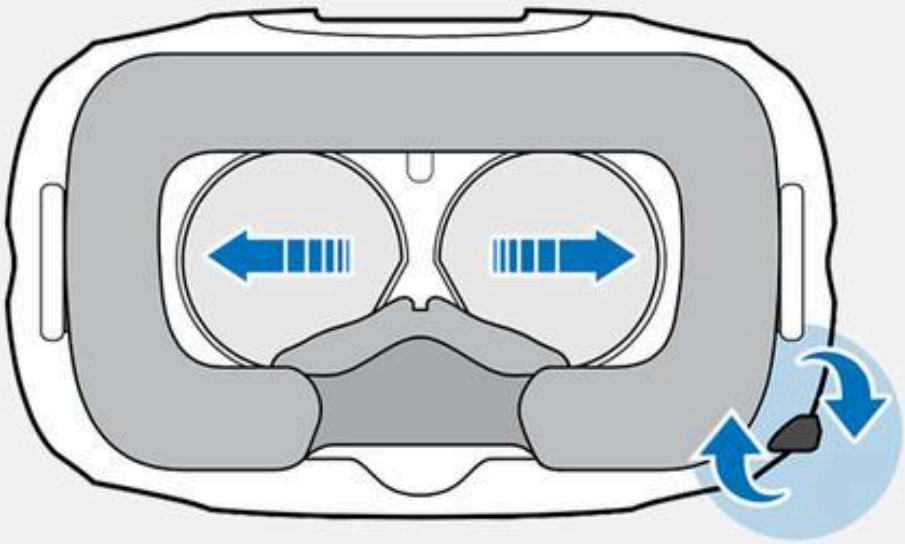
$$\text{Samples} = \text{Sampling rate} \times \text{Time}$$

$$= 250 \times 4 = 1000$$

So the answer is **D**.

Remember: Convert centimetres to metres before calculating lens power.

Question 5: What happens when IPD does not match?



Correct answer: A. The eyes look through peripheral regions of the lenses, increasing optical aberrations.

IPD means *Interpupillary Distance* — the distance between the centres of your two pupils.

In this question:

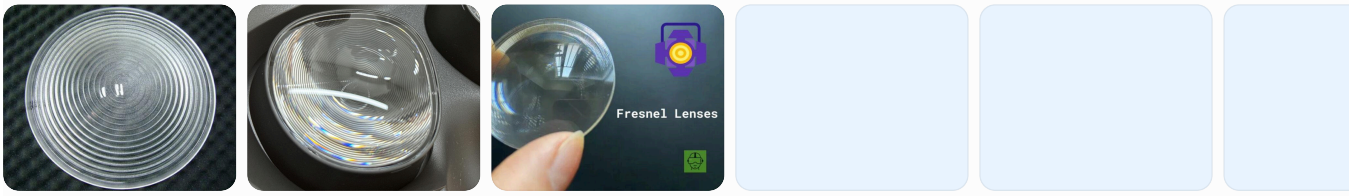
- Headset IPD = 63 mm
- User's IPD = 68 mm
- Difference = 5 mm

The lenses and eyes do not align properly. The eyes may look through less-optimal parts of the lenses, causing more distortion and poorer image quality.

Remember: Wrong IPD → poor eye–lens alignment → more distortion.

Question 6: Fresnel lenses

Correct answer: B. Reduced weight but increased glare and reduced contrast.



A Fresnel lens has small circular grooves instead of the full thickness of a conventional lens.

Advantages:

- Thinner and lighter.
- Useful for compact VR headsets.

Disadvantages:

- Grooves can scatter light.
- Bright streaks called *god rays* may appear.
- Image contrast can decrease.

Fresnel lenses do not eliminate vergence–accommodation mismatch.

Remember: Fresnel = lightweight, but possible glare.

Question 7: Pancake optics

Correct answer: B. They use two optical paths with different focal lengths that are rapidly alternated.

The important idea in the lecture is that pancake optics **fold the light path** using multiple optical elements and polarization-dependent reflection and transmission.

Think of folding a long road into a compact shape. The road remains long, but it occupies less space.

Similarly, pancake optics allow a headset to have a longer optical path in a smaller physical space.

Advantages: Thinner and more compact headset.

Disadvantage: Some light is lost, so a brighter display may be needed.

Remember: Pancake optics = folded light path = thinner headset.

Question 8: How can chromatic aberration be reduced using software?

Correct answer: C. Pre-distort the individual color channels before rendering.

Chromatic aberration happens because different colours of light can bend differently through a lens.

For example, red, green and blue may not line up perfectly, producing coloured edges around objects.

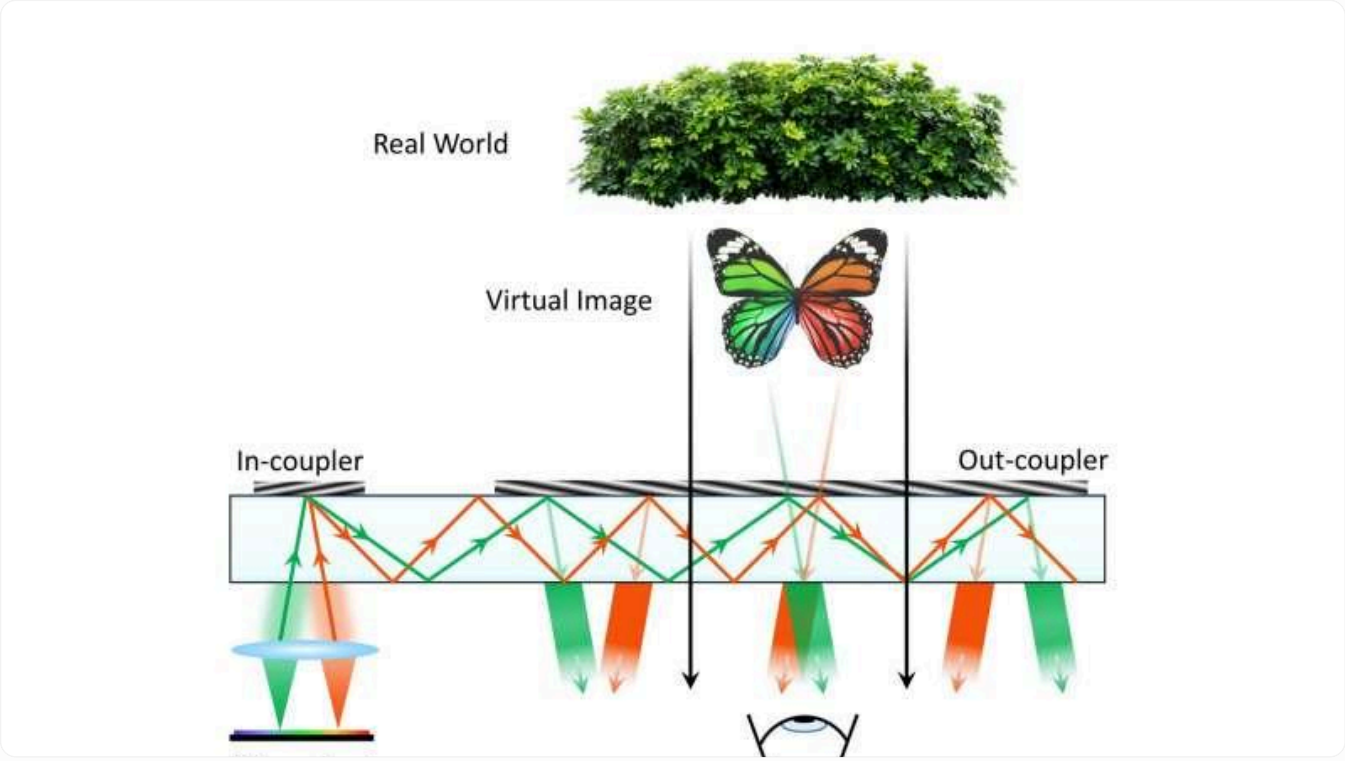
A software solution is to adjust each colour channel before displaying the image.

The lens then bends the light, helping the colours line up more closely on the retina.

Remember: Chromatic aberration = colour mismatch; correction = adjust RGB channels.

Question 9: Waveguide optics

Correct answer: B. They guide light using optical structures and reflections before reaching the eye.



A waveguide is a structure that guides light along a particular path.

In some AR headsets, light travels through an optical structure using reflections before reaching the eye.

Examples discussed in the quiz include HoloLens and Magic Leap.

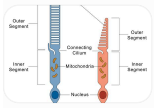
Remember: Waveguide = guides light towards the eye.

It does not remove every optical aberration.

Question 10: Rods and cones in different lighting

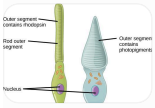
Correct answer: C. Cones → Mixed rods and cones → Rods.

Your eyes have two main types of light-sensitive cells.



Rods

Work well in dim light. They mainly help you see brightness and shapes, rather than colour.



Cones

Work best in bright light. They help you see colours and fine details.

Lighting condition	Type of vision	Main photoreceptors
Bright daylight	Photopic	Cones
Twilight	Mesopic	Both rods and cones
Moonless night	Scotopic	Rods

Remember: Day = cones; twilight = both; night = rods.

Quick revision sheet

Q.	Topic	Correct answer	Key memory point
1	VR eye discomfort	A	Vergence changes, focus stays nearly constant
2	Thinner headset	B	Stronger convex lens, shorter focal length
3	Perfect tracking	B	Tracking cannot eliminate optical mismatch
4	Lens power and samples	D	$1/0.04 = 25$ D; $250 \times 4 = 1000$
5	IPD	A	Wrong alignment increases aberrations
6	Fresnel lenses	B	Light and thin, but may cause glare
7	Pancake optics	B	Folded light path
8	Chromatic aberration	C	Pre-distort RGB channels
9	Waveguides	B	Optical structures guide light
10	Rods and cones	C	Day: cones; twilight: both; night: rods

Test yourself

1. Which phenomenon can cause discomfort when eye vergence changes but accommodation stays almost constant?

- A. Chromatic aberration
- B. Vergence–accommodation mismatch
- C. Astigmatism
- D. IPD tracking error

2. A lens has a focal length of 4 cm. What is its power?

- A. +4 D
- B. +10 D
- C. +25 D
- D. +40 D

3. An IMU samples at 250 Hz for 4 seconds. How many samples are collected?

- A. 250
- B. 500
- C. 750
- D. 1000

4. Which optical design folds the light path to make a headset more compact?

- A. Pancake optics
- B. Fresnel optics only
- C. Waveguide-free optics
- D. Concave-only optics

5. Which photoreceptors dominate vision in very dim light?

- A. Cones
- B. Rods

C. Both equally in all conditions

D. Neither

Try again

Study the **formulas, optical trade-offs, and rods-versus-cones table** carefully. These are the main facts to remember from this quiz.

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[Music] So whatever um whatever that size is and whatever that magnification factor is due to the lens it would have to be right now they are about 2,000 Sorry, let me start it over. For VR displays, as we'll see in a bit, you would have to have 2,000 pixels per inch. But what we have now is not that because 2,000 uh pixels per inch would be equivalent to 8K * AK per I. But the best that we have so far is 4K times 4K per I but even less so for regular high-end HMDs. So um here this is 4K the what we have the best ones 4K* 4K per eye. Okay. So what are we concerned in the quality of visual displays in virtual reality? So there are three factors that determine how good a display is. First is the spatial resolution that we just talked about. So how many pixels are enough so that there's no aliasing anymore, right? So that you don't perceive an individual pixel structure or so that you have this what's called retinal display. So we just did this calculations and 8K * 8K would be corresponding to 30 cycles per um per degree in virtual reality displays. And um that's not what we uh have so far. Those kind of displays are not available yet. But for younger adults, you would have to have double of that. So the second factor is light intensity, resolution, and range. So this could be called like color resolution, right? because we have the the displays have this RGB sub pixels and all of them have some kind of intensity. So you could think of this requirement of how good a VR display is is how many light intensity values need to be in VR to reproduce natural light conditions. And remember that um our photo receptors can spend many orders of magnitude of light intensity. So we just had we just talked about it in the last lecture. However, modern displays will only provise 256 intensity values per per each channel RG&B. Right? So um there might be so even though it's possible that we can enter scootopic and misopic u modes in VR it may be that there will be not enough content that we can recreate with this uh few intensity levels in VR displays. So there may be not much to do there. So we cannot recreate night conditions in real world in VR right it will be very very limited. So finally the third um parameter of display quality would be temporal resolution right so that's extremely important to trigger our motion detection and perception so how many frames per second is enough to reproduce motions of real world on the VR display and this will be the goal of uh what we'll start now we'll start this lecture and next lecture so we will talk about eye motions. So in order to understand how is it that we perceive motion on displays which are not moving right our phones are not moving no content on it the phones are not really moving just the pixels change right so how does it trigger perception of motion as so so that we perceive or we have this illusion of things moving on the screen while they're really not it's just pixels turning on and off and changing the intensity so we need to understand eye emotions for that that becomes extremely important. We need to understand neurohysiology or how what are the brain structures that will let us make this conclusion that something is moving in the world. So neural structures that detect motion across retina and finally we need to talk about motion perception as well. So that's what we'll talk about next. We have 15 minutes so we can start on that. So we'll start with eye movement. So let's have a little bit of a look of this at this move um at this video and you were talking about eye tracking. So start thinking about the challenges of eye tracking as we go along. >> The eye sits within the bony orbit. The orbit has a roof, a floor, a medial wall and a lateral wall. It is therefore somewhat pyramidlike in shape. The two orbits have medial walls that run parallel to one another. The lateral walls are at about 45° to the medial wall or 90° to one another. The eye sits within the middle of this space with the four recti muscles converging on the orbital apex where they insert into the annulus of zin, a fibrous ring of tissue that encircles the optic nerve. These recti are superior rectus, medial rectus and inferior rectus all supplied by the ocular motor or third cranial nerve and lateral rectus supplied by the abdusens or sixth cranial nerve. The inferior oblique is also supplied by the ocular motor nerve while the superior oblique is supplied by the tleia or fourth cranial nerve. Contraction of the lateral rectus abducts the eye while contraction of the medial rectus adducts the eye. The two opposing muscles are described as being antagonists. >> So, uh you can you can if you're interested to uh learn more about the physiology of the muscles and how the eye moves, this is an excellent video. But let's see what we can already use uh for our purposes. So the eyes uh all this video explains the eye motions and they are they performed using the six eye muscles which work in pairs. So uh these eye the six muscles are the ones that enable the eye movement and you can think of the movements then as yuppet roll for each

eye right resulting in three degrees of freedom. Of course, in most cases, you just want you can just approximate it as two degrees of freedom only yaw and pitch. So, we don't really talk about the role of the eye too much. However, it's really important to know that it's there because again for for various purposes, it will be important. So, why is it important? So, what are the implications for virtual reality? Well, we if we want to develop realistic avatars, so we we will want to model proper eye movements. Otherwise, the avatars might look creepy or unreal, and that's called uncanny valley. Also, to develop more realistic fiated rendering, as we were talking about it a little bit, you might need to track your eyes. And it's important to understand how they move. And sometimes they move unpredictably or they move in a way that are difficult to def detect or they move extremely fast as you'll see in a bit. So also as the eye move we end up looking through the periphery of the lens inside of the HMDs and this results in more distortions and aberrations. So that will affect how we view the content. Um so in general as you'll see our eyes the mo the motions of our eyes really interfere with what we really perceive in virtual reality. [Music] [Music] So let's look for let's continue. Notice that when the light enters the surround region on this on center ganglen cell, the level of activity recorded in the cell decreases. Conversely, a spot of light in the center of the receptive field increases the firing rate. A maximal response in an on center ganglen cell is achieved when the entire center of the receptive field is illuminated. Likewise, if we illuminate only the surround with a ring of light, the ganglen cell is maximally inhibited. Note that if we illuminate both the center and surround region, the response is just above baseline. This occurs because center effects are slightly stronger than surround. Now let's look at the response. So again in textbooks you can see this the following illustrations and they just show where on the same receptive field depending on where the light hits the neuron will be activated in different ways and this on center ganglen cells that are maximally activated when the light is right in the middle of the receptive field. uh and they are massively inhibited if the light hits the periphery or the surround but not the center. So now similarly there are also offcenter ganglen cells and they respond to kind of the opposite effect. So let's see how is what what >> of an off-center ganglen cell to see what happens when a spot of light enters the receptive field. Use the mouse to position the beam of light so that it falls on either the center or surround of the receptive field. Notice that when the light enters the surround region of this off-center ganglen cell, the level of activity recorded in the cell increases. Conversely, a spot of light in the center of the receptive field decreases the firing rate. If we illuminate the entire center of an off-center ganglen cell, the cell is maximally inhibited. A maximal response is achieved when the entire surround of the receptive field is illuminated. So now let's look at an example and what this model of um or what this allows us to perceive or humans to perceive with having on center and offc center um uh gangland cells is to um achieve edge detection. Right? So we we're really good at detecting edges between uh different contrasts. So if this is what the raw image that our retina receives. So this is just the photons that would hit our photo receptors. Then the neural image will be something like this. So again think of it as your retina with corresponding um with the receptive fields for ganglen cells. And I have a question for you. If I start drawing receptive fields, so which photo receptors actually um are responsible for which part of the image and then they'll be connected to one ganglen cell. So my question to you this red with red and green what they would would correspond to? Which ganglen cells would they correspond to? Would they correspond to um on center or off center? So based on the videos that we just saw the maximal activation of on center and off center um gang cells. So can you can you infer which one is green and which one is red? Yeah, green on center. No, green on center, red off center. Right. So let's look at it very closely again. So if we are looking at the receptive field of green so it will be maximally it will be maxim probably raw image would be much better. So this um what corresponds to green would be maximally activated because there is light in the center. Yes, there is a little bit of light on the periphery as well or surround, but there's most light in right in the center. While for these guys on both sides, there's a little bit of light on surround, but nothing in the center. So, they'll be maximally uh maximally activated then. So, the red ones have to be off center and the green ones have to be on center. Yeah. So, this is the correct answer. So now I have a question for you. So if let's have a little bit of a quiz. So we have this on center receptive field. So the the on center um ganglion what will be the what will be the highest activation? What will which one of these will result in the highest activation rate? So can you tell me? So if the light covers the entire receptive field. Yeah. Yeah. So this is just by it follows immediately from what we saw. So this example, right? When the light fills the center portion. That's good. This is a little more complicated question. So what if we have an off center receptive field it will maximally respond to which one here? Can you guess? So rectangle-shaped light that covers 25% of the field, 50% of the field, 75 or circle. So I asked this question, it probably will be hard for you to figure out, but there is this app that I found which is a really cool illustration of how the ganglion cells work. So if this is the receptive field of a ganglian cell, let's talk about edge detection not dot because dot at the center we kind of know that okay that the on center will be maximally activated with the uh bright when the center is covered but the off center will be maximally inhibited but we're talking about edges so we and the off center neuron. So what if the edge will be circle? We don't want circle. We just want to be more occupying 25% of the receptive field, 50% of the receptive receptive field or 75 of the receptive field. And then the correct answer will be 25. Right? So for the off center it really has to be 25 the rectangle that covers 25% of the receptive field so that it doesn't get the center then it will be maximally u activated. Right? So, what did you guys answer? Aha. So, the correct answer is this rectangle shaped light that covers 25% of the receptive field, right? And we just saw it on this demonstration. So, if you increase if you get a little

bit of the center, it will be minimal. So, you really want to get as much of the off center without touching the on center. And for rectangle, that's really hard to achieve. Of course, if you had the circle for the off center, then well, you just really don't want to touch any center for this to be active. So, no, entire receptive field will be wrong too. As you can see, if you cover entire receptive field, it's still minimally activated. So no because you really don't want to touch any of the center essentially. So no center should be uh perceiving receiving any light. Okay cool. So then we'll go to I want to finish the I have very little time so I want to finish some thought on this neuroscience. So how these experiments are done but before that let me tell you about hierarchical processing. So you already saw that multiple photo receptors will send the information to uh a few of the ganglen cells or bipolar cells and then ganglen cells. So uh some filtering already happens there and one ganglen cell might have um more photo receptors involved in its activation than a bipolar cell. But then that information gets sent to the visual cortex and by the time it reaches a neuron here this the receptive field of this neuron may be much more of the retina or even the entire retina or even information from both retinas because of thisalamus that will combine this this is kind of a relay station that will combine information from both retinas. So there is this kind of um hierarchical processing that's going on right. So further and further neurons will be responsible for more and more of the um information from the retina larger part of the retina. So how are these experiments done? Some you you guys asked how it's done. So here is the illustration. You insert an electrode next to some neuron and then you show some images in front of the eye. put some images in front of the eye and uh you can start with very basic shapes like here there is an illustration that there was a neuron that was found to be responsive only to lateral and actually just to one side moving sideways to the right lateral motions of a rectangle. So yes there are respon there are these very simple detectors in our brain or neurons that are responsible for something very simple in terms of shape or direct orientation or direction of motion. So uh then that you can only imagine how tedious these experiments are and how long they take to uh to to complete because then you have to insert this electrons to various neurons across the brain or maybe across the different between spear between species or for one species there may be inter speeies variations as well. But anyway, in order to understand what the brain uh how the brain works, you need to map the brain to the to the function of those neurons. And in order to understand what the neurons do, you have to do this kind of experiments. So um designated neurons were found to respond to v various things. Here is an example of a neuron that's responsible or activates only when a a horizontal A rectangle is oriented vertically only. So everything else is not good. So then there are neurons that are responsible to orientations like that to basic shapes like circles again rectangles, dots, lines, movements like horizontal or vertical and but then as you go up in the hierarchy then higher and higher level concepts can be responsible for activation of certain neurons. But before I go to those, let me show you very simple experiments very simple. They demonstrate how these um detection neurons work. Dete detection neurons neurons that are detectors for very simple concepts. But the experiments they got Nobel Prize in 1981 but the experiments were connected in 1959. So here is so these are actually very famous experiments one of the first ones uh figuring out how detectors in the brain work. So here here the experiment is work done on cats. So these cats are under anesthesia. Their eyes are open and these signals or these illustrations of light was placed in front of their eyes and you can hear for this particular I think this is the neuron that's responsible for circles. [Music] Okay. Circle at this particular place. What what gets the response? Yeah. Um, so here's another I think this one is maybe not as clear as the second experiment. What's going on? Here's the the this neuron is responsible for orientation of a rectangle and rotation is mapped like this. So if you change the position of the rectangle then the neuron is not firing anymore activated. this particular orientation position of the rectangle triggers the or activates the neuron and you can increase the size of the rectangle but it will not be active anyway unless the light hits this part. So this orientation is what trigger this orientation position will trigger the neuron. So then you can start moving this rectangle in different orientations positions. They will not trigger the neuron. And so one orientation position, one configuration of this rectangle will trigger uh the neuron. There you go. So you can only imagine how tedious these experiments are because you have to clean the skulls. You can prepare you have to prepare the animals. You have to sedate them. It's uh it's not that easy to keep this animals uh healthy under such conditions for a long time. So these are these are pretty tedious. Okay, let's go see. I have one minute. So what did I want to mention? Yeah. So one more thing. And so this one slide is that um I showed you how very simple um shapes have their designated regions in the brain that detect those. So there are detectors for those. But then as you go in the higher level processes of our brain, so there are still dedicated neurons to even higher level concepts such as faces and bodies and houses are the object are mapped here on the brain. So the job of neuroscientists is to map the brain is to understand what are different regions responsible to for and um so so there's a lot of experiments that can be done but it's this is a very slow moving process due to the fact that it's very tedious to to do conduct this experiments. Therefore, we are very limited in what we know about the brain um at this point and therefore the next topic of our um of of our class will be about perception. So, human perception as we do not know much more about the brain that will be helpful for us. But there's a lot in perceptual psychology that has been uh discovered about how we see things. But they were discovered by other means due with the help of um questioneries and human subject research and that's very still very important and relevant for virtual reality findings. But we are done with neuroscience. I think I have just one slide for next time but

that's maybe related to other so that's okay. [Music] lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

Worked for 13s

ChatGPT said:

NPTEL Introduction to Virtual Reality — Easy Notes

This lecture has many new words, but don't worry. We'll learn them like a 7th-class student, using simple examples, memory tricks, and MCQs.

The lecture mainly covers **three topics**:

1. Display quality — pixels, resolution, and brightness.
2. Eye movements — how our eyes move when we read or follow objects.
3. Brain vision processing — how our eyes and brain detect edges, shapes, and movement.

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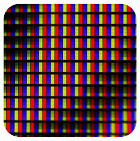
1. Display quality in VR

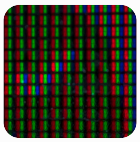
Imagine looking at a picture on your phone. A picture looks good when it is clear, has good brightness and colours, and shows movement smoothly.

There are three important factors.



1. Spatial resolution

How clear and detailed is the picture? More pixels can show finer details.



2. Light intensity and range

How many brightness levels and colours can the display show?

3. Temporal resolution

How smoothly does the display show movement? Frame rate and pixel response time matter.

Memory trick: Spatial = sharpness, Light = brightness, Temporal = time and movement.

What is a pixel?

A pixel is a tiny dot on a screen. Many pixels together form a picture.

- Fewer pixels → the picture may look blocky.
- More pixels → the picture can look clearer.
- If you look very closely at a screen, you may see individual pixels.

What is a Retina display?

A Retina display is a marketing term for a display whose pixels are difficult or impossible to distinguish at a typical viewing distance for a person with normal vision.

It depends on two things:

- Pixel density (PPI = pixels per inch).
- Viewing distance.

For example, a phone held close to your eyes needs more pixels per inch than the same phone viewed from far away.

What are cycles per degree?

This is a way of measuring how much detail the eye can see within a one-degree viewing angle.

A **cycle** means one repeating light-dark pattern. Think of alternating black and white stripes.

Your lecture uses approximately **30 cycles per degree** as a benchmark for normal 20/20 vision.

That corresponds to approximately 60 alternating light and dark bars per degree, using the lecture's convention.

Important: Cycles per degree measure visual detail by viewing angle, whereas PPI measures pixels per physical inch. They are related, but they are not the same thing.

Easy calculation: iPhone resolution

The lecture assumes a phone is viewed from 12 inches away.

$$s = d\tan(\theta)$$

Here:

- $d = 12$ inches
- $\theta = 1^\circ$

$$s = 12\tan(1^\circ) \approx 0.21 \text{ inches}$$

If about 60 pixels must fit into that one-degree viewing angle:

$$\text{PPI} \approx \frac{60}{0.21} \approx 286$$

So the lecture estimates roughly **286 pixels per inch**.

Exam trick: The same screen looks more pixelated when it is closer because each pixel takes up a larger angle in your vision.

2. Brightness and colour resolution

Your eye can handle a very wide range of brightness, from bright sunlight to a dark night.

But a typical display has **256 intensity levels per RGB colour channel** in the lecture's example.

RGB means:

- R = Red
- G = Green
- B = Blue

Each channel can have 256 possible intensity values, from 0 to 255.

For example, a screen can show dark red, medium red, and bright red. Combining red, green, and blue creates many colours.

However, real-world darkness and light are much more complicated than a screen can reproduce perfectly.

Exam point: A VR display may struggle to reproduce natural night vision because of its limited brightness range and available content.

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3. Eye movements — the most important part

Our eyes are not cameras that remain still. They constantly move to collect information.

The lecture discusses several types of eye movements.

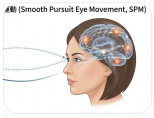
1. Saccades — fast jumps

When you read a sentence, your eyes quickly jump from one word or group of letters to another.

- Very fast.

- Voluntary in many everyday situations.
- Help you explore different parts of a scene.

Memory trick: Saccades = sudden jumps.

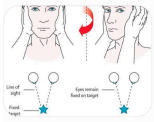


2. Smooth pursuit — slow following

Imagine watching a slowly moving bicycle. Your eyes smoothly follow it.

- Tracks a moving object.
- Helps keep its image near the fovea, the sharpest part of your vision.

Memory trick: Pursuit = follow.

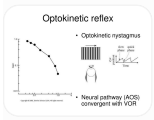


3. Vestibulo-ocular reflex (VOR) — stabilising vision

Hold your gaze on a fixed object and turn your head. Your eyes automatically move in the opposite direction to keep looking at the object.

- Involuntary reflex.
- Uses information from the vestibular system in the inner ear.
- Helps reduce blur when your head moves.

Memory trick: Head moves, eyes compensate.

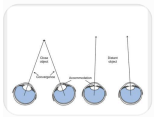


4. Optokinetic reflex — following fast-moving scenes

When a large scene moves quickly across your view, your eyes alternate between smooth following and quick resetting movements.

This helps keep visual information stable.

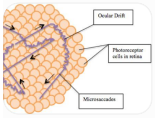
Memory trick: Fast moving scene = follow, reset, follow again.



5. Vergence — eyes move together or apart

- Near object → eyes turn inward (convergence).
- Far object → eyes turn outward (divergence).
- Very distant object → eyes become nearly parallel.

Memory trick: Near = eyes come Near each other.



6. Microsaccades — tiny jumps

These are very small, quick, involuntary eye movements, even when you try to stare at one point.

Scientists study their possible roles in maintaining vision and preventing visual fading.

Their full functions are still being investigated.

Memory trick: Micro = tiny; saccade = jump.

Quick comparison table

Eye movement	Speed/type	Example
Saccades	Very fast jumps	Reading a message
Smooth pursuit	Smooth, relatively slow	Following a bicycle
VOR	Fast, automatic reflex	Turning your head while staring at a target
Optokinetic reflex	Automatic follow-and-reset pattern	Watching scenery pass from a moving train
Vergence	Eyes turn inward/outward	Looking at a nearby or distant object
Microsaccades	Tiny, involuntary jumps	Staring at a dot

Important exam distinction: Saccades are fast, while smooth pursuit is smooth and relatively slow. Both can be voluntary. VOR is a reflex.

4. Why are eye movements important in VR?

Imagine a virtual ball moving across your screen. Your eyes follow the ball, but the headset lens and display stay in place.

This creates several challenges.

- When your eyes turn, you may look through the **peripheral part of the headset lens**.
- Peripheral viewing can increase optical distortion and colour fringes.
- Poor tracking or delayed display updates can make the virtual world appear to swim or move unnaturally.
- Eye tracking helps a headset estimate where you are looking.

What is foveated rendering?

The fovea is the central region of your retina that provides your sharpest vision.

Foveated rendering uses this fact:

- Render the area you are looking at in high detail.
- Render peripheral areas at lower detail when appropriate.

This can reduce the processing required by the VR system.

Memory trick: Sharp centre, simpler sides.

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5. The six muscles that move your eyes

Each eye has six main extraocular muscles. They work together to move the eye in different directions.

Muscle	Simple function
Superior rectus	Helps move the eye upward
Inferior rectus	Helps move the eye downward
Medial rectus	Moves the eye inward, towards the nose
Lateral rectus	Moves the eye outward, away from the nose
Superior oblique	Helps with downward movement and rotation
Inferior oblique	Helps with upward movement and rotation

The exact movement of the oblique muscles depends on the direction the eye is looking.

The lecture describes eye rotation using three degrees of freedom:

- **Yaw:** left and right.
- **Pitch:** up and down.
- **Roll:** rotation around the viewing direction.

For many simple explanations, yaw and pitch are the main movements considered.

Memory trick: Yaw = sideways; pitch = nod; roll = tilt.

Eye movements also matter when designing realistic avatars. If an avatar's eyes move unnaturally, it can look creepy or unrealistic. This is related to the **uncanny valley** effect.

6. How does the retina process images?

Your eyes do not send a perfect photograph directly to the brain. Some processing already happens inside the retina.

Think of it like a computer that processes an image before sending it to another computer.

Light enters the eye

↓

Photoreceptors: rods and cones

Detect light

↓

Bipolar, horizontal and amacrine cells
Help process and combine signals

↓

Ganglion cells
Send processed signals onwards

↓

Optic nerve → Brain

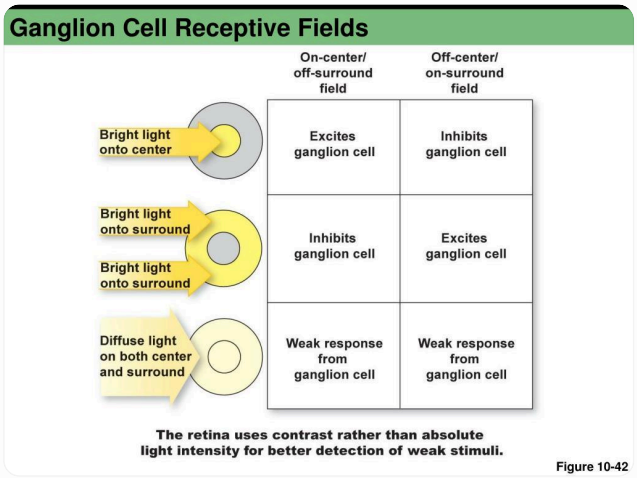
The lecture gives approximate figures of **126 million photoreceptors and 1 million ganglion cells per eye**. Many photoreceptors contribute to fewer output channels, so the visual information is combined and filtered.

What is a receptive field?

A receptive field is the region of the retina whose light affects the activity of a particular neuron.

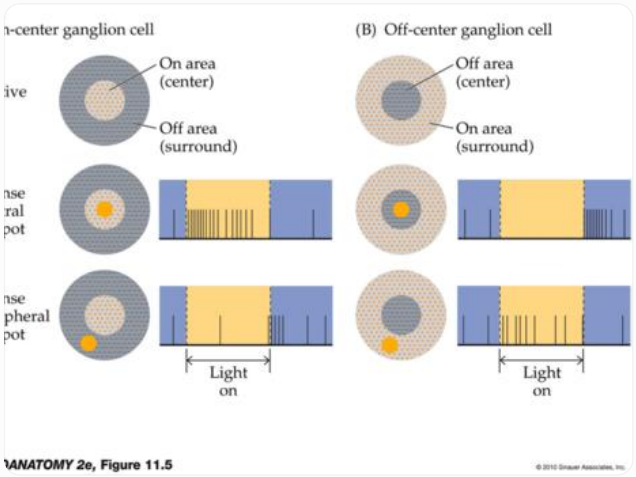
Imagine a small circular area on the retina. Light falling in different parts of that circle can make a neuron more active or less active.

Two important types of ganglion cells are **ON-centre** and **OFF-centre**.



ON-centre cell

- Light in the centre increases activity.
- Light in the surrounding ring decreases activity.



OFF-centre cell

- Light in the centre decreases activity.
- Light in the surrounding ring increases activity.

The centre and surround have opposite effects. This is called **antagonistic receptive fields**.

Easy way to remember

- ON-centre: light in the centre switches activity up.
- OFF-centre: light in the centre switches activity down.

The lecture's demonstration also shows that an OFF-centre cell responds strongly when its surround receives light while its centre remains dark. A small rectangular patch covering about 25% of the receptive field was the intended answer in that particular example.

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7. What is edge detection?

Imagine a black object placed on a white background. The boundary between black and white is an edge.

Your visual system is very good at detecting such boundaries.

ON-centre and OFF-centre cells help emphasise changes in brightness, so the signals sent to the brain highlight edges more strongly than uniform regions.

The lecture calls the processed information sent from retinal ganglion cells a **neural image**.

Remember: Raw image = light detected; neural image = signals processed by the retina.

This is one reason the information reaching the brain is not simply a copy of the light pattern entering the eye.

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8. Hierarchical processing in the brain

The brain processes visual information in stages.

1. Photoreceptors detect light.
2. Retinal neurons process brightness changes and edges.
3. Signals travel through the optic nerve and visual pathways.
4. Neurons in the brain respond to increasingly complex patterns.

Some neurons respond strongly to simple features, such as a line with a particular orientation or movement in a particular direction. Other brain regions contribute to recognising complex things, including faces, bodies and objects.

Think of it like learning to read:

- First, recognise lines.
- Then, recognise letters.
- Then, recognise words.
- Finally, understand sentences.

The lecture describes experiments in which researchers record the electrical activity of neurons while showing images to an animal or human participant. These experiments help researchers discover which visual features activate particular neurons.

Exam point: Visual processing is hierarchical, from simpler features to more complex recognition.

9. Field of view (FOV)

Field of view means how much of the world you can see without turning your head or changing your gaze.

The lecture estimates the combined horizontal field of view of human vision at about **270°**, based on its simplified calculation:

$$100^\circ + 35^\circ + 35^\circ + 100^\circ = 270^\circ$$

This is an approximate physiological estimate, not a claim that every person sees exactly 270°.

Many VR headsets provide a smaller field of view, so the user may see a dark boundary around the virtual scene.

Remember: Human vision has a wide field of view; VR headsets usually show a narrower window.

10. Display timing and frame rate

A VR display must update its pixels quickly and at the right time.

The lecture mentions several factors:

- **Frame rate:** How many frames are displayed each second.
- **Pixel response time:** How quickly a pixel changes from one state to another.
- **Flicker:** Unwanted perception of brightness changing repeatedly.
- **Motion blur:** A moving image looks smeared or unclear.
- **Scan-out:** The display may update one line at a time rather than the entire image simultaneously.
- **Photoreceptor response:** The eye also takes time to respond to changing light.

If the display updates too slowly while your eyes are moving, the image may become blurred or appear in an unexpected position.

OLED pixels generally respond faster than many LCD pixels, but display quality also depends on the panel, driving method, frame rate and other factors.

Memory trick: Good VR needs fast pixels, fast frames and accurate tracking.

Practice MCQs

Try answering before opening the explanations.

14-question quiz

0/14 answered

1. Which three factors determine VR display quality in this lecture?

- A. Size, weight, cost
- B. Spatial resolution, light intensity/range, temporal resolution
- C. Sound, touch, smell
- D. FOV, IPD, yaw only

2. What does PPI stand for?

- A. Pixels per inch
- B. Pictures per image
- C. Pixels per input
- D. Points per intensity

3. What happens when the same display is moved farther from your eyes?

- A. Pixels always become more visible
- B. It usually becomes harder to distinguish individual pixels
- C. Its PPI increases
- D. Its frame rate doubles

4. Which eye movement is commonly used when reading text?

- A. Smooth pursuit only
- B. VOR
- C. Saccades
- D. Accommodation only

5. Which movement helps your eyes follow a slowly moving bicycle?

- A. Smooth pursuit
- B. Microsaccades only
- C. Optokinetic reflex only
- D. Roll

6. You turn your head but keep looking at a fixed object. Which reflex helps?

- A. VOR
- B. Saccade
- C. Chromatic aberration
- D. OFF-centre inhibition

7. What happens during convergence?

- A. Eyes turn inward to look at a near object
- B. Eyes become permanently closed
- C. Eyes turn in opposite vertical directions
- D. Pupils stop responding to light

8. Which part of the retina gives the sharpest central vision?

- A. Blind spot
- B. Fovea
- C. Optic nerve
- D. Lateral wall

9. In an ON-centre ganglion cell, light in the centre generally does what?

- A. Increases activity
- B. Always stops all neural activity
- C. Has no effect
- D. Changes the eye's IPD

10. What is edge detection useful for?

- A. Highlighting boundaries between different brightness levels
- B. Increasing the physical size of a screen
- C. Measuring head rotation
- D. Changing the lens focal length

11. What does hierarchical visual processing mean?

- A. The brain processes every feature in one step
- B. Simple visual features are processed and combined into more complex information
- C. The eye does not process information
- D. All neurons respond identically

12. What does FOV mean?

- A. Frames of vision
- B. Field of view
- C. Frequency of voltage
- D. Focus of vergence

13. What can slow pixel response contribute to?

- A. Motion blur
- B. More photoreceptors
- C. A wider natural field of view
- D. Automatic convergence

14. What is the main goal of the VOR?

- A. Keep the image of a target stable on the retina during head movement
- B. Make the pupil larger permanently
- C. Increase screen PPI
- D. Detect only red light

Check answers | Reset

Last-minute revision: remember these 12 facts

1. Spatial resolution = image detail.
2. Light intensity range = brightness and colour levels.
3. Temporal resolution = changes over time.
4. Retina display = pixels not individually noticeable at the intended viewing distance.
5. Saccades = fast eye jumps.
6. Smooth pursuit = smoothly follow a moving object.
7. VOR = eyes compensate for head movement.
8. Vergence = eyes turn inward or outward.
9. Fovea = sharpest central vision.
10. ON-centre = light in centre increases activity.
11. OFF-centre = light in centre decreases activity.
12. Edge detection = emphasising boundaries between brightness levels.

If you remember these facts and understand the examples, you'll have a strong foundation for the lecture's MCQs.

You said:

VR_NPTEL_Quiz_... PDF

explain easily

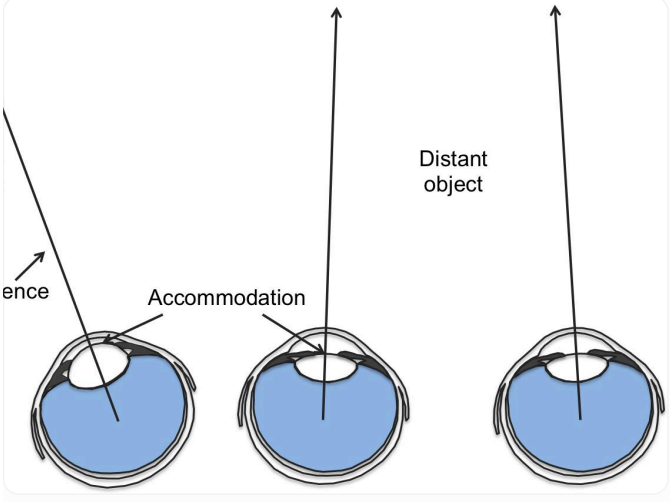
ChatGPT said:

NPTEL VR Quiz 6 — Easy Explanation

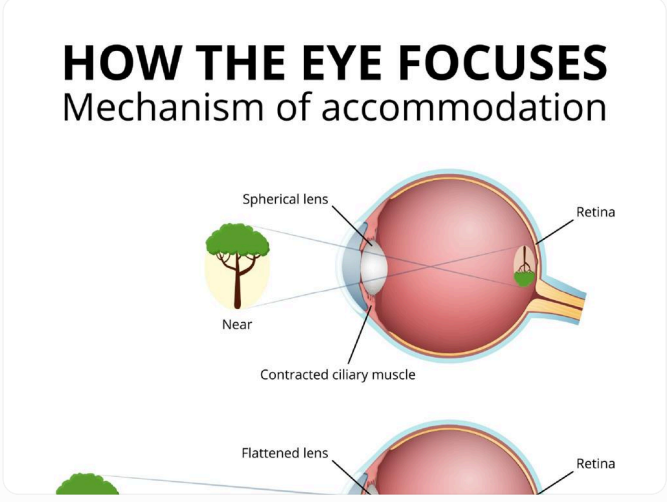
I'll explain all 10 questions from your PDF in **very simple English**, with examples and memory tricks so you can remember the answers for your exam. VR_NPTEL_Quiz_6_26.docx.pdf

1. Vergence vs. Accommodation

Imagine bringing a pen closer to your nose.



Vergence



Accommodation

Both eyes turn inward to look at a nearby object. The eye lens changes shape to focus on a nearby object.

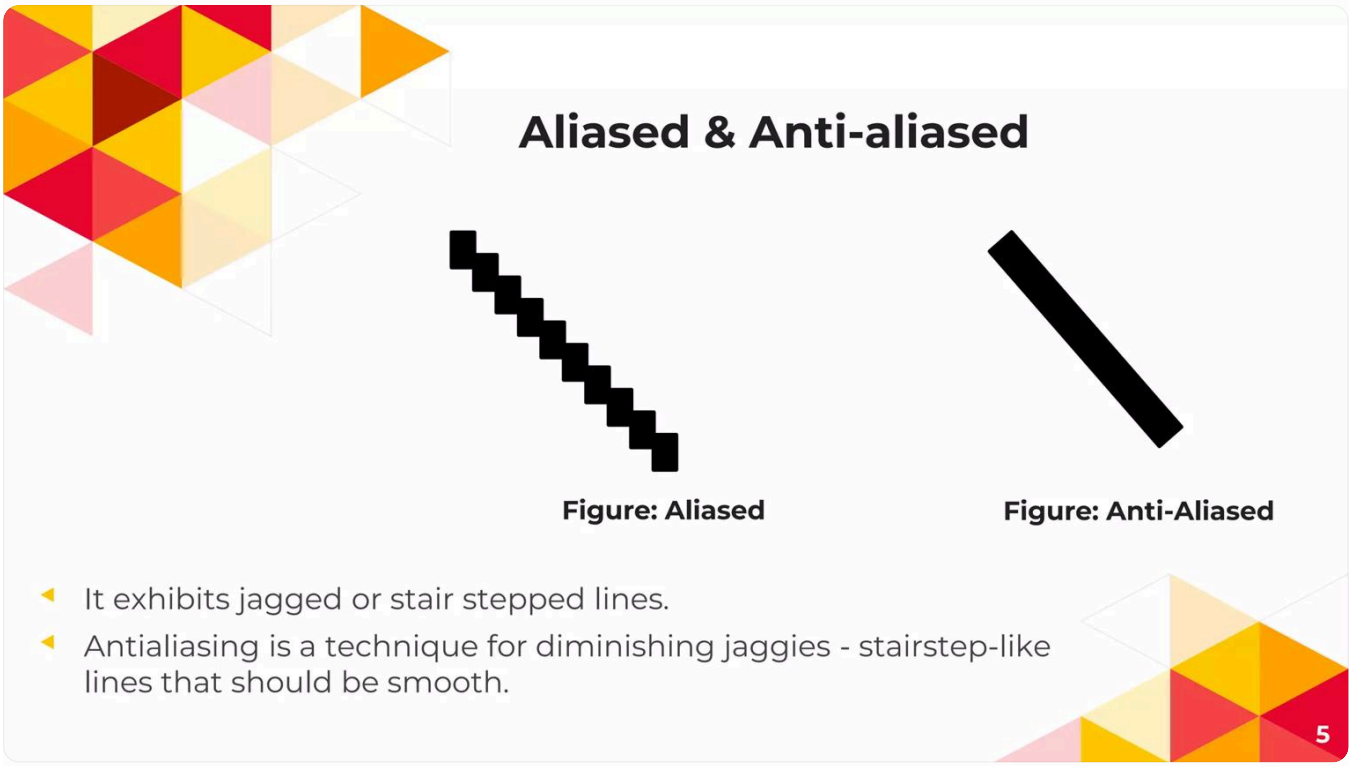
- **In the real world:** Both vergence and accommodation increase when an object comes closer.
- **In VR:** Your eyes turn inward to look at the virtual object, but the headset's display stays at nearly the same optical distance. So accommodation stays approximately fixed.

Correct answer: C — Vergence changes, but accommodation remains approximately fixed.

Memory trick: **Vergence = eyes turn; accommodation = lens adjusts.**

This difference in VR is called *vergence-accommodation mismatch*.

2. Why is aliasing distracting in VR?



Aliasing means that smooth lines look like **jagged stairs** because of pixels.

Imagine drawing a diagonal line using small square blocks. The line looks like a staircase. In VR, when you move your head, these jagged edges can appear to move, making them distracting.

Correct answer: B — Head movement makes pixel staircases appear to move.

Memory trick: **Aliasing = jagged edges.**

3. What is a Retina display?

A Retina display is a screen where you **cannot easily see individual pixels with your naked eye** under normal viewing conditions.

For example, when looking at a phone normally, you see text and pictures rather than separate tiny squares.

Correct answer: B — Individual pixels cannot be resolved by the naked eye.

Important: It does not simply mean 4K or one pixel per photoreceptor. Viewing distance also matters.

Memory trick: **Retina display = pixels are not noticeable.**

4. What can a person with 20/20 vision resolve?

The lecture gives approximately **30 cycles per degree**.

- A *cycle* is one black-and-white pair of stripes.
- One degree means a small angle in your field of view.
- 30 cycles per degree means roughly 30 black-white pairs within one degree.

Correct answer: B — 30 cycles per degree.

Memory trick: **20/20 vision → remember 30 cycles per degree** for this lecture.

5. Which VR headset needs higher PPI?

PPI means *pixels per inch*. It tells us how tightly pixels are packed on a screen.

Display closer to the eye

Pixels look bigger across your field of view, so more pixels per inch are needed to make them invisible.

Display farther from the eye

The same physical pixel generally takes up a smaller viewing angle.

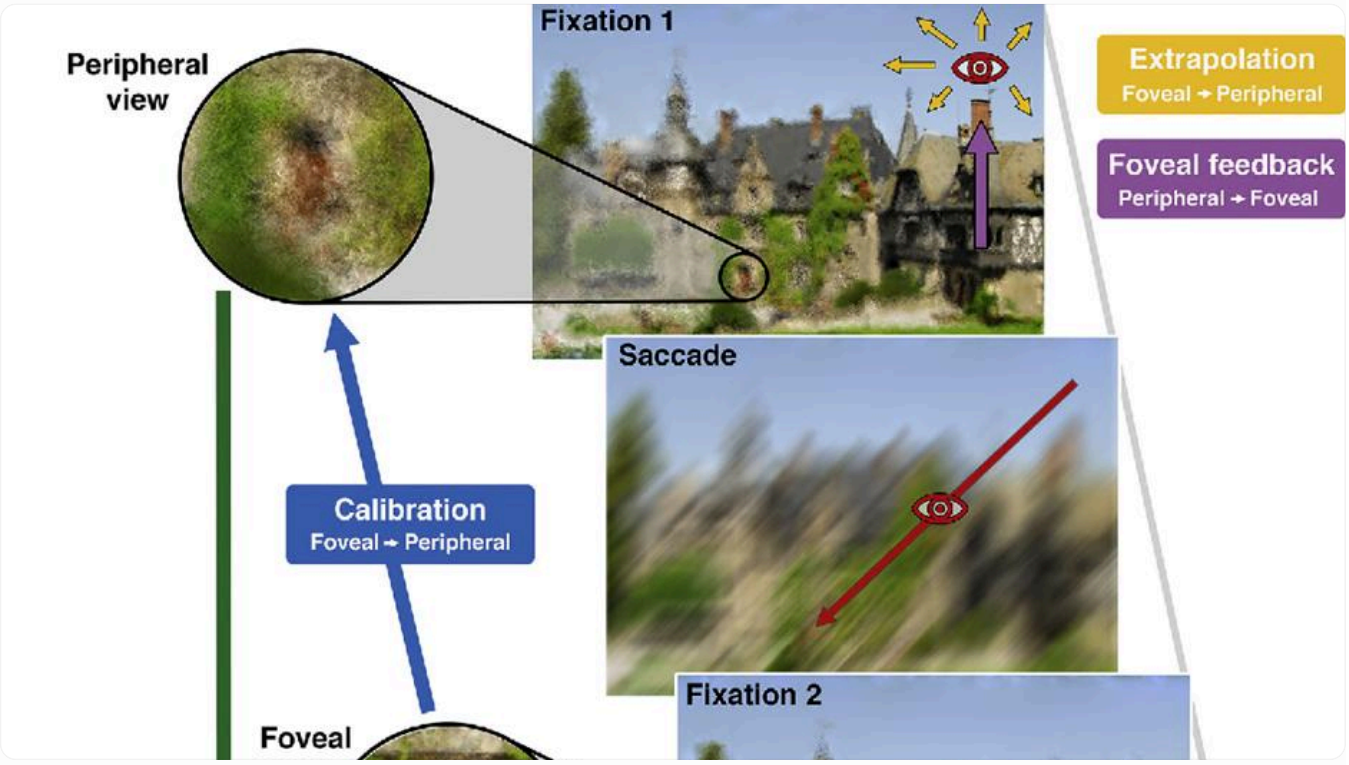
Correct answer: B — The headset whose display is closer to the eye.

Memory trick: **Closer screen → higher PPI needed.**

6. Which eye movement helps us see a wide scene sharply?

Our eyes have a small area called the **fovea**, where vision is very sharp. The rest of the eye's view has less detail.

So, how do we see the whole scene clearly? Our eyes quickly jump from one object to another.



These rapid jumps are called **saccades**.

For example, when reading this sentence, your eyes jump from word to word rather than moving smoothly over every letter.

Correct answer: D — Saccades.

Memory trick: **Saccades = sudden jumps.**

7. How do we follow a moving football?

Imagine watching a football slowly moving across the ground while keeping your head still.

Your eyes follow the ball smoothly so that it stays in your sharp central vision.

This movement is called **smooth pursuit**.

Correct answer: B — Smooth pursuit.

Remember these differences:

Eye movement	Simple meaning
Saccades	Eyes jump quickly
Smooth pursuit	Eyes smoothly follow a moving object
Vergence	Eyes turn inward or outward to look at different distances
Vestibulo-ocular reflex (VOR)	Eyes compensate for head movement

Memory trick: **Follow smoothly = smooth pursuit.**

8. Why do virtual objects appear to swim during head movement?

Imagine wearing a VR headset. You turn your head, but a virtual wall should remain in the same position in the virtual world.

If the wall appears to drift, swim, or shake, the headset's tracking or display timing may not be working accurately.

Your eyes normally compensate for head movement using the **vestibulo-ocular reflex (VOR)**.

Correct answer: B — Vestibulo-ocular reflex adaptation.

Memory trick: **Head moves, eyes compensate = VOR.**

9. Why do vergence and accommodation normally happen together?

Imagine holding a pen far away and then bringing it close to your nose.

Two things happen:

1. **Vergence:** Both eyes turn inward toward the pen.
2. **Accommodation:** The eye lens changes shape to focus on the pen.

In the real world, these two actions normally work together.

In VR, the eyes may turn inward to look at a virtual object that appears close, while the lens continues focusing at the display's optical distance.

Correct answer: B — Objects moving closer require eye convergence and increased lens curvature.

Memory trick: **Near object = eyes turn + lens adjusts.**

10. What are microsaccades?

Even when you try to look at one fixed point without moving your eyes, your eyes make tiny, involuntary movements.

These are called **microsaccades**.

For example, try staring at a small dot on a wall. Your eyes will not remain perfectly motionless.

Scientists study these movements, but their exact function is not fully understood.

Correct answer: C — Tiny involuntary fixational movements whose exact function remains uncertain.

Memory trick: **Micro = tiny; saccades = jumps.**

Quick revision table

Q.	Topic	Correct answer	Remember
1	VR focus mismatch	C	Eyes turn; lens stays nearly fixed
2	Aliasing	B	Jagged pixels move with head motion
3	Retina display	B	Individual pixels are not noticeable

4	20/20 vision	B	30 cycles per degree
5	PPI	B	Closer screen needs higher PPI
6	Saccades	D	Rapid eye jumps
7	Smooth pursuit	B	Smoothly follow a moving object
8	VOR	B	Eyes compensate for head movement
9	Vergence + accommodation	B	Eyes turn and lens adjusts
10	Microsaccades	C	Tiny involuntary eye movements

Test yourself

8-question practice quiz

0/8 answered

1. When a virtual object appears closer in VR, which response changes while accommodation stays nearly fixed?

- A. Accommodation only
- B. Vergence
- C. Neither
- D. Pupil colour

2. What does aliasing look like?

- A. Smooth curves
- B. Jagged, staircase-like edges
- C. A blurry background only
- D. A change in colour

3. What does PPI measure?

- A. Eye movement speed
- B. Pixels per inch
- C. Frames per second
- D. Field of view

4. Which movement follows a slowly moving football?

- A. Saccades
- B. Vergence
- C. Smooth pursuit
- D. Microsaccades

5. Which reflex helps stabilize vision when your head moves?

- A. VOR
- B. Accommodation
- C. Scotopic vision
- D. Aliasing

6. Which eye movement consists of rapid jumps?

- A. Smooth pursuit

B. Saccades

C. Accommodation

D. VOR

7. What happens when a nearby real-world object is viewed?

A. Only the lens changes

B. Only the eyes turn

C. Vergence and accommodation both increase

D. Neither changes

8. What are microsaccades?

A. Large voluntary movements

B. Tiny involuntary movements during fixation

C. Movements that occur only in sleep

D. Head movements

Check my answers

Exam tip: First memorize the five key differences: *vergence vs. accommodation, saccades vs. smooth pursuit, VOR, aliasing, and PPI.* They are the main concepts behind these questions.

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[Music] So now let's look at the standards for computer monitors and phones and VR displays. So so far we talked about movie theaters mostly but now let's move to TVs and monitors and phones. So notice that all of these devices are usually much closer to your eyes than the screens in a movie theater or TVs, right? So which means that the problems of flicker and jutter become more noticeable as you move the screen closer. Uh so as you get to 60 hertz, this is the most common frame per second for PCs and smartphones. So for for example, for our cardboards, you'll have 60 Hz. frame rates most most likely because your phone is most likely have 60 Hz frame rate. Uh now for classical TV though it's 25 and 30 depending on voltage also which country you're using. So that's um that's also I skipped on that. I skipped talking about TV's requirements actually. So we mostly talked about movie theaters but not much about TV. It's slightly closer the display and now we'll talk about phones, monitors and VR HMDs which are even further closer. So let's see uh so when at some point they were CRT so I don't know if you ever experienced cathode ray tube monitors so they had low persistency. So although it was already the frame rate standard for monitors was 60 Hz these uh display CRT displays they had low persistency and suddenly the flicker became a problem. So at 60 they could not be really used because many people would have anno flicker and headaches. Even at 72 it was the minimum ergonomic recommendation. Many people would still have a headache after working o on this kind of displays for a while. So 85 hertz was comfortable for most and that we're talking about monitors, right? Monitors, PC monitors. So if they are low persistency, they have to have 85 Hz uh frame rate. However, if it's not CRT display, so if it's um always on display, then it can be 60. But there will be maybe some kind of blurring depending on the speeds of objects. But again, we are not interacting with monitors or phones in the same way as we are interacting with virtual worlds. First of all, in virtual reality, the screen is even closer, but also there is this fast motions of the head that triggers um tracking and rendering updates. So that that's what affects the frame rates. So now for LCD displays that we have mostly now uh they are always on and they have 60 frames per second and DK1 was the so Oculus DK1 um development kit one was running on this phone so it was always running in 60 and of course um the problem was a lot of jutdder. it was not very clear and what you have in your cardboards it's pretty jutddery so there is a lot of blur due to a slow uh frame rate however on this cheaper LCD displays or LCD displays at the time the pixel refresh rate was so high that you couldn't push the frame rate higher than 60 until OLEDs came in so DK2 Rift S they had

higher frames per second and they had OLED display and OLED that has OLED pixels. They um their refresh rate is really really small. So that allows to increase the frame rate and u but in but you have to put something in between the frames. So usually there is a blackout in between the frames and that introduces the flicker. So that's why 75 and 80 was was not as good as um later devices that had 90 or higher frames per second. So the current standard for consumer VR headsets is actually 90 Hz. 90 Hz is the recommendation. However, recently um so the the recent uh headsets they introduced this fast switching LCD displays. that are still cheaper than OLEDs. That's why the switch happened. So, LCDs are usually cheap cheaper to produce than OLEDs. And Oculus Quest 2 has 120. Uh, however, because it's LCD, it's always back lit because that's how the pixels work. You have to have the back lit light to trigger the reaction. So, there is no true black in LCDs and latest headsets, which is also a problem. So again, you fixing one problem, you introduce another one. However, there's no careful or finalized standard for for these devices because none of this is enough. There is still jut jutdder. There is still blurriness or there is still flicker that is perceived. So none of this is enough. we're not really at uh the best technology yet or the there is technological limitations to have some finalized standards for virtual reality. So uh so I'll show you in a bit if time allows today or maybe next time I'll show you exactly the situations where jutter will occur independently of the um frame rate that we have so far. Also I already showed you the situation where flicker will be perceivable. Um if you remember in our one of the first lectures I was talking about an OLED um LED um light that would have different frequency different number of frame rates and I started moving it sideways which is very possible in virtual reality. So if you have some kind of object that's moving really fast depending on the speed you will perceive this zipline. So flicker can be per perceivable even at very high frame rates. So we had that already. We had that proof in the beginning of the lectures. Similarly to shadows is shading and reflection. So just based on the shadings we can infer even the shape of the object. Right? Of course, you can trigger lots of illusions like for example, it's not very clear if this is a convex or concave object because some of the visual information is missing here. So, this is not the entire picture, right? And that's why when you emit some cues and leave some other cues, you can um create illusions. So this is an example of that. Yes. Yeah. It makes it more realistic. Very good. Yeah. Very good. Can we use panoramas in indoor scenes or can we use it only for nature? I think so. I wanted to try panoramas in indoor scenes. And if I have time in the end of this lecture, I'll show you a project idea that I would love to see it see it happening. It's a very interesting project idea of how you can use panoramas and interact with them using twisters in cardboards which has never been done before for cardboard. So it would be fantastic. But let's wait until that. So yeah, and I think that would involve uh panoramas not only with outdoor scenery but indoors as well. In fact, I really wanted to try it. We even bought with Steve the panoramic camera just to see if we can put in Unity a couple of panoramic um spheres and use twisters to jump from one to another and we could perceive that it's our home, you know. So that would be wonderful. There aren't shadows for virtual. Yeah. Well, no, there are no shadows of virtual objects in panoramas. But um I don't understand this question. Of course, if there's shadows in the natural environment, the capture will the the camera will capture it. But if you put some virtual objects, they will not cast shadow on your panorama and that's another problem. Yeah. So, how do you do this interactions? It's similar to like throwing an object. Where would it land on your panorama? Because there's no 3D structure. There is no floor in panorama. It's a sphere, right? It's a it's a wallpaper that was painted on your sphere. So, it's really difficult. That's a good That's a good point. Lighting is also yeah it is a factor. So another perception is enter position. So nothing is closer or further away sometimes in your uh in your painting or picture. But this interpositions or T-shapes see just this T-shape will trigger the brain to infer which object is in front of another. And it's a moninocular cube present in panoramas. It's very good. So here for example, this daughter and father figure, it's not clear which one is closer or further away, but the T or T shapes or interposition is the only Q that's present and you'll make the conclusions and there's a lot of T- shapes here and which helps us figure out which one is in front of which object is in front of another. Similarly here, so how do we know where the spare is going to hit? Well, uh there's not many cues. There is height in the visual field which will tell you that maybe the elephant is further away. Um but there's interpositions, right? The interpositions would really make it kind of clear. But these objects are also in the same height. So, okay. So, let's see how to throw things in panorama. See, I put it here on the slide explicitly because I wanted to make to make sure that you understand this problem. We do not know how to interact with images with the objects in panorama. How to cast shadows on them or from virtual objects inside or how to interact with them or how to throw things. Here interposition is creating this illusion again like you can see that the T- shape will trigger the perception. Yeah, lighting is is the cues it's related to shadow. We had shadows and lighting. Okay. So now accommodation is another cue. So it's a moninocular cue. You close one eye as objects move closer to your eye the eye will accommodate. That is the silary muscles will press the lens. the lens will increase increasing the diapter and optically you'll be able to focus on this object but just this this action is the cue the fact that you squeeze the lens the brain will remember it and it will infer just from that information of how much you squeezed uh the lens how close must be the object. Therefore, first of all, usually accommodation in real world works only for objects that are under 2 m uh in depth from us. Therefore, Oculus guide, the best practices guide recommends not placing any virtual objects closer than 2 m, which is of course is a strange advice because of course we want to place objects closer. But why is this advice there? is because the brain would have to accommodate on real objects closer than 2 mters. But in virtual reality accommodation is missing because you know the objects are always

optically at the same distance which is infinity or so. Therefore this cue would be absent and it would lead to virgin accommodation mismatch which may be uncomfortable. So hence the recommendation. So is accommodation present in in panoramas? No. Again for same reason nothing is 2 m or 5 cm nothing is very close in panor in virtual reality experiences to us everything is at the same optical distance which is infinity due to the optics okay so there's no accommodation nevertheless what I wanted to say it's a moninocular queue and it's not present in panoramasia so the next is is paral motion paral this is a difficult one so it's a moninocular cue you close one eye and try moving your head sideways. The objects will move with respect to each other wildly. And the more objects are separated in depth, the more they will move with respect to each other. So here you can see just an example of these dots because they move faster than those object those dots in the in the far away far away which is the Q. So we infer that objects that move slower are further away and objects that move uh faster are closer and that's a Q. It's not present in panoramas, right? You move your head sideways, nothing will be uh moving with respect to each other. You'll just feel that you moved over the surface of the panorama and the objects will not be interacting with each other uh on this image panorama image. So that's very unfortunate because it's really hard to to fix this this depth information between objects that are captured in panorama. It's missing. So it's really really challenging task without doing scientific computations to mimic parallax. Okay. But it's a moninocular cube so you could perceive it with one eye. However, what would trigger paral if you have two panoramic images and you walk from one to another? So if you teleport from one to another, if there's close by, it may still trigger a little bit of pan paral. So you could um you could play with that with the like twisters and panoramic images and see if you it wouldn't be continuous paral, but it would be consistent with what you expect. So something to explore. Again, something that we talk in research group in our research group about virgin's angle. So that's an interesting one, right? If your if your object gets closer, your eyes verge together. So this is the first example of a binocular queue. If you close one eye, there will be no virgins. You look around in panoramas or in VR, you you see objects, but there will be no virgin. So that queue will be missing. And um well, it may or may not cause the problem. Of course, perceiving things with one eye is difficult. Like um manipulating objects that get too close to each other with one eye are difficult. For example, this task is difficult. Putting the fingers really close to each other. So similarly in VR, uh this is an important cue and it helps us perceive distances to objects. This is present in VR. However, in panoramas, it's not really present the way you would expect it. Right? If you have this image of objects that were captured with panorama, they're all at the same virgin angles that is on the sphere. Right? So if if these two objects are on panorama since they're painted on the sphere, you will verge similarly on this and this object while in reality you wouldn't. So that is a very important cue that is missing in panoramas. Okay, I have very ambitious game idea for final sign, but I'm scared if Unity shows Gradel. Yeah, that's actually if you have ambitious things, Unity may be an obstacle. So, run it by us, please. And on Discord, I would love to see your ambitious ideas or maybe you can implement part of it. That will be fine because if it's too hard to implement, we know sometimes the simplest things are really difficult to implement. Head tilt can create motion parallels and panorama like switch to spheres. Yeah, if you if you do to if you switch in between spheres in some way that is an interesting project I haven't seen it done. So banara is being projected on other surfaces other than spheres. Yeah, I think it has been done something like this. Of course then it can help but then the whole point of emitting that information that depth information will be lost. So there's a trade-off but an interesting idea maybe you can still save some um use some optimization but like only only have simpler simplified surface or so to speak. Now let's talk about binocular disparities very important important topics topic and it will lead to uh dipopia. So something that is sometimes difficult to understand. So it takes my students traditionally several tries. So if you don't get it today, it's fine. Just watch lecture again and and try to understand. So we'll talk about binocular disparities. Of course, you know that if you close one eye and and one eye at a time, you'll see completely different images, right? So here's an example of two objects that a person is looking at, they are quite far in depth. They they're further away in depth than uh these two. These are approximately at the same depth, but these two are not. And then the further they are from each other in depth, the more different the right and left eye view may be. Right? So here this thumb and uh and that um house are very far away from each other. And you can see that the left and right eye views are quite dramatically different because of that. So So that's very interesting. So it's a it's a Q. Diplopia is a cue. So the fact that the images are different and some objects will appear as double images is a cue for for for perception. That is the further away this um images on the right eye and left eye for some objects are the further that object must be. So that's again a cue that the brain will use. So what is dipopia? So far we talked about binocular disparity just the difference in what the right and left eye sees that's called binocular disparity. Of course it's a binocular cue right and of course it's not present in panoramas because it's very similar to parallax right very similar to parallax. If there's some objects on captured by panorama, they're different depths. You see, look at them from left and right eye. There will not be differences in um in how you see them on your retinas from left and right eye. There may be only binocular disparities looking at panoramas from the distance to the sphere itself but not relative positions of objects in panorama. Okay. But what is dipopia? So diplopia is perceiving a double image of some object that's related to binocular disparities because diplopia is an example of binocular disparity. So let's look at this um situation where a person we're looking at the head from above. So this is nose. He's looking at this yellow duck, right? And the yellow duck must be then at phobia. And what happens to the duck that is behind the yellow duck?

Well, in one eye it will be it will appear to be to the left. In another eye it will appear to be to the right from the closer object. So if you do the same experiment with two fingers, in one eye the far away finger will look like it's on one side of the front finger and in another eye it's on another side. But when you're looking at this front finger, when you're uh focusing on that, you don't even notice that there are double image of the other object. In fact, it's not only of this one finger that there's double image. There are double images of all of the objects that are behind this finger. But we don't even notice, right? The brain masks it from us. [Music] So let's look at this um another binocular Q another binocular Q and depth perception Q is horopter. So horopter is the space in real world where objects are not double. So we don't perceive them double and they usually related to our fixation point. So if you if I'm looking at this finger there will be an entire shape in the whole world and it's not even it's not surface it's a 3D surface kind of a 3D shape uh which is called horer and in that region other objects will not be in diploia they'll fall on corresponding points of both of my retina so they will not appear as double while any object away from horopter so in front of this region or behind will be in dipopia. So you'll perceive them as double images or not corresponding images on your two retinas. But we don't perceive that. We don't notice it. In fact, it's so hard to notice that this is happening that students don't even get this idea uh as I see again and again from year to year in the final exams. So it's a it's a a little difficult concept to um understand. However, if you know that there is this curve monitors and TVs, they actually try to mimic this horopter region so that the text that you're trying to read will be uh will not appear as double uh image on your right eye. So if you look at one letter or word on the screen, all of the others will also appear as one to your brain, not as double image. So if you're okay with this, let's see what would be a cool project for you to implement. So a demonstration that would be very useful for education. So for me, if I wanted to explain what diplop is, you could actually create this in cardboard. I would if in fact if you do this I'll really appreciate it because I'll be remembering about your contribution to this course forever as I'll be showing it and I will be showing it to my future students. So this project was implemented in Oculus Rift but not on cardboard yet. So a cardboard demonstration would be really cool but I've seen it working on Oculus. So it is true. Uh I think the first time I proposed this project I wasn't even sure it's going to happen. But let me demonstrate how you could show diplopia to people who are not aware aware of this notion. So let's see how could I demonstrate that this ducky is actually a double image and it appears both to the right and to the left of the object that you are sating on which is in in front of that ducky. How would I show it in virtual reality? Well, in virtual reality, we have so much control about what you can show to left and right eye. So, I could show the uh so if you're looking at my screen in virtual reality, well, a perception perception of screen. So, uh the screen will show something else. But let's see the perception. So, if you try to render a ducky in front of the person right here, then what uh so so let's draw the screen actually here. So you will this this is not good. Sorry I wasn't prepared to draw the screen. So let me try again because this is very important. I want to get it right. So how would I do it? You will render the right view and the left view for each eyes. But instead of showing in the for the left eye view the two duckies the way they're supposed to look. So the one ducky would be here. And for the right eye view, the rendering on the screen would have to be this ducky and this gray ducky to the right. So instead of doing that, I'm going to render one eye at a time this image of the gray ducky. So in one frame I'll show it only to the right eye in to the left eye. In another frame I'll show it only to the right eye and so on. And maybe it will not be one frame at a time. Maybe it will be 10 frames or 20 frames. I don't know exactly what that interval is, but I want to trigger the photo receptors. So, so essentially I'll always be rendering the yellow ducky, but the gray duck will appear only in one eye for some time followed by only in another eye for another time and so on. So that's what you do. So then perceptually what you'll feel you what you'll see is that when you look at the yellow duck the the right duck will be moving and changing its position and it will look very strange but that way you demonstrate dupia. So in fact if the world or real world would would work like this when you look at some object all other objects show one eye at a time you would feel like everything is moving all the time because most objects are not in your horopter which means that they appear in two different places on your retinas with respect to the object that you're looking at but we never notice it. So I hope I explained it well and this is a really cool project. Why this is important? Well, horopters seem to be important. Horopters are the surfaces where there's no this kind of motion would not be triggered or would not be perceived because they because it's a perfect reading surface. So that's why they're important because they provide a perfect reading surface. In fact, in virtual reality, it's really hard to read. So maybe if uh test was displayed on horopter menus that would be easier to read because there would be no diplopia for images on the reading surface. So uh the these are possible applications. Now let's see what we have done so far. So we talked about moninocular cues, binocular cues. So binocular cues are the the cues that require two eyes. moninocular the ones that um will be inferred from image on one eye only. Of course, the brain uses combination. So all of them are working together and then the brain will have to fall on some priors and bias of course based on what it learned so far. What are the sizes or distances that uh we incur encountered through our lifetimes but then maybe some will be contradictory cues and the brain will disregard it. But general in general it's some kind of biasian probabilistic kind of inference model about what should be the distances two objects stand in the world that we are looking at that's what the brain uses fusing all of this together. Okay. So now since I have a bit of time I want to talk about of course there is also scale perception. So depth and scale perception are very related. So for example, one interesting uh result for VR is that if you change your virtual IPD in in VR. So think of the size of your virtual head and how far the cameras

that see the world are and you can you could it used to be that you could put it you could uh adjust it in Unity. Now it's not so easy because Unity now is very constrained. So it will only allow you for whatever fix parameter for your real IPD. Uh however if you in real world is however if you could change the distances between these virtual cameras and show the world in that way that would affect your perception of a scale. So suddenly you would feel like either a giant or like you're really really tiny in the world. So just that perception how put putting this virtual cameras would affect that perception. Very interesting um effect. But anyway, here is an illusion. So it's really hard to see that this is not a bigger van because this illusion that actually these cars all are the same size. So they have the same retinal size. They trigger the same amount photo receptors. However, because of all of the cues that we already discussed in the height in the visual field, the interposition, it's really hard to see this car as the same size. You really perceive it as bigger, right? Because our brain is just doing what it's usually doing. And you would only see this kind of situation if this car was bigger, right? But this is nothing but uh click copy and paste. Copy and paste. Photoshop. So now I want to jump to project proposals since we talked about panorama. So here is walking in panorama project. So let's see. Imagine that you have several panorama images that are taken by panorama by panoramic camera and you want to put them in your unity project. And then what you do to engage interaction or at least engage engage in walking through this world or through this experience that you want to create is that if you are positioned in one panorama then through twisters depending on where you look on each panorama you would be teleported instantaneously to the corresponding panorama. And all of these panoramic spheres could be then thought of as being placed on some kind of grid. And then of course any direction of the twist would meet some some teleportation point. So this hasn't been done yet for panoramas. The only way to lock to do locomotion in panoramas that I've seen was to look on the bottom on the floor and wait for one or two seconds to get to the next panorama. And uh it's very annoying and it's very slow. So I think with uh twisters you could create much more natural locomotion in this panoramic experiences. And I think it would be amazing to experience it. Like I would love to see this working because I think it would be seamless. In fact, given that in cardboards we have only three degrees of freedom tracking. We don't have six degrees of freedom tracking. So if you even try to move in this panorama, move your head, there will be no paral just because there's no tracking. So this constraint of panoramas not having paral is not even there because there would be no tracking mechanism to to even track your head. So we and we are fine with that right the experiences that you create the APKs that you create in your homework three and four there's no parallets either because you cannot there's no positional tracking which means that it's really interesting to see how would this experience compare to APKs that you create with building 3D models in Unity and of course then the next step would be can you even put objects then inside of 3D objects in panorama of course they would not cast shadows like you said it It would be hard to place them to match the floor. I don't know. But that would be an amazing experiment. So that's something that is really cool to tinker with. So I hope this um motivates you to try projects or panoramas. But uh as I said before, you can start thinking of project ideas and you can do whatever you want. We don't constrain you. Professor Manivan is preparing a big list of uh other more supervised projects as well that he'll recommend. Uh however you're welcome to as I said just do anything you want and create any kind of fun games and uh let's discuss it on uh discord. So I'll be happy to discuss it on discord with you. [Music]lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

NPTEL Introduction to Virtual Reality — Easy Notes

This lecture has two main topics:

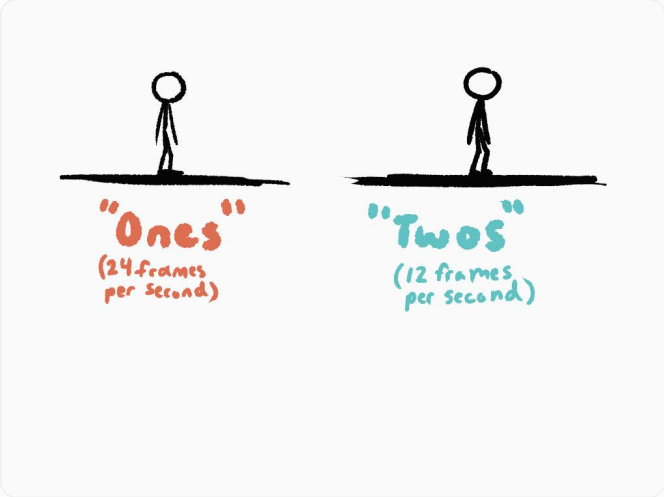
1. **Display frame rates:** Why VR screens need fast, smooth images.
2. **Depth perception:** How our brain understands which objects are near or far, and why panoramas have limitations.

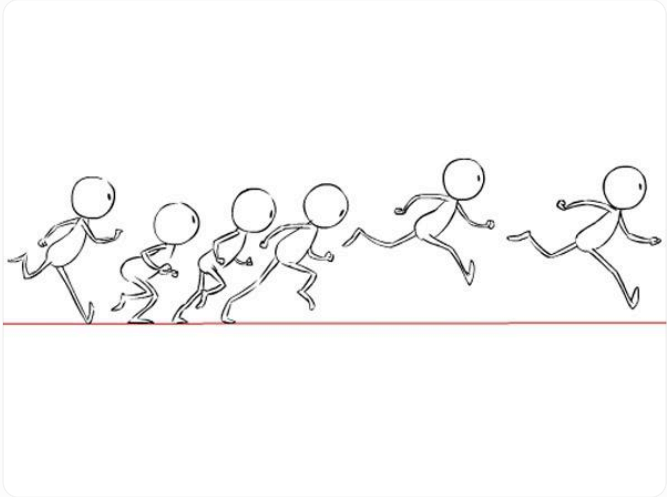
I'll explain them using simple examples, memory tricks, and MCQs.

Part 1: Frame rate and motion in VR

1. What is FPS?

FPS means **Frames Per Second**. It tells us how many pictures a screen displays in one second.





Low FPS
Fewer pictures per second. Movement can look jumpy.

High FPS
More pictures per second. Movement usually looks smoother.

Imagine a flipbook. When you flip its pages quickly, the pictures look like they are moving.

Remember these lecture numbers:

Frame rate	What the lecture teaches
2 FPS	Motion can begin to be perceived, but it looks discontinuous
10–12 FPS	Individual frames become harder to notice; motion looks more continuous
16 FPS	Common historical silent-film rate
24 FPS	Standard cinema frame rate
60 Hz	Common monitor and phone refresh rate
90 Hz	A commonly used VR refresh rate and lecture recommendation
120 Hz	Used by some VR headsets, including Quest 2 in supported modes

Important: These are teaching examples, not universal limits. Perceived smoothness also depends on movement speed, display technology, and viewing conditions.

Memory trick: **2 starts motion, 12 blends motion, 24 cinema, 90 VR.**

2. What is refresh rate?

Refresh rate is how many times the screen updates per second. It is measured in **hertz (Hz)**.

For example:

- 60 Hz = 60 screen updates per second.
- 90 Hz = 90 screen updates per second.
- 120 Hz = 120 screen updates per second.

FPS and Hz are related but not exactly the same: FPS is how many frames the computer produces, while Hz is how many times the display can refresh.

3. What is flicker?

Flicker means the screen or light appears to flash on and off.

Imagine a bulb that keeps switching on, then off, then on again. You may notice the flashing.

In early display technology, low refresh rates could cause eye strain or headaches for some people. The lecture explains that film projectors used shutters to flash each frame multiple times, reducing the perception of flicker without needing extra film frames.

Memory trick: **Flicker = flashing.**

4. What is judder?

Judder is an uneven, jumpy appearance of movement.

Imagine a ball moving across a screen:

- The ball should move smoothly.
- But the screen shows it at one position, then suddenly at another.
- Your eyes follow the ball, but the image updates in steps.

The movement can look jumpy or blurry.

Memory trick: **Judder = jumpy movement.**

5. What is retinal image slip?

This is one of the most important concepts in the lecture.

Imagine following a moving ball with your eyes. In the real world, the ball moves continuously. But on a screen, the ball may only appear in a few separate positions.

Your eyes move smoothly while the screen updates in steps. Between updates, the ball's image can slip away from the sharp central area of your retina, called the **fovea**.

This is called retinal image slip and can contribute to perceived blur or judder.

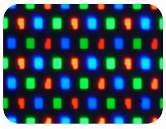
Memory trick: **Eyes move continuously; screen images jump between positions.**

6. Always-on displays vs. low-persistence displays

Always-on display

Each frame remains visible until the next frame replaces it.

- Can cause retinal image slip and motion blur.
- Increasing the frame rate usually helps reduce judder.



Low-persistence display

Each frame is shown briefly, then the display goes dark before the next frame.

- Reduces retinal image slip and motion blur.
- Can introduce noticeable flicker if the flashing is too slow.

The key idea is that fixing one problem can create another.

Display problem	What can help?
Judder / retinal image slip	Higher frame rate
Motion blur from a frame remaining visible too long	Shorter display persistence
Flicker from flashing frames	Higher flashing frequency

Exam point: On an always-on display, increasing frame rate mainly helps reduce judder. Low-persistence displays can reduce blur but may introduce flicker.

7. Why is VR more demanding than watching a movie?

In a movie theatre, you usually sit still and watch a screen far away.

In VR:

- The screen is very close to your eyes.
- You turn your head quickly.
- The headset must track your movement and update the image.
- Delayed or uneven updates can make the virtual world look unstable.

That is why a VR headset needs good tracking, fast rendering, and a suitable refresh rate.

8. What is scan-out?

A display does not always update every pixel at exactly the same instant. Many displays update the image in a sequence, often from the top of the screen to the bottom.

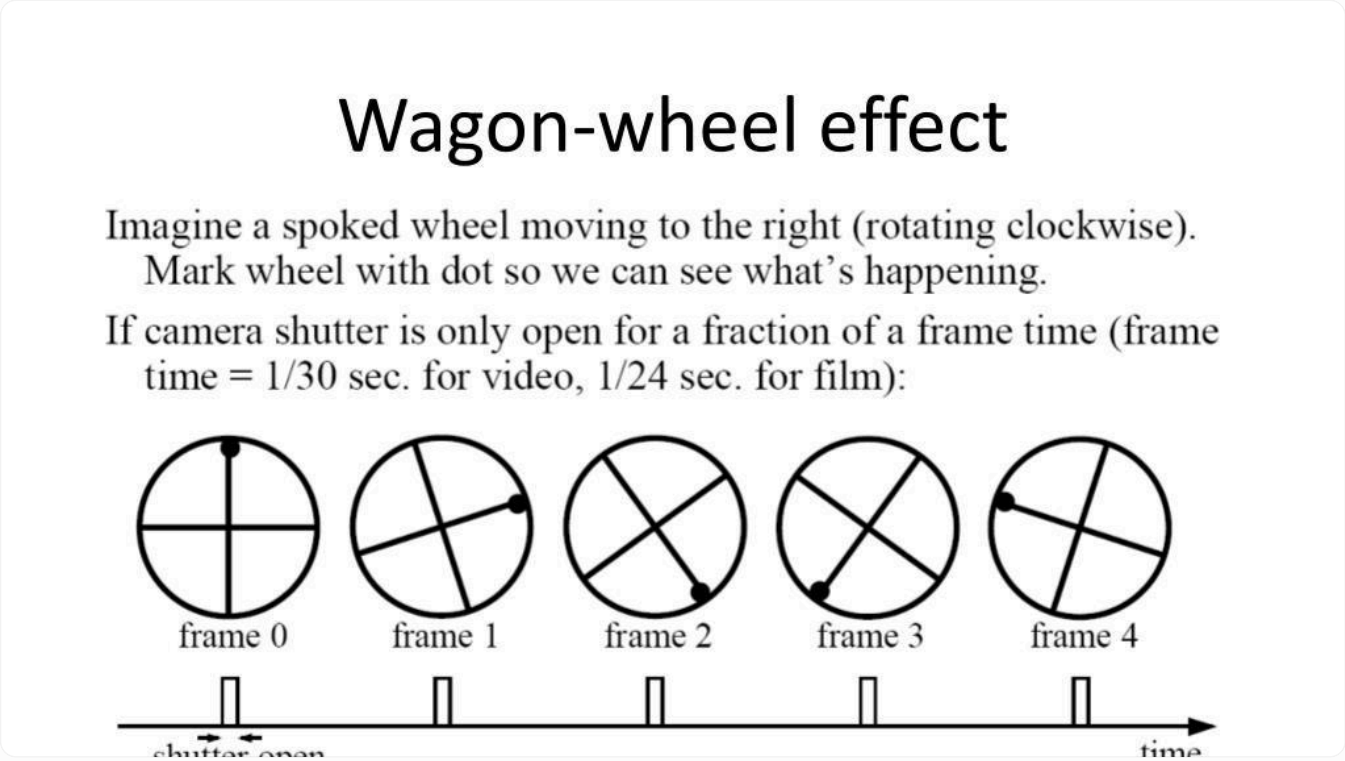
This is called **scan-out**.

Imagine printing a picture one line at a time. If the paper moves while the printer is working, the picture can become distorted.

Similarly, when you turn your head in VR while the display is updating line by line, a straight virtual object can appear tilted or wobbly. This is called a **wobble or waggle effect**.

Memory trick: **Scan-out = drawing the screen line by line.**

9. What is the wagon-wheel illusion?



Imagine a wheel with several identical spokes rotating very quickly.

Sometimes it appears to:

- Stop moving.
- Rotate more slowly than it really does.
- Rotate in the opposite direction.

Why? The brain sees separate images and may incorrectly match a spoke in one frame with a different spoke in the next frame.

This is related to the **correspondence problem**: the brain must decide which part of one image corresponds to which part of the next image.

Memory trick: **Fast wheel + separate frames = wrong direction possible.**

10. What are Reichardt detectors?

The lecture discusses hypothetical neural circuits called **Reichardt detectors** that may help explain how the visual system detects motion.

Think of them as tiny motion-detection systems that compare visual information arriving at different times or places.

Scientists study motion processing in the retina and visual brain, but the full biological process is more complicated than one simple detector.

Memory trick: **Reichardt detector = proposed motion detector.**

Part 2: Depth perception

11. What is depth perception?

Depth perception means understanding **which objects are near and which are far away**.

For example, when you look at a road, you can tell that a nearby car is closer than a car near the horizon.

Your brain does not measure every distance directly. Instead, it combines different clues called **depth cues**.

There are two main types:

Monocular cues	Binocular cues
Clues that can be used with just one eye.	Clues that depend on information from both eyes.

Memory trick: **Mono = one; bino = two.**

12. Monocular depth cues

These are the main clues you should memorize.

A. Retinal image size

Imagine two identical balls. One looks big and the other looks small.

Your brain often concludes that the bigger-looking ball is closer and the smaller-looking ball is farther away.

This works best when you know the objects are roughly the same real-world size.

Memory trick: **Bigger image often means nearer object.**

B. Height in the visual field

Objects near the horizon often appear farther away, while objects lower in the visual field can appear closer, depending on the scene.

For example, the bricks on a road appear farther away as they approach the horizon.

Memory trick: **Position relative to the horizon gives a distance clue.**

C. Linear perspective

Railway tracks are parallel in real life, but they appear to meet at a distant point.

Your brain uses this pattern to estimate distance.

Memory trick: **Parallel lines appear to meet far away.**

D. Texture gradient

Look at stones on a pathway:

- Nearby stones look larger and more separated.
- Distant stones look smaller and closer together.

This change in size and spacing is called the **texture gradient**.

Memory trick: **Near = larger texture; far = smaller, crowded texture.**

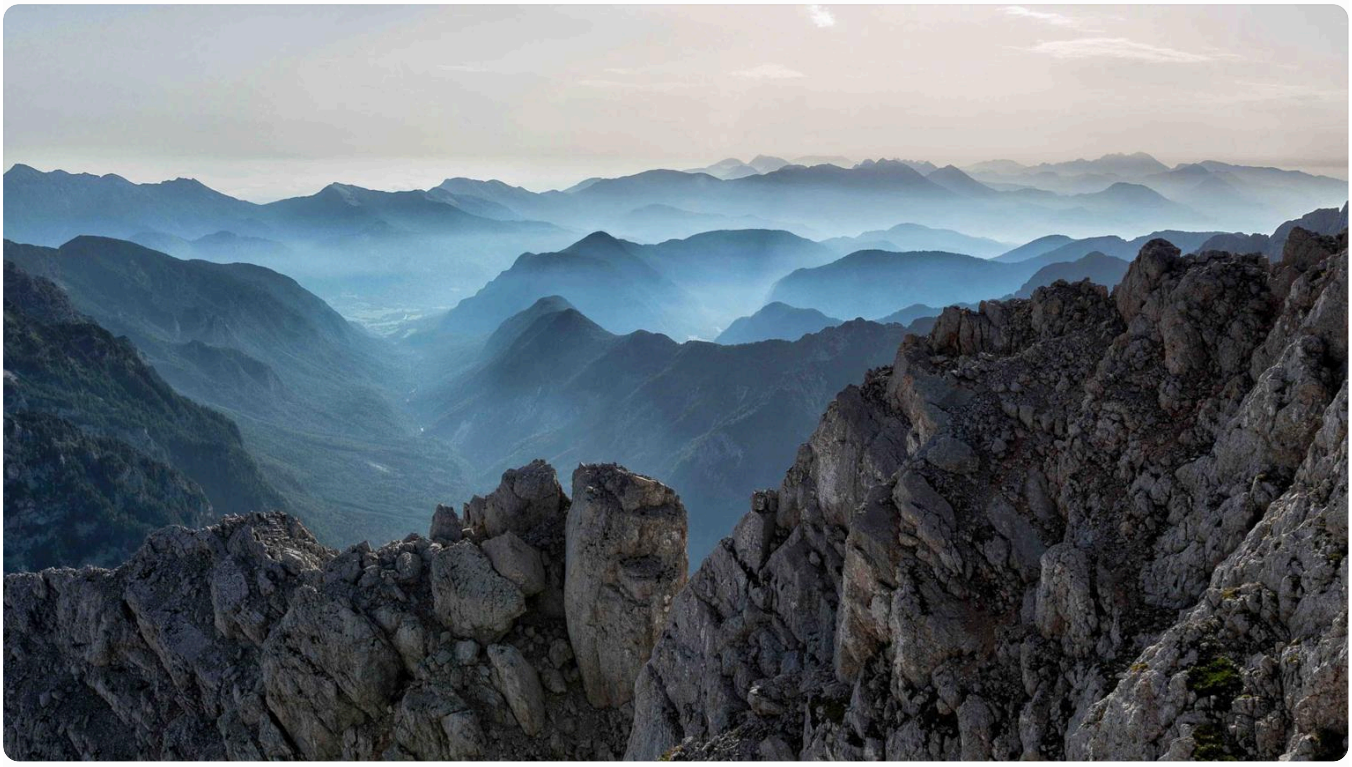
E. Image blur

Objects that are out of focus often look blurry, while objects in focus look sharper.

Your brain can use this difference as a depth clue.

Memory trick: **Blur can indicate distance or being out of focus.**

F. Atmospheric perspective



Distant mountains often look lighter, hazier, and less detailed because of the atmosphere between you and them.

Memory trick: **Far mountains = hazier appearance.**

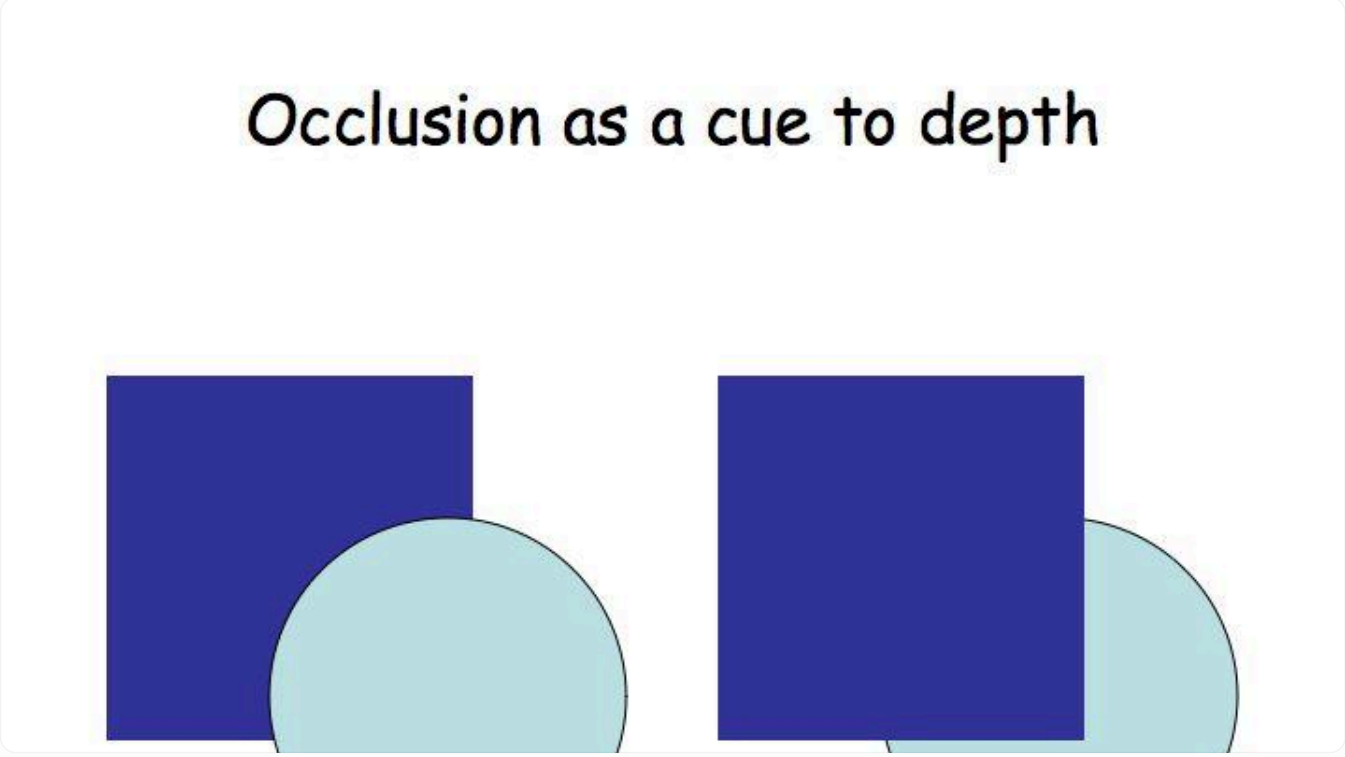
G. Shadows and reflections

Shadows help the brain understand an object's shape, height, and position.

For example, a ball's shadow on the ground can help you judge whether the ball is resting on the ground or floating above it.

Memory trick: **Shadows help show shape and position.**

H. Interposition (overlapping)



If one object covers part of another, the covering object is usually perceived as being in front.

For example, if a tree hides part of a house, you normally perceive the tree as closer.

Memory trick: **The object that covers another is usually in front.**

I. Accommodation

Accommodation happens when the eye's lens changes shape to focus on an object.

When you look at something nearby, the lens becomes more curved to focus the image.

Your brain can use this focusing effort as a clue to distance, especially for relatively close objects.

Memory trick: **Accommodation = lens adjusts focus.**

13. Which depth cues are present in panoramas?

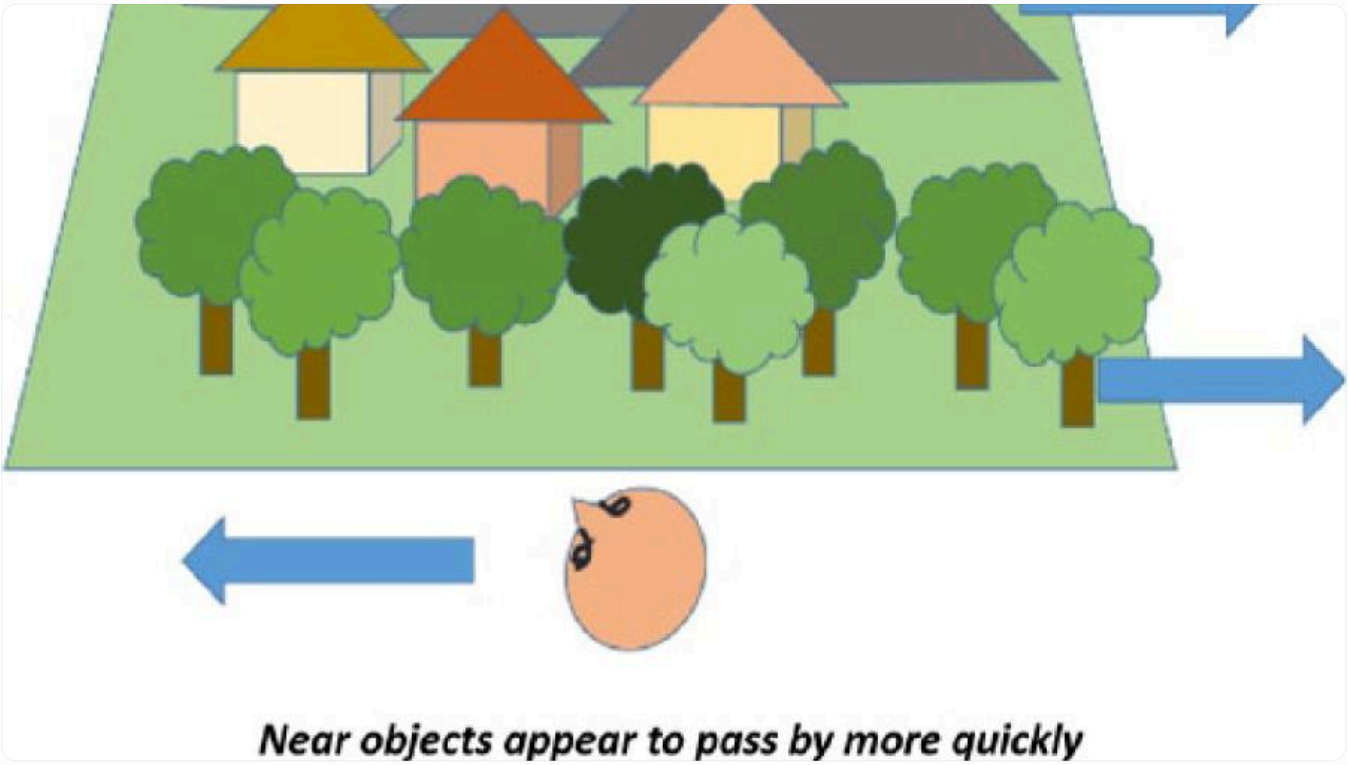
A panorama is a wide photograph that can be displayed around you inside a virtual sphere.

It can look immersive, but the scene is essentially a captured image rather than a fully modelled 3D environment.

Depth cue	Usually present in a panorama?
Retinal image size	Yes
Height in visual field	Yes
Linear perspective	Yes
Texture gradient	Yes
Image blur	Yes, if captured in the photograph
Atmospheric perspective	Yes, if captured in the photograph
Shadows already in the photograph	Yes
Interposition / overlapping	Yes
Accommodation for different scene depths	No, not naturally
Natural motion parallax when you move sideways	No, not from a single fixed panorama
Correct binocular disparity between objects at different depths	No, not from a single ordinary panorama

The last three are particularly important. The panorama contains visual clues about depth, but it does not contain the full 3D geometry needed to reproduce every depth effect correctly.

14. What is parallax?



Imagine looking out of a moving bus:

- A nearby tree seems to move quickly across your view.
- A distant mountain seems to move much more slowly.

This relative movement is called **motion parallax**.

Memory trick: **Near objects shift more; distant objects shift less.**

In a single fixed panorama, moving your head sideways does not naturally recreate this correct relative movement because all the objects are painted into the same image.

15. What is vergence?

Vergence is when your two eyes turn in different directions to look at an object.

- Looking at a nearby object: both eyes turn inward.
- Looking at a distant object: the eyes become more nearly parallel.

This helps your brain estimate how far away an object is.

Memory trick: **Near = eyes turn inward.**

Vergence can occur in stereoscopic VR, but a single panorama does not correctly represent different vergence angles for every object at its captured depth.

16. What is binocular disparity?

Your left eye and right eye see slightly different views of the world because they are separated by a small distance.

The brain compares these views to help estimate depth. This difference is called **binocular disparity**.

For example, hold one finger close to your face and look at it with each eye separately. The background appears to shift relative to the finger.

Memory trick: **Two eyes, two slightly different views = binocular disparity.**

17. What is diplopia?

Diplopia means seeing a double image.

Imagine focusing on your finger close to your face. A more distant object behind it may appear as two images, one to either side of the finger, when you examine the views from both eyes.

Normally, your brain combines visual information so that you are not constantly aware of all these double images.

Memory trick: **Diplopia = double vision.**

18. What is a horopter?

A **horopter** is the set of locations in space that correspond to matching points on your two retinas when you fixate on a particular point. Objects on or near it can be perceived as single rather than double under the relevant viewing conditions.

Imagine focusing on a finger. There is a particular geometric region associated with that fixation where objects have matching retinal positions.

Objects away from this region can produce binocular disparity and, if the disparity is large enough, double vision.

Memory trick: **Horopter = locations associated with matching retinal points.**

The lecture suggests that understanding horopters may help researchers design VR reading surfaces that are more comfortable.

19. How can virtual IPD affect perceived size?

IPD means **interpupillary distance**: the distance between the centres of your pupils.

In a VR system, changing the spacing between the virtual cameras that represent the eyes can change the apparent scale of the scene.

- A smaller virtual camera separation can make the world appear larger.
- A larger virtual camera separation can make the world appear smaller.

This is a perceptual effect, and the exact result depends on the scene and viewing conditions.

Memory trick: **Virtual eye spacing can change perceived world scale.**

20. Why use panoramas in VR?

Panoramas have an important advantage: they can create an immersive experience without modelling every object in 3D.

Advantages:

- They can be relatively inexpensive to render.
- They capture realistic scenery, lighting, and shadows.
- They can run well on less powerful devices.

Limitations:

- Virtual objects do not automatically cast correct shadows onto the photograph.
- Throwing or placing objects on the pictured floor is difficult because the floor is not a real 3D surface.
- Moving sideways does not naturally create correct parallax.
- Different scene depths do not naturally produce correct vergence and accommodation cues.

A possible student project is to capture multiple panoramas and let users teleport between them using a simple head-controlled interaction. This creates a way to explore a scene without building every detail as 3D geometry.

Part 3: MCQs for exam practice

Try answering these questions before checking the answers.

16-question MCQ test 0/16 answered

1. What does FPS stand for?

- A. Frames Per Screen
- B. Frames Per Second
- C. Focus Per Second
- D. Frequency Per Scene

2. What is flicker?

- A. A flashing appearance
- B. A 3D shadow
- C. A blurry photograph
- D. An eye lens

3. What is judder?

- A. Smooth motion
- B. Jumpy or uneven motion
- C. Colour blindness
- D. A type of lens

4. What can low-persistence displays reduce?

- A. All depth cues
- B. Retinal image slip and motion blur
- C. The number of eyes
- D. The field of view

5. What does scan-out mean?

- A. The eye scanning a book
- B. The display updating pixels in a sequence
- C. Turning off the headset
- D. Measuring IPD

6. Why can a wagon wheel appear to rotate backwards?

- A. The wheel actually reverses
- B. The brain incorrectly matches spokes between frames
- C. The wheel has no spokes
- D. The screen is always black

7. Which is a monocular depth cue?

- A. Binocular disparity
- B. Vergence
- C. Linear perspective
- D. Stereoscopic difference

8. What is texture gradient?

- A. A lens changing shape
- B. Textures appearing smaller and closer together with distance
- C. A screen flashing
- D. Two eyes turning inward

9. What is motion parallax?

- A. A lens focusing
- B. Nearby and distant objects shifting differently as the observer moves
- C. A display's refresh rate
- D. Double vision

10. What is binocular disparity?

- A. A difference between left-eye and right-eye views
- B. A change in screen brightness
- C. A type of shadow
- D. An image blur effect

11. What does diplopia mean?

- A. A single image
- B. Double vision
- C. A blurry lens
- D. A fast eye movement

12. What is accommodation?

- A. Eyes turning inward
- B. The lens changing shape to focus
- C. The head moving sideways
- D. A display updating

13. Which cue tells you that one object is in front of another?

- A. Interposition
- B. Flicker
- C. Refresh rate
- D. IPD

14. What is a horopter?

- A. A type of VR display
- B. A set of locations associated with corresponding retinal points
- C. A screen refresh cycle
- D. A motion detector

15. What is a major limitation of a single fixed panorama?

- A. It cannot show colours
- B. It cannot naturally reproduce correct motion parallax
- C. It cannot show shadows in the photograph
- D. It cannot create immersion

16. What does IPD stand for?

- A. Image Processing Display
- B. Interpupillary Distance
- C. Internal Pixel Definition
- D. Inter-frame Picture Delay

Check answers

Last-minute memory sheet

Learn these one-line meanings first:

Term	Remember it as
FPS	Pictures per second
Hz	Screen refreshes per second
Flicker	Flashing
Judder	Jumpy motion
Retinal image slip	Image slips across the retina as the eyes move
Scan-out	Display updates in a sequence
Wagon-wheel illusion	Wheel appears to stop or reverse
Monocular cue	Depth clue from one eye
Binocular cue	Depth clue using both eyes
Retinal image size	Bigger image often means nearer
Linear perspective	Lines meet in the distance
Texture gradient	Distant patterns look smaller and denser
Interposition	Covering object is usually in front
Motion parallax	Near objects shift more during movement
Vergence	Eyes turn inward or outward
Accommodation	Lens changes shape
Binocular disparity	Difference between two eye views
Diplopia	Double vision
Horopter	Locations associated with matching retinal points
IPD	Distance between pupil centres

Best way to study: First learn the memory sheet, then take the MCQ test without looking at the notes. Pay special attention to the difference between **flicker and judder**, and between **monocular and binocular depth cues**.

You said:

Quiz-7-2026.pdf
PDF

explain easily

ChatGPT said:

NPTEL Week 7 Quiz — Easy Explanation

I'll explain all 10 questions from your PDF in very simple English, with easy examples and memory tricks to help you remember them for the exam.

1. Reichardt detector

Correct answer: B — It detects motion by comparing the timing of neighboring receptors.

Imagine two sensors, A and B, placed side by side.

- A detects an object first.
- A short time later, B detects it.
- The brain uses this timing difference to detect movement and its direction.

Easy example: If a moving ball passes sensor A and then sensor B, the timing helps detect that the ball is moving from A to B.

Remember: Reichardt detector = **motion + timing**.

2. Why do two still images look like motion?

Correct answer: B — Activation of motion detectors by appropriately timed successive frames.

Imagine a cartoon. Each drawing is a still picture, but when you show the drawings quickly one after another, the character appears to run.

Nothing is physically moving in the individual pictures. Your brain interprets the changing images as movement.

Remember: Still pictures + correct timing = apparent motion.

3. How does the brain distinguish self-motion from object motion?

Correct answer: A — Proprioceptive efference copies.

Imagine you are sitting in a chair and turn your head to look at a door. The image moves across your eyes, but you understand that **you moved your head**, not that the whole room moved.

Your brain combines information from:

- **Retinal motion:** how the image moves across your eyes.
- **Motor-command information:** signals related to your own movements.
- **Vestibular information:** information about head movement from the inner ear.
- **Optic flow:** the overall movement pattern of the scene as you move.

Remember: The brain compares visual movement with information about your own movement.

4. What suppresses motion signals during rapid eye movements?

Correct answer: C — Saccadic suppression.

A **saccade** is a very fast movement of your eyes from one point to another.

For example, when you quickly look from your book to the blackboard, the image moves rapidly across your retina. But you do not normally experience the entire classroom as sweeping across your vision.

Your brain temporarily reduces awareness of this motion.

Remember: Saccade = quick eye jump; saccadic suppression = reduced perception during that jump.

5. At approximately what frame rate does motion first become perceptible?

Correct answer: A — 2 FPS.

FPS means **frames per second**.

- 2 FPS means two images are shown every second.
- At around 2 FPS, motion may become perceptible, but it looks very discontinuous.
- Higher frame rates generally make motion look smoother.

Remember: FPS = pictures shown per second.

This is the approximate value given in your lecture, not a universal threshold for every situation.

6. What does increasing frame rate mainly reduce on an always-on LCD?

Correct answer: B — Judder caused by retinal image slip.

An always-on display keeps each frame visible until the next frame appears.

Imagine looking at a moving car while your eyes follow it. Your eyes move smoothly, but the screen updates in separate steps. This mismatch can make the image appear blurred or uneven.

Increasing the frame rate makes these updates more frequent and can reduce this effect.

- **Flicker:** the display appears to flash.
- **Judder:** motion appears jumpy or uneven.
- **Retinal image slip:** the image shifts across the retina as the eyes move while the displayed image updates.

Remember: Always-on LCD → think about image slip and judder.

7. What do low-persistence OLED displays reduce?

Correct answer: A — Motion blur caused by retinal image slip.

A low-persistence display shows each frame for a short time and then briefly turns it off before the next frame.

Imagine taking a photo of a moving fan. If the image remains visible for too long, the moving blades may look blurred. Showing the image for a shorter time can reduce this blur.

However, flashing the display too briefly or too slowly can cause flicker.

Remember: Low persistence = shorter display time = less motion blur.

8. Why are VR headsets commonly recommended to operate around 90 Hz?

Correct answer: B — It is a compromise between flicker, judder and hardware limitations.

Hz means **how many times the display refreshes per second**.

A 90 Hz display refreshes 90 times per second.

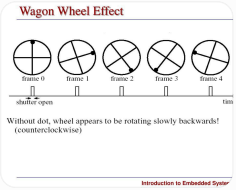
VR needs smooth visual updates because the display is close to your eyes and changes in head position can make display problems more noticeable.

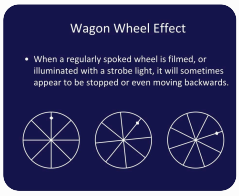
The lecture presents 90 Hz as a practical recommendation that balances visual comfort and hardware limitations. It does not mean that human vision cannot perceive higher refresh rates.

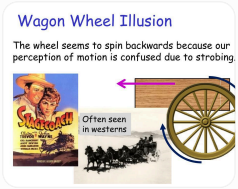
Remember: 90 Hz = a practical VR balance, not a perfect limit.

9. What does the wagon-wheel illusion demonstrate?

Correct answer: C — Correspondence ambiguity between successive frames.







Imagine watching a wheel with several identical spokes in a video. Sometimes the wheel appears to rotate backward, even though it is actually moving forward.

Why?

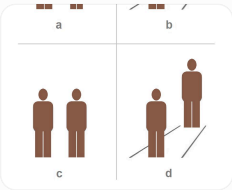
The brain tries to match a spoke in one frame with a spoke in the next frame. Because the spokes look similar, it can match the wrong spokes and interpret the movement incorrectly.

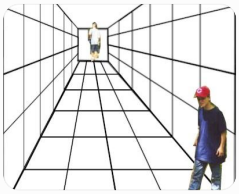
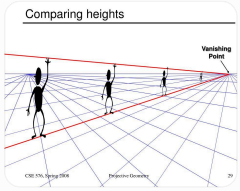
This is called **correspondence ambiguity**.

Remember: Wagon wheel illusion = the brain matches the wrong features between frames.

10. Why does a farther person look smaller in an image?

Correct answer: B — Retinal image size.



Imagine two people who are approximately the same physical height. One stands near you, and the other stands far away.

The farther person usually occupies a smaller area in your image and forms a smaller image on your retina. Your brain interprets this as a sign that the person is farther away.

This is a **monocular depth cue**, meaning you can use it even with one eye closed.

Remember: Smaller retinal image of a familiar object often means farther away.

Quick revision table

Q.	Topic	Correct answer	Remember this
1	Reichardt detector	B	Motion + timing
2	Apparent motion	B	Still images shown in sequence
3	Self-motion	A	Motor-command information
4	Saccadic suppression	C	Reduced perception during quick eye jumps
5	Motion perception	A	About 2 FPS in the lecture
6	Always-on LCD	B	Retinal image slip and judder
7	Low-persistence OLED	A	Reduced motion blur
8	VR refresh rate	B	Practical compromise around 90 Hz
9	Wagon-wheel illusion	C	Wrong matching between frames
10	Retinal image size	B	Smaller image often means farther away

Test yourself

10-question MCQ practice

0/10 answered

1. Which detector uses the timing of neighboring receptors to detect motion?

- A. Reichardt detector
- B. Horopter
- C. Vergence detector
- D. Texture detector

2. Why can a cartoon look as if it is moving?

- A. The paper moves
- B. The brain interprets successive still images as motion
- C. The eye lens changes shape
- D. The images become larger

3. What is a saccade?

- A. A depth cue
- B. A slow change in focus

C. A rapid eye movement

D. A display refresh

4. What is judder?

A. Jumpy or uneven motion

B. A change in colour

C. A depth illusion

D. A lens defect

5. What does FPS stand for?

A. Focus per second

B. Frames per second

C. Frequency per screen

D. Frames per sensor

6. What is a key benefit of low-persistence displays?

A. They remove all depth problems

B. They reduce motion blur

C. They eliminate binocular disparity

D. They remove the need for refreshes

7. What does 90 Hz mean?

A. 90 pixels per image

B. 90 colours per second

C. 90 refreshes per second

D. 90 seconds per frame

8. What causes the wagon-wheel illusion?

A. Correspondence ambiguity between frames

B. Accommodation

C. Atmospheric perspective

D. Retinal image size alone

9. Which is a monocular depth cue?

A. Binocular disparity

B. Vergence

C. Retinal image size

D. Two-eye comparison

10. What does a smaller retinal image of a familiar person often suggest?

A. The person is farther away

B. The person is definitely shorter

C. The person is moving faster

D. The display has lower Hz

Check answers

Final memory trick: Motion questions usually involve timing, changing frames or eye movement. Depth questions usually involve clues such as image size, perspective or information from both eyes.

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[Music] and um vestibular system is one part of our perception. So now I'm going into chapter 8 vestibular system uh topics and a little bit of chapter 12 that talks about evaluation how to evaluate VR systems and experiences on human subjects and this is a new slider.com. So starting today and the next time this is the road map of how we will study vestibular system. So we will distinguish the elements of vestibular system. So essentially um physiology we'll talk about physiology of the vest vestibular system. Then we talk about vection so and identify factors that affect it. So we'll talk about perception, human perception um related to the vestibular system and then we'll talk about the uh effects of your sick sickness and design of virtual reality experiences for user comfort. So how to maximally improve comfort and after understanding the effect that vestibular system has on perception or and comfort and we'll talk about scientific method if we have time and how to design research experiments for this kind of um problems. So uh we are going to start talking about the vestibular system and how the human body determines motion or self motions and self changes in or self-orientation and self position. So changes in position and orientation of our heads. So our vestibular system is located inside of our inner ear. So right next to our sound system and it's responsible for the sense of balance. But essentially and ultimately it measures. So the vestibular system measures the head position orientation and then it sends that information to the brain so that the brain then can maintain the posture, the balance, the equilibrium. Right? So this system is important to understand in the class because it is so relevant to virtual reality. It's relevant to comfort. And if you're not careful, if you don't understand this how it's relevant, then we may design experiences that lead to ver sickness. So maybe it's the most important part of our class, right? >> The human ear is divided into three compartments. The external ear, middle ear, and inner ear. The inner ear contains the spiral-shaped cookia where sound waves are transduced into neural signals and the vestibular complex which contains the receptors for our sense of equilibrium. The central egg-shaped cavity is the vestibule which contains a pair of membrinous sacks the sacule and the utricle. Inside the utricle and sacule are hair cells similar to those in the organ of corti. The hairs are clustered in the macula where their processes are embedded in a gelotinous mass and lie under a thin layer of crystals called ottoliths. When the head tilts, gravity moves the crystal mass and distorts the stereocyia of the hair cells. This is how the sacule and utricle provide information about position with respect to gravity. >> Position. So position utricle and saculle are essentially like accelerometers and the brain can infer position. So on this picture then there is uh this is where the inner ear is located within our skull. So this is the right inner ear, the left inner ear if you're looking at the head from above. So this video that we just watched described the ottoith organs. So they they have the ottoith organs have actually very interesting geometry. So the utricle and sacle are are located in two orthogonal planes. So if I were to draw the axis here, so say Z goes up, X goes forward and Y goes sideways. So it's a little different from what we did in our math courses, but that's okay because um I just want to be consistent with this videos that we're talking about later. So after that we'll reverse to where uh to when when we talk about your patrol we'll talk we'll we'll redefine aces but this is just so that the next video is okay for you to watch. So then um the utricle is located in y z vertical plane and um the cycle is located in s and y horizontal plane right here. And then even more interesting that the second part of the visib system is called the simmer circular canals and they consist of this three canals. They're all orthogonal to each other. And you can think of these canals as gyroscopes or well they're more like angular acceleration uh sensors because there's rotation of liquid happening in each of these canals around x y and z axis as well. So this should remind you very closely then um if you remember from very early uh lectures on bird's eye view and we talked about MEMS's gyroscopes and MEMS's accelerometers. So the gyroscope also was measuring um rotations around three axis. So this is what we have inside of our brains and it evolved over millions of years and it's hardwired in us. So very similar computations are happening within our heads. So the so you can think then kind of like roughly that the organ is kind of like an accelerometer. So these two plane organs in each plane and then the semicircular canals are kind of like a gyroscope except that they really do integration double integration of angular accelerations but um it's kind of like a gy gyroscope in the sense that it will allow estimates of orientations or the yaw pitch and roll of the head. So now let's look at similarities kind of and differences between vestibular system and our engineered sensors. >> The vestibular system has important sensory and motor functions contributing to the perception of self motion, head position and spatial orientation relative to gravity. The function of the vestibular system can be simplified by remembering some basic terminology of classical mechanics. All bodies moving in three

dimensions have six degrees of freedom. Three of these are translational and three are rotational. The translational components may be given in terms of movements along the X, Y, and Z axes of the head. Rotations about the X, Y, and Z axes are commonly referred to as roll, >> pitch, >> pitch said that they call pitcher and y. So this is really >> and yaw >> different they they name the differently which is fine. >> The positive direction of head rotation follows a right-hand rule. That is if the fingers of the right hand are curled in the direction of the arrows, the thumb points in the positive direction of the axis. Buried deep in the temporal bone. The main peripheral component of the vestibular system is an elaborate set of interconnected chambers, the labyrinth that has much in common and is in fact continuous with the cookia. The labyrinth consists of the two odtoolith organs, the utricle and sacule, and three semic-ircular canals. The vestibular hair cells, which like cclear hair cells, transduce minute displacements into behaviorally relevant receptor potentials, are located in the utricle and sacule and in three jug-like swellings called ampule, located at the base of the semic-ircular canals next to the utricle. In the utricle and sacule, the sensory epithelium or macula consists of hair cells and associated supporting cells. Overlying the hair cells and their stereocyia is a gelatinous layer. Above this layer is a fibrous structure, the odolithic membrane in which are embedded crystals of calcium carbonate called odiconia. The crystals give the odolithic organs their name. So yeah, I wanted to finish this video so that we start already with having all of these definitions next time and we'll go into yeah into perception. So we we'll finish with the physiology of the vestibular system and a little bit of neuroscience. So what is it exactly? How do we know how the signals are sent to the brain and what the brain does to compute these values? But then we don't really understand a lot about the human brain yet. A lot of things are unknown. So instead perceptual psychologists learned a lot about how we maintain our balance, how we compute our balance which comes from vision. So we we'll again come back to vision because vision is the ultimate um drift character and a very important source of information about our uh self orientation and positions in the world. So uh so we will go through all of that and learn everything that we know that may help us design uh good VR systems at this point. So see you uh then on Tuesday for that. [Music] So now let's look at this object flow. So imagine that you have a scene where objects or features move like that. And what I want to do is instead I want to crop out the scene and zoom in. So if your retina originally was like this then I just want to I will not shrink your retina. I will just crop and zoom in. So I have a question. What will be the change in perception for this scene in comparison to this one? How will it look like? Those are potential answers. So try guessing and then I'll show you exactly the answer in an animation. It's difficult to guess, right? Because it's about perception. So without trying it, it's really hard to tell. So look what happens. So it just looks like things move slower, right? Yeah, maybe there's zooming in effect as well, but still we don't care as much because it seems like the same scene. It just things move slower. There's zoom in as well. But okay, so this is the perception. That's what happens. So cropping, changing the field of view, but also zooming in. Very interesting, right? Okay. So perceptual psychologists did various experiments about how optic flow affects our sense of motion and balance. So this example is a study and it's done by house and and zanker in 1951. So these experiments were done on uh how the optic flow affects a common uh fly common house fly. So the researchers then taped this poor fly uh to some location, taped it so that it couldn't fly away and changed the optical flow around the fly in a cave system. So they projected some kind of a moving world, right? And then what happened is that the fly changed its direction and flight patterns according to the optical flow. So if you think that flies are just silly and they did it, look at this experiment with humans. So it also works. The same thing works for humans. Let's see. So this video was done in 1974. >> How important then is this visual information in maintaining balance. In most adults, even when doing a difficult task, the sway is small and balance is easily maintained. But if visual information is important, then it should be possible to upset a subject's balance in an experimental room by moving the room gently forwards or backwards. Infants learning to stand are less stable than the adult and far more easily put off balance. The infant in this experiment has only been walking for a month. When we move the room backwards, the child falls. And again, in 92 trials of this type with seven infants, balance was clearly disturbed in the predicted direction in 82% of the trials. It seems then that for the infant learning to maintain balance, visual information is far more potent than mechanical information. But the adult, however, is less sensitive to conflict between visual and mechanical information, and movements of the room do not, as with the infant, knock him over. But let us give him an unfamiliar balancing task, such as standing on a narrow beam. With his eyes closed, he depends solely on mechanical stimula. quickly becomes unstable and falls off or with his eyes open. If the room is moved even slightly, he will lose his balance just as the infant did. So this is really fascinating. This experiments show how amazing virtual reality can be for this kind of studies. And in fact when you talk about no killer app for VR in VR um we always forget that something where VR will stay for sure for for for a long time is actually these studies um in uh human perception and neuroscience because you can create all sorts of controlled stimuli and then study the um reactions of people or their perception which will help us understand more about ourselves and I hope this lecture will show how much we don't understand for very simple things such as um our vestibular system and how brain analyzes this information. So you'll see in a bit how little we understand about relationship between optic flow vection and resulting VR sickness cinematography is related. So this brings us to vection. So what is vection? Vection is related to optic flow but at the same time it's a completely different notion. So vection happens when a large part of the visual field moves. So if there is a lot of optic flow but the viewer is not moving. So uh but so if there's a lot of optic flow and the viewer is not moving

but the viewer feels like he did. So that is vection. ation is the illusion of self motion when there was no selfm motion which was produced by optic flow or pure visual stimulation. There are other by the way um that's the definition that we'll use in class because there's vet that is produced not only by visual stimulation but so if you if you're really into this research you'll see that maybe uh definition of vction shouldn't be just like this maybe there could be other stimulation but it's okay to use for our purposes this definition. So what are examples? So when does it happen in real life? Well, you just saw uh possibly in this video that you just saw, maybe people experienced self motion there maybe. So that optic flow of the room moving is very unusual in real real world scenario. So it's very likely that the infant and that human felt it. But other example would be sitting in a stationary train and then you're looking outside the window, another train is moving and you feel like you're not sure which one moved. Was it the train next to you or your train? Right? So that feeling of self-motion when you actually didn't is vion. So that illusion of self motion. So then maybe you look then at another window in the train and see that okay the actually the train station is stationary. Your train didn't move. So that was the illusion that was the next train to you moved. So that that's fashion in real life. Right? So we can see that in this situation where the brain is not sure because the vision tells you that you're moving. There is motion with respect to the next train. So possibly you're moving. So the vision says you're moving but the vestibular system will send this the signal that you're not moving because your train actually didn't move. Then the vision wins. Often the vision wins because it's our primary sense and uh visibility system loses. So optic flow has a lot to do with vision because it's the optic flow that will trigger this sensation or this illusion. And this sensation is very distinct that I'm moving when you're not and you'll feel it in VR. So we'll have this homework 7 is exactly the homework on vection. So it's not released yet but I can show it to you. It's the sensation of vection. So we'll show you various optic flows in this um in this homework. and we'll ask you whether you feel like you are actually moving. Of course, you'll be just sitting still, not moving in reality, just putting the headboard, the cardboard, and experiencing the these object flows, but you might have a sensation that you're moving. So, that's what that's the questions that we'll ask in this homework. So, why is this important to virtual reality? So, what's the relevance? Well, in virtual reality, we are doing this over and over. So, we might be experiencing vection when we move our avatars with a mouse or a controller. So, the images that generate it then has a lot of object flow and may trigger the sensation of action or self motion. Um, and uh this can lead in long term to fatigue or discomfort. It may may lead in short term to nausea and discomfort or in long term it can lead to fatigue and nausea and just feeling of instability. So it has a profound impact on VR comfort. So vetin optic flows leading to vetion may lead to uh VR discomfort and surprisingly we do not know the real relationships in between. We do not know which optic flows are guaranteed to cause vection and we do not know whether vection is guaranteed to lead to VR comfort and we are not certain that certain optic flows that do not cause vection may still lead to VR comfort. So all these three things there is no clear there we the scientists do not know clear relationship between these three and this is fascinating just fascinating how much we don't know. This is why uh this homework is so important because we actually will ask you to consent to analyzing your data because we want to understand more. We want to see if we can use this data for the hypothesis that we have and maybe we can even publish it if um if the data is uh supporting our hypothesis. So I'll walk you in the end of this lecture through the scientific process and we'll show you exactly what we're doing in this homework. So you might be actually participants in a valid human subject experiment that will help us to understand a little bit more about this relationship between vion optic flow and VR comfort. Okay. So in virtual reality um we have two options to fix the discomfort. So if we know that there is a relationship between object flow and vection and um and comfort maybe we could just not show optic flow at all or not cause vection at all so that it's all comfortable. So how could we do that? Well maybe we could have some kind of hardware solution to this problem. So um with action and optic flows we are presenting the stimula of self motion only to vision right with visual display but there are other display or other senses that we do not really present the same stimulation. So there are more senses that would detect self motion such as our vestibular system. Uh but we do not really stimulate the vestibular s system. So we do not fully immerse uh ourselves in this VR system in the sense that if we want to pres present motion of the avatar then we would have to present it on all of the displays to all of the senses that would otherwise uh sense the motion or self motion. But we don't uh we could so there is actually there are vestibular system displays that is uh displays that would send a signal artificial signal to the vestibular system as well about the self motion but they are just R&D and I've heard that they're quite painful because it involves direct stimulation of the vestibular nerve which is painful. Therefore, we are only left with software solution or best practices guide on how to eliminate potential side effects of fashion or uh the discomfort associated with that. So that's why we study uh best practices guide and maybe the most important part of it is to actually create comfort related to this self motion illusions. Okay, so things are worse because of lens distortion. And uh you've seen this slide before where I talk about our ability to detect optic flow is strongest around 20° of phobia. So if you have this kind of optic flow, wait, I need to turn off pen. So if you have this kind of optic flow then you detect your optic flow mostly around 20° of a phobia where there's the highest concentration on retina of rods and anywhere on periphery. So that's where we really are good at detecting optic flow. It's the strongest and uh what does it mean? So think of the implications for virtual reality. Well vion is not a good thing in VR headsets. It is made worse on the periphery due to lens aberrations. Right? So periphery is where we perceive optic flow the most but that's where lenses make it worse. So this is

quite a problem. Right? So we saw something like that with aliasing. Right? Lens aberrations make it worse make it look worse but it's a flaw that you don't want people to see. Similarly optic flows. We don't want them to be changed due to the aberrations or less distortions. We don't want them to get worse. this optic flow due to these lens severations but they are made worse and we see them we perceive them more there. So that's kind of very unfortunate. So let's look at more examples of mismatches between vestibular system and vision. So vion was one but let's look at more. So here are here's a table of when vestibular system and vision system receive contradictory information about self motion. So for example vestibular system will report that there's accelerations associated with self motion. So it will say yes there's motion but maybe the vision will say no. So the the picture the view is stationary nothing is moving on our retinas or the other way around. So let's go I'll ask you to look at each example each of these cases and give me an example that happens either in real life or virtual reality. So what is an example of a situation where vestibular system says there is motion and vision or self motion and vision says no there is no selfm motion. What is a scenario for this situation? Give me both scenarios in real life and virtual reality. Sitting in an airplane. Very good. Sitting in an airplane is a real world scenario. What about virtual reality bike? Right. No, I would say sitting. Yeah. In she so inside of a ship, right? Or inside of a car where you're looking only in the interior. Then the interior will move together with your head and if you're looking still, of course. And then uh so nothing is moving really, but you have a sibler system will detect accelerations associated with the motion of the vehicle that you're in. So it's very good. Also, another case is when you are in virtual reality, imagine that the track tracking system is broken and it's off. So, you have your VR VR world and as you move around, it's stationary, right? Nothing moves. So, it's actually very uncomfortable. U we don't have a homework like this anymore in this course, but we used to have a a demonstration of such situation, which is uncomfortable. Okay. Now situation is the other way around. So vibable assistant says no there is no motion. Vision says yes there is self motion. Self motion of course roller coaster VR. There you go. So any kind of optical flow corresponding to large scale self motions. Right. So large scale uh optical flow will have that. So moving train passing by. Exactly. So exactly what we just discussed related to vection. So this is what usually triggers a vection sensation or illusion of self-motion. When vision says yes, vestibular system says no, there's no self motions. It happens in virtual reality all the time. If there's self motions in VR, uh the the or the optical flow corresponding to self motions. Okay. Uh what's the next one? Can you think of an example where vestibular system says yes but vision has delayed or mismatched information? So not really matching it. Maybe it detects some self motion but it's kind of wrong. I don't know if you can guess this one. Maybe it's a little obscure. Starting walking treadmill. That's very interesting VR game. So that's something I have to think about. Yeah. H but what I meant is um so you can simulate this dropping frames or latency in VR. We do have a simulation of dropping frames in homework five was it? I think homework five right so you can experience it. It's not comfortable right so something dropping frames is not comfortable and latency in VR is not comfortable. So until latest advances in IMUs that made latency or perceived latency almost zero, VR before that was very nauseating and uncomfortable for most newcomers. Okay, now the other way around. So it's the vestibular system that's delayed and mismatch and vision says yes. So what is it then? Uh-uh. Nuh-uh. So, vestibular system or maybe if you had an accident and you um disturb your vestibular system, but usually it's either vestibular system disorders or alcohol is a very good example of that. So, what happens with uh when you consume alcohol is that if you have enough of it in the blood, it actually penetrates those organs that I just showed you. And once it penetrates those uh gelatinous uh liquids, it will change the viscosity of it. So which means that when it moves, it will send completely wrong signals or mismatching signals and that will result result and that's that's a hypothesis by the way. So it's not really proven. It's a hypothesis. It's called buoyancy hypothesis. And um so essentially these wrong signals from vestibular system cannot be really interpreted by the brain which results in this spinning sensation of the that you feel. Similarly for vestibular organ disorders. By the way I have to warn you that both Steve Lavel and I have vestibular organ disorders. So we have once in a while we have sensation of spinning and that makes uh trying various VR experiences even worse because of that. So please try to have comfortable VR experiences so that we are not really dying after trying all of the final projects. Okay. So please please consider all of this comfort measures that we'll talk about in this class. Okay. Yeah. in the deep ocean. I bet it changes some the vestibular organ might might be changing the perception, but um yeah, usually what I know about deep ocean is that you lose sense of where up is where is up. So, it's really hard sometimes to uh go back to the surface because um visual system doesn't really provide as much information anymore to help understand where up is. Okay, so let's talk about the symptoms, right? So there are a lot of not fun symptoms associated with vion or um sensation of um self motion or other mismatches. So this is called VR sickness. And what you may feel is fatigue, headache, nausea, dizziness, drowsiness and ice strain. And these are very subjective feeling. Right? So these are you can you can report that I'm not feeling well but it's a subjective measure. And of course we want to make we don't want to make people experience this thing. So we want to avoid any of these symptoms because otherwise they won't do VR anymore. Right? So sometimes it takes for some people just getting sick once and they'll never want to try it again. So we need to control as much as possible for in software as developers. So when you develop your final projects, you want to control for this paints because we do not have hardware that will help us to control for this.[music] Right? So [snorts] we are stuck with the self motion illusions and we have to make sure that we provide comfortable experiences. So although these are subjective, there's also objective sensations associated with VR

sickness and this is increased salivation. So you might might feel like that you your salivation increases or you might experience cold sweating or palar. So your face becomes unhealthy and pale looking or warmth and flushing or eye accommodation issues. So all of these although they're not comfortable they're actually good in the sense that we can measure them objectively and then we can quantify the experiences and see which one which VR experiences are actually okay and which ones are not with respect to these. So what is the reason for VR sickness? The most prominent theory again there's no answer there's just theories so we don't know uh but one is the most prominent is this sensory mismatch theory. So because mismatching stimulus are conflicting to each other this leads to nausea and emmesis which is the um the process of vomiting. So that's one theory. Another theory is that when all of this mismatch stimuli get integrated or need to be integrated by the brain the brain has to work harder. So it leads to fatigue and headache and eye strain. So slightly different set of symptoms as you can see not necessarily vomiting. So maybe maybe I like this one the most the poison hypothesis. So this kind of symptoms of conflicting stimuli usually happened in prehistoric times or we evolved in a way that this would usually happen if we injected something poisonous just like alcohol because then it changes the actual structure of the balance system. So which means that it's a good idea to eject. So we evolved to react to this kind of stimulus then to with this kind of symptoms that vomiting is one of them. So I like this one the most but again these are hypothesis. So another thing that is not known is even how to term it this VR sickness. Where does it belong? Because people studied various sicknesses with respect to visually induced motor sickness or flight simulation sickness or cyber sickness. Cyber sickness is just any kind of video game or any kind of digital device that can provoke this feeling of nausea. uh also there's actually sicknesses associated with actual physically uh physical um stimulation to the vestibular organ as well. So scientists really fight over where VR sickness belongs to and these arguments are pretty heated to the point that I don't even like this slide because I in some sense I don't like the discord that there is happening in science lately about not really or arguing where it should belong. So until we know for sure where VR sickness belongs, I just refer you to these terms and you're welcome to investigate further, but I'm just waiting for the consensus. Um, in our lab, we used to use flight simulation sickness questioner to ask subjects about their VR sickness experiences, but we no longer even use that. So we use just the scale from 0 to 20, how sick you are, and that's it. because it's not really clear if any of these questioners are relevant. So I hope it kind of shows you the state-of-the-art in this um scientific field and how far we don't know and um how difficult it is to to understand all of these issues. Why is vetin not if it does not felt on 2D screens? It is it is absolutely. So vection is studied on 2D screens a lot. Ultimately your screen is a 2D screen. So you yes you can feel vection in theaters. If you get very very close to a screen large 2D screen and all of your vision is occupied by the screen then you'll feel vection. You can uh simulate sensation of action. And I'll show you slides where if you zoom in on them quite a bit. So later we'll see optic flow slides. If you zoom in on them, you might experience vion. Another concern about this question though is that maybe you particular the person that you ask this question may not ever feel v. So there also it's very subjective. So some people may never even feel it. Um so there's always outliers. Should we use vection in games like swimming in river? No. So the answer is try to avoid it as much as possible and that will be better and I'll show you how we can do this. Usually teleportation is your friend. So avoid vet at all costs. Use teleportation for any kind of self motion. So let me go with slides a little bit because we have only 15 minutes and then we uh we go I'll show you some examples of vion of optic flows that may or may not lead to vion. So this one is an optic flow. See there's a lot of things a lot of features will be moving in your retina. This is very unlikely to trigger any vection because most of the features are stationary which is very reasonable right this optic flow most of the features will be moving. So these ones will are likely to trigger vection right? So there's most of the field of view is moving. So look at these slides and zoom in on them. So make try to maybe um what is it um display them on some big monitor and and get your eyes very close to them and see if you experience fashion. So that's a good example. So here are canonical optical flows. So canonical optical flows associated with yaw patrol and we're talking about constant velocity. So constant angular velocity up patrol and constant uh lateral movements lateral vertical and forward. So constant linear velocity movements with along uh three axis XYZ. So all of the canonical rotations and translations optical flow. So this image is from your textbook st. So let's look at all of them. Let's look at an example how it would look like in virtual reality. In fact, your your homework seven will exactly walk you through this with different tapes. There will be eight tapes. The first six of them will be this canonical optical flows and we'll ask you to tell us whether you feel first of all which one to identify it so that you really try it. We want to make sure that you really try the right one. And the second question will be do you feel vion? That is do you really feel like you're moving? Do you feel this illusion of self motion? Although you're not moving it's only the object flow right. So first we'll ask the question and second we'll ask how uncomfortable you are on the scale from 0 to 20. And then what we'll try to figure out is a secret because our hypothesis is something that you never should reveal to participants. So we'll reveal it after uh after the uh after you all submit your experiments and after we analyze the data then we'll release the results with the hypothesis. >> Okay. So then um but let's look at this canonical optical flows and let's do some assumptions. So assume that you're sitting still and looking straight. So no vestibular system uh stimuli at all and you're only provided visual stimuli the optical flows are just visual right and uh then let's look which ones should be maybe theoretically more comfortable or not and that will be related to your question that you asked accelerations versus constant velocities and making the slides did

make me sick. So looking at this kind of and selecting this kind of animations for a long time for a long exposure time made me sick. I don't think I experienced vection as you said but it made me sick. So there is again what is the relationship between optic flow vection and sickness which is not known. We don't know. So okay don't look at this guy because you're not supposed to really move. So imagine that these optical flows are presented to your eyes only while you're just standing still and you see this in VR. So one is yaw. So that's how the world would rotate. But we look at the constant angular velocity. Here you can see that there's acceleration deceleration stop. Acceleration deceleration stop. So the optical flow that we want to analyze now is actually constant um angular velocity. your constant angular velocity roll and constant angular velocity pitch. So I want to ask you what do you think is the result of these three optical flows? Can they be ever made comfortable in virtual reality? So what's your guess? Can any of these rotational optical flows with constant angular velocity so no angular acceleration just constant constant angular velocity can they be made comfortable for majority of people again and the answer should be no I I don't think so and again of course there is no known answer but I don't think so and let's see why so let's see boy. And let me demonstrate now that any of this yaw pitch and roll optical flow will always result in vestibular system and vision system mismatch. So let's look at the vestibular system and vision. So this is vision and what will they sense right? So vestibular system will sense zero linear acceleration and zero angular velocity because while you're presented with this optical flow you're actually sitting still looking straight. So there is no motions whatsoever. So the vestibular system will say I don't sense any linear acceleration with the ottolith organ. I don't sense any angular acceleration with my semicircle canals. So all of these are zeros and it will sense them as zeros because your head is not moving. But the vision provides you with this optical flow which is say with constant angular velocity optical flow. So for constant angular velocity optical flow angular acceleration is zero. So this kind of optical flows they suggest that acceleration is zero. But we just discussed last uh last lecture I showed you how if a point moves with constant angular velocity on a circle. Remember that picture then the linear velocities are never constant and there is even centrial linear acceleration component that is also not constant. So linear acceleration is not zero which means that vision tells you that linear acceleration is not zero while vestibular system detects zero linear acceleration. So there is a mismatch right here. There is no mismatch with angular accelerations but there is a mismatch with linear acceleration which means that this kind of optical flow for yaw for pitch the same thing linear acceleration is not zero and for roll linear acceleration not zero the same argument they'll always result in a mismatch between the stabil system and vision system which means that it's very likely to trigger infection and it's very likely to be extremely uncomfortable. [snorts] Now the next question is optical flow in with linear constant velocities forward backwards vertical lateral. So here they are. You move laterally, you move forward backwards with constant linear velocities or you move up up or down. So I have a question for you. Can any of these translational optical flows made comfortable in VR? What is your guess? [snorts] So it turns out that maybe this can be done and the idea is the following. So let's again compare our vestibular system when you're presented with this optical flow. What does it measure? Vestibular system will measure zero linear accelerations and zero angular accelerations because it's just visual information. You're sitting still looking straight. Your head is not moving. So the vision detects 0 0 just like in the previous set of slides. But the vision in this case for this optic flows it sees the optic flow with constant linear velocity. Right? So that's constant linear velocity which was not the case for angular rotations. We never have constant linear velocity. But this is translational motions like this with linear velocity with constant linear velocity. they will result in zero linear acceleration because if the velocity is constant, linear velocity is constant then linear acceleration is zero. So there is no mismatch between vision and vestibular system and for this kind of constant linear velocity translations angular velocity is zero. Therefore angular acceleration is zero. So look there is no sensory mismatch. So theoretically if there's no sensory mismatch then what can happen? We don't know if you still you may still experience fashion. You may still feel like you're moving, but it may not be as uncomfortable as it was when there was a sensory mismatch.[music] Again, it's not clear what causes what or if there is even causality between optical flows, vection and nausea and sensory mismatch. So those four there there's no clear causal relationships yet and the scientists are actively working on it to figure out what causes what and how and when. However, this homework 7 will be a little step in that dete in that direction for us because we'll be investigating all of this canonical optical flow and we'll ask you to report on your subjective feelings about it. and then we can u analyze the data and see as an example of a um as an example of a human subject experiment. So how would you even test any of the hypothesis about severe? So I have 5 minutes actually. So let this is a very good question and let me explain it really well. So uh so similarly I just wanted to [clears throat] make sure that similarly there will be no contradiction between vibular system and vision system for um for vertical and forward backwards constant linear velocity optical flow so this means there is a chance to make them comfortable right now I want to mention something since I'm out of time I'll ask this questions later but I want to mention something about The again this I'll have to do next time. So what are the factors that affect your sensation of vion or your discomfort and there's a lot of factors but let's go through them very carefully next time and this is essentially best practices recommendations for you in your final projects or any VR experience whatever killer app that you want to release you better stay together with or on top of the developer recommendations for vetin because vetin can be extremely uncomfortable and we don't want you to make people uncomfortable when they play your games. But let me go through to the last slide which is very important and that is

um developers or scientific method. So this slide but I'll have to leave the others for the next time. So when you test whether your experiences are comfortable or not or any other hypothesis about anything. So say an example is relationship between vection optical flow and sensation of nausea. You don't know what causes what but maybe you have an idea maybe you want to test a hypothesis and our homework seven is an example of such hypothesis that we want to test. Uh so how do you do that as an experimentter is some this is the process that you would go through as a subject homework seven is the process that you would go through in the end with this homework we want to demonstrate how a scientific process is done to you. So that's part of this homework is for you to learn how this scientific process is done and therefore since you're both subjects and later you learn about how we conducted this scientific method. First you should not know the hypothesis but let me uh show you what we are doing on our side to demonstrate the scientific process that we'll walk you through after we are done collecting your data. So first you have to identify the problem. So something that is burning like I don't understand what is comes first what causes what vection optical flow and uh VR sickness so which comes first what causes what so then you have to do some kind of literature and in our team we had to conduct a neuroscientist that studies version uh to help us understand and go through all of the literature to identify if people already figured out maybe this problem and the answer is no they haven't they haven't figured out this Nobody knows what's the causality between the two. So then you have to set up the variables independent, dependent and controlled. So independent is something that we will uh change. So we will give you several optical flows. So we tell you which we we give you we provide you the subjects with various um with various um values for the optical flow. So we'll give you different uh scenarios or different um experiences and those are independent variables because depending on this experience you'll have different maybe um feeling of fashion and feeling of comfort but these are independent optical flows or canonical optical flows. So they independent variables the dependent ones will actually will be your subjective feelings. So dependent variables will depend on what we set up as independent variables. So they will depend on the experience that we provide to you. Controlled variables is something that is always controlled. So we want to make sure that the subjects that will take um or will conduct will be use doing our studies that say they have the same experience or exposure to VR previously or maybe uh we want to control for the duration that they will be exposed in this experiment. So something that we want to keep tight and control and that is the hardest with respect to uh having subjects use students because we cannot really control for a lot of variables. So this is a big problem with our studies. So this may never lead to publication because of this limitation. We cannot really control for a lot of things that you will be doing because we will not be observing our subjects. However, you have to just then imagine that in real experiment, we will actually be controlling for a lot of these questions. We'll make sure that you are really answering properly and you really understand the questions and you really are participating in the experiment the way we want to, but that will be a little bit hard. So, I guess that answers your question whether this will be ever published. Maybe not. Maybe not. So how do you then what should we proceed with this? We make a hypothesis and of course we do not tell the subjects about the hypothesis because that will affect your judgment and your answers. So you should not know neither the hypothesis nor the variables but then we should also kind of know have a prediction of what we expect from this experiment. Uh we designed the experiment really well. So APK is our experiment. We'll provide it for you. You try it. Hopefully, you do it all right. You follow the instructions and you're very responsible. And then we collect the data from you through love lace, of course. Then we will analyze the data. Uh-oh. And I just erased what I really wanted to do. So, we'll analyze the data and make conclusions. So, the conclusions could be two that this data will support our hypothesis. Then, we'll accept the hypothesis, but then we'll have to repeat. And that's very important. We have to repeat the experiment. So we'll never just say, "Wow, what our hypothesis was is true." You should be always very critical of any results. And I'll show you why. And that's what does the best science. You have to be critical. When you talk to neuroscientists and perceptual psychologists, they'll be always negative. They will not believe any of the data that you present to them right away. They'll want to redo the experiments many times to really believe that the hypothesis is probably true, but I think they'll never be fully convinced because there may be something else that will explain the data. So you'll have to always question as an experimentter always question what else could have explained this data, not just your brilliant hypothesis. Um so then if the data does not support the hypothesis then you have to reject it and then you have to revise or modify it and can do and start over again. Um so that's um that's how it's done. So a good science is the essence of good science is to take a piece of data, take an argument and think how else might those data arise or explain um not just your conclusion but maybe in some other way. So that's the that's the essence of good science. So it's very very important not to just take any data and believe that the hypothesis is true. You should always critically critically evaluate it. So it's very important for science. Anyway, I'll talk about best practices tomorrow on Friday then because it's very important that you follow best practices and you minimize vashion as much as possible in any VR experiences and that seems to be the best way to avoid discomfort in virtual reality and I hope you remember that. So no flying, no roller coasters unless you really want to make people sick, but otherwise try to avoid it. Try to always use teleportation. That's the best practices that we know of off so far. And there's a lot not known. So maybe there is better better practices. Maybe there's some other ways to um eliminate discomfort while still preserving vection. A lot of this is not known but yet still the best practices that we know so far is to eliminate fashion, use teleportation. >> Okay, so let's

stop here and um I'll see you on Friday.lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

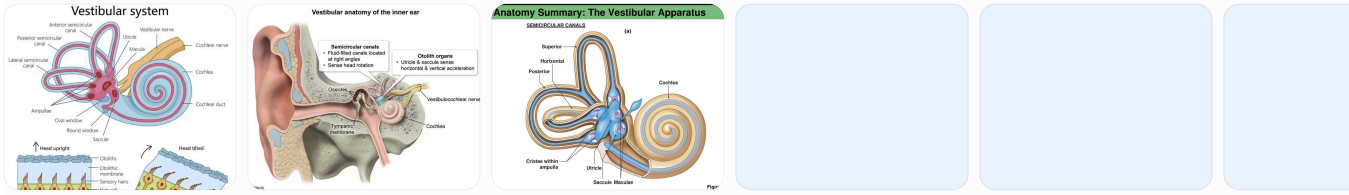
NPTEL Introduction to Virtual Reality — Easy Notes

Chapter 8: Vestibular System, Vection, VR Sickness and Scientific Experiments

Let's learn this like a 7th-class student, using simple examples, memory tricks and MCQs. The main idea of this lecture is:

Your eyes tell your brain what you see, and your inner ear tells your brain how your head moves. When these two disagree in VR, you may feel uncomfortable or sick.

1. What is the vestibular system?



The **vestibular system** is a balance-sensing system inside your inner ear.

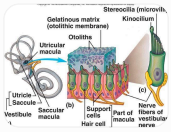
Imagine riding a bicycle. You need to know whether your head is tilted, turning, or moving. Your vestibular system helps your brain understand these movements and maintain balance.

It helps you with:

- Balance and posture.
- Head movement and orientation.
- Knowing whether you are tilting, turning or accelerating.
- Keeping your eyes stable while your head moves.

Remember: Vestibular system = the body's balance sensor.

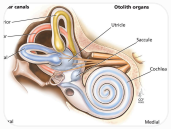
2. The two main parts of the vestibular system



A. Utricle and saccule — linear movement

These contain tiny calcium carbonate crystals called **otoconia**. When your head tilts or accelerates, the crystals and a jelly-like layer move, bending tiny hair cells. The hair cells send signals to the brain.

Think of them as tiny **accelerometer-like sensors**.



B. Three semicircular canals — rotational movement

These are three loop-shaped canals arranged in different directions. Fluid inside them moves when your head rotates, bending hair cells and sending signals to the brain.

Think of them as sensors that help detect **head rotation**.

Easy comparison

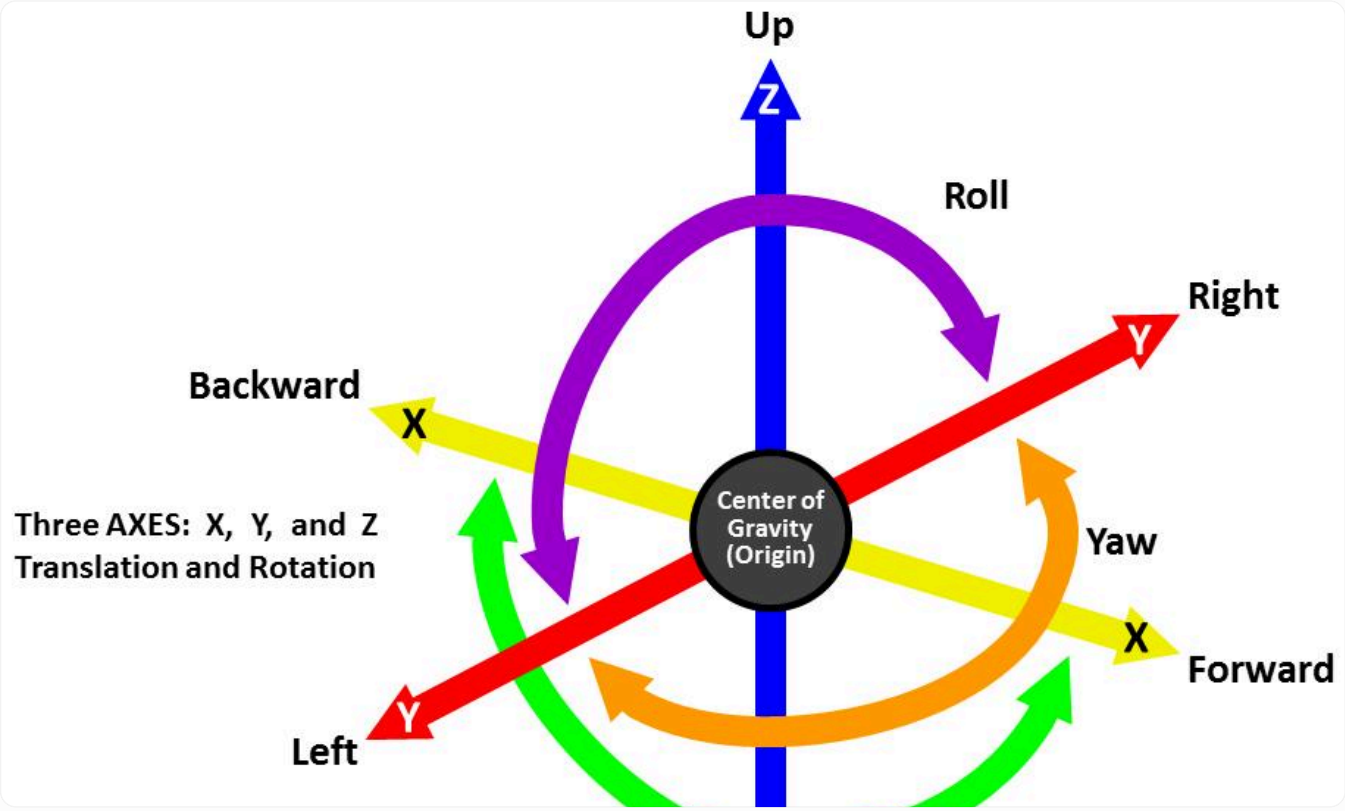
Part	Main job	Easy example
Utricle	Mainly senses horizontal linear acceleration and head tilt relative to gravity	Moving forward in a car
Saccule	Mainly senses vertical linear acceleration	Moving up or down in a lift
Semicircular canals	Sense rotational movement	Turning your head left or right

Memory trick: Utricle and saccule = straight-line movement; semicircular canals = turning.

A small scientific detail: these organs detect movement-related forces and accelerations, not position directly. The brain combines their signals with vision and other information to estimate head position and orientation.

3. What are the six degrees of freedom (6DoF)?

A head can move in six basic ways: three translations and three rotations.



Type	Movement	Example
Translation X	Move left or right	Slide sideways
Translation Y	Move forward or backward	Walk forward
Translation Z	Move up or down	Jump
Yaw	Turn left or right	Say “no” with your head
Pitch	Look up or down	Nod “yes”
Roll	Tilt your head sideways	Tilt your ear toward your shoulder

Axis names can differ depending on the coordinate system used.

Remember: 3 translations + 3 rotations = 6DoF.

Why does this matter in VR? The headset must track your head's movement and update the virtual scene accordingly.

4. How does the vestibular system work?

Imagine your head is a small car.

1. Your head moves or turns.
2. Fluid or tiny crystals inside your inner ear shift.
3. Tiny hair cells bend.
4. These hair cells convert movement into nerve signals.
5. The brain combines these signals with vision and other body information.
6. Your brain estimates your movement and helps maintain balance.

Memory chain: Movement → fluid/crystal movement → hair cells bend → nerve signals → brain → balance.

5. How does the brain use these signals?

The vestibular signals travel through nerves to different brain areas.

Brain part	Simple job
Vestibular nuclei and related brainstem areas	Process balance signals and help control reflexes
Thalamus	Relays sensory information to other brain regions
Vestibular cortex	Helps us become aware of movement and spatial orientation
Cerebellum	Helps coordinate movement, posture and balance
Motor areas	Help control muscles and body movement
Occipital lobe	Processes vision
Parietal lobe	Helps process spatial information and movement

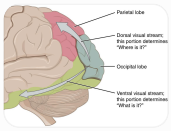
The brain also uses the **vestibulo-ocular reflex (VOR)** to keep your eyes relatively stable when your head moves.

Example: When you turn your head to the right, your eyes automatically move left enough to help keep your gaze on the same object.

Remember that the brain doesn't depend on the vestibular system alone. Vision, muscle and joint information also help it understand where the body is.

6. What are the dorsal and ventral visual pathways?

Your brain processes visual information in two important ways.



Dorsal stream — WHERE and HOW

Helps process where things are, how they move and how to guide actions.

Example: Catching a moving ball or reaching for a cup.

Ventral stream — WHAT

Helps recognize objects and identify what you are looking at.

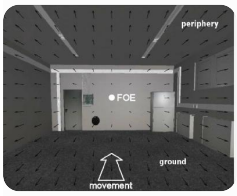
Example: Recognizing that the object in your hand is a cup.

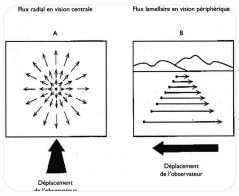
The dorsal stream connects visual processing with spatial information, including information from the vestibular system.

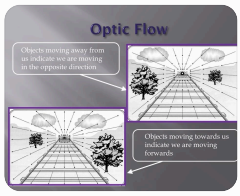
Memory trick: Dorsal = where; ventral = what.

7. What is optic flow?

Optic flow is the pattern of apparent movement of objects and features across your vision when you or the scene moves.







Imagine riding a bicycle:

- Nearby objects seem to move across your vision quickly.
- Faraway objects seem to move more slowly.
- When you move forward, the scene often appears to expand outward from a point ahead of you.

Your brain uses these patterns to estimate how you are moving.

Two important types of optic flow

Type	What you see	Example
Rotational flow	The visual scene appears to rotate	Turning your head while looking at a room
Translational flow	Nearby and distant objects move by different amounts	Looking out of a moving train

This difference is important:

- During rotation, fixed scene features can have similar angular movement, depending on their location in the visual field.
- During translation, nearby objects generally show greater motion relative to distant objects. This is called **motion parallax**.

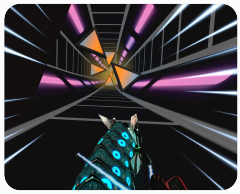
Remember: Optic flow = movement pattern across your eyes.

8. What is vection?

Vection means the illusion of self-motion when you are actually stationary.

Imagine you are sitting inside a train. Another train next to you begins moving. For a moment, you may feel that your own train is moving, even though it is still.

That feeling is vection.



Other examples include:

- Watching a large screen that fills much of your vision and makes you feel like you are moving.
- Seeing forward-moving scenery in a VR headset while sitting still.
- Watching a virtual roller coaster from a stationary chair.

Optic flow versus vection

Optic flow	Vection
A visual movement pattern	A feeling that you yourself are moving
Describes what appears to move across your vision	Describes what you feel
Can occur without vection	Can be triggered by optic flow

Memory trick: Optic flow is the visual signal; vection is the feeling.

Vection is not automatically unpleasant. But visual movement that conflicts with the body's movement signals can contribute to VR discomfort.

9. Why can VR make us feel sick?

Your eyes might tell you that you are moving forward in a virtual world, while your inner ear tells you that your head and body are not accelerating.

This disagreement is called **sensory mismatch**.

Vision says: "We are moving!"
The virtual scene flows past you.

Vestibular system says: "I don't detect that movement!"
Your body is actually sitting still.

Possible result: VR discomfort

Some users may experience nausea, dizziness or headache.

Four mismatch situations

Situation	What happens
Body detects movement, vision appears stationary	Sitting in a vehicle that accelerates while looking at its interior
Vision suggests movement, body detects none	Sitting still while seeing forward motion in VR
Body detects movement, but vision is delayed or incorrect	A VR headset has high tracking latency or dropped frames
Vision is updated correctly, but vestibular signals are abnormal or delayed	A vestibular disorder or certain effects of alcohol can disrupt balance and movement perception

The last example is a simplified illustration: alcohol can affect balance and eye movements through several mechanisms, not just one inner-ear process.

Symptoms of VR sickness

Subjective symptoms — what a person reports feeling:

- Nausea
- Dizziness
- Headache
- Fatigue or drowsiness
- Eye strain

Observable physical signs can include:

- Increased salivation
- Cold sweating
- Paleness
- Warmth or flushing

Not everybody experiences the same symptoms, and researchers are still studying the exact causes of VR sickness. Sensory mismatch is an important theory, but it is not the only proposed explanation.

10. Which virtual movements are more likely to be uncomfortable?

Your lecturer compares two types of virtual movement.

A. Rotational optic flow

Examples: virtual yaw, pitch and roll — the world appears to rotate around you.

When you are actually sitting still, your inner ear detects no corresponding head rotation. The visual scene may nevertheless suggest movement and acceleration. This can create a mismatch.

B. Translational optic flow

Examples: moving virtually forward, backward, sideways, up or down.

If the visual motion represents **constant linear velocity**, the implied linear acceleration is zero. That means there may be less acceleration-related mismatch than in the rotational examples discussed in the lecture.

Virtual movement	Main point from the lecture
Constant angular velocity rotation	Can imply linear accelerations that your stationary vestibular system does not detect
Constant linear velocity translation	Can imply zero linear acceleration, so there may be less mismatch
Acceleration, deceleration or abrupt stopping	Can make the visual and physical signals disagree

This is a theoretical comparison, not a guarantee. Even constant-speed translation can cause vection or discomfort, and individual responses vary.

Remember: Constant rotation can still imply linear acceleration; constant straight-line velocity implies zero linear acceleration.

11. How can developers make VR more comfortable?

Your lecturer's main recommendation is to avoid unnecessary simulated self-motion when possible.

- **Use teleportation:** The user selects a destination and moves there without seeing the whole journey.
- Reduce sudden acceleration and deceleration.
- Avoid unnecessary rapid rotations and camera movements.
- Keep head tracking accurate and reduce latency.
- Avoid dropped frames and unstable visual updates.
- Let users take breaks and stop if they feel unwell.

Why is teleportation helpful?

With smooth virtual movement, your eyes see continuous movement while your body may remain stationary. Teleportation avoids showing much of that continuous visual motion.

Teleportation does not guarantee comfort for everyone, but it is often a useful comfort option.

Exam memory line: To reduce VR sickness, minimize sensory mismatch, maintain accurate tracking and offer teleportation.

12. What is the scientific method?

Scientists use the scientific method to investigate questions rather than simply guessing.

Imagine your question is: “Does virtual rotation cause more discomfort than virtual forward movement?”

1. **Identify the problem**
   What are we trying to understand?
2. **Review existing research**
   Has someone already studied this question?
3. **Set a hypothesis**
   Make a testable prediction.
4. **Identify variables**
   Decide what changes, what is measured and what stays the same.
5. **Design and conduct the experiment**
   Give participants controlled experiences.
6. **Collect and analyze data**
   Measure and compare their responses.
7. **Draw a conclusion**
   Check whether the results support the prediction.
8. **Repeat and verify**
   Run more experiments to check whether the result is reliable.

Three important variables

Variable	Simple meaning	Example
Independent	What the researcher changes	Type of optic flow

Dependent	What the researcher measures	Reported discomfort score
Controlled	What the researcher tries to keep the same	Exposure duration and instructions

The lecturer describes asking participants to rate discomfort on a scale from **0 to 20**.

- 0 = no discomfort
- 20 = extremely uncomfortable

The exact interpretation depends on the instructions given to participants.

Why shouldn't participants know the hypothesis?

If participants know what result the researcher expects, it may influence their answers. Keeping the hypothesis hidden from participants when appropriate can reduce this bias.

What if the results support the hypothesis?

Scientists should not immediately declare it proven. They should repeat the experiment, check for other explanations and see whether the result holds up.

Remember: Good science tests, measures, questions and repeats.

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13. Extra important topic: Gyroscopes and VR head tracking

Your transcript also includes a tracking-systems lesson. Here are the important points in simple English.

What is a gyroscope?

A gyroscope in a VR headset measures **angular velocity** — how quickly the headset is rotating.

For example, it can detect how quickly you turn your head left or right.

An accelerometer measures **linear acceleration**, such as acceleration forward or upward.

Sensor	Measures	Example
Gyroscope	Angular velocity	How quickly your head turns
Accelerometer	Linear acceleration	How your head accelerates forward
Camera or other external reference	Visual position information	Where the headset is relative to the room

How does a gyroscope estimate orientation?

Suppose a gyroscope reports that you are turning at 30 degrees per second for 2 seconds.

If the rotation rate stays constant:

$$\text{Angle change} = \text{Angular velocity} \times \text{Time}$$

$$30^\circ/\text{s} \times 2\text{s} = 60^\circ$$

So the estimated rotation is 60 degrees.

Real tracking systems perform these calculations very frequently.

What is drift?

A gyroscope's measurements contain small errors. When those errors accumulate over time, the estimated orientation can gradually become inaccurate. This is called **drift**.

Sensor fusion combines information from multiple sensors to improve the estimate and reduce drift.

Remember: Gyroscope = turning speed; integration = estimate angle; drift = accumulated error; sensor fusion = combine sensors.

What is a quaternion?

A quaternion is a mathematical way to represent 3D rotation. VR systems often use quaternions because they handle combined rotations well and avoid problems associated with some other rotation representations, such as gimbal lock with Euler angles.

You do not need to memorize the full equations for this simple revision. Remember:

- Euler angles describe rotations using angles such as yaw, pitch and roll.
- Quaternions represent 3D orientation using four numerical components.
- Gyroscope readings can be integrated to update the orientation estimate.
- Sensor fusion helps correct accumulated errors.

What is collision detection?

A physics engine must determine whether two objects touch or overlap.

For example, in a VR game, the system must know when a virtual ball hits a wall.

A common approach is to check simple bounding shapes first. If those shapes do not overlap, the detailed objects do not need to be checked for a collision. If they overlap, the engine performs more detailed checks.

Remember: Collision detection = checking whether objects hit or overlap.

14. MCQs for exam practice

Try answering these before checking the answer key.

15-question practice test

0/15 answered

1. What is the main function of the vestibular system?

- A. Hearing speech
- B. Maintaining balance and sensing head movement
- C. Recognizing colours
- D. Controlling digestion

2. Which organs mainly detect linear acceleration and gravity-related head tilt?

- A. Semicircular canals
- B. Cochlea only
- C. Utricle and saccule
- D. Optic nerve

3. What do the semicircular canals mainly detect?

- A. Colour
- B. Rotational movement
- C. Smell
- D. Sound frequency

4. How many degrees of freedom are used to describe general 3D head movement?

- A. 2
- B. 3
- C. 4
- D. 6

5. What is optic flow?

- A. The flow of blood
- B. The pattern of visual movement across the field of view
- C. A type of lens
- D. A hearing signal

6. What is vection?

- A. The real movement of the body
- B. The illusion of self-motion while stationary
- C. The bending of a lens
- D. A type of eye disease

7. You sit still in VR but the virtual world moves forward. What might happen?

- A. Vection
- B. The inner ear stops working permanently
- C. The display loses all pixels
- D. Nothing can be perceived

8. What is sensory mismatch in VR?

- A. All senses report exactly the same thing
- B. Vision and body-motion signals disagree
- C. The screen has too many colours
- D. The sound is too quiet

9. Which can be a symptom of VR sickness?

- A. Nausea
- B. Improved hearing
- C. Increased height
- D. Permanent colour vision improvement

10. Which technique is often recommended to reduce discomfort from virtual travel?

- A. Sudden camera rotations
- B. Frequent dropped frames
- C. Teleportation
- D. Unnecessary acceleration

11. What does the dorsal visual stream mainly help with?

- A. Where things are and how to act on them
- B. Recognizing object names only
- C. Producing sound
- D. Detecting taste

12. What does a gyroscope measure?

- A. Angular velocity

B. Blood pressure

C. Colour intensity

D. Linear distance directly

13. What is drift in a tracking system?

A. A gradual buildup of estimation error

B. An increase in screen brightness

C. A sound delay only

D. An increase in object size

14. Which variable is the type of optic flow selected by a researcher?

A. Dependent variable

B. Independent variable

C. Controlled variable

D. Random conclusion

15. Why should experiments be repeated?

A. To guarantee the first hypothesis is correct

B. To check whether results are reliable and reproducible

C. To avoid collecting data

D. To remove all participants

Submit and check answers

15. Final one-minute revision

Term	One-line meaning
Vestibular system	Balance and head-movement sensor
Utricle	Mainly horizontal linear acceleration
Saccule	Mainly vertical linear acceleration
Semicircular canals	Rotational movement sensors
6DoF	Three translations + three rotations
Optic flow	Visual pattern of apparent movement
Vection	Feeling of self-motion while stationary
Sensory mismatch	Conflicting information from different senses
VR sickness	Possible nausea, dizziness and discomfort from VR
Teleportation	A useful way to reduce continuous visual travel
Gyroscope	Measures angular velocity
Accelerometer	Measures linear acceleration
Drift	Accumulating tracking error
Sensor fusion	Combines sensor information
Independent variable	What the researcher changes
Dependent variable	What the researcher measures

Controlled variable	What the researcher keeps consistent
Scientific method	Test, measure, analyze, repeat

The five most important things to remember for your exam:

1. The vestibular system is located in the inner ear and helps maintain balance.
2. Optic flow is a visual pattern; vection is the feeling of self-motion.
3. Conflicting visual and vestibular information can contribute to VR sickness.
4. Teleportation and accurate, low-latency tracking can improve VR comfort.
5. A good scientific experiment defines variables, tests a hypothesis, analyzes results and repeats the experiment.

You said:

Quiz-8-2026.pdf
PDF

explain easily

ChatGPT said:

Quiz 8 – 2026: Easy Explanation

I'll explain all 10 questions from your PDF in **simple English, like you're in 7th class**, with examples and memory tricks so you can remember them for the exam.

1. Euler integration and position error

Correct answer: A — Decrease Δt while maintaining the same total duration.

Imagine you are tracking a moving person in VR. The computer calculates their position using small time steps.

- Δt means the time between two measurements.
- Smaller Δt means more frequent updates.
- Smaller time steps can reduce numerical integration error in the lecture's model.

Example: Walking a long distance using many small steps gives you more detailed position updates than using a few large steps.

Memory trick: Smaller time step = smaller numerical error.

2. Why is double integration of acceleration unsuitable for long periods?

Correct answer: B — Numerical error accumulates quadratically.

An accelerometer measures acceleration. To calculate position, the computer integrates acceleration twice:

Acceleration

↓

Integrate once → Velocity

↓

Integrate again → Position

Even a tiny measurement error can grow into a large position error over time. This is called **drift**.

Memory trick: Acceleration → Velocity → Position. Two integrations can cause big drift.

3. Circular motion with constant angular velocity

Correct answer: A — Angular acceleration is zero, while linear velocity changes direction.

Imagine a toy tied to a string and moving in a circle.

- **Angular velocity:** How quickly it rotates around the circle.
- **Angular acceleration:** How quickly its angular velocity changes.

- **Linear velocity:** The speed and direction of the toy as it moves along the circle.

Even if the toy rotates at a constant rate, its direction of travel keeps changing. Therefore, its linear velocity changes direction.

Memory trick: Constant rotation rate does not mean constant direction of travel.

4. Gyroscope and quaternion rotation

Correct answer: B — Axis = normalized ω; angle = $|\omega|\Delta t$.

A gyroscope measures how quickly something rotates.

- ω is the angular-velocity vector.
- $|\omega|$ is its magnitude, or size.
- Δt is the time interval.

The computer calculates:

$$\text{Axis} = \frac{\omega}{|\omega|}$$

$$\text{Angle} = |\omega|\Delta t$$

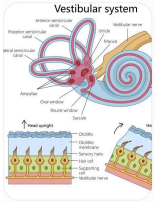
Example: If rotation speed is $30^\circ/s$ for 2 seconds, the estimated rotation angle is 60°.

Memory trick: Direction comes from the normalized vector; angle = speed × time.

5. Human balance system and engineering sensors

Correct answer: A — Otolith organs → accelerometer; semicircular canals → gyroscope-like sensor.

Your inner ear helps your brain understand movement and balance.



Otolith organs (utricle and saccule)

Detect linear acceleration and help sense head tilt relative to gravity. They are similar to accelerometers.

Semicircular canals

Detect rotational head movements. They are similar to gyroscope-based motion sensing.

Memory trick: Otolith = straight-line movement; canals = turning movement.

6. What is vection?

Correct answer: B — Vection.

Vection is the **feeling that you are moving when you are actually stationary**.

Example: You are sitting in a stationary train. The train beside you starts moving, and you feel as if your own train is moving.

In VR, the virtual world might move forward while your real body remains still. You may feel that you are moving forward.

Memory trick: Vection = virtual feeling of motion.

7. Why can rotational optic flow cause sensory mismatch?

Correct answer: A — Vision indicates nonzero linear acceleration while the stationary vestibular system indicates zero linear acceleration.

Let's understand it simply:

- Your eyes see a VR scene that suggests circular or rotational motion.
- Circular motion at constant angular velocity involves **centripetal linear acceleration**.
- But your body is stationary, so your vestibular system does not detect that corresponding physical acceleration.

Your eyes and inner ear are giving different information. This is called a **vestibular–visual mismatch**, which can contribute to VR sickness.

Memory trick: Eyes say “moving”; body says “not moving” = mismatch.

8. Constant linear velocity versus rotational motion

Correct answer: A — Constant linear velocity implies zero linear acceleration.

Compare these two situations:

Motion	Linear acceleration
Moving straight at constant velocity	Zero
Moving in a circle at constant angular velocity	Nonzero, directed toward the centre

Why? In straight-line motion at constant velocity, neither speed nor direction changes. In circular motion, the direction keeps changing.

For a stationary VR user, simulated straight movement at constant velocity avoids that particular linear-acceleration mismatch. However, it does **not** guarantee that the user will avoid VR sickness.

Memory trick: Straight and steady = zero acceleration; circular motion = changing direction.

9. Independent and dependent variables

Correct answer: A — Optic flow is independent; subjective feelings are dependent.

Imagine researchers want to find out whether different VR movement patterns make people feel sick.

- **Independent variable:** What the researcher changes.
- **Dependent variable:** What the researcher measures.
- **Controlled variables:** What the researcher keeps the same.

Type	Example
Independent	Type of optic flow shown
Dependent	Participant's discomfort rating
Controlled	Duration of the test and instructions

Memory trick:

- Independent = I change it.
- Dependent = Data I measure.
- Controlled = Keep it constant.

10. What happens after a hypothesis gets support?

Correct answer: B — Repeat the experiment and critically consider alternative explanations.

A hypothesis is an idea that researchers test.

Suppose a researcher thinks that a particular VR movement pattern causes more sickness. The experiment supports that idea.

Should the researcher immediately declare it proven? **No!**

The researcher should:

1. Repeat the experiment.
2. Check whether the results are consistent.
3. Consider other possible explanations.
4. Collect more evidence before reaching a strong conclusion.

Memory trick: One experiment is not enough—repeat and recheck.

Quick revision table

Q.	Topic	Answer	Remember this

1	Euler integration	A	Smaller time steps reduce numerical error in the lecture model
2	Double integration	B	Error can cause large position drift
3	Circular motion	A	Direction changes even at constant angular velocity
4	Quaternion update	B	Axis = normalized ω; angle = $\|\omega\|\Delta t$
5	Vestibular system	A	Otolith = accelerometer; canals = gyroscope-like sensor
6	Vection	B	Feeling motion while stationary
7	Rotational optic flow	A	Visual and vestibular information can mismatch
8	Constant linear velocity	A	Zero linear acceleration
9	Experiment variables	A	Change optic flow; measure subjective feelings
10	Scientific method	B	Repeat and check alternative explanations

Quiz-8-2026.pdf

Test yourself — without looking at the answers

10 MCQs 0/10 answered

1. What does decreasing the time interval Δt generally do in the lecture's Euler integration model?

- A. Increases numerical error
- B. Reduces numerical error
- C. Removes all sensor noise
- D. Stops position updates

2. What is drift in positional tracking?

- A. A type of display
- B. Accumulated error in the estimated position
- C. A camera lens
- D. A rotation axis

3. What happens to linear velocity during circular motion at constant angular velocity?

- A. Its direction changes
- B. It becomes zero
- C. Its direction never changes
- D. Angular velocity disappears

4. How do you calculate the incremental rotation angle from angular speed?

- A. Angular speed ÷ time
- B. Angular speed × time
- C. Time ÷ angle
- D. Angular speed + time

5. Which inner-ear structures detect linear acceleration?

- A. Semicircular canals
- B. Otolith organs

C. Retina

D. Cochlea only

6. What is vection?

A. Actual acceleration

B. A tracking error

C. An illusion of self-motion

D. A type of integration

7. Why can VR movement cause sensory mismatch?

A. The eyes and vestibular system may give conflicting motion information

B. The screen is always too bright

C. The user must be physically moving

D. The gyroscope measures sound

8. For constant straight-line velocity, linear acceleration is:

A. Always increasing

B. Nonzero in every case

C. Zero

D. Equal to angular velocity

9. In an optic-flow experiment, what is the independent variable?

A. The participant's measured discomfort

B. The optic-flow condition changed by the researcher

C. The final conclusion

D. A variable kept constant

10. What should researchers do after one experiment supports their hypothesis?

A. Call it proven immediately

B. Ignore the data

C. Repeat the experiment and examine alternatives

D. Change the results

Check my answers

Final memory line: Small time steps reduce numerical error; double integration causes drift; circular motion changes direction; otoliths detect linear acceleration; vection is the illusion of movement; and science requires repeating experiments.

You said:

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[Music] So once you have a camera then you can do double integration of accelerometer which drifts it incredibly fast. So we we know from last time that the drift is quadratic. So you can only integrate it maybe for 16 milliseconds. So exactly the time between camera readings but after that the drift is so bad that you don't want to use it. So you really want this extra sensor then to correct for the drift and

you didn't if you didn't have it then you would just drift really really fast and far away. So so it's impossible to use accelerometer without an extra sensor which is camera. But if you have a camera then you do double integration for a very short amount of time and then you need to use camera. So the question is how do you use camera to correct for the drifting accelerometer? Uh so let's see what camera can do. Well camera from a single image you could already estimate 3D pos both position and orientation. So why do you need that imuse? Well, it you only can get these estimates every 16 milliseconds and that's way too long to wait for updating you. It's kind of like what you asked like why don't we drop frames? Well, you don't want to drop frames. You really want to have a lot of frames and you want them to be consistent with your motion so that you don't have this extra unexpected velocities and accelerations in your optic flow. So, you want to update the world accordingly really really fast. Otherwise, there's a lot of discomfort. So, tracking that has zero latency is the best and with high frames um per second as well. So, you really don't want latency and you don't want um any delays and you don't want drop frames. Therefore, cameras uh estimates of 3D position and orientation every 16 milliseconds is just not good enough for what we uh for comfortable VR systems. Nevertheless, it can estimate position orientation. So, let's see how uh it's done. So, there are different camera arrangements. You can put cameras in the outside world, a fist with in the in the physical world, and then if you have a headset with some kind of markers on it, which was done in earlier Oculus versions, then the camera would take pictures of these markers and you could estimate where what's the position orientation of them. Or the other way around, you have camera on your headset and then you take pictures of the world around the physical world and there's some kind of markers. They can be artificial or they can be natural features. So uh these are terms outside in and inside out tracking. Inside out tracking is what we normally have nowadays. This is early valve prototype where the markers were actually in the physical world but they were QR codes. So this was one of the earliest prototypes of positional tracking that we saw in Oculus and the val V valve uh employees would come to Oculus and set up these QR codes in a specific room where they would demo their positional tracking. So the headset would have a camera on it and um this was the valve room that we called it. So this was one of the earliest prototypes of >> this first room >> which um we don't have time to go through. Oculus DK2 CV1 and S had um infrared um LEDs on on the on the headset themselves and there would be an external camera that would track it. So this is HC5 a little different technology with this infrared light drums kind of sending the sheets that are rotating and then this receiving photo dots whenever they hit they would send information about it. So it's kind of equivalent to a camera, very similar. But anyway, this is the setup. You have a camera. It takes pictures of your features. And from this pictures, from this picture alone, you could make an estimate of what is the orientation and position of the uh world is. So here's an example, right? If you have some features on the headset itself and you take a picture of those features just from the size of these features, the spread the um you could estimate it using very similar technique to the calibration that I showed you. So you do correspondence problem, right? You know what is the physical uh coordinates of these features in the real world. you know what is the coordinate on the picture in the on the picture and then if there's enough features so if there's many many features in the world then you can just do correspondence obtaining the coordinates on the image from the world coordinates using the transformation parameters that you want to estimate. So these are the only unknown values. The orientation and position of this object is the unknown. But then the knowns are the feature um physical positions and the uh image positions. So if you have enough of these equations, so if you have enough features then you can again set up this um linear list least squared problem and solve it by inversing the big big big matrix. But we don't have time to go through this because we have one minute left. So um so let me just just go back just go to the final uh formulas and kind of just have I hope you'll have some idea about how it's actually done then. So you have a camera, you have an MU, you you double integrate the accelerometer. So this is your accelerometer raw values. You integrate it to obtain velocity. You you you obtain velocity estimate. You integrate that to obtain the position. So this is just oiler integration. Oiler double integration of accelerometer values to obtain position estimate it will drift. So how do you do the sensor fusion with camera? How do you correct for the drift for the position? Well, if you take a picture at some point with the camera, you can estimate a actual true position and velocity from that from the camera, but maybe it will be too far away from your estimate. So, you don't want to snap back right away because you don't want people feel uncomfortable. So, instead then you can take your estimate from the accelerometer and do what's called complimentary filter. So make a little step in the direction of this arrow that was estimated from the uh camera to get this updated better estimate of position and velocity that integrates both camera values but also the uh accelerometer integration. And then you do this every 60 milliseconds. And of course, drift correction should be I think I have a slide here. Um, where was it? Yeah, fast enough to eliminate error but slow enough to avoid VR signal. So, you want to make this correction steps uh far enough to to not have accumulated drift to to to but but slow so slow so that there's no extraction due to that. Anyway, I think this is a good overview. I hope you got some general ideas. Clearly we cannot test in details any of this because we went so fast but I hope you got some idea general idea about how tracking system are implemented in these devices and um I'll see you on Tuesday that will be my last lecture I'll I'll talk I'll start talking about rendering techniques and rendering pipeline but um so yeah have a good Friday and have a good weekend. Bye. Let me see if I have any questions because I can answer some questions maybe. I think these are Oh, yeah. Here. Okay. No questions. So, I'll see you on Tuesday. Yeah. Thank you guys. [Music] [Music] So good evening

and welcome all. So we'll be continuing u this course on uh virtual reality mastering no XR principles. Professor Anna has taught you lot of fundamentals in the last uh uh uh many classes. Hereafter we are going to go know more faster. We are going to go to the go in the know second gear or third gear or fourth gear or whatever it is. We are going to talk about many advanced topics in the next uh maybe about eight classes or 10 classes. Okay. Today we are starting a new topic rendering pipeline right. So um first of all we are going to now learn why rendering pipeline we have to learn as a XR developers or XR designer or XR engineering. Now why should we learn rendering pipeline? What are the different components of the rendering pipeline? what are the different rendering pipelines we are looking at it and what are the advances in the rendering pipeline how AI actually is is knowing being used in rendering pipeline is what we are going to look at it so in today's class we will cover the the most fundamental concepts for rendering pipeline and in the next class we look at the the very recent very advanced now rendering pipelines. Okay, particularly we are going to look at the now chapter 7 of our textbook virtual reality that will give you the know all the basics and know that is that will go with this you know today's lecture we will start with the definition and fundamentals of uh we are rendering then we will look at the graphics pipeline common for know uh all the all the know graphics know gaming and animation or virtual reality right then we are going to look at the stereoscopic rendering uh various uh uh recent uh rendering techniques for particularly for virtual reality that is the focus of this particular course texture mapping and material rendering VR specific rendering challenges and advanced topics so today we will have time for now first two uh topics the other things we'll look at it in uh the next class right even before looking at uh what is rendering I wanted to want you to think about why we should study visual rendering okay why should we as a developer should be studying the rendering at Okay, rendering is something that converts your 3D model into your screen pixel, right? Why should we look at it anyway? You know the the Unity game engine is there or Unreal game engine is there. Now we only have to know model and then give the lighting and all the things. Why should we know worry about is you know rendering how Unity does it or Unreal does it, right? We need to learn the entire process how unity is doing it because only then we can control how of each of this pixel is actually know seen in the in the either the visual display right and particularly it is very important for all the exact developers right whether you know uh unity or You should know uh this rendering pipeline because XR or in VR we are looking at 90 frames per second. 90 frames per second means every frame we have how much time we have about uh no uh 1,000 frames uh and uh divided by 90 it is going to be about know 1.1 uh 111 no milliseconds for every frame we have only 1111 milliseconds within this time I want you to know know What are the different and now things are happening? How this is uh model is is uh taken through the graphics engine and then how it is transformed from translated transformed and then added color texture is added every stage. Right? So you would appreciate at different transformation each pixel is going through and you will know in case of of uh now optimization is required where to optimize it which stage you should optimize it right and the second thing is the the color in the ARVR is actually is going to be different you should know that now there is a ambient color there is a diffused color there's specular color. All how all these colors are now put together and then how this final image is actually rendered and therefore how to control the final rendering of the image by adjusting each of these components of the light ambient color or diffused colors. Right? By I this we have already seen it. Rendering comes into picture when now we have this uh uh uh the world the 3D world and finally it is going to be rendered in the display and uh how it is translated from 3D world into the into the display is the the visual rendering we are talking about. Right. And then this uh display is artificially stimulating the eyes and that is what we have now professor Anna has taught you in the last classes as in the neuroscience of it the physiology point of view and uh and the now cognitive point of view and then how know we can track and how this 3D model can be interacted all those things we have seen in the previous classes. In this class what we are going to see how this 3D world is converted into the pixels in the display that is a rendering process. Okay. This is a crucial bridge I would call it as between the complex 3D model and the visual representations we observe on our screens. Here the visual representation is in terms of the pixels. We know what is in our resolution in our display. We need to create that many pixels from this node 3D model. Correct. And that is is this rendering is a crucial bridge. We are looking at it. Yes. So this is one know uh one of the sample images we are now showing you. You can see that this complex image has you know even the glasses and then reflections metallic surfaces right non-metallic surfaces plastics right and there is a specular light and then diffused light all those things right it's a complex scene what you're seeing it the goal in the computer graphics is to generate 2D image from the 3D scene right 2D images because we are talking in the display and it is full of know pixels right the input data is the scene description scene description is not just the know 3D model but it the lighting as well the material as well right so we have to input as much as description as possible of the scene the output is an image on the display so as of now since we are talking about the VR display it is the 2D image There are 3D displays are coming up. In the 3D display then it is going to be a 3D image. It is not a no 2D image. Right? The output can be a no 2D image. But in in our course we are going to restrict the output as a no 2D image. Right? In the scene there are objects lights and the viewer and the materials. Correct? This is the input we are talking about. The output is the output is the image. So rendering is a image synthesis process. It's an automatic process of generating photorealistic like or nonphoto realistic image of the virtual world from 2D or 3D model. Input can be also 2D model. Output is going to be 2D or 3D. And the visual representation as of now you're looking at only the pixels but the visual representation need not be

pixels. It can be no charts. There can be a uh rendering uh device instead of pixels. It will can it can only show the charts and graphs or maps or timelines or tables which are specialized displays. Right? So the displays can be not necessarily not pixels uh in the now many places uh airports and all the things now you can say light based displays correct instead of pixels based right so our visual rendering process should take care of now what type of the the display it is right but in this class we are talking about only the pixel based visual display but uh I just wanted you to be aware that the the dis pixel is not the only way of visual representation. Okay. So we all we use this jargon called visual rendering pipeline. Pipeline is a one which you know uh there is a typical now pipe you all imagine. Whatever we give input we can know uh expected output in between know we cannot uh know uh manipulate it whatever we given it comes out output right so like this so uh since we give 3D model as an input and at the exit we get the pixels in between what is happening now we don't have any control we are going to see what is happening but now we don't have a control of it right and that is why it is named as the pipeline. So in the application layer that is in the know 3D computer model it supplies a geometric data we convert it into triangle and then the geometry is actually no transformation is happening and then shading and clipping is happening and then finally there's a rasterization process which converts triangles into pixels and then interpolates lighting and texture is added. These are the three you know uh very basic stages of uh rendering pipeline. This is what we call as a graphics pipeline. There are lot of jarens or nicknames given to this pipeline. You have to get used to it. Right? So application level and the geometry level and then raster resization level. This is the three stages as of uh any typical pipeline. Right? So in the uh uh in the uh rendering pipeline we will focus on the transformations and from from the now uh from the 3D model to the the pixels. What are the different ways in which each of this vertexes if it is a 3D model how it is actually transforming? We are going to see the different levels of the transformation from the world of uh world with objects and lights. How we are going to generate the uh viewers's perspective is what we are going to see right single matrix can accomplish it right but this matrix is composition of so many levels of so many matrices I want we want you to understand what are the different uh matrices that composes this now one single matrix so by when you understand it you will appreciate what this GPU the graphics processing unit is actually doing. Right? So uh this is a four different levels of uh transformations we are looking at it. One is the uh the initially the the model 3D model whatever we have it is in the object space and then this object space 3D model has to be mapped to the world space right and that it maps local object coordinates to the world coordinates this one and know uh transformation is there and then from the world space to camera space we are going to do it. We are going to set up a camera and then we are going to look at the the 3D model in the world space through a camera and therefore this is the camera space we are looking at it. So the second transformation is uh now mapping the world coordinates to the camera coordinates and then once we have the camera coordinate we are going to convert this is into a canonical model. uh this camera model to canonical view. What is canonical view in the sense if you look at this camera model this is in the perspective you can see that now this is not a regular rectangle whereas in the final screen if you look at it it's a regular rectangle or it can be a square or some known space. So we are going to map this irregular rectangle to the regular rectangle using this canonical uh view volume. Basically it is a it is making into a rectangular no shape mapping into a rectangular shape. That is what we call as a canonical view volume. Map camera coordinates to a canonical view volume. Right? and then finally map this canonical view volume into a computer screen in 2D. That is the final way which we are going to look at it. So there is a four level of transformations I want you to uh appreciate. So one is the from the local to world and then world to camera and then camera to canonical canonical to the the screen space. So we will have a matrix for each of these transformations. We will go through each of these transformations. how to set it up and then how we can how each of these uh uh matrices are actually helping. Okay. So um this each of those matrix is called the this is a model matrix this then this is a view matrix and then there is a canonical matrix and then viewout matrix four matrix are there and this as you know we have already studied we will do the reverse of it now model matrix view matrix canonical matrix and v matrix multiplied right in the reverse ordered to get one single transformation matrix. So the transform mat transformation matrix consists of this four stages of transformations but in the reverse order in order to get you the the the pixels in the right space. All right. Um so if uh this is the input we are talking about for example one triangle in the in the in the local space right and then now finally it gets you to into the the the in the you know screen space in the textbook this is what is called uh viewport matrix canonical matrix um and then this is a view matrix and then this is the model matrix. Okay. So uh let us look at each of this uh matrices because it is very important to understand the the implification of uh each of these matrices. Here is a 3D mesh object from Unity. how how this u model matrix is set up it and uh how it can be controlled. Okay. So um as we told you this uh model matrix is the transforming from the the local object space to know this uh world space right the transformation is applied to all the vertices of the mesh. Let us say the input matrix is a mesh and it has all the vertices and each of the vertices has to be transformed from local space to know uh the world space and this is a rotation matrix and this is a translation matrix and uh therefore it is u uh it is called the the model matrix. This model matrix in the unity it is set it up in a transform matrix over here. Correct. And um you can see the the position rotation and then uh the the scaling as well. So uh you you must have been already know setting up all the things when you are making your assignments. I want to just now uh now see that where this no model matrix is actually coming up. Right. Yeah. Now let us look at the the view matrix which is mapping the now world space to the the

camera space. Camera space is the uh is the where we are keeping a camera and then looking at the the object in the world space. The I is a rigid body as far as the the entire rendering is concern is concerned. Right? So that is going to come somewhere over here. Um again in the unity we are going to see uh the camera from the unity 3D model it generates a view of the virtual world and this is accomplished by no view matrix in the geometry we see the information and in the transform uh transform box is used to create the the view matrix. So here is the transfer matrix for the main camera right and this is where we are setting up where this camera is and what it is going to look at it. So this projection and field of view, clipping plane and view port rectangles are all used for now further transformations which we will see now later and uh the camera as I told you this is going to it's a rigid body and it is going to be placed in a particular point and not only the position is important but also the orientation at which which uh it is looking at the object is very important that is why now we use a vector called the I know look at vector right and this camera is positioned within the now world space right now because we have already now have this model matrix implemented right so this is the object we are going to look at it and uh we have to define what is the position of the eye and what is the look at matrix as well. The so the easy way to construct a I matrix is a create an orthonormal basis with the i at the origin and the vectors x i y as a basis vector. Right? This is the no uh basis vector we are looking at it. Right? And this is the vector we need to know set it up. Yeah. So the I is at the origin. So we know this is know E1 and E2 E3 and this position we are talking about and the orthonormal vectors are are um XC we can look at. So the I matrix we can construct it. you all now familiar with know how to construct the vector given this position. So we will quickly go through it. We'll not spend too much time on this since you're all familiar. So this is is this a correct I matrix transformation matrix right? Let us look at it again. Is this the correct I matrix? Now I want you to know uh you know check whether this is a one. So given the know four different matrices can you identify what is the correct I matrices. Okay let us take a small pole of it. I want you to know look at where this basis vectors are there. For example this is the I positive xc this is the y positive and this is the z positive. Whereas the look at is going to be no in this way. All right. Therefore, no you need to choose the correct matrix. Okay. Can you uh can you choose the correct I matrix? Now again and there are two eyes in the virtual reality and therefore we need to know u be little careful we have this IPD which is already measured and then uh now uh this is the center where the the camera is there and therefore this has to be now t by 2 t is the IPD and this is t minus 2 for the next view right So then each I will have a know two different I matrices. Now you know how to construct each of those I matrices. This IPDS has to be multiplied by the T uh two T view to get the I matrices. Okay. So we have this model matrix done. We have this view matrix also done. And then how about the canonical matrix? So here before going to the canonical matrix so I want you to look at uh now what this camera has done this camera when it is looking at it it's the know the uh projection it's a it's not a orthographic projection it is the perspective projection it has done perspective projection in the sense now nearby plane there is a near plane there is a far plane so we have to set up the near plane and the far plane as well while setting up the camera as well. So um how do we set up the near plane? How do we set up the far plane? Why should we know choose this near plane and far plane? Why not we uh know avoid it? These are the questions we need to answer. Right? That is the answer we will look at it in the the next stage in the canonical model canonical transformations. Right? So, uh I'm going to skip, uh skip it, uh here. Yeah. While defining um this uh look at I vector suppose if you have if you know take this is a now point P and we know this is the no point E small E and look at vector is going to be P minus E and suppose if you define yeah vector up vector is a U vector or then what if the vectors XC and Y and ZA are not known. Instead, we know only the up vector, P vector and V vector and therefore P minus E vector also know from this can we construct the the the orthonormal basis that is the now uh the point we are talking about. So again, no it's a it's a very straightforward thing. I will leave it to you to construct this up vector from u sorry the the orthonomal bases from these vectors. I'm going to skip it. Uh these are the four options. Again I would like you to now take this as an exercise and then and get it done. Right? I'm going to skip it. So what is XC here? Think of a vector product. Again, I'm going to skip it because we all we all know. Okay, we'll go to the now canonical model. Canonical model as I told you it maps camera coordinates to the canonical volume. Canonical volume is the normalization normalized volume. Normalized volume is a regular rectangle. Right? From the irregular rectangle it is going to the regular rectangle. Correct? So in a perspective projection from the camera point of view this is going to be the now irregular volume right perspective projection. But from this irregular volume we are going to construct a regular volume. So that from once we have the regular volume then converting into a regular rectangular is very easy. Then that is the step we are actually doing it. Okay, understood. So entire this irregular volume has to be mapped into the the regular volume that is a canonical conjunction and this is the actual chart we are talking about right and then field of view also is very important and that is what you are setting it up in uh the unity or the the camera view also right and if it is know 60° there is the know uh there are several I'll Now the perspective volume you are looking at it depending upon the FOV and this will be now going into this canonical projection. [Music] [Music] And again in the unity you'll be setting it up projection and the field of view and clipping planes to create a canonical projections volume. Correct. So clipping plane again we are going to discuss it. V port V port rectangle is used to create the V port matrix over here. So where is a V port rectangle and then there's a projection matrix field of view and clipping plane. These are all know creating this uh canonical projection matrix. So uh again this is a near plane we are talking about right and the other far away is the the far plane. This is a clipping

plane. Out of this clipping plane all the 3D model world is actually is not visible to the camera which need not be worried about for rendering. within this now view volume we have to know take care of it and whatever is in the view volume that has to be mapped to the canonical volume that is the the step this canonical projection transformation is doing it correct so after the canonical projection transformation canonical volume cube is 2x2 and xy screen coordinates and then z is the depth we are looking at so Again know uh the projection and arthographic at this stage I want you to be familiar with. I think most of you are familiar but if not now you better get used to it. The projection uh or perspective projection is different orthographic projection is different. Now uh uh the image in this this left one is the incorrect perspective. it uh it appears to be now mostly know uh autographic whereas in the right one is the correct perspective. The same size objects are looking smaller if they are further with the depth they are smaller. So here the uh in the image you can see that both the persons are same height probably almost similar height but their size is is smaller because the the the smaller person is now much behind further away right so this is one of the uh one of the one of the cues of the perspective projection depth cues we have right that has to be simulated and that is what this uh view uh that's what the canonical projection is actually now trying to do it. What is wrong with the perspective in this particular painting in this uh it is not you know actually capturing the uh it is almost doing the know kind of u autographic projection. It is not the perspective projection right. So in the autoraphic projection and the p the perspective projection when you are doing the view volume what is going to happen is the further away objects are going to look smaller. In this example the red ball which is behind the yellow ball and it is further away from the camera that is looking smaller. And this green ball which is out of the view volume that is not there in the final image that is not there. Whereas in the orthographic projection what happens the behind the ball the red ball and the yellow ball although there are different distance the size of the two balls are the same. Correct. This is a major difference between the autographic projection the perspective projection and this projections. Now again you can choose in the unity and set it up with the orthographic perspective and you'll find the the differences but now this depth cues you'll have to get used to it. Okay. So again and from this volume we need to map this into know the canonical volume. How are we going to map it? to map it. Again we use a simple analogy of if you have a intermediate plane and the same object will look at no smaller size over here. Okay. Can we use this this idea to get the the the plane in autographic projection? This is going to be the the same right the size is not going to know reduce it. So the projection and the field of view and the clipping planes ends are are used to create a canonical projection matrix over here. Right? The view port rectangle is used to create a uh the the view port matrix. Right? So uh again we are going to talk about why we should create near plane. If you have this volume, we are going to set up near plane. This is a near plane. This is a near plane. This is a far plane. Why should we get the near flame? Why should we set up the far plane? We are going to look into little bit of u uh uh perception engineering later after now we are done with the transformation. There is a now volume where in between this near plane and far plane which is now sweet sweet depth or sweet spot we call it as if it is too near there is going to be something know different or we are going to have some difficulties I want you to know uh think about what will be the difficulty I'm going to explain it in the later classes today but I want you to think about it why why We keep it keep the near plane very close. What will be the challenge right in the UX design? The content placement has to be within this now near plane and far plane and then choosing of this near plane and far plane is extremely critical for having a good experience of virtual reality and that is a part of the the canonical projection transformation. Right? When you're looking at u now uh uh the different field of view of course uh now there is a sweet spot this is a sweet spot correct and this is most comfortable if it is beyond this we are going to have no strenuous experience if it is near also that's going to be a strenuous experience and people suggest after experience they talk about 1.25 and 25 m near plane and then 5 m far plane. That is a best practice the industry standard and UX ex you will learn all those things but I want you to relate that to the the canonical projection transform where we are setting it up. Right? So we have this camera spectrum over here. This is a canonical volume over here. The job is to you know map this canonical spectrum to the the sorry camera spectrum to the canonical volume. So after the canonical transformation canonical volume is 2x2x2 and then xy are the screen coordinates z is the depth. Okay so this we have seen it. Let me go through the next one. Yeah. So this is a projection matrix and this is the near plane and then far plane we are going to talk about yes this is the where this is the origin the near plane is uh n + f n minus f why should the this be n + f why should this be no * f that we will see okay so tp This transformation is not a homogeneous is not a homogeneous transformation matrix. Can you tell me why it is not a homogeneous matrix at this point of time? No, you should be very familiar. I want you to look into it. It is not a no projection either. Again, can you tell me why it is not a projection matrix? It is a hack to make the calculation work. Okay. So, what is the hack? we have done is what we are going to see to understand the the TP let us draw a cross-section of it let us say it's a small cross-section right so this is a cross-section in the 2D we are looking at this is the object we are talking about this is a now the height of the object let us say YP this at the distance of ZP over here if we place a cross-section screen let us Say at n distance distance we know what is the this height it is know easy to calculate the goal is to scale down the further away objects what should the yp become um the tp this is a screen we are talking about right it's a simple uh linear scale we will do it correct now to scale down the further object what should the y become yp P will become N into YP divided by ZP. Correct? This is straightforward. Similarly, XP will become N into XP divided by YP. And this is what we are going to talk about it. YP and XP we put it in

the matrix N and N multiply by the ZP of X and P and YP. Right? This matrix will be now like this. So our goal is to find out what is this. So this is the now XP we are talking about. This is the YP we are talking about. This is the scaling uh the Z we are talking about. This is a up vector. This is a canonical model both kind of uh TP projection as well as the the the the ST matrix. This is matrix we are talking about. So the the ST is the scaling and and uh uh translation matrix. S is the scale and the T is the scale and we will multiply TST and that is the ST matrix we are talking. Okay. So I'm rushing up because you all already know have lot of uh assignments and homeworks you have done. Therefore you are all familiar with it. And the final one is the the V port matrix. Vport matrix now um maps this canonical volume into a square. Correct. So window to view port. Right. Now what uh here what what we first will do is first translate the lower left corner to Z uh 0.0 zero, right? And then now scale the upper right corner from two to 2A 2 to this uh window is uh uh minus one and height minus one. That is a typical height of the number of resolutions uh in the width as well as the number of resolution the height that minus one and that will give you this matrix to get finally this matrix. So we have all the matrix done. We have looked at the model matrix, view matrix, canonical matrix and then view matrix and then all these four matrix together is what uh we can multiply together make it to one single matrix to transform the input to the the output pixel input vertex to the output pixel. Now I wanted to now uh take you back now one and step further and uh and tell you why we should look at um the near plane and then um and then the um far plane why should we set it up right so industry practice we've told you know 1.5 m and 5 m that's a sweet spot we are looking at it right but why should that Right. Can we know reason that out from the uh perception engineering point of view? Correct. uh to reason it out I'm going to uh recall or add some of the terminologies what we have already looked at in the neurohysiology I as well as the um uh as well as the the neuroscience and of the know vision which professor Anna has already taught you in the last few classes. Okay. There is one particular you know uh you know process of I is called the stereopsis. how we actually take two images from the eye and then merge into one single image in the the the visual cortex. Correct? That process is called the stereopsis. That's the ability to use the binocular retinal disparity. So when two images coming, there is a small disparity between those two images. Disparity in the sense differences. It is the two images from the two eyes are not going to be the same. And the differences in these two images are going to tell how far the images is. That's a retinal disparity, right? That information to extract the relative depth information from the retinal image pair. That is a stereopsis. This process is inherent and it is we are doing it now million times a day. Right? This retinal mismatch can be used to reconstruct the much of the missing uh third dimension from the retinal images. This teopsis process is going to tell us where we should keep our 3D models in the VR images and why should we keep it. Right? So retinal disparity is understanding the understanding begins with a consideration of the geometry of the corruptor. Um I think uh professor Anna has now briefly mentioned about the harapter in the aerial classes. Now I want you to relate the harapter to the the the canonical matrix what we are talking about the near plane and the far plane. So to recall what a horter is you know it's a Greek word it has you know two phrases horo and opter right horo means it's a boundary boundary and then opter means it is the observer from the observer what is the boundary suppose if this is an observer I of the observer I'm going to show you another object So this observer what is he going to what is the boundary of the his observation that is the horopter horopter is an imaginary surface to his points are all at on the at the same distance as the fixation point. Okay, if this is the fixation point now all the similar fixation points we can connect and then get the horopter right points on the horopter project to the corresponding locations on the temporal and nosal retinas respectively we are going to look at what is the temporal nosal retinas as if professor Anna has not mentioned we'll look at it these corresponding locations exhibit zero retinal disparity So all those points on the no the harapter they are not going to have a disparity at all. They are at the same distance as far as the the the um the stereopsis is concerned. Essentially the disparity is actually zero. This is the temporal disparity and then this is the nosal disparity and difference is actually zero. That is what the anaropter is. So this is the nozzle side of the retina. This is the no temporal side of the retina. So when uh now when your nose is over here near the nose, the retinal side is called the nosle retina and the the the retina near the temporal side is called the the temporal is the side of your head. Right? That is called the temporal retina and and uh the haropter the disparity from the nosal retina and then temporal retina are zero and that is going to give you the the haropter. All right. So this is the left hemisphere. This is the right hemisphere. In the right hemisphere also we'll have the temporal retina and the nosal retina and that is going to have a right hemisphere. And then together is what we are going to have a the entire know vision field right. And in a practical situations like this when a observer is looking at it the the horopter is at the same depth as far as the observer is concerned. Right? That's a that's where the disparity is zero. Okay. And we are talking about the the corresponding locations. Now uh closer to the if it is a closer what happens? If it is a further what happens? If it is a closer object then there is a disparity. You can see that now this nasal disparity is this much and temporal disparity is this much. There is a difference between the nasal and temporal disparity. U whereas here this is a nasal disparity is this much whereas further away is going to be now this much. So the corresponding locations for the closer green stimulus exhibit positive retinal disparity. This is a temporal disparity is now greater than the nasal disparity. Therefore it is positive. It is called the crossed disparity. the uh the corresponding location for the further red stimulus exhibits a negative disparity temporal disparity is know lesser than the nasal disparity that is negative this is called the uncrossed disparity so what what is the point in know learning about the crossed and uncrossed u know disparity right I'm

going to show you a small example or small experiment you all you can all do and then know experience yourself right so before I tell you what is the experiment I'm going to explain what is this now one more concept of now fusion area panum's fusion area is one now once you know this haraper this is the harapter we are talking about and this is the no uncrossed disparity this is a crossed disparity and within this the uh fusion area. This is the fusion area. Within this area, we fuse two images into one images, one image. Right? So within the uncrossed and crossed, we are going to fuse. This is the panam's fusional area. If it is outside of this fusional area, we are going to have we are not able to know combine these two images into single image. Right? And that's the experiment I want you to do uh in the next few minutes. Right? [Music] [Music] So again to emphasize it the fusion of topetic topetic images to um images panum's limiting case is the most simple example. to illustrate the problem of the combining different images. There's a fashionist paradox also is coming. When both eyes are open, is the room brighter than with only one eye open? Can you check whether that is true? Does the brightness in each of the eyes add together? That is now know one experiment, one simple experiment you can do. Therefore is is know is the fashion paradox is it true or how do we explain is a question right that's one experiment. There is another experiment I would like you to know do this is called the binocular depth cues. It is called the finger sauce. So you keep your fingers very close very close your eyes. Okay. So hold your two index fingers about 5 in in front of your eyes with your tips half an inch apart. Right? Now look beyond them. Don't look at the finger. Look beyond them. Note the weird result. What is going to happen is you will have a floating finger appearing in the middle. You will have two fingers coming aside but you will also see a floating eye. That is called the finger sauce. Can you all try it out? And and if you uh can get it, move your fingers out further and the retinal disparity will shrink. Okay. So I hope everybody would get it right. Now the question is that why do we have this floating eye? It is called a finger sauces. Can we explain this floating eye with a haropter concept with the you know panam's no fusion area right so this is where no you are you know this is where this our canonical near plane and far plane is coming into picture if it is too close I want to show you that the two images are not fused properly If it is far away also it is not going to fuse properly. There is a sweet volume in which two images are fused properly and that is why we need this you know 1.5 m to 5 m fusional area right so when focusing on distance object the nearby fingers fall outside the range of the panum fusional area panum fusial area you know we saw that you know if it is too close it is not you know fusing fusing two images so they are seen has a double images one from each. Normally the brain suppresses one of the images to avoid the double vision. But here because both the fingers are very close your brain attempts to fuse parts of the two images together creating the appearance of the the finger sausage. Now let us go back and then see the panamis fusional area. Right? And your is your finger is very close in this region and you are focusing and fixating in this particular area and panam's fusional area is within this any object within this area it will be fused properly. If it is too close or too away it is not uh um not fused properly. particularly know if it is too close it will not be fused further away there are now other problems we'll discuss in the later class I hope um now uh it makes sense why this now suits volume is defined as a best practice in the virtual volume virtual reality best practices Right now let us uh now speed up a little further because we need lot of other topics to be discussed. Whatever we have discussed this now transformation is the very basic graphics pipeline which is uh know defined some you know 30 40 years ago and still it is a very basal one for all the graphics pipeline but it has transformed so much in the last uh know uh 20 years particularly in the last five years there's so much of change over here and that is what we are going to talk about in the next classes. Okay. So in the next few uh minutes what we are going to talk about it is there are other computations done in the pipeline. It is not just the transformations. So rasterization also we have done it right. So um in the application this is the now data we are giving input. Okay. So geometry transformation is what we uh uh talked about and then rasterization light is now to be done and then image assembly is to be done. So we are going to talk about uh all the things in the coming classes. What we have done is only the geometric transformation. Correct? So how are we going to rasterize this? How do we add light to it and image assembly? How are we going to do it? We'll see in the next classes. Rasteration is only one way of rendering. There are other ways of rendering which again we are going to know uh see in the uh the next class. There are very advanced way of particularly this uh rendering pipeline what we have discussed is common for all the um all the graphics related thing. But what is specific for virtual reality? Is there a is there a no something which we have to take care as far as the rendering pipeline is concerned? Again we will discuss in the next class. Can you imagine now um uh among all the things we are using the same pipeline for now movie producing we are using the same pipeline for gaming. We are using the same pipeline for VR as well, AR as well, right? But what is so special requirement for virtual reality? Among all these uh applications, virtual reality is the one which is demanding the most one. Why? Movie means for every frame you can spend half an hour. It doesn't matter because it's all done offline. gaming also the the the fidelity doesn't really matter the same object again and again you're going to show correct so again uh it is is not going to be no very demanding the number of frames per second is is actually requirement is very less now 60 frames or even 30 frames and sometimes 10 frames also right but for virtual reality we have to go for 90 frames or 120 frames. Higher the better, right? Therefore, this pipeline has to be now optimized in every stage actually to get to the virtual reality. Okay. So, where all things can be optimized is what we are going to see in the next class. I would like you to come with the know thinking about or the suggestions of how this can be optimized. Okay. And uh that is where the in the last 20 years lot of advancements has

happened. We are going to see now several advancements and then I hope you know it will give you an idea of uh appreciate how this no GPU is actually. We are also going to see what will happen in the next 10 years for the the rendering pipeline. How this is evolving? What is know future rendering pipeline is going to look like is al we going to look at it. Okay. So um uh here is a very simple renderer where you can implement it yourself. It's a now we have the uh simple wireframe graphics a 3D object with a wireframe which defines all this u vertices and the edges. It's a four lines of codes. Uh there is a link which we are going to give you. You can download it and then you can render it yourself and then see how this transformation is taking place. Right? And this code is over here. This is how it will it will uh now appear in your u mobile phone right. So this this is a link uh we are going to talk about this link we will share it in the uh in the um uh discord. So you can download the link and then get it useful. So get a input from the IMU project it onto a now phone screen and add a camera for position tracking and all those things. So rendering plus now those things also can be done. Now we wanted to see whether the wireframe model produces less vection. That is a hypothesis you want we would like you to test it. If it is yes why if it is not again explain why. Right? So, so I will end this u class saying that there is lot more things to be done. What we have have seen is only there 40 years ago. But these are the fundamentals which is absolutely important for everybody who is learning the virtual reality. But it's also equally important to know what is the recent developments and also to know what is going to happen in the next 5 years or 10 years. That's where we will see in the next class. We'll stop here. Thank you for listening. We'll all see you in the next class. See you. Thanks. Bye. [Music] lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

Virtual Reality: Easy Notes + MCQs

Let's learn this lecture in simple English, as if you are in 7th class. I'll divide it into small topics with examples, memory tricks, and a practice quiz.

This lecture mainly covers **two topics**:

1. **VR tracking:** How a headset knows where your head is and corrects movement errors.
2. **Rendering pipeline:** How a computer changes a 3D world into the picture you see on the VR screen.

Part 1: VR tracking and sensor fusion

1. Why can't we use only an accelerometer?

Imagine you are walking while holding a phone. The phone has an accelerometer that measures acceleration.

The computer can calculate movement in three steps:

Acceleration
How quickly velocity changes

↓

Integrate once → Velocity

↓

Integrate twice → Position

The problem is that the accelerometer has tiny errors. When we integrate its readings twice, these errors can grow very quickly. This is called **drift**.

Example: Suppose the computer thinks you moved 2 cm, but you actually moved 1 cm. If small errors keep accumulating, the computer may eventually think your head is somewhere completely different.

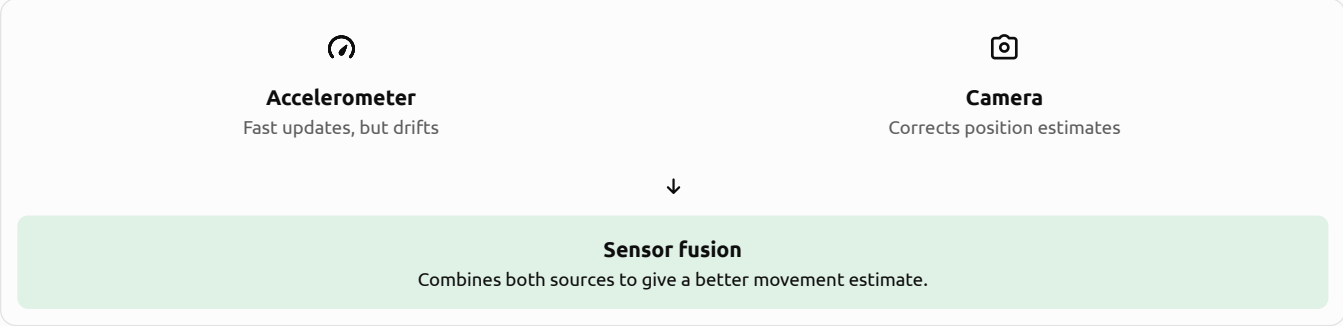
Remember: Accelerometer alone = position drift.

2. How does a camera correct the error?

A camera looks at the real world and identifies features, such as corners, patterns, or markers. Using these features, it can estimate the headset's position and orientation.

But a camera may only provide a new estimate every 16 milliseconds in the lecture's example. The headset needs frequent updates to keep VR movement smooth.

So, we combine the camera with the accelerometer. This is called **sensor fusion**.



What is a complementary filter?

A complementary filter combines two kinds of information to make a better estimate.

- The accelerometer gives frequent movement updates.
- The camera helps correct the accumulated drift.
- The filter blends the estimates rather than suddenly jumping to a different position.

Memory trick: Accelerometer = fast; camera = correction; complementary filter = combination.

The correction should be fast enough to reduce errors but smooth enough to avoid sudden, uncomfortable jumps in VR.

3. Why are low latency and high frame rates important?

- **Latency:** The delay between your movement and the headset showing the corresponding change.
- **Frame rate:** How many images the display shows each second.
- **Dropped frame:** An expected image is not displayed on time.

Imagine turning your head, but the VR world turns a little later. This delay can feel uncomfortable.

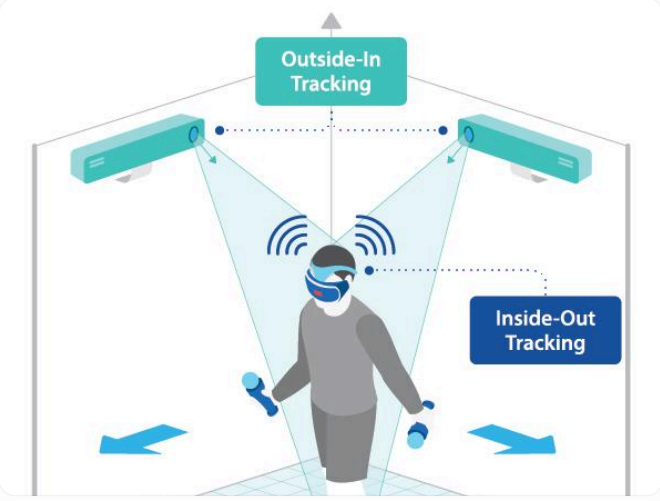
For a comfortable VR experience, we want:

- Low latency.
- Consistent frame delivery.
- Accurate tracking.
- Smooth, frequent updates.

Memory trick: Move your head → VR should respond quickly.

4. Inside-out versus outside-in tracking

Inside-out tracking
Cameras are on the headset and observe the surroundings.



Outside-in tracking
External cameras or sensors observe the headset.

Remember: Camera inside, looking out.

Remember: Cameras outside, looking in.

Both methods estimate where the headset is and which direction it faces.

How does a camera estimate position?

It matches features seen in an image with features whose real-world positions are known or estimated.

For example, if a marker looks smaller, it may be farther away. By matching several features and their image positions, the system can estimate the headset's position and orientation.

Memory trick: Find features → match features → estimate pose.

Pose means position plus orientation.

Part 2: Rendering pipeline

5. What is rendering?

Rendering means **converting a 3D virtual world into an image that appears on your screen**.

Imagine a 3D chair inside a computer game. The computer must decide how the chair looks from your camera's viewpoint, which parts are visible, what colours it has, and how light falls on it.

↓

Rendering pipeline

Transforms, projects, shades and converts the scene into an image

↓

A 3D model becomes a 2D image on the display.

Why must VR developers understand rendering?

Because VR needs images to be produced very quickly. If the image is delayed, jerky, or incorrect, the experience may become uncomfortable.

The lecturer uses an example of **90 frames per second (FPS)**.

$$\text{Time per frame} = \frac{1000}{90} \approx 11.1 \text{ ms}$$

That means the system has only about **11.1 milliseconds per frame** at 90 FPS.

At 120 FPS, it has only about 8.33 milliseconds per frame.

Memory trick: Higher FPS = less time available for each frame.

6. The three basic stages of the graphics pipeline

Remember these three stages in order:

1. Application stage
Provides objects, shapes, vertices, lights and materials.

↓

2. Geometry stage
Transforms objects and determines how they appear from the camera.

↓

3. Rasterization stage
Converts shapes into screen pixels and helps produce the final image.

Let's understand the important words:

- **Vertex:** A point on a 3D object.
- **Mesh:** A collection of vertices connected to form shapes.

- **Shading:** Deciding how bright or dark surfaces look.
- **Texture:** An image or pattern applied to a surface.
- **Rasterization:** Converting geometric shapes into pixels.

Memory trick: Application → Geometry → Rasterization (A-G-R).

7. The four transformations

This is one of the most important topics for exams.

A 3D object passes through four coordinate spaces before reaching the screen.

1. Object space
The object's own coordinates

↓

2. World space
Where the object is placed in the virtual world

↓

3. Camera space
Coordinates relative to the camera

↓

4. Screen space
The final 2D screen coordinates

The lecture describes the transformations using four matrices:

Matrix	What it does	Easy example
Model matrix	Object space → world space	Put a chair into a room
View matrix	World space → camera space	Choose where the camera is looking from
Projection matrix	Camera space → canonical view volume	Make objects appear with perspective
Viewport matrix	Canonical volume → screen space	Map coordinates to screen pixels

The lecture calls the projection stage the canonical transformation or canonical projection.

How to remember the four matrices

M → V → P → V

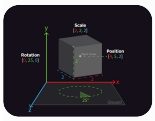
- **M**odel
- **V**iew
- **P**rojection
- **V**iewport

Although we describe them in this order, the exact multiplication order of the matrices depends on the mathematical convention being used. In the common column-vector convention, the combined transformation is:

$$p_{\text{screen}} = V_{\text{port}} PVM p_{\text{object}}$$

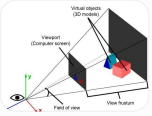
The rightmost transformation acts first.

8. What does each matrix do?



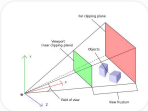
Model matrix

Moves, rotates, or scales an object. For example, it can move a chair from its own origin to a position in the room.



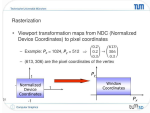
View matrix

Represents the viewpoint of the camera. Moving or rotating the camera changes what we see.



Projection matrix

Converts the camera's view into a standard view volume. It determines perspective and depth mapping.

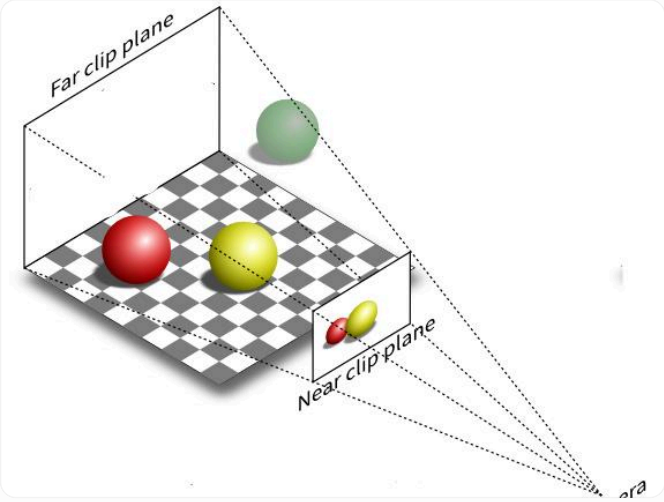


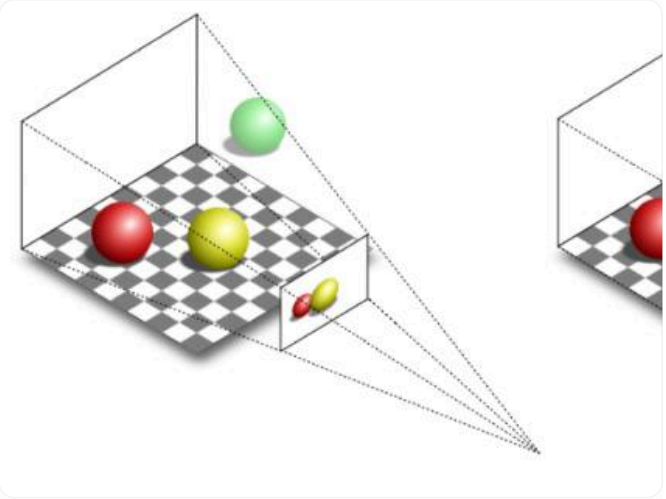
Viewport matrix

Maps the projected coordinates onto the actual screen area.

9. Perspective versus orthographic projection

These two terms are frequently tested in MCQs.





Perspective projection

Faraway objects look smaller. This resembles how we normally see the world.

Orthographic projection

Objects of the same size remain the same apparent size even when their depth changes.

Easy example: A railway track looks as if its sides meet far away. That is a perspective effect.

Remember:

- Perspective = farther objects look smaller.
- Orthographic = size does not shrink just because an object is farther away.

10. What are the near plane and far plane?

The camera does not need to render every object from infinitely close to infinitely far away.

- **Near plane:** The closest boundary of the camera's viewing region.
- **Far plane:** The farthest boundary of the camera's viewing region.
- **Clipping:** Excluding geometry outside the selected viewing region.

Imagine looking through a rectangular window. Objects outside the window's viewing range need not appear in the rendered image.

The lecturer also discusses a comfortable depth range for VR content. The exact suitable distances depend on the headset, optics, content, and viewing conditions; the numbers mentioned in the lecture should not be treated as universal rules for every VR system.

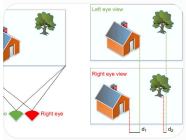
Memory trick: Near = closest boundary; far = farthest boundary; clipping = remove what is outside.

Part 3: How our two eyes understand depth

11. What is stereopsis?

You have two eyes, and each eye sees a slightly different image. Your brain combines these images to help you understand depth.

This ability is called **stereopsis**.



Example: Hold a finger in front of your face. Close your left eye, then your right eye. Your finger appears to shift relative to the background.

Your brain uses differences between the two views to help estimate depth.

Memory trick: Two eyes + two slightly different images = depth perception.

12. What is retinal disparity?

Retinal disparity means the difference between the images formed on your two retinas.

Term	Simple meaning
Zero disparity	The images fall at corresponding retinal locations
Crossed disparity	Typically associated with an object closer than the fixation point
Uncrossed disparity	Typically associated with an object farther than the fixation point

The **fixation point** is the point your eyes are looking at.

Memory trick: Crossed = closer; uncrossed = farther.

13. What is the horopter?

The horopter is an imaginary set of points that have approximately zero binocular retinal disparity for a particular fixation point.

Imagine focusing on a flower. Other points on the horopter project to corresponding locations in your two eyes.

Remember: Horopter = zero-disparity reference for a particular fixation.

14. What is Panum's fusional area?

Your brain can combine the two images from your eyes into one image, even when they are not perfectly identical. There is a limited range of disparity within which this fusion works.

This range is called **Panum's fusional area**.

- Inside the fusion range: the brain can generally combine the images into one.
- Outside the fusion range: the images may appear double, or fusion may fail.

Memory trick: Panum's area = range where two images can fuse.

15. The finger-sausage experiment

Try this simple experiment:

1. Hold both index fingers in front of your face.
2. Keep the fingertips close together, with a small gap.
3. Look at a distant object behind your fingers rather than directly at your fingertips.
4. Notice whether you see a floating, sausage-like image between the fingers.

This effect can occur because your eyes are focused on the distant object while your nearby fingers produce different images in the two eyes. Your brain combines parts of these images in a surprising way.

The exact appearance depends on your finger spacing, viewing distance, and eye alignment.

Why does this matter in VR? VR presents separate images to the two eyes. If their depth differences are unsuitable, the brain may have difficulty fusing them comfortably. This is one reason depth placement and stereoscopic design matter.

16. Why is VR rendering more demanding than movies?

Application	Main challenge
Movies	Frames can often be rendered offline, even if one frame takes a long time to produce
Games	Need reasonably fast, interactive rendering
VR	Needs fast, consistent rendering and tracking to respond to the user's movement

VR commonly targets high frame rates, such as 90 or 120 FPS, depending on the device.

Memory trick: Movies can wait; VR must respond.

The lecturer ends by previewing other rendering topics, including rasterization, lighting, image assembly, VR-specific optimization, and newer rendering methods.

Part 4: Important exam revision

Topic	One-line answer to memorize
Drift	Position error that accumulates when integrating sensor readings
Sensor fusion	Combining multiple sensors to improve estimates
Complementary filter	Blends estimates that have different strengths
Inside-out tracking	Headset cameras observe the environment
Outside-in tracking	External sensors observe the headset
Rendering	Converting a scene into an image
Rasterization	Converting geometric shapes into pixels
Model matrix	Object space to world space
View matrix	World space to camera space
Projection matrix	Camera space to canonical view volume
Viewport matrix	Projected coordinates to screen coordinates
Perspective projection	Farther objects look smaller
Orthographic projection	Apparent size does not shrink with depth
Stereopsis	Using the two eyes' images to perceive depth
Horopter	Reference surface for approximately zero disparity
Panum's fusional area	Range of disparities that can be fused into one image

Part 5: MCQs for practice

Try these questions before looking at the answers. They cover the most important ideas from the lecture.

15-question quiz

0/15 answered

1. Why does double integration of accelerometer readings cause problems?

- A. It changes the screen colour
- B. Errors accumulate and cause drift
- C. It increases camera resolution
- D. It removes all motion

2. Which sensor helps correct accelerometer drift in the lecture?

A. Microphone

B. Speaker

C. Camera

D. Keyboard

3. What is sensor fusion?

A. Combining information from multiple sensors

B. Removing all sensors

C. Increasing screen brightness

D. Making a 3D model

4. In inside-out tracking, where are the tracking cameras usually located?

A. Only on the walls

B. On the headset

C. Inside the user's eyes

D. On the floor only

5. What does pose mean in VR tracking?

A. Only colour

B. Only distance

C. Position and orientation

D. Only screen size

6. What does rendering do?

A. Converts a virtual scene into an image

B. Measures heart rate

C. Records sound only

D. Charges the headset

7. Which is the correct basic pipeline order?

A. Rasterization → Application → Geometry

B. Geometry → Application → Rasterization

C. Application → Geometry → Rasterization

D. Application → Rasterization → Application

8. Which matrix moves an object from object space to world space?

A. View matrix

B. Model matrix

C. Viewport matrix

D. Projection matrix

9. Which matrix maps world coordinates to camera coordinates?

A. Model matrix

B. Viewport matrix

C. View matrix

D. Projection matrix

10. In perspective projection, what usually happens to farther objects of the same size?

A. They look larger

B. They look smaller

C. They disappear in every case

D. Their colour must change

11. What is the purpose of clipping planes?

A. To define the visible depth range

B. To increase the number of eyes

C. To create sound

D. To measure acceleration

12. What is stereopsis?

A. Rendering shadows

B. Using two eyes' images to perceive depth

C. Measuring rotation speed

D. Converting shapes into pixels

13. What is the horopter associated with?

A. Zero retinal disparity at a fixation point

B. Maximum screen brightness

C. Accelerometer drift

D. The far clipping plane only

14. What is Panum's fusional area?

A. A camera lens

B. The range of disparities that can be fused

C. A tracking algorithm

D. A type of matrix

15. Why does VR need fast and consistent rendering?

A. To avoid using a camera

B. To respond smoothly to head movements and reduce discomfort

C. To make all objects the same size

D. To remove the need for a display

Check answers

The easiest way to remember the whole lecture

Learn these four lines first:

1. **Tracking:** Accelerometer measures acceleration; camera corrects drift.
2. **Rendering:** 3D scene becomes a 2D image.
3. **Four matrices:** Model → View → Projection → Viewport.
4. **Depth perception:** Stereopsis uses two eyes; crossed means closer, uncrossed means farther.

If you remember these four lines, you already have the foundation of the lecture.

You said:

Quiz-9-2026.pdf
PDF

explain easily

ChatGPT said:

Quiz 9 – 2026: Easy Explanation

I'll explain all 10 questions from your PDF in **very simple language**, with examples and memory tricks so you can remember the correct answers for your exam.

Part 1: VR motion and comfort

Q1. What is vection?

Correct answer: A — Vection may occur because visual information suggests self-motion despite the user being stationary.

Imagine you are sitting in a stationary train. The train next to you starts moving, and for a moment, you feel that your own train is moving.

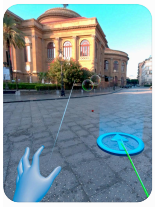
Something similar can happen in VR.

- **Optic flow:** The visual pattern of things moving across your eyes.
- **Vection:** The feeling that *you* are moving, even when your body is still.
- **Vestibular system:** The balance system in your inner ear that senses head movement and acceleration.

Example: You stand still wearing a VR headset, but the virtual road moves backward. You may feel that you are moving forward.

Remember: Optic flow = visual movement; vection = feeling of self-movement.

Q2. Which technique reduces VR discomfort when moving to another location?



Correct answer: C — Teleportation

Instead of smoothly moving the avatar from one location to another, the user selects a destination and instantly appears there.

Why does this help? Your eyes do not see a long period of virtual acceleration while your body remains stationary.

- Gradual acceleration can create a mismatch between vision and the inner ear.
- Increasing the field of view can expose you to more visual motion.

Remember: In many VR experiences, teleportation can reduce motion discomfort.

Q3. Why can a large menu attached to your head cause discomfort?

Correct answer: B — It occupies much of the field of view and behaves differently from the surrounding environment.

Imagine a large board stuck in front of your face. You turn your head, and the board stays in exactly the same position relative to your head while the virtual world moves around it.

This can feel unnatural, especially when the menu covers a large part of your vision.

Remember: Small head-attached menus may be fine; large ones can be uncomfortable.

Q4. Which is NOT a recommended VR comfort strategy?

Correct answer: C — Turning off head tracking.

When you turn your real head, the virtual scene should update to match that movement.

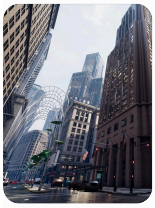
If head tracking is turned off, your head moves but the virtual view does not respond properly. This mismatch can cause discomfort.

Other strategies that may help include:

- Minimizing acceleration.
- Reducing the field of view during virtual movement.
- Reducing exposure to conflicting optic flow.

Remember: Do not turn off head tracking to improve comfort.

Q5. Why can detailed virtual environments cause more discomfort?



Correct answer: D — Detailed environments generate more retinal features that contribute to optic flow.

When you move through a detailed virtual street, buildings, windows, trees and road markings all move across your vision. These features provide strong visual information about movement.

A nearly empty environment has fewer visible features moving across your eyes.

Remember: More visible details can mean stronger optic flow and potentially more discomfort.

Q6. Why might a developer feel fine while first-time VR users feel sick?

Correct answer: B — Adaptation and experience.

A developer may test the same VR application many times. Their brain gradually becomes accustomed to its movement and visual effects.

A first-time user may not have adapted yet and can experience more discomfort.

Example: The first time you use a VR roller coaster, you might feel uncomfortable. After repeated exposure, you may become more accustomed to it.

Remember: Repeated exposure can lead to adaptation, but it does not guarantee that everyone will feel comfortable.

Part 2: IMU, acceleration and tracking

Q7. If angular velocity is constant, which quantity is zero?

Correct answer: A — Angular acceleration.

Let's understand the difference:

- **Angular velocity:** How fast something rotates.
- **Angular acceleration:** How quickly its rotational speed or direction of rotation changes.

Imagine a fan rotating steadily at the same speed.

- Angular velocity is not zero because the fan is rotating.
- Angular acceleration is zero because its angular velocity is constant.

Remember the formula:

$$\text{Angular acceleration} = \frac{\text{Change in angular velocity}}{\text{Time}}$$

If angular velocity does not change, angular acceleration is zero.

Q8. Why can't an accelerometer provide reliable long-term position by integrating twice?

Correct answer: D — Double integration causes accumulated drift that grows very rapidly.

An accelerometer measures **acceleration**, not position.

To estimate position:

Acceleration
What the accelerometer measures

↓

Integrate once → Velocity

↓

Integrate again → Position

Even a tiny measurement error can accumulate over time. After double integration, the estimated position may be far from the real position.

Example: Your headset is actually stationary, but its estimated position slowly drifts away from its true position.

Remember: Acceleration → velocity → position, but errors accumulate.

Q9. Why combine an accelerometer with a camera?

Correct answer: C — The accelerometer provides rapid updates while the camera can correct accumulated positional drift.

Each sensor has a different strength.

Sensor	Main advantage	Main problem
Accelerometer / IMU	Fast motion updates	Position estimates drift
Camera	Helps estimate position using visual features	Updates may be less frequent
Sensor fusion	Combines information from both	Requires careful estimation

A tracking system uses the IMU to estimate rapid movement and uses camera information to correct the accumulated error.

This combination is called **sensor fusion**. A complementary filter is one technique for combining sensor estimates.

Memory trick: IMU = fast updates; camera = correction.

Part 3: Rendering pipeline

Q10. What is the correct sequence of coordinate transformations?

Correct answer: B — Object → World → Camera → Canonical → Screen.

A computer creates a 3D scene and then converts it into the 2D image shown on your screen or VR headset.

1. Object space
Where the object is defined locally

↓

2. World space
Where the object is placed in the virtual world

↓

3. Camera space

Coordinates relative to the camera

↓

4. Canonical / projected space
The scene is mapped into the viewing volume

↓

5. Screen space
Final screen coordinates and pixels

The four transformations are:

Transformation	What it does
Model matrix	Object → World
View matrix	World → Camera
Projection matrix	Camera → Canonical
Viewport matrix	Canonical → Screen

Memory trick: M → V → P → V = Model → View → Projection → Viewport.

Quick revision: All 10 answers

Question	Correct answer	Remember this
1	A	Vection = illusion of self-motion
2	C	Teleportation can reduce discomfort
3	B	Large head-attached menus may feel unnatural
4	C	Do not turn off head tracking
5	D	More visual details produce more optic flow
6	B	Adaptation through experience
7	A	Constant angular velocity means zero angular acceleration
8	D	Double integration causes drift
9	C	IMU gives rapid updates; camera corrects drift
10	B	Object → World → Camera → Canonical → Screen

Test yourself

8-question practice quiz 0/8 answered

1. A stationary person sees a virtual world moving forward. What may happen?

- A. Angular acceleration
- B. Vection
- C. Rasterization
- D. Teleportation

2. Which VR movement method can reduce visual-vestibular mismatch?

- A. Teleportation
- B. Turning off tracking

C. Increasing acceleration

D. Adding a large head-attached menu

3. Why can turning off head tracking cause discomfort?

A. It removes all textures

B. The virtual view may not follow real head movement

C. It increases camera resolution

D. It stops the avatar from existing

4. A fan rotates at a constant angular velocity. Its angular acceleration is:

A. Always increasing

B. Equal to its angular velocity

C. Zero

D. Always negative

5. Why does double integration of acceleration create tracking drift?

A. It removes position

B. Measurement errors accumulate

C. It measures only images

D. It prevents all movement

6. Which sensor combination helps correct position drift?

A. Two screens

B. Camera and IMU

C. Two textures

D. Keyboard and monitor

7. What is the first coordinate space in the quiz's transformation sequence?

A. Screen

B. Canonical

C. Object

D. Camera

8. Which matrix transforms world coordinates into camera coordinates?

A. Model matrix

B. View matrix

C. Viewport matrix

D. Texture matrix

Check answers

You said:

[Music] So let us uh continue from where we left. We are looking at the visual rendering chapter seven from the you know our textbook by professor Steve Lavele. uh we have been uh learning several concepts in the last class. We learned about the the the definitions and the fundamentals of the virtu components. Now specifically we looked at the the the basic components of a graphics engine, right? Graphics pipeline, right? the the more advanced concepts we look at in today's class, right? Particularly all the graphics pipeline whatever we learned in the last class how it is modified for virtual reality rendering, right? So uh in a graphics course we will talk you know very detail of each of the components but since the focus of this group this course is not the graphics related concept but VR virtual reality related concepts. Therefore we will not get into the details of the graphics components but rather how each of these components are modified for virtual reality. Right? particularly we are going to talk about virtual reality specific rendering challenges and then certainly some advanced topics. So in today's class we are going to discuss the basic components of the graphics engine how it is modified in the last 30 years or so. I'm going to u I'm going to give you a chronological order of the the the unfolding of different events that led us to the advanced virtual reality rendering as of today and then we will end with uh end this class with uh our you know our projection of how this the developments are going to unfold in the future. We are projecting to the now future and let's see whether the the projection is correct or not. Some of the topics may be inspiring some of you to know take uh uh uh some proposals the future futuristic proposals as your VR project. Okay. So I will quickly review on what we have done in the last class. We started with uh know studying why visual rendering is first of all necessary particularly for virtual reality. There's a so much of requirement particularly 90 frames per second right and that puts a very hard constraint and therefore we need to look at all the components of the graphics and therefore we can optimize things. We'll we'll get to see in the later part of this class what component we can tweak in order to achieve this 90 frames per second. Right? And we started with uh know uh where rendering is actually playing a role um in the overall know overall the skeleton of know virtual reality. And uh we saw that there are three different know uh layers of v visual rendering pipeline. Application layer uh and the geometry layer and then rasterization application. We looked at all this as properties of the the geometry or 3D data model and then uh where the geometry we apply lot of transformations and then shading and clipping and then rasterization we converted this geometry often transformations into know pixels right and interpolated lighting and the textures right those are the three uh uh course divisions of the rendering pipeline and then we looked at the know several u uh several transformations. We looked at the now model transformation then view transformation now canonical transformation and then vo transformations right and we specifically look at uh know discuss more in detail about the canonical or projection transform which transforms the first term of the cone first term of the the camera into a regular cube that is a canonical volume right and we reasoned out why we needed this um near plane and far plane. We looked at the the best practices. We looked at the depth and uh some of the UX design content placement as well. We also reason a little bit of on the physiology point of view and a psychology point of view. Also we uh uh re reminded ourself or correlated ourself this concept of horopter and now panom fusion area uh uh and place the objects in the sweet area or sweet spot right there are several other computations of the rendering pipeline we looked at it right so in today class we are going to take forward of this basic graphs graphic uh uh pipeline to to know uh to present day and we are going to see several stages in which it has progressed. Right. The graphics pipeline which we saw in the last class is actually a fixed pipeline. The name pipeline means it is fixed. You input something and we can take something output. We input the 3D model at the application side. The output is the pixels at the at the display. In between we cannot now make any modifications, right? It's all hardware implemented. That is why we call it as a fixed function pipeline. Right? And there were uh now APIs were uh written. This was in 1991 to 2000. Now this is the time now we all had a uh now good time with SGI mach silicon graphics machines which are very very costly machines right that time was the time at which this Microsoft window was also now coming up I still remember how you know we had a very nice time with uh know silicon graphics machine and this time at around 2000 this silicon graphics has no made standards of you know Open standards it is called the openg that was the API 1.0 zero. Um where uh know the couple of functions you can call and uh know you can uh know draw triangle or rasterization any of the things whatever the functions we the graphics pipeline we talked about it can be done using this no openg APIs so there's a software interface right it can be done at the hardware but also software also can be done this is a standard industry standard and therefore it's a consortium of industries together they have come up with this uh standard called openg but uh Microsoft also has defined its own standard it wanted to know develop uh it want standard it didn't want to start follow the openg standard although openg is the industrial standard right so uh in this one you can see that um uh the the green one it's all uh no software and the the blue one it's all hardware functions right and then 2002 1 to 2005 the whole pipeline become programmable it is not fixed anymore right so what do we mean by programmable the hardware what we are talking about there the graphics hardware Right, GPUs, graphics processing unit, whatever we talking about, right? That has become programmable. The programmable means we can load a very small program called a shader program, right? It's a small uh snippet of the program and we can compile it and then load it in the hardware the GPU hardware and then it can do no lot more functions now that is customized to know particular interest

or application right for example movie industries they were talking about now giving a very finer shading of colors or you know the geometry right both of them they have used the The movie industry started using this called now vertex shader or pixel shader. Vertx shader what does it does? It does take a vertex as an input and then and apply the position transformation and then apply the texture and the colors etc and then outputs the the shaded vertex. Okay. Similarly pixel shader what does it do? It takes a pixel and then color information and depth or alpha and then and outputs a shaded no pixel right. There are several other shaders were added later but in 2001 to 2005 these were the now first two shaders added it it itself has now transformed the pipeline not so much right here is again some example of how vertex and pixel shaders were now useful for example here input at application level we are giving set of vertices and then It's all in the second stage it's all transformed into the world coordinate and then the third stage is all made into triangle and then and it's all rasterized rasterized and then the pixel shader it added you know colors to each of those pixels that's a shader right so vertx shader it's all colored and then transform and then and this is a shape assembly and then rasterization ation and then pixel assembly. Okay. So um uh this uh vertx and pixel shading it was program programmable. It is now shown here. It is the in the hardware called the now vert.x shading and this is a the pixel shader also is called the fragment shader. The blue color hardware which is you know fixed. It cannot be changed. Whereas the the pink color it's all can be no programmable using the sheddder language. Okay. And by then now OpenGL 1.4 has come. So OpenGL it's again uh right now we are talking about 4.6 six from 1 in 1990s or 2000 to no uh around know we are talking about 35 years now it is at the 4.6 six level. This gives you a tremendous flexibility. Right now it has much more flexibility. Although all of you are now using the Unity, Unity itself is using this OpenGL. OpenG itself has a 3D computer game engines uh no AP are written over here. It does most of the know the work which Unity is actually you know needing. Yeah. So it is not visible directly to you but you know Unity is using all the functions OpenGL which is necessary for knowh doing the game engine for Unity right so there are some advanced Unity developers who can directly access the the OpenGL functions and then improve the the the delivery of the the pixels. If you are interested you can now go through the OpenGL web page and then make it useful. Now let us look at the another 10 years. Next 10 years it was again programmable pipeline. Um but now it is the unified uh shader. So know the developers had difficulty in knowh using the vertex shader separately and then pixel shaders separately. They the next five years saw the unified shaders. Again you use this small snippet of the program. Now uh the same program can use can can be used as a vertx shader as well as the the pixel shader. So sometimes pixel shader will be more used more sometimes vertx shader may be used more but you know the unified shaders gives you much more flexibility and all those things. Right now open shell 3 has been open shield 3 yeah 3 has been developed and directx is has seen the 10th version and the next five years the tessellation and compute shaders were added. There were more or components of the hardware the GPUs are made programmable. For example, tessellate Jad shader. It takes a co geometry on the distance from the camera and then know outputs the outputs the what do you call the final geometry. So the input is going to be the coarse geometry depending upon the distance from the camera you wanted to see whether you wanted a no finer geometry or the coarse geometry. If the the if the triangle is very very far away, we don't need to divide into finite geometry at all. Right? If it is very close, then we need to divide into the finite geometry. All this decisions is done in the teslate shader. Okay? So the compute shader is another uh hardware or programmable hardware which does this physics engineering non-graphics related behavior. So if you want to know uh change the the the the movement of the 3D objects in a particular no physics right so all those things can be programmed in the the computer shader so all the physics based simulation uses the the computer shader the computer shader is going to be in this one this is an additional hardware added to the graphics pipeline again And if you look at it only the the rasterize component alone is the the fixed one right and all others had made were made into the programmable right that is between 2011 to 2005 now we made into no openg4 and direct access is is 11 right and then 2016 to 2020 the more components were are um u added uh and the API itself has been uh removed. The API now has become Wulcan. The OpenG, OpenGL API now is called the Wulcan. And this time there was another API also another API called Metal has come. Metal is from Apple. So um Apple programs all this graphics engine and uh using this uh metal and similar industries industrial standards have now evolved. Now we can see that there is no you know demarcation of API compared to the earlier one for for example this is where the the API starts below this we'll see application layer right after this it's all hardware correct this was the demarcation now in the uh 2016 to 2020 that demarcation has been removed it can make into completely now hardware Right. That was the the idea completely know programmable API level which uh you can convert anything into you know either uh programmable or application level or the the hardware but still you can see that this rasterization is still a fixed functions. This is called the generalpurpose GPU. Yeah, there were new APIs added for the general purpose GPUs. Now 2021 to present that is the last five years we are talking about. Now the general purpose GPUs now they uh they have uh added little more component to it. For example, um where is it? There is something called no direct storage has been added again at the GPU level. And then there is also no RTX. This is the you know rate tracing has been added and then there is the AI has also been added. The latest GPUs when you talk about it it's all all added over there. So what is now direct storage? This uh this is a small what do you call memory in the graphics hardware that is programmable that is accessible for you as a developer. Why do we need it? uh if you keep uh data shuttling between the CPU to GPU that takes a lot of time right and in order to avoid

that know you know transition no they have given you small storage in the GPU itself. So you can read the data from the GPU itself and then does some programming or some changes and then write it back to this no storage itself. Correct. that kind of know flexibility the the latest hardware has given correct and then now the AI the last five years there's a lot of AI component has come and that has revolutionized the the graphics engine itself correct so particularly I'm going to uh just mention about the Nvidia's as DLSS deep learning super uh Super sampling. It's a super resolution. No, super sampling. Super sampling. Right. So, um uh you can talk about now your uh the the lower resolution to higher resolution how deep learning method can be uh used. uh you don't have to generate a high resolution pixel. You can generate a low resolution pixel and therefore you spend lesser effort in generating one image but use the AI to generate the super resolution and image or super sampling. Correct. So uh a higher resolution image can be done by the AI. That is what the Nvidia has done. Similar things is about the DAA. It is a deep learning based anti-aliasing. I'm not going to go into the details of it. I'm going to give you the link and uh since you're all familiar with the AI, I want you to to know learn about this DLSS and DA as a reading assignment and then get more details. We have lot of other things to talk about now like Nvidia doing the uh anti-aliasing and the super resolution. AMD also has now similar thing called there fidelity super resolution and then gaming super resolutions. All those things again based on the AI I which has made the the the recent graphics engine now super fast. Right. I will not talk about more than uh this about the AI. I want you to read it as a now homework. We will talk in some details about the this RTX, right? RTX is the real time graphics engines. So some of think Nvidia's latest hardware are all you must have seen RTX series. Arctic series is about this u rate tracing and um rate tracing series but I I also want you to know look at the specifications of the GPUs what you have what are the different components and of this latest graphics um pipeline and your hardware as it and how much of is each memory for example direct storage how much memory it has RTX know What are the different components of the optics? It is uh it is supporting AI different what are the different AI algorithms it is supporting you just looking at the specification you should be able to find out more details of it. Now let us look at rate tracing a little more closely because that is an important topic advanced uh uh graphics rendering right. So uh in order to improve the the the the quality of the image the graphics rendering pipeline is actually now um u outputting right. People have used uh rate tracing technique. This technique uh know is very old technique. It starts with the 19 you know 60s onwards. It's about now 60 years we have been using this techniques but only recently it has come to the the hardware stage particularly Nvidia RTX series has made into hardware in the last now uh five years right so one small trick has made uh this rate tracing possible even in 1960s earlier if you look at it uh the real world what is happening it is very very difficult to implement in the graphics but when very small trick has made it possible in the computer graphics. So what happens is now uh if there is a light over here and then there's a lot of light coming from know the light sources and it might fall on objects and then it reaches the the the eyes right and it may be reflecting and it may be no uh it may not be reaching the eyes as well. there be millions of the race we may have to trace it which is which is computationally so expensive right um so there is somebody called Vitner we are going to see he made an observ observation in which he saw that there is a forward rate tracing which is happening in the the real world right this is what is happening in the real world in order to make into graphics we came up with called now backward rate tracing. What is backward rate tracing? In the forward rate tracing the the incident ray is coming from the light source and then the reflected ray is reaching the eye. There may be many objects and it might reflect some rays which is not reaching the eye at all. may go off of the uh light. But in the real world, no all there are rays are there which may we are we are not receiving in our eyes. But in the backward rate tracing we are starting with the I and uh now we are finding out what are the now uh what are the ways which is emanating from the eyes reing reaching the the objects as well as the light source. This is called the now backward rate tracing right and this is what made rate tracing possible in the graphics and there are a lot of improvements has been made from this particular rate tracing. [Music][Music] Let us look at what are the increments. So 1980s right witted ray tracing algorithm somebody called turner witted or apple he has created what he has done is that he used the backward rate tracing plus rays to the light not only from eyes to the object but also the rays to the the light also he has rendered. So his algorithm is very pretty simple. For each pixel he trace the I ray in the direction. Now V we are talking about. This is a pixel. I connect the eyes to the pixel and that gives you a ray. I extend this ray until it hits an object. That is the the V the first visible surface or the ray casting. This is called the ray casting. For each intersection trace the secondary rays. From here there are secondary rays possible. There is a shadow ray in the direction L to the light sources. For example here uh there is a shadow ray in the direction of the light and all this objects uh from for this uh light this kind of areas is the shadow. Correct. And then reflected ray in the direction of the R. This is the direction of the R. and then refracted ray ray or transmitted ray in the direction of t and this is the transmitted ray. These are the three ways we have to follow for each of the ray. And this is what we have done in 1990 and it was the big success as now 50 years ago uh 20 uh 45 years ago right and yes so here in this diagram it is made into very simple. So we have this A ray coming out here it is reaching the I the A dash whereas this ray B which is in the shadow because it is not reaching the the the uh light. Therefore this is falling in the the shadow region and that is of the shadow rays and reflected rays and refracted rays are all all know rendered. You can see that even the object that are are are no reflecting or refracting that can be rendered using the rate tracing. Right? And then Nvidia recently which has now further speed written algorithms to speed up the things they have come up

with something called the bounding volume hierarchy. Right? So bounding volume hierarchy in the sense now you cast array and then check whether the the bounding box of all the objects is hit or not. If it is not hit we don't have to follow we don't have to follow the the the array at all. If it is hitting then again look at another smaller bounding volume hierarchal bounding volume. For example this is this is called the C volume. This is called the B volume and this is another volume. So as a hierarchy we can keep dividing it and then find out whether there is a collision traction. So initially we start with A. If there is a collision and with A we can check whether B and C is colliding. If it is colliding then we can check out the following two branches of either of the objects. we can find out which object is actually colliding. when we find out the object really colliding then and know we can find out the the shadow and reflection and retraction finally otherwise now if it is not colliding at all we don't have to now take it forward right this way the entire rate tracing has become now much much faster and then we also use the den noising filter at the end once the image is formed uh we can use the den noising filtering to make it uh make the image much more appealing and also right now the latest RTX cards they use both rasterization asless rateracing rasterization is the the common or the general uh rendering pipeline what we have seen in the last uh now 40 years or so right the rate tracing is the alternative to RA rate tracing you know right now know Nvidia supports a hybridization of both the the rasterization and rate tracing rendering. Correct. Now this rate tracing let us look at uh uh let us add little more reality to it. In the rate tracing we assume that the object on which the light is falling that property is actually known. Right? But um no can we have more or details of the surface on which the light is falling that is called the the birectional reflectance distribution function for each of the surface in rate tracing this BRDF is actually defined. What is BRDF? BRD is a now function of four or variables. It defines a ratio between the reflected to the incident way. So why why it is a fourth? Because uh uh if you look at this um from the eyes, this is a normal we are talking about. This is the incident light. The incident omega I omega I since it's a solid angle we have two variables and the reflected wave also is in the now in the 3D position right is a 3D angle. Therefore it is the another two variables theta and gamma sorry theta and s are whatever. So two variable over here two variable over here. Therefore, this BRD BRDF is a function of four variables and uh it is actually a ratio between the the reflected and light intensity to uh incident light intensity and and and this is ratio is actually tells you about in the same direction of the no incident light how much of the the the the reflected light is actually now reflected right so since it is mapped onto this one we get a now cos theta term which we'll see uh later so basically if this is any surface any good surfaces in the all the the latest movies where for each frame we are spending about half an hour to generate it all the movie makers they spend lot of time in defining the BRDF to get a very good quality of the the surfaces right so so for example here there's a grass over here at this point we define the BRDF this BRDF for different directions it may be different reflections and let us say u uh this is where we are looking at it this is actual surface this can be now idealized into a lambian surface means the reflection is uniform in all the directions whereas here in the actual surface the reflection may not be uniform. will right. So uh uh this function what we are talking about is is is actually comparing the reflection of the real surface is to the lambian surface. How much it it is related to the ideal lambian surface and this BRDF properties make sure that two important properties one reciprocity. reciprocity in the sense suppose if you now if you reverse the the light and the eyes the the BF properties are the same the the ratio is actually the same so instead of ice put the light and instead of light put the eyes this reflection and the ratio is actually the same there is a reciprocity and then conservation of the the uh energy right the reflected and energy in is is is the same as the incident energy. There is no the in no way the reflected energy can be no more than the incident energy. That is the conservation of energy. Right? So this is what is shown over here. The BD of incident wave and the outgoing wave if it is uh uh if it is uh inverted this is going to be the same. It is equal right. This is equal. Whereas the uh this conservation of energy the incident energy is the same of the the sum of all the deflected energy and this is what is mentioned is that now integral vdf over all the uh outgoing and is actually less than or is equal to one. Yeah, there are a lot of BRDF variants are there and there is something called a specially varying birectional reflectance distribution function which is nothing but a six of six dimensional function instead of four dimension this is actually varying over a 2D area therefore this extra two-dimensional know two dimension the function is coming into picture right and then there is something called a birectional texture function texture function is Again it is modeling for the non-flat surfaces and the and has the same parameterization of the SVB or D right and then birectional surface scattering reflection distribution. This is a eight dimensional function which includes um uh which is includes the what do you call um the position as well to not only the 2D uh function but also uh birectional it is not this uh the light entering the surface may scatter internally and then exit in another location. It need not be no uh exiting at the same point. It may be exiting another point. Therefore, no two more dimensions are added. Okay, these are there are several other more BRDFs or variants are there. I I'll give you a small reading material on this if you are interested. And then there are many BRDF databases are there. All this BRDF research has started with 1970s or 19 no 60s right and still people are trying to find out the BDF bases there is a MIT material database is there that's a Mitsubishci database is there bond and database is there and several other databases are there so how BRDF of material properties are measured and then model is again is a big research and there are several instruments have been developed which now I will not go into the details right and this BRDF material is not only useful for rendering in the movies and other things but also they have know several other uses modeling the light trapping in the solar uses so if you know the BDF of

solar cells we can model how much light is getting trapped there also now NASA uses BRDF uh to characterize surface reflections anotro anisotrophy. For example, all these moon surfaces or Mars surfaces, they can actually, you know, uh model using the the BRF. They measure the the the moon surfaces or Mars surfaces using the BRDF and the that data is actually available in the NASA website. You can take take a look at it how this all measured. There is a big history of N last 60 years that is available in this uh link and I'm going to share it with you and then you can take a look at it how it is going to be useful right so there are several models of the BRDF BRDF is a it's a measured from the real surfaces but there are also modeled after the the real data yeah lambarian model is a simple constant BRDF you have seen it and there are lunar and Martian reflection models H half K scattering model which is porous irregular and particulate surfaces all the maybe uh u astronomical surfaces right pong reflections model like plastic like specularity there are overview of the B of metals also available these are the developments as far as the rate tracing is concerned right we are going to now further develop this PRDF in the u in the later classes when we talk about the the light field rendering right so one of my colleagues Mr. Koshik Mitra is going to talk about the light field rendering in the one of the upcoming classes later. [Music][Music] So far we have looked at the the common graphics engine how it can be know what is the latest advancement and all the things right that again common graphics engine is common for know movie making game engines. But in a virtual reality, the the the requirements are much much more stringent, right? So the most challenging, you know, all the requirements is the virtual reality requirements because we have this 90 frames per second, right? Requirement. So we have only 11.11 now milliseconds requirements within this we need to render all those things, right? This is a very stringent requirement. There is a lot of of optimization has to be done has to be done. So given all the components of the graphics engine which one we can tweak that is a question we are going to answer in the next few slides right first one is meta has come up with a concept called time warping so it has come up in probably some some 8 years ago right so what is time warping we will see so uh essentially in a virtual reality not only this FPS requirement is more stringent because uh uh the the headset is actually keep changing and according to the headset we have to keep changing the keep rendering the know different images right so the we have the HMD tracker and then this HMD tracker it is going to input position is input to the application this application is going to use the rendering this is where the the visual rendering which coming and the rendering engine is going to instead of directly giving to the display it is going to give it to the time warping right. So uh if it is directly giving to the display there is a typical typical now cycle right but what meta has done is it is given to a time warping and then time warping it is warped the frames and then give it to the display. What do we mean by warping the plane? Basically depending upon by the time it rendered what happens is is the position let us say uh the HMD is sampled at this point of time time let us say it is some position now P1 but by the time we render it takes about let us say 11 seconds 11.1 milliseconds right when it comes over here the position would have changed, right? Or so what we needed to do is we needed to know we needed to display what is the the current position what is the the frame as far as the current position is concerned not 11.11 seconds milliseconds before right it is going to be slightly awkward right so what meta has done is using this time warping it called the no asynchronous time warping it warps the the image page for the current headset position. Okay. So, right now sample it and according to the current sampling you w the the rendered image and then give it to the the display that is the the the time warping right. So what it does it's it uses the not only the GPU but also the GPU but also the DSP yeah digital signal processing hardware in the you know graphics engines also uses all right and when the the first frame is developed it uses the CPU GPU and some part of the DSP as well. So when the next frame comes we don't have to use the uh uh synchronize and we know sample the HMD and then use the the rendered frame and now warp it to know suit the the current HMD process. This is called the uh time warping. there are more or uh uh there are more uh uh details I'm going to give it in the recent papers and then u and then reading materials. So let me give you a little more uh uh details of how this is done. This is how this modern virtual reality you know uh visual rendering is happening right. So suppose if there is a particles vertices or triangles over here and then we have this shadow names or rasterization can be done and then the display is at the 90 Hz and the head head is tracking over here the head tracking is given to the time warping and then that's given to the rasterization and then rasterization again and gives backs to the time warping and the time warping is all given to the display right then next uh optimization can be done is that in the fixed or programmable graphics pipeline the entire resolution is the same can we know change the resolution depending upon where we are looking at it so nowadays all this headset comes with the no eye tracking so we know where we are actually looking at it depending depending upon where we are looking at it. We know since we have learned the the eye physiology since we have looked at the I neurohysics neuros right we know that wherever our phobia is focused that uh resolution needed is more and where the peripheral vision is there no our uh requirement for resolution is very less. So if we know where actually we are looking at it, we can increase the resolution there and the other places we can actually reduce it. Right? So that is called the no fiated rendering. So eyetracking enables this fiated rendering possible. Stereo reprojection is the third technique. So for virtual reality we have to now develop two different frames for one for the left eye another for the right eye. Right? If you develop one full frame for one eye can I uh repro for the next eye from the uh the left from the the other eye. What is going to happen is that there's going to be slight difference because IPD we know what is the IPD and uh uh we can use this IPD and do some reprojection to get the the entire know uh image for the other eye. Of course there's going

to be some challenge right? Suppose if the the red one is the left eye and the blue is the the uh the right eye the object what we are looking at it is going to be shadowed by the left eye. Right? For example, this point is going to be shadowed by the left eye. If we generate the image for the left eye and if you do the reprojection and and we may not get the the uh the this point for the right eye. Therefore we need to do little bit of uh now fixing that uh that is why in this repro uh stereo reproedic image you can see all this now green shaded area it's all no gaps are there this needs to be filled up because this gaps are all all no shadowed area as far as the left eye image is concerned. So if you can fill up this one then uh the right image we can get it with less cost or with less time right. So using the information from the depth buffer that is the whole idea I can find out how much is the depth of this east one right using the depth buffer can we reproect the the first eye is rendering result in the second eye you know frame buffer there's going to be some misalignments but now this may not be noticeable for you know each eye this is called the stereo reprojection right and then there is a hybrid monos oscopic rendering this fourth uh trick we are all going to use it for virtual reality right so uh stereo and monoscopic some places we need stereo some places we don't need the stereo where do we need we don't need the stereo the when the object is very very far away we don't need the stereo so if you look at all this know small small walls which is very very far away they don't need the stereo at all those places we'll do the only the mono whereas the object which is very close it needs the it needs the the stereo right there we can put the uh the stereo and this is called the monoscopic stereo and then cloud rendering we may not have to have any of this hardware GPUs in our headset at all right in our uh no mobile phone at all. Everything can be rendered in a know remote uh GPU and then it can all all can all shift shifted or communicated through 5G or know 6G in the future to the the current hardware right so typical multiplayer games they use this kind of engine for example at the open J level they communicated and this is the local device this is the remote device. The lo uh remote device communicates the OpenGL level to this local device and then OpenGL at this local device which know renders locally. This is the typical multiplayer games. But in the cloud rendering VR rendering what we can do is at the frame buffer level itself we can render it. All the know computation is happening rendered rendering is happening at the the remote device and and the rendered buffer they can take it and then communicate to the the frame buffer at the local device. There are several rendering cloud rendering options are a available now there is know Nvidia and Amazon and there are all the cloud service providers they're all competing for the cloud rendering right so um I don't want to go into the details but those of you who are interested know search for the cloud XR or by the the cloud for the by the you know Nvidia and similar Amazon on cloud rendering options as well. Right? So given this now several uh no tricks and no improvements for virtual reality. So what is going to happen in the future rendering V we VR rendering right? Can we project it into five years down the line right? So instead of having all the components separately it can all be combined together and therefore instead of 11 milliseconds it can be just 1 millisecond we can render it in the feature systems right so uh with the with the time warping and the lens distortion doising fiated rendering and uh no stereo reprojections all these things can be combined in the feature VR system as of now there is no such VR system at all but uh it is likely to happen in the maybe couple of years, right? I also wanted to think about what is that you can do for this feature virtual reality systems. What is that this the the industry has not looked at it? Now having gone through the details of this course, this course virtual reality course we looked at the nuts and bolts of the entire VR systems. Now you should be in a better position to know come up with the feature rendering systems and improvement in the feature uh rendering systems. I'm going to give a very short uh introduction to what I mean by uh what you can do as a part of this course. Right? I'm not going to directly tell you. I'm going to indirectly hint you know by explaining something called the the gamma encoding. I'm going to ask you this question. You all are using this mobile phone for know we are rendering. You all has been using this camera for taking the pictures in the mobile phones. Have you seen have you noticed have you observed that the the the photo what you have seen on your mobile is far different from the real the real scene. Your eyes see that the the real scene and but know when it is recorded in the uh camera mobile camera it is all different. Have you observed it? Can we reason it out? Why? And that reasoning can you come up with an idea to know design your feature virtual reality system that is what I'm hinting at. I'm going to introduce a concept called gamma encoding. This is just an example. I'm not asking you to know telling you that this is the uh uh this idea which you have to implement it but it is just an idea that it might it would um hint you uh to design the future virtual reality systems and here is the reason why the camera image is far different from what we see right so u uh uh in uh this This is where the perception engineering comes into picture. So uh professor Anna has introduced the concept of the perception and visual perception in the earlier classes. I'm going to introduce or emphasize in the coming classes as well. Perception is a single most concept I would like you to uh take as a take-h home message from this course. Right? So particularly visual perception is uh is different for the eyes and the cameras and that makes the difference. I'm going to tell you why. Yes. So suppose if you have this input x-axis as a actual luminance that's a light intensity we are talking about right and then a predicted light that's a perceived light in the y-axis right in the camera that is going to be a linear relation the incident light and the trajectory light that is the linear relation but not in the eyes The eyes is going to have a slightly curved one and this makes the difference. So what you can see that at a low very low luminance very low luminance very low intensity we have a now we have a very sharp rise it means what for a small increase the the detected percentage is increasing a lot. It means what? Our eyes are

very good in the low light. We can see a lot of variations in the low light. We cannot see lot of variations in the highlight. Now high illumination. And that is what this graph is saying. And that exactly is what making the the camera and the eyes see differently. Right? And this has been observed and this has been engineered in many different ways. In fact, you know, all your JPEG and M impact images actually encode using this. The eyes are much more sensitive to darker tones. That is because of uh the sharp right the lower illumination. This increases the range of the illumination luminance of our eyes. Again to make it much more uh much more I'm going to give an example. This is called the gamma. So v out is equal to v in power gamma. This that this curve can be modeled as a small power function. This power function is called the gamma. Right? So if gamma is equal to one that is the linear relation for the camera. If gamma is less than one it is called the now arching upward like this. If gamma is greater than one, it is arching downward like this. Right? So the gamma u is a very important factor and in all our images this gamma is used to encode jpeg images is actually no encoding the uh encoding with the gamma. Why do we need to encode with gamma? Here is the relation. Suppose if we have a virgin image the shading is changing from continuously from very low to very high like this right suppose and we have a linear encoding like in the camera five bit means it is 32 level right to power 5 32 level and it is linear encoded so the whole range is divided into 32 now small small now segments you can see that now at the very low um uh okay there is you know certain variations we can see the same thing if you are encoded with a gamma same 5 bit level 32 level you can see that at the lower darker regions we can find more variations right so uh this gamma encoding lets you have more variations in the darker region and that is the whole idea. If you encode the JPEG images with a gamma, we have no uh we have more ranges in the darker regions. This is evident in the images uh evident in the images like this. Right? So this is the linear encoded and this is a gamma encoded. Can you see the difference between these two things? And if you look at all the image processing Adobe or Blender and all this all there the options also may be there but you may not understand the you may not have know noticed it or gone into the details of why gamma is actually necessary. It is coming from the the human vision point of view right and how is it happening in the display. Now we have to do the gamma correction when you project or when you display in your mobile phone or in your know TV or in your projection screen. Gamma encoding is for the recording images but gamma correction is for the displays right. So image gamma is is the upward for example this are raw image to now JPEG. We will do the image gamma for the display gamma in the video graphic card. We have to compensate the image gamma. So we will do the opposite of it. So the combining this image gamma and uh display gamma we will have the system gamma which is actually linear. Correct. So ideal gamma is actually one. So this is how the perception is actually engineered in all our uh displays. Correct. I would like you to know think about how powerful this vision as visual perception and is engineered. Correct? And uh your different displays have different gamas actually. Right? So you can see how the uh display gamma can be changed. You can actually uh set different gamma uh uh uh in a different displays or in your cell phone basically uh there is by default there is a gamma you can change it to different gamma and changing the gamma will let you see the images differently that is because so this is the image gamma let us say by changing the now uh the gamma this is actually 1.8 1.2 two now four and you can see how for the same image gamma how changing the display gamma varies the images right so earlier all the CRT BL based displays had a very high gamma but nowadays TFTs have no nearly no one as a native gamma so uh understanding this visual perception is very important to come up the the featured virtual reality systems. As of now, the the VR rendering systems have not exploited the perception. I would like you to now imagine how this perception based improvement of the visual rendering is possible. Some of you can take like this as a project come up with ideas. We'll be very happy. So both uh know professor Steve Laval and professor Anna and myself we all be very happy to know discuss with you this futuristic uh visual rendering pipeline for VR systems and this is where the future is. This can also be made into now very high impact now papers as well. [Music] lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

VR Rendering – Easy Notes for a 7th Class Student

Your lecture covers **five main topics**:

1. How graphics engines developed over time.
2. How ray tracing creates realistic images.
3. How VR makes rendering faster using special techniques.
4. How our eyes see brightness differently from cameras.
5. How gamma encoding and gamma correction improve images.

Let's understand each topic with simple examples, memory tricks, and MCQs.

1. What is a graphics engine?

Imagine you are playing a 3D game. You see a car, trees, roads, shadows and sunlight on your screen.

The **graphics engine** is the system that takes information about these objects and turns it into the picture you see.

The basic rendering pipeline has three main stages:

1. Application stage
Decides what objects, lights and materials are in the scene.

↓

2. Geometry stage
Positions objects and prepares their shapes for viewing.

↓

3. Rasterization stage
Turns shapes into pixels to form the image.

A **pixel** is one tiny coloured dot on your screen. Millions of pixels together form an image.

1.1 How did graphics engines improve over time?

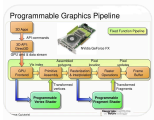
In the lecture, the professor explains how more parts of the graphics pipeline became programmable.

Think of an old calculator versus a modern computer. An old calculator performs a limited set of operations, while a computer can run many different programs.

The graphics pipeline evolved in a similar way.

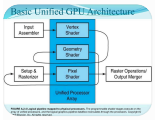
Stage 1: Fixed-function pipeline (roughly 1991–2000)

The hardware followed mostly predefined operations. Developers had limited control over how individual stages worked.



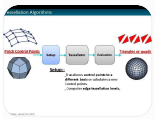
Stage 2: Programmable shaders (roughly 2001–2005)

Developers could write small programs called shaders to control vertex processing and pixel colours.



Stage 3: Unified shaders

More flexible hardware could use similar shader resources for different graphics tasks.



Stage 4: Tessellation and compute shaders

Hardware became more programmable, supporting finer geometry and general-purpose calculations.

Stage 5: Modern GPUs

Newer GPUs include hardware and features for ray tracing, AI-based image enhancement and faster data handling.

These periods are approximate, as the lecture presents them as a broad historical timeline.

Important terms to remember

Term	Easy meaning
API	A set of functions that lets software communicate with graphics hardware.
GPU	The processor designed to handle graphics and many parallel calculations.
Vertex shader	Processes vertices, which are points that define 3D shapes.

Pixel/fragment shader	Calculates the colour and other properties of image fragments.
Tessellation shader	Helps subdivide coarse geometry into finer geometry.
Compute shader	Runs general-purpose calculations on the GPU, such as parts of a simulation.

OpenGL, DirectX, Vulkan and Metal are graphics APIs or graphics-related programming interfaces. Unity can use graphics APIs to communicate with the graphics hardware.

Memory trick: Fixed → Programmable → Unified → More advanced GPU features.

2. What is ray tracing?

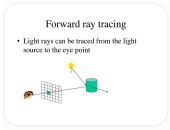
Imagine shining a torch at a mirror. The light hits the mirror and reflects into your eyes.

Ray tracing is a computer graphics technique that follows the paths of simulated light rays to create realistic images.

It can help produce:

- Shadows
- Reflections in mirrors and shiny objects
- Light passing through transparent materials
- More realistic lighting

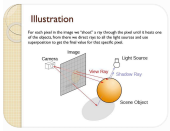
2.1 Forward ray tracing vs backward ray tracing



Forward ray tracing

Start at the light source and follow light rays toward objects and the environment.

Problem: many rays never reach the camera, so calculations can be wasted.



Backward ray tracing

Start from the camera or eye and trace rays through image pixels into the scene.

This focuses calculations on the objects that can contribute to the image.

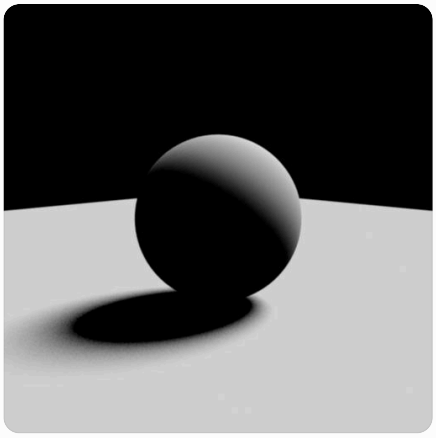
Memory trick: Forward = light to scene; backward = eye to scene.

2.2 What happens during ray tracing?

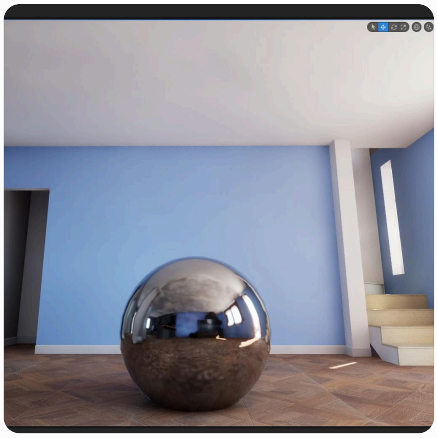
For each image pixel, the system can trace a ray from the camera into the scene.

When that ray hits an object, it may trace additional rays:

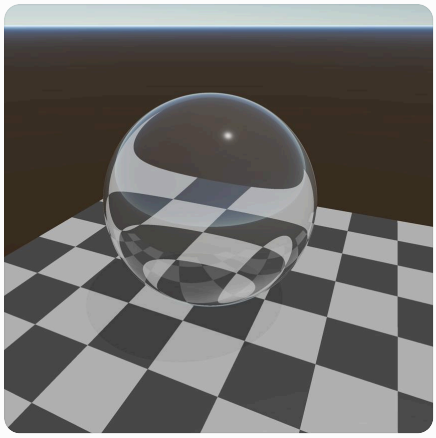
Camera ray hits an object



Shadow ray
Checks whether light reaches the object.



Reflection ray
Helps show reflected objects.



Refraction ray
Models light bending through materials.

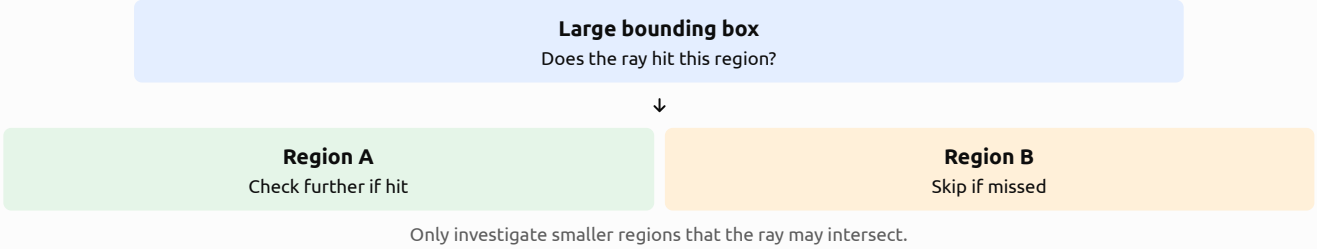
These extra rays help calculate the final appearance of the surface.

2.3 What is a Bounding Volume Hierarchy (BVH)?

Imagine searching for one book in a huge library.

You would not inspect every book one by one. You would first check the correct section, then the shelf, then the book.

A BVH works similarly.



A **bounding volume** is a simple box or other enclosing shape around an object or group of objects.

A BVH speeds up ray tracing by avoiding unnecessary intersection checks.

2.4 What is denoising?

When only a limited number of rays are traced, the rendered image may look grainy.

A denoising filter reduces that noise to make the image look cleaner.

Memory trick: BVH saves calculation time; denoising cleans the image.

Modern GPUs can also combine **rasterization and ray tracing**. This is called hybrid rendering: rasterization handles some tasks efficiently, while ray tracing adds realistic effects where useful.

3. What is BRDF?

BRDF stands for **Bidirectional Reflectance Distribution Function**.

The name looks difficult, but the idea is simple.

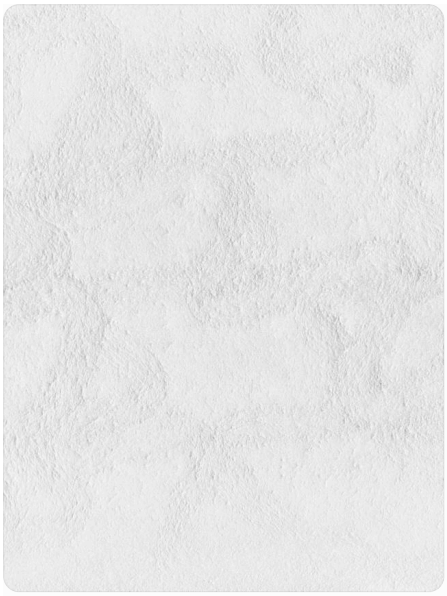
Imagine shining light on three things:

- A piece of chalk
- A mirror
- A shiny metal spoon

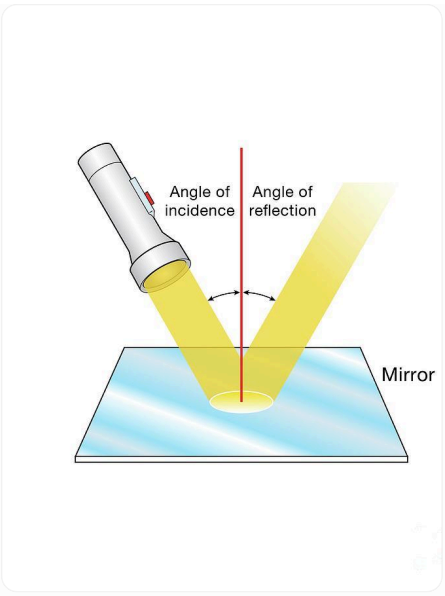
Do they reflect light in the same way? No!

- Chalk scatters light in many directions.
- A mirror reflects light in a particular direction.
- A metal spoon may produce bright highlights and reflections.

BRDF is a mathematical function used to describe how a surface reflects incoming light toward different outgoing directions.



Matte surface
Light spreads broadly.



Mirror
Light reflects directionally.

Shiny metal
Strong highlights and reflections.

3.1 Why does the lecture say BRDF has four variables?

To describe light reflection on a surface, we need two directions:

- Incoming light direction: two angular coordinates.
- Outgoing viewing direction: two angular coordinates.

So, $2 + 2 = 4$ angular variables.

The BRDF describes how much light is reflected in an outgoing direction for light arriving from an incoming direction.

You do not need to memorize all the mathematics first. Remember its purpose.

Memory trick: BRDF = how a material reflects light.

3.2 Two important BRDF properties

1. Reciprocity

For ordinary reciprocal materials, swapping the incoming and outgoing directions gives the same BRDF value, under the usual BRDF assumptions.

Easy way to remember: reverse the light and viewing directions; the reflection function remains the same.

2. Conservation of energy

A passive surface cannot reflect more total energy than it receives.

For example, if a surface receives 100 units of light energy, it cannot reflect 150 units of that same incoming energy without another energy source.

Memory trick: Reciprocity = swap directions; conservation = don't create extra light energy.

3.3 What is a Lambertian surface?

A Lambertian surface is an ideal diffuse surface. Its BRDF is constant with respect to incoming and outgoing directions.

Think of a perfectly matte surface: its reflected brightness is spread diffusely rather than concentrated in a mirror-like highlight.

The lecture also mentions more advanced material models:

Term	Easy meaning
Spatially varying BRDF (SVBRDF)	Reflection properties change from place to place across a surface.

Bidirectional texture function (BTF)	Describes appearance changes across surface positions and viewing/lighting directions, including complex surface appearance.
BSSRDF	Models light entering a material at one point, scattering inside it, and leaving at another point.

BRDF research is useful not only for movies and games, but also for studying planetary surfaces and how light behaves on solar-cell materials.

---

4. Why is VR rendering more difficult?

A normal movie can take a long time to render each frame before it is played.

A VR headset must render images quickly while tracking the user's changing head position.

At 90 frames per second:

$$\frac{1000\ \text{ms}}{90} \approx 11.11\ \text{ms/frame}$$

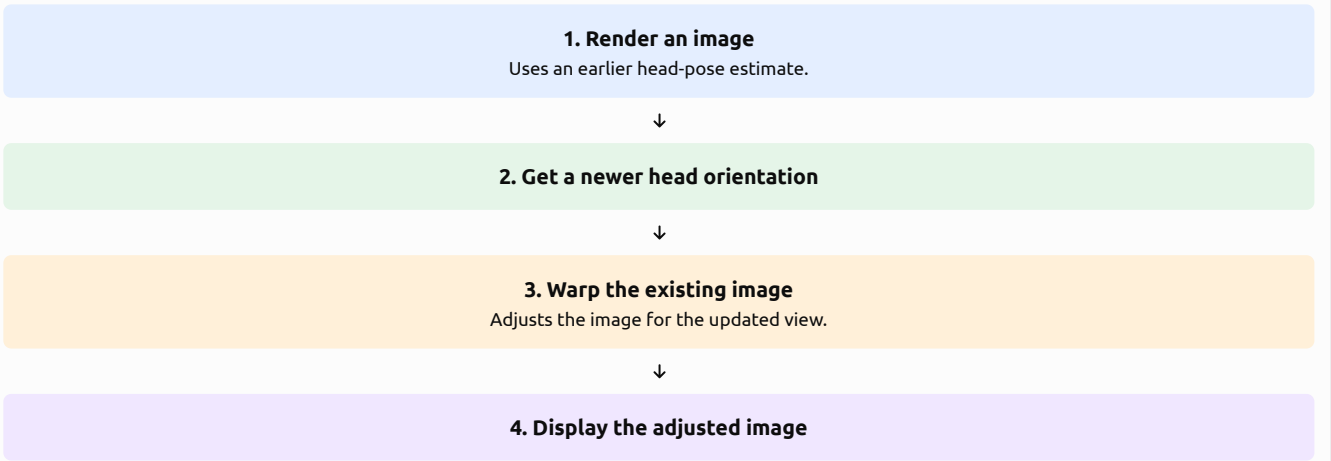
That means each frame has about **11.11 milliseconds** in the overall frame-time budget.

The lecture introduces several ways to reduce the work required.

4.1 Asynchronous timewarp

Imagine you draw a picture of the room. While you are drawing, you turn your head slightly.

The picture may now show the view from your old head position. Timewarp adjusts the already-rendered image to better match the newer head orientation before displaying it.



Why useful? It can reduce the apparent effect of head-motion latency without requiring the entire scene to be rendered again.

Important limitation: ordinary timewarp mainly corrects changes in viewing orientation. It cannot perfectly recreate all changes caused by translation, moving objects, or newly visible surfaces.

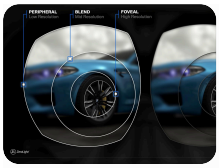
Memory trick: Timewarp = adjust the old frame for the newer view.

4.2 Foveated rendering

Our eyes have a small central region called the **fovea**, where visual detail is perceived most clearly.

Peripheral vision is less sensitive to fine detail.

Foveated rendering uses this difference to save processing time.



- Where you look: render more detail.
- Away from where you look: render less detail.
- Eye tracking helps the system identify where you are looking.

This can save GPU work while keeping the important part of the image sharp.

Memory trick: Foveated rendering = sharp where you look, less detail elsewhere.

4.3 Stereo reprojection

VR normally needs separate images for the left eye and right eye because our eyes see slightly different views.

Stereo reprojection attempts to use an image already rendered for one eye to help create the image for the other eye.

The system can use depth information to reposition image points.

However, some areas visible to one eye may be hidden from the other eye. These newly exposed areas are called **disocclusions**, and they may need to be filled or rendered separately.

Memory trick: Stereo reprojection = reuse one eye's image to help create the other eye's image.

4.4 Hybrid monoscopic and stereoscopic rendering

- **Monoscopic:** Uses the same view for both eyes, without separate eye-specific depth views.
- **Stereoscopic:** Uses different views for the left and right eyes to provide binocular depth cues.

A hybrid technique uses stereoscopic rendering where separate eye views matter and monoscopic rendering in selected regions where the loss of stereo detail is acceptable.

The lecture gives distant objects as an example where stereo information may be less important.

Memory trick: Mono = one view; stereo = two eye-specific views.

4.5 Cloud rendering

Imagine your headset is not powerful enough to render a complex VR world.

Instead, a powerful remote computer renders the scene and sends the resulting images to your device over a network.

That is cloud rendering.

Remote computer / cloud GPU
Performs most of the rendering work.

↓↑

Network
Transfers rendered frames and other required information.

↓↑

Local headset
Receives and displays the images.

Main advantage: The local device may need less rendering power.

Main challenge: Network delay, unstable connections and data-transfer time can make VR uncomfortable.

VR optimization revision table

Technique	Main purpose
Timewarp	Adjust a rendered frame for a newer view

Foveated rendering	Spend detail where the user looks
Stereo reprojection	Reuse one eye's rendered image to help create the other
Hybrid mono/stereo	Use different rendering methods in selected regions
Cloud rendering	Render on a remote GPU and send images to the headset

Memory trick: Timewarp, Foveated, Stereo, Hybrid, Cloud.

---

5. Gamma encoding: Why do camera images look different from what our eyes see?

This is one of the most important concepts in the lecture.

Imagine looking at a dark room with your own eyes. You can notice some details in the shadows. But a camera photograph of the same room may show those details differently.

Why? Human vision and cameras do not respond to brightness in exactly the same way.

5.1 Human eyes vs camera

According to the lecture:

- A camera's basic light measurement can be represented approximately as a linear response.
- Human vision responds nonlinearly to brightness.
- We can be very sensitive to brightness changes in darker regions, depending on viewing conditions.

Imagine a picture containing dark, medium and bright areas.

If the available digital values are divided equally across the entire brightness range, some dark details may not be represented as effectively as they could be.

Gamma encoding helps distribute digital values in a way that better suits human visual perception.

5.2 What is gamma?

The lecture introduces this simplified formula:

$$V_{\text{out}} = V_{\text{in}}^{\gamma}$$

Here:

- V_{in} = input brightness value, usually normalized between 0 and 1.
- V_{out} = encoded output value.
- γ (gamma) = controls the shape of the brightness curve.

There are three cases to remember:

γ = 1 — **Linear response**
The output follows the input directly. For example, an input of 0.5 produces an output of 0.5.

γ < 1 — **Curve bends upward**
For values between 0 and 1, the output is larger than the input. For example, $0.25^{0.5} = 0.5$.

γ > 1 — **Curve bends downward**
For values between 0 and 1, the output is smaller than the input. For example, $0.5^{2} = 0.25$.

These examples explain the mathematical curve, not every detail of real camera or display processing.

5.3 Why is gamma encoding useful?

Suppose you have only **5 bits** to store brightness.

Five bits can represent:

$$2^5 = 32 \text{ levels}$$

That means you have just 32 possible brightness values.

With linear encoding, these values are distributed evenly across the input brightness range.

With a suitable nonlinear encoding, more of the available levels can be devoted to darker tones, where our eyes may notice small changes more easily.

This can help reduce visible banding and preserve useful dark detail.

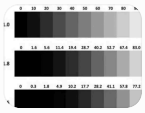
Important: Gamma encoding does not create additional bits or increase the total number of levels. It changes how those levels are allocated across brightness values.

5.4 Gamma encoding vs gamma correction

This distinction is important for exams.

Gamma encoding

Transforms image values into a nonlinear representation, helping use digital levels effectively for visual perception.



Gamma correction

Applies a compensating transformation so that the displayed image has the intended brightness and appearance.

A simplified processing chain is:

Scene / image brightness

↓

Gamma encoding

↓

Storage or transmission

↓

Decoding / display transformation

↓

Intended displayed brightness

Real image systems may use more complicated transfer functions, colour processing and tone mapping, but this is the basic idea.

Memory trick: Encoding prepares the values; correction compensates for the display or system response.

---

6. The biggest lesson: Perception-based VR rendering

Your professor's final message is that **future VR systems should take human perception into account**, instead of simply trying to render every pixel with the same effort.

You have now seen three examples:

- Foveated rendering uses differences between central and peripheral vision.
- Gamma encoding uses differences in how we perceive brightness.
- Timewarp uses newer tracking information to reduce the effect of latency.

All three take advantage of something about how people see or experience images.

A possible future research question is:

Can a VR system use human visual perception to decide where to spend rendering effort, how to represent brightness, and how to reduce latency?

That is the kind of idea the lecturer encourages students to explore for future projects.

---

7. Final revision sheet

Topic	One-line answer to remember
Graphics engine	Converts scene data into a displayed image.
Fixed pipeline	Uses mostly predefined processing stages.
Shader	A small program used to control graphics processing.
Ray tracing	Simulates light paths to produce realistic effects.
Backward ray tracing	Traces rays from the camera into the scene.
BVH	Organizes bounding volumes to reduce intersection checks.
Denoising	Reduces noise in a rendered image.
BRDF	Describes how a surface reflects incoming light into outgoing directions.
Reciprocity	Reversing incoming and outgoing directions preserves the BRDF under the usual assumptions.
Energy conservation	A passive surface cannot reflect more total energy than it receives.
Timewarp	Adjusts an existing frame for a newer viewing orientation.
Foveated rendering	Renders more detail where the user is looking.
Stereo reprojection	Reuses one eye's image to help generate the other eye's image.
Cloud rendering	Uses a remote GPU to render and sends images over a network.
Gamma encoding	Nonlinearly maps image values to make better use of available levels.
Gamma correction	Compensates for a system's brightness response.
Perception-based rendering	Uses how people see to improve image quality and efficiency.

---

8. MCQ practice test

Try answering these questions before checking the answers.

12 MCQs 0 of 12 answered

1. What is the main purpose of a graphics engine?

- A. To store passwords
- B. To convert scene data into images
- C. To measure heart rate
- D. To connect a keyboard

2. What is a shader?

- A. A type of headset strap
- B. A small program that controls graphics processing
- C. A camera lens

D. A network cable

3. Which technique traces rays from the camera into the scene?

A. Forward ray tracing

B. Backward ray tracing

C. Gamma encoding

D. Foveated rendering

4. What is the main purpose of a BVH?

A. Increase the number of pixels

B. Reduce unnecessary ray intersection checks

C. Make every object transparent

D. Correct display gamma

5. What does BRDF describe?

A. How a surface reflects light

B. How fast a headset battery charges

C. How sound travels

D. How a camera stores files

6. Which technique adjusts a rendered image for a newer head orientation?

A. Tessellation

B. Timewarp

C. Gamma encoding

D. Cloud storage

7. What does foveated rendering use to save processing effort?

A. The user's viewing direction

B. The headset's colour

C. The battery percentage

D. The size of the keyboard

8. Why can stereo reprojection require extra processing?

A. The other eye may reveal previously hidden regions

B. It removes all depth information

C. It disables the display

D. It turns pixels into sound

9. What is one challenge of cloud rendering for VR?

A. Network delay

B. Too many keyboards

C. The absence of pixels

D. Excessive paper usage

10. What does gamma encoding help achieve?

A. More efficient allocation of digital values across brightness

B. More physical pixels without new hardware

C. Higher network speed

D. More vertices in every object

11. What does gamma = 1 mean in the simplified power formula?

A. A linear mapping

B. A completely black image

C. A reflective surface

D. A ray intersection

12. What is the key difference between gamma encoding and gamma correction?

A. They are exactly the same operation in every system

B. Encoding transforms image values; correction compensates for system response

C. Encoding is for sound only

D. Correction creates extra bits

Check my answers

Last-minute memory line:

Shaders process → Ray tracing lights → BRDF reflects → VR techniques optimize → Gamma improves brightness representation → Perception guides the future.

You said:

[Music] Welcome back. There are few announcements before we start our class today. I hope all of you have completed the the homeworks um seven and eight. There are no pending homeworks so far. We are going to post the next homework in the next one or two days. So please um now uh be active in discord right? So we are going to continue our discussions most of the questions now in today's class and a couple of last classes as well in the discord. So please be active in discord and uh we have also sent you one feedback form for last classes. So I would request you to know fill up those forms and uh give a feedback on how our lectures can be improved right and next class onwards the zoom link which is going to be sent for NPEL classes is not going to be the same. So we the the the uh NP team wants us to now compressorally take this feedback form only then know you'll be given uh now the zoom link. So please fill up the forms so that the system can be improved. Right? We've been looking at the know visual rendering in the last two classes and today we will be completing the the visual rendering. We are going to now talk about know advanced topics in the visual rendering. In the last class itself we started the advanced topic but today we will complete it and particularly now uh we are going to talk about the last topics of uh no visual rendering. we have uh done and we have uh discussed the VR specific rendering techniques in the last class and uh most of the other things in the previous classes as well. So just to remain now we have looked at the now the transformations and in the in the the typical visual rendering and particularly model and uh now view matrix and know canonical matrix and view port matrix all combined together all at one and then we looked at the now the latest development as far as the this graphics rendering pipeline is concerned. It is completely programmable is what we have seen in the last class right there is no fixed pipeline like you know some 30 years ago or 40 years ago right and every stage in the does rendering is actually programmable right and there's a lot of component of AI coming to picture now then we have also looked at the rate tracing algorithms and we also looked at the the direct storage and uh now um and and the what do you call compute now shader as well right we spent quite some time on no rendering

pipeline and we looked at in some details of of now BRDF of birectional reflectance distribution function and uh we looked at the different models different um now the history of it how do we collect it how are the different uses of the BR RDF all those things we saw in the last class right. So in this class uh in the last class we also looked at the how this traditional graphics pipeline is actually know uh modified to suit the requirement for a VR engine. For example, meta time warping is one of the asynchronous time warping is one of the uh most important technique we have uh discussed that's about 10 years ago. then space warping and then fiated rendering and then stereo reprojection, hybrid monoscopic rendering, cloud rendering and we also looked at a little bit of peing into the now feature VR rendering techniques, right? And we looked at now very shortly how perception is going to play a very important role in the coming days in shaping the VR virtual reality or visual rendering techniques. Right? We particularly look at um now this uh uh uh this particular uh title called the gamma right gamma display right so how gamma display has been now phenomenal in shaping the the display part of it. Therefore, some such perception parameters can and shape the uh can and push the the uh the rendering visual rendering techniques to the new height in the coming days is what we looked at it right now. uh we look at we will look at some of the know uh very uh recent advanced techniques. In the recent advanced techniques, one of the phenomenal things are happening so far in the graphics pipeline, the traditional graphics pipeline in the last 30 years or so or 40 years or so. We started with the we started with the 3D model and then that 3D model is converted into a 2D image at the computer screen. This is called the the forward rendering. Correct? Forward rendering. We started with a 3D model but now recently we have new technique called neural rendering where we don't have to start with the 3D model. We don't need a polygon model to render an object. How else can we render an object? There's a neural rendering we are talking about. Now this is called the inverse rendering. Inverse rendering in the sense we take an image set of images of a scene and from this set of images we are trying to reconstruct the scene. This is actually inverse right. So from the 3D model we constructed the uh in the old process traditional method from the 3D model we reconstructed the image. Now from the image we are reconstructing the the the scene that is the neural rendering. it is actually a inverse rendering. Essentially what is happening is that we have the neural network model and then we have the the radiance which is a five parameter model uh five parameter function uh like the uh radians we discussed BRDF we discuss here also in the BRDF it's a four parameter model but here is a fivep parameter model including the color of it and this is actually a field it is called the uh nerf Right. This is neural radiance function is what which is coming up. Neural radiance field is what is is coming up. We will quickly look at the the the main idea of uh this nerve and then go into the the the rest of the topics. The technique is called the ray marching. We looked at the ray tracing. We looked at the uh slightly complicated technique than the rate tracing called the now path tracing. Now rate tracing is only one and uh one ray from the eyes to the now first object first hit object. But what happened to the the rays which is uh coming from all other directions right that is a path tracing much more complex than ray tracing is the uh uh path tracing then we have the know march tracing basically essentially we have this ray and u this is actually a now five parameter model each point is a five parameter uh model this is the position plus direction. Each one is a thread off. This is a thread off of and this is the direction also is a thoff basically. Now fight off and then output we wanted to look at the the color plus density at every point every position and direction we wanted to look at the now color and density. So um basically when uh the neural rendering lets you synthesize as a new image from the the the the nerve basically right so it's a new image synthesis is called the nerve when you change the direction of looking from where you are pointing your uh camera you need the same point in a uh it is going to look differently. That is where the color and the density for every position and direction we need a different color. That is what is now the ray marching is going to be correct and this is also called the differentiable volume function and this is also called the different differentiable now rendering. Correct. So essentially if there is a scene let us say this person is there we have multiple now projections onto the image basically now from the images we are trying to get this is seen and uh rendering. So there is a input and then that uh given the input it is actually u encoder and decoder differentiable now model is made and this can be a now volume basically a know field and it can be a volume that's a ray marching volume from this ray marching volume we can look at the the field from any direction that is the neural rendering right enough and uh this neural rendering it's a differential rendering or the know inverse rendering it is useful in a know very different uh know situations for example I I don't know how many of you have observed when we put on the headmounted display for example Oculus the eye tracking is looking at the eyes looking at the eyes of the for the emotions and all the same thing one of the recent topics research topic is that just from the images of the eyes from the eye trackers can we reconstruct the the entire face. Right? So this is where the inverse rendering is actually useful. This is where the nerve is actually useful. So this is the the image of the eye tracking. Can I use this two input images to reconstruct the entire face? And this is what is happening in the Oculus. This is a very very nice paper. I would like you to now uh I would like you to now take a look at it. Okay. Um so I'm going to also introduce is uh now another topic called the know uh point based rendering in the point based rendering. So unlike the nerve here or know point based rendering we look at only the point as a you know cloud point cloud is what we we are looking at it there is no geometry there right so only the point as the now uh part of the now the entire structure is available that is the point based rendering that is the next level of next level of rendering in the nerves we have the neural network which is is now reconstructing the the scene from a new point of view. But

here in the point based rendering there is no neural network at all. It is just the no set of point clouds that point clouds has to be now reconstructed. Right? So uh uh the uh that is actually called the the gshian splatting we are talking about. I'm going to now give a very brief introduction of the caution splatting as well. Right. So, initially there are photoggramometry uh techniques which you know I I I'm sure you're all aware of it. Photoggramometry is a technique to construct 3D model from the images. So, we are constructing the 3D model itself. It's a mesh model from the photo photos several photos one photo or several photos right there's a photoggramometry uh uh structure from motion the are all now very established techniques in the computer vision field those techniques have been borrowed to now virtual reality as well. The advantages of such method is that there are accurate measurements, detailed surface textures, realistic content can be done using the photoggramometry. But the disadvantages processing time, computational resources, we need many photos to accurately model it. Right? There's a nerves what we have talked about is novel view generation is what is good about realistic lighting effects are possible. Creative freedom is possible. But the the challenge is that it needs a very high training time. Neural network training time. Competition resources is very high and accessibility also is very very difficult. So the nerves to to create a simple scene of a a small room for example it will take hours of knowing hours of synthesizing the nerve. Right? So uh then comes the the point based rendering that is also called the gshian splatting. We are going to do this realtime rendering interactive explorations accurate representations and it is lesser than the less photorealistic than the photoggramometry or the nerve but it is much more real time. Okay and that is what we are going to see in the next know few classes. Okay. So if you have the nerf like this in array and each array we are going to sample this points and that is the the function we are sampling over here in the gshian plotting since it's a gshian plotting is about the point based rendering each point we have a what do you call gshian represented into it each gshian can have a different know mean different variance different know shape right there's gshian splatting. So each gshian has the position and now coariance matrix now opacity color and harmonics all related to this one. This now five different now values associated with each of the gshian helps us to uh do the the uh novel view synthesis. If you change the camera parameters, we should be able to now synthesize a new image that is coming from this uh no gshian splatting. The real advantage of gshian splatting coming is is that it doesn't need the training time. It doesn't have the neural network at all. Right? Instead it is using something called um uh gradient descent but it is called the stochastic gradient descent. It uses only know some random positions and then use a gradient descent method to know reduce the error between the target and the input. That way it becomes very very real time. Correct. And this gshian splatting has been the the the most important topic as of now in the the in the know computer vision and literature'ses and hopefully it'll catch up in the virtual reality as well in the last three last one year alone and there are many variants of the gshian splatting which has been published for example June 13 and and this gshian splatting was introduced August 13 that's a dynamic 3D gshian and then dream gshian now physics gshian no uh vitable gshian editable gshian right and then vr go gshian uh and uh no gshian probe and several others are still going on this year I attended CVPR one of the most prestigious computer vision conferences computer vision pattern recognition conference right in Seattle there were 2,000 papers totally and among them no at least 100 papers are no gshian splatting related there's so much of interest as of now in the gshian splatting soon that will be no that will be taken into our virtual reality as of now know since it's a point cloud it needs a lot of memory space the the VR headset doesn't have that much memory to handle the gshian flat but no soon there will be no modification done to cautious platting to adapt to the virtual reality headset. Here is a graph that talks about the number of papers coming in the last few years. So you can see that December 23 there are about no 50 papers in in February 24 February 24 there are no uh papers but in uh maybe uh June there may be about 100 papers coming up. there's a lot of uh uh a lot of interest in gosh planning. So please keep a watch on this and let's uh let us uh uh let us keep our fingers crossed how it is evolving. Correct. So there are obviously there are some problems with the gshian splatting as well. It is real time but it's not real time as we expect in the in the 90 Hz time. In virtual reality we want 90 Hz that kind of real time it is not happening probably it will happen later this has to be editable there are some versions of the now goshian splatting in the literature which are being discussed that are editable right and there are no methods to make it real time and editable there are now people trying to combine traditional 3D model and gshian splatting to make it much more real time and editable Right. So, uh is this all we talked about the the nerf, we talked about the G-splat, we talked about the traditional graphics rendering pipeline and the latest development in the graphics pipeline. Correct? Is that all? Where is the trend in the next five years? How is it going to be evolving? Right? So uh let's uh before I discuss it, I wanted to see if there is any um any uh questions on this. Any questions before I go to the next topic? If there are no questions, we'll go ahead. Okay. So um again I before I discuss the feature of uh you know how where how this the visual rendering pipelines are going to be evolving and what is the probable way of evolving. I wanted to discuss with some of the questions some of you have raised in the last classes. Now uh sometimes you will come across this uh terminologies know rendering pipeline as far as the unity is concerned. What is the rendering pipeline pipeline the Unity you know is what is a rendering pipeline of the Unreal Engine or God do right. So the basic virtual reality or graphics pipeline is what we have already discussed but above that Unity will have its own rendering pipeline particularly Unity has three different pipeline depending upon the requirement. So there is a lightweight uh it is also called the now URP universal rendering pipeline. So this is a default if you don't change any of the settings um this universal

rendering pipeline is what is assumed and uh it doesn't assume any of the no special programmable hardware in the in your mobile phone. It assumes the basic uh necessities is for rendering. Right. There is a scriptable we looked up all the shaders scriptable pipeline is there and then now finally HDRP uh high definition and rendering pipeline. So whenever you run this Unity in a desktop with the latest graphics pipeline it has a lot of hardware and accessibilities. So HTP will be able to access those hardware and then make it much more accelerated. Right. Similarly uh Unreal Engine and you know it is is uh it doesn't have different three different pipeline like Unity but it has only one unified rendering pipeline. It is mostly written for high-end systems. Um for mobile phone it is not much appreciable but again and in the mobile phone also it is actually knowing well but not like know Unity. Unity 3D is for mobile phone whereas Unreal Engine is for the uh for the desktop. The com the output of the Unreal Engine is very very know again and is a free uh game engine completely opensource game engine which again it has its own u uh its own pipeline. So uh there is a before or um now uh before uh uh considering which uh game engine to use now probably you can look at the the uh rendering pipeline again whether it is a open source or paid source or 3D rendering pipeline or 2D rendering pipeline export options all those things. Now there are a lot of comparison charts available in the you know internets you know you need to consult it before knowing this. Similarly the blender also it has its own rendering pipeline. Now the uh movie production pipeline it has its own movie production pipeline that's a different uh uh pipeline and movie uh animation and pipeline is different and then movie uh production pipeline is different and uh no gaming pipeline is different. So but all the uh pipelines whatever we discussed now is sitting over the the basic graphics pipeline which we discussed in the previous classes right apart from this now we have the rendering haptic rendering and we will discuss in the the next classes right okay so let me take a question how different is rateracing from ray marching Okay. So, um yeah, rate tracing is is going to hit the object and then look at the now reflection or refraction or the transmission like what we discussed in the other classes. Right? So, it uh the rate tracing assumes the BRDF of the surface objects. Correct? But here in the ray marching we are assuming a assuming a radiance field right. So each point is a radiance field. It's a five different radiance field right position and then the direction and the color and then opacity all this every point whether it is the transparent object or interior object it it is assuming it's a it's a volume rendering basically whereas this is going to be rate tracing is the now surface rendering correct should I keep it this way is go dot considered good for VR go dot is a uh open source completely open source game engine and there are significant um um um online support is therefore a lot of people are know using it. In fact, know for our course we were initially planning to use the go dot instead of uh no unity but uh the support needs to grow a little bit. Probably next year we can consider it. [Music] [Music] So the future of visual rendering I believe that it can benefit lot from the perception engineering. um the concept of p perception engineering I uh I'm sure and you all would understand from the professor Anna's earlier lectures but I thought I will emphasize it from my side as well because that is the single most important concept uh that I would like we would like you to take it from this particular course. So I will walk you through the what we mean by perception engineering and then and we will relate to the uh the visual rendering pipeline. Um I would like you to imagine how this perception engineering can be applied to the the featured visual rendering pipelines. You all know that virtual reality is a completely interdisiplinary. It is about the experience the human experience how immersion presence all those uh quality factors we discussed about it for uh for being for to to become an expert in XR we need to know a little bit of robotics meatronics computer graphics neurohysiology biomechanics and also another subject which I'm going to introduce because this is about the human perception we need to understand the human perception. So please understand that XR AR VR is not about the hardware, it is not about the software. Many people they think that no Unity or Unreal Engine if you know you are an expert in ARVR. You are expert in using one particular tool. Okay. So if you know programming in Oculus you become an expert. It is just one know small component of it. But what is more important is that now understanding the human perception component of the XR to understand the human person perception of the XR we have introduced another subject called psychopysics in IIT medas there's a separate one semester long course on psychopysics what is the psychophysics psychophysics is about the human perception and how do we measure the human perception so XR is about the experience XR about the human perception. Can we measure the experience? Therefore, can we measure the human perception? What do we need to measure the human perception? What do we need to do the uh measure the experience? Only then only when we measure it, we can have a science or engineering or technology of the XR. Only then we can have innovation. Therefore, measuring perception, sensation and manipulations is very important. Measurement itself is very important. We all know you're all know engineers, you have done now bachelor's, masters or some of you are having PhD, postocs and some of you are faculty as well. We know the importance of the measurement. Measurement is very very uh uh important that basic understanding in science depends on the measurements. The reduction of observation to numbers is what measurements. This is what know the Kelvin has been know keep saying it right only when we measure it we can improve on it. If you want to make a small improvement in your visual rendering display rendering pipeline, we need to do measurements, perception measurements. Only then we can improve on it. Capacity to innovate depends on the measurements. Only when we measure it, we can innovate it. Therefore, measurement becomes very very important. So, we stress on the reliability and accuracy in our measurements and checking and re-checking our results. as far as the perception is concerned only then we can master the VR if we

don't measure it we can't master it right that is why we are emphasizing again and again the perception measures so only when we measure it we can convert that measurement into application that process is what we call it as the perceptual engineering the first step is the per perception measurement right then the similarly We can extend this concept of perception to algebra as well. So the mathematics how two perceptions are added, subtracted, divided, integrated, how perception is actually influencing the physiology that's a perceptual physiology. Right? So since this this concept is very important, this is a single most concept. I would like to know uh spend some time on it and see if there are any questions out there. What are the advantages of G-splatting and nerve over traditional 3D pipeline? Okay. Um it's a good question. I have mentioned it uh when I explained the slides. 3D pipeline assumes a mesh. We start with a 3D scene with a mesh. Whereas G-splat or Nerf, they don't need any mesh at all. No need of meshing at all. The scene itself is a no radiance field in the not the scene itself is a point cloud in the G-splatting. Correct. We don't need a mesh at all. That is the advantage or it can be also a disadvantage. The advantage is that this is completely photorealistic. The disadvantage is that it is so huge data. Correct. Uh do we have a transformer-based encoder decoder architecture in no nerve? Um uh do we need a transformer? Why do we need a transformer? I think there are several variants of the nerves there. Again last two 3 years there's a lot of research on uh nerf maybe. No you can look into it. Yes definitely there is a transform based nerve. Realtime rendering is still a research problem in modern rendering techniques for VR. Realtime rendering is possible with a traditional mess based rendering but real time rendering with the nerf or the gsplat is still a research problem. Okay. So that clarity I would like you to be for the nerf we provided five dimensional input. So it is RGB image along with other two. It is not RGB image. It is a position and orientation. That's where the the ray is coming up. A point in the ray. That is the uh position and orientation and a color image. Okay. So therefore it becomes a fivedimensional input. Any question on the no perception engineering? Uh what kind of camera is needed for nerf? Again it is uh it is a image is given as an input. And then there is a neural network is learning the uh learning the the radiance field and then constructing the radiance field. Right? It is a normal camera. It doesn't need an is different camera. It is RGB image but the neural network is constricting the the radiance field. So if there are no questions on the perception engineering we are going to look into it. How do we measure the perception? Okay. What is the challenge in measuring the perception? Anything a subject or somebody is measuring, human is measuring, then it is subjective. It is called subjective. But what we need is the objective, right? What we want to say objective parameters. Anything subjective, any measurement is called subjective because it has a bias, it has a memory or it has an attention. So we need to avoid all this biases, memory, attention. There are no specific techniques are there to avoid the memory attention and bias and that becomes the objective and that is how no uh we are are going to measure it and particularly I I would like to point out that there are several SI units you all know particularly seven basic you know fundamental SI units. I'm going to add perception to it. All the seven units are no physical units. Perception is going to be a psychological unit. We are going to talk about the SI unit for not SI, it's a unit. How do we measure? What is the unit of the measurement of the perception? That is what I want I would like you to now pay attention to. Right? So this uh the subject that I teach entire course the psychopysics it starts with no uh connecting the stimuli to perception. How do we measure perception? This says stimuli is a physical input. The output is a perception at the mental level. Correct? So connecting the physical to the mental level is there a relation that relation is what is psychophysics finding the relation is there a law between the stimulate to the perception is there a physical law is what we are going to look at it if there is a physical law then and and is it a linear nonlinear or is that a logarithmic all those questions we can arise about 200 years ago Somebody came up with the idea that the threshold is is a concept that can be used as a measure of the perception. Measuring the limit of sensitivity is what the threshold is. We are going to talk about two types of the threshold. Absolute threshold and difference thresholds. Right? So uh Fner Gustoau Theodor Feser 18 and 01 to 1887 about 200 years ago uh he came up with this idea and he is called the the father of psychopysics uh psychopysics right? Why is he called so father of psychophysics? We'll find out. But before finding it out, I wanted to understand the concept of threshold. The smallest stimulus that can be perceived is called the threshold. Right? Suppose if you have a now intensity in the x-axis this is a physical unit and the y-axis is the proportion correct in the sense suppose if I say yes or no then it can be taken up um suppose if you have now perception is uh is plotted in the y-axis in the form of the perception how many times know I'm correct Right? Then if you are an ideal human being, so we'll start with a small intensity of the light. For example, we are keep increasing it. We are asking whether you can see the light or not. It'll continue to increase until at this point now uh we'll start seeing it. Right? This is the threshold. Below the threshold, we don't see no seeing. After this now uh fully seen this is a 0% this is the 100% visibility okay this is the threshold this particular point is called the threshold this is the intensity at this particular intensity is what the threshold is called correct that is for the ideal human being but a normal human being they will not behave like ideal human being the the response is going to be a smooth sigmodal curve yes kind of curve sometimes they will say yes sometimes they will say no for the same threshold the same threshold if you ask them 100 times here they'll say 20% of the time they say yes 80% of the time they say no similarly here this particular threshold they say 40% of the time they yes and 60% of the time they say no. Similarly here it is uh 70% and 30%. Here it is now 80% 20%. And then finally above a certain threshold it is 100 totally 100%. This curve sigmoidal curve is called the psychometric function. Correct. So the stimulus that produces an arbitrary but defined level of performance is the

threshold. This threshold is the absolute threshold or the difference threshold. We can say absolute pressure is again physical intensity and then sensation magnet is the perception in the y-axis and we keep increasing the intensity until we start now uh start increasing sensing it right until then and there is no sensation. This point is called the absolute threshold and after that there is a difference threshold. This is a linear assumption, right? The rate at which the slope, the rate at which the sensation grows as we intensity increases is called the difference threshold. So it means absolute threshold means uh intensities below the absolute threshold are unpredictable. Intensities above the absolute threshold are predictable. Difference threshold means uh a change in the intensity that is smaller than the difference threshold is unpredictable that is larger than the difference threshold or predictable. This difference threshold is also called the just noticeable difference or J&D. This is called the J&D. Just noticeable difference. this name this jargon we will be uh we will regularly use in our you know courses and the literature and the know reading material. So I want you to be you know familiar with this absolute threshold or difference threshold J and D are the the uh jarens I would like you to be uh familiar with right so uh here is an example of know sound and observer able to predict the difference uh between know intensity of 100 and 110 hertz for example Right? And also we'll be able to def uh uh uh find the difference between 1,000 and,0 this is not the case. So uh uh um trying to say the the linear psychophysical equation means the difference between 100 and then 100 110 right? uh at the at the at the at this level 100 level is not going to be the same as I know 1,00 uh,0 level. Okay. Now at a perceptual level we will not be able to now predict this difference whereas we will be able to predict this difference because as uh no psychologically we are going to talk about now Weber's function which is going to take about this is a 10% difference the same 10% is what has to be increased over here instead of 10 it has to be 10% increase is going to be 1,00 only then we can actually perceive there is a difference threshold. So the Weber's law is in you know difference threshold we are talking about deli what is the change in the intensity that I can perceive divided by the basic uh intensity that is actually a constant okay so u this is actually now in in this case the case what we discussed earlier it is maybe now 10%. 10% is is uh the 100 to 110 or 1,000 to,00 the 10% change is what can be perceivable and that is the Weber's law right this is the per law this Weber's law holds only approximately which we are going to [Music] [Music] So Weber is before the fashioner and he was the guru of ser and he studied with the weight of the proportions. He asked people to carry weight and then differentiate between and the weights difference weight and absolute weight. He did a lot of research and then he came up with this law called the Weber's law and uh he also coined the term just noticeable differences as but at but the fer is called the father of psychopysics for one reason I'm going to tell you in the later slides. So J&D is a very important concept which I'm going to emphasize on this. How much must a stimulus change in order for a person to sense the changes. What should be the know the smallest difference that can be now perceived that is the J&D. So J and D can be expressed as a ratio. This is what we have done. This ratio is called the verber's fner law. He um uh he uh did a something uh very phenomenal and he connected this relation between the the sensation magnitude and the and the uh intensity as a logarithmic function. Okay. So suppose if this is a physical intensity and then this is a sensation unit. He defined something like like a SI unit. He define he used J&D as a unit. Okay, this is a zero J&D, first J&D, second J&D, third J&D and so on. And for every J&D, you can see that the change in the uh stimulus intensity required is actually a lot and this del I right divided by you know the I that is constant. This is a fetchness law and this curve is actually a logarithmic curve and this is a contribution and for this single most contribution the um uh this fner is called the father of father of the psychopysics. Then about about 150 years later and there was a Harvard professor called Stevens. He came and said that now logarithmic law is only approximately true for only few uh what do you call modalities or few sensations whereas now power law he generalized the logarithmic into a power law which is holding good for most of the sensations. The perceived strength of the stimulus increases in proportion of the now physical intensity rise to some power. So this is a sensation magnitude and this is some constant proportional proportional constant stimulus intensity and then power factor right for loudness sound perception loudness it is the power factor is going to be 67 right so uh for brightness this is uh no it's about now 3 or 6 whereas for electric shock this This is about uh know 3.3 power factor right there that also can be linear it is one power factor if it is zero 1.0 zero it is a linear that way Stevenson power law takes care of almost all the know uh sensations in fact now if you if you um if you yeah if you search in Google for now power law Stevenson power law for many stimulus the exponential values are given okay loudness we saw it is a 67 Seven there is no electric shock you can say it is a 3.5 and then you know cold and cold perception it is almost one right and small small is a 6 there's a lot of of no stimulus you can see in fact now um uh there are several researchers still doing experiment and finding the power law power factor for several stimulus right time the threshold is only one of the measurements. The J&D unit uh is one one of the you know measurements. The power factor is only one of the measurements. In psychopysic there are several other measurements are there. Now threshold is one and then accuracy how accurate you can actually perceive it right reaction time can be another measure. Now uh point of subjective equality which I'm going to show you scales ratios or another one there are many other measurements also is possible right okay so let us do a very small experiment of finding the threshold because the concept of threshold absolute threshold and the difference threshold is very important I'm going to emphasize on it uh let us measure the absolute threshold of the visible brightness. Right? So, what I'm going to show you is in the next couple of slides, I'm going to show you a black screen with or without with actually this uh spot and of different intensities. I want you to say either yes or

no. There are no in between no u answers. Okay? So, u and then we are going to plot it in a in this scale. Now stimulus intensity because I'm going to present one 2 3 4 5 6 7 different stimulus intensities and uh I can I'm going to present it in many number of times and find out how many times you know uh how many times how percentage of times this is all you know uh you are correct right let us do it let us uh start it uh are you able to see any uh Uh, are you able to see the the the spot? Okay. So, could you see the spot of light now? Now, now now. How about now? How about now? How about now? How about now? How about now? Now, now now yes. So um this in fact in a real experiment it is repeated many number of times and this percentage u is calculated much more accurately. Right? So in this case let us say we have calculated like something like this. you would find a very nice uh smooth sigmo graph. Correct? This smooth sigmoid graph is called the psychometric function. We have seen it. Psychometric function. And then from this function we can now say at 50% level this is a 50%. This is 50% level. Whatever is the intensity that is the threshold. So some researchers say that we should make it as a 75% some say 50% but for our course or majority of the users take 50% as the threshold right. So that is the absolute threshold uh sig model functions. Now we can also find out how do we do the the difference threshold. So difference threshold is the only the same experiment but there will be a reference stimulus. We'll find out this is a standard stimulus along with this one of this will be given one of the comparison stimula will be given. So you'll have to tell whether this two uh light spots are same or different. Right? The standard stimulus has a fixed intensity. This is not going to change and only the comparison stimulate will be changing. The intensity of the comparison stimulate um uh bracket uh uh will be the will be changing. Right? So these are the different sets. So this different sets can again be know shown to the user many number of times and the response in terms of the percentage can be now um uh can be obtained and uh and that can be again plotted in the sigmodel graph right. So in the sig model graph then again and there can be a point of subjective equality. So uh because we are trying to compare it this is going to be a point of subjective equality PSC and uh uh this is a 50%age and 25%age is going to be now um it it's going to be now u what do you call uh lesser and 75% that is going to be stronger right so this is called the point of subjective equality and this is going to be dw W in German language and then DS in the the other language weaker or smaller like this the experiments have been conducted and uh these numbers are plotted. This is for the now tactile systems whereas the experiment what we have done is for the visual systems. Okay. So uh this numbers is what the the perception parameters are and this numbers are are are is what now used to engineer several objects. Okay. Uh how this numbers can be used to design several engineering uh object is what we are going to take few examples. Okay. So the measures of the perception and its applications is what the perception engineering. Now we have measured it. We have got a know a specification and of our human and uh skin over here in this case right the earlier case is a human brightness perception. So we got the numbers. How do we use this numbers? How do we make a engineering application is what the perception engineering is. We'll take couple of examples. The gamma example is what you have already seen it in the last class. Now I hope you remember uh this particular curve. The x-axis is the physical stimuli. This is the actual luminance. It's a physical intensity of the light. And then yaxis is the the perception. Yaxis is a perception. Percentage time it is perceived correctly. for our human perception it is know uh bending upward it means as we have seen in the now Steven's power law this power factor is about you know 6 or 7 or.3 some numbers right so this gamma corrections we can do in the display in the image compression image compression we have to do and the image um what do you call um know uh encoding and decoding we have done it in order to know make this accommodation of uh very high sensitivity of our eyes in the last class we have seen it and that is one of the engineering applications all the impacts JPEGs they're all using this encoding of of this particular gamma corrected signals your displays are there right your smartphones can you find out what is a gamma uh factor of your smartphones. Correct. Let's see now uh application of this is particular uh uh power factor of uh now visual perception here gamma factor right. So similarly the tactile factor which we have seen uh let me take uh uh let me take let me ask you one question. We have we are all using these smartphones correct and we all have felt the vibration in the smartphones. Can you tell me what is the frequency at which your vibr your cell phone is vibrating can you uh can you can you respond it in the now slider? What is the frequency at which your cell phone is vibrating? Can you type in your numbers? What is the frequency at which your cell phones are vibrating? Make a general guess. You don't need to know search in the Google. Just make a general guess. 150 to 280. Okay. 200. Okay. Don't search in the Google and tell me the answer. Just guess it. Okay. 100 Hz, 90 Hz. Do you think that is a 90 Hz? Have you ever felt a 90 Hz? Okay. So, those of you now who have not looked at the Google's, now let me ask you questions. Why should um uh the answer is 250 Hz around 250 Hz. Now the question is why your cell phone should be vibrating at 250 Hz or around that Hz. Okay, that's the question. Can you answer why it should be now vibrating at 250 Hz or 200 Hz to feel? Okay, that is distin distinctively felt. Okay. Any other answers we may not feel below the threshold. Okay. To activate our feeling. Okay. So just because you know we learned our concept of threshold. Now you you're using it. Okay. All right. Fine. Haptic range of palm skin. Okay. Any other answers? forms in it's both cured and filled. Okay. Any other answers? So I'm trying to explain now where this perception engineering is coming into picture. Okay. All right. Let us go back to the numbers. So this this is a sheet of of a tactile perception system. Correct? So I'm talking about know absolute deduction threshold at uh no static we can perceive 20 micron dots. Yeah. Dot rise to 20 micron we can predict. If it is below we will not be able to predict. At 10 hertz we can predict 10 microns right at 250 htz we can dictect.1 microns. So

what does it mean? If I vibrate your phone cell phone at 0.1 micron you'll be able to feel the vibration. So the the the sensitivity of our palms is highest at 250 Hz. It means the amplitude of our vibration can be minimum if it is at 250 Hz. If it is any other hertz, the amplitude required for you to feel the vibration is going to be more. What does it mean? More amplitude means more power you need to spend. more powers you need to spend means no more uh battery no exhaust it will be exhaust right battery will be drained very quickly so at 250 Hz it will take the the minimum battery and uh it will vibrate at um it it will have a no less amplitude and therefore it'll be now optimum okay I hope uh now u uh this makes a uh know good application of engineering principles of the perception parameters to know the cell phones. So this is uh this is only the vibration but our real our kind of u the key press is actually not only vibration but also small stretch right. How much vibration and stretch is needed in order to feel the key press is what we have used in the design of the Apple phones. And as I told you in the one of the earlier lectures, the Apple phone and haptic feedback is designed by our students from our lab and and it is from now uh using this numbers what we are looking at this page that is what I call it a sub perception engineering. Okay, let us give uh let us look at you know several other uh numbers. Similarly, I can talk about know each of the numbers and its applications. Okay. Uh I run a separate course called haptics engineering. There we go through each of these numbers and their applications. Right? So this is called the tuning curve. Although I given you the this number of uh know.1 micron at 250 Hz. This is that number is coming from this particular graph. So this graph x-axis is a frequency of vibration, y-axis is the displacement of the the vibrators and this graph tells you what is the threshold. You can see that the threshold is minimum at about uh 250 htz, right? And then after 250 Hz it starts increasing it. Okay. So uh this is about um uh if you look at what is the what is the you know displacement required in terms of the dB uh at 1 micron peak that is about minus 20. Therefore it is equal to.1 micro m vibration. Similarly you can look at the you know 10 and hertz what is the what is the peak needed what is the threshold needed all those things this is called the tuning curve similar tuning curve we can look at the visual we can look at the you know audio and that again can be useful for now designing our engineering objects. Okay let us look at the audio perception tuning curve. Okay, again this is the x-axis is the frequency, y-axis is the sound pressure and then this graphs tells you what is the threshold. Again you can look at it is that this uh around 4,000 is where our sensitivity is maximum. We need very less sound pressure in order to hear. Right? So we can find out how much is the sound pressure required. It goes into negative. Negative in the sense since we are talking about the dB the reference is about the uh reference is about the I think uh know uh some 20 uh 20 pascal or whatever it is that's a reference. So negative means it's below the the reference as uh Pascal reference sound pressure level right you can see that this is the range of no maximum perception so we have about 1,000 to th00 4,000 is where we have the very good sound sensitivity the best is at 1 to 4 kHz all our mobile phones are designed for or no this particular hertz your air plex or headphones this are all designed to give you this particular ones okay so you can see that 0 dB sound pressure level at 3 kohz uh at 3 uh 3 kHz yeah 3 kHz is uh 3,000 it's about here this This is at the uh 3 kHz 40 dB sound pressure level at 100 kilo 100 Hz. 100 Hz comes at around over here. This is about 40 40 or 30. This is about 100 30 uh or 40 dB. Okay. The DL what is DL? You remember the difference or the J&D, right? Difference threshold. Different threshold is always mentioned in percentage the ratio we talked about remember del L by L or D I by I right the ratio this is a ratio is instead of a percentage so DL is.3 percentage at 3 kHz what is the know smallest change in the frequency you can differentiate is actually 3 percentage Okay. So at 3 kohz what is the one person that is going to be uh 30 hz what is the.1 person is going to be no three three htz and uh maybe about 10 htz difference is what we can actually perceive at 3 kohz we can differentiate 3,10 hz below that we may not be able to now differentiate that is what uh this means so at using this we can design the encoding and decoding a better encoding we don't have to know worry about the uh the signal which is know 3 kohz and 33 kilo 3,00 and 3 herz right because it is not perceivable therefore it is not needed only above know.3% we need to encode it otherwise we can keep it constant. So perception based encoding and decoding is possible. That is what no this this is actually yielding it. DL difference limit is 3% at 100 Hz. It is not.3% 10 times is higher at 100 Hz. Right? So 1% is going to be now one. Uh but this is three person. So 100 and 103 we can differentiate. We cannot differentiate 100 and 101 hertz. Okay. The piano keys are uh differentiated at 6%. The neighboring keys the difference in the frequency is about uh no 6%. That is just to make sure that their frequencies are within this DL level. Okay. that the piano itself is designed to make sure that this DL is actually taken care. Okay. So you can see that how a single perception tuning curve like in the audio it can lead to so many applications. This is this is a perception engineering. Okay. So uh DL for loudness is no 12%. The uh difference threshold for loudness is no 12%. In the sense um um change in the loudness lesser than know 12% we cannot uh perceive it. Steven's power factor for loudness is 33 that is less than one. So suppose if you have a sound which is very very loud and if you increase the amplitude by 10 times the loudness increases only by two times that is because of the Steven's power factor that is because of the deal for the loudness is 12%. reducing the amplitude of a sound by 99% if you reduce it the loudness reduces is to only you know 20%. That is because of again the Steven's power law because the amplitude you have to raise it to the power of 33 the reduction in the amplitude you have to raise to power of.33 therefore it reduces to only 20%. Right? audio perception localization some of the concepts you are going to use it in the next classes so I would like you to emphasize this again the uh x-axis is frequency y-axis is the error of localization localization in the sense how what is the knowh the sound which direction the sound is coming okay what is the error in localizing the sound the

direction of the sound and is coming right so it's about uh about 3,000 htz we have the maximum error our localization is very poor at 3,000 htz whereas at the lower hertz our localization error is now much lesser again this factor we are going to use it in tomorrow's class in the next class when we talk about the the spatial audio okay so there are no several methods we are going to know use it when we talk about the spatial audio we have of uh our our perception sound perception is using the interoral now level difference interoral time difference interoral phase difference. So there are three different techniques we are using in order to find out where the the source is where the uh the direction of uh the sound is coming from right and we are going to talk about how the faces we have a uh we have a less as uh less what do you call a sensitivity to the face of the the signals than the uh time of the signals. We are going to talk about the the cone of confusion and we are going to talk about the um the the localization errors through around the around spherical space of the the human and and we are going to know uh based on this we are going to design our spatial audio in the in the the coming classes and particularly we are going to talk about the HRDF head related transfer functions. How this spatial sound can be modeled, can be captured, can be personalized for virtual reality uh spatial sound. We will look at it in the next class in more details about the the the audio perception. The point I'm going to emphasize is that the perception measures are very important for all our All our now virtual reality engineering the entire course we have been discussing about lot of hardware and software every stage it is using one or other perception measures although we may not have been now explicitly talking but there are perception measures and this understanding the perception measures is the the single most important take-home message from this particular course. If there are questions I can pick it. Can we have any course or material to study haptic and perceptual engineering? Uh there are online courses. Um our one NPEL course we will uh release it soon. Okay. So we will um right. So uh we will summarize what we have done in the whole class. We talked about the the last three classes actually visual rendering. We looked at the traditional visual rendering. We looked at the the the latest now visual rendering pipeline and we looked at how know other than the meshbased 3D model you're talking about the nerve and uh G-splatting is coming up and uh we looked at how now virtual reality rendering pipeline have been now modified and then finally we looked at how the perception measures um can be used for for know feature know virtual uh reality rendering and uh improvements. So so far know this parameters are are not exploited for improving the visual rendering pipeline. I hope some of you will be able to use this perception parameters to innovate on the virtual visual rendering pipeline in the future. Each innovation can be a now paper at the cigar level. SIG graph is one of the best conferences as far as the graphics is concerned, VR is concerned. So I would like all of you to aim very high using this basic uh principles of this perception engineering. I will stop here and uh we will meet in the next class. Till then stay safe. Bye. [Music] lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

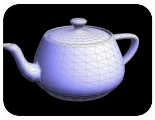
ChatGPT said:

Visual Rendering, Neural Rendering & Perception Engineering — Easy Notes

I'll explain this lecture as if you're in 7th class, using simple examples, memory tricks, revision tables, and MCQs with answers. You don't need to memorize the lecturer's long sentences — understand the main idea and remember the keywords.

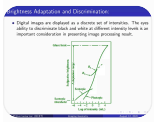
Part 1: The big picture

This lecture mainly teaches us **three important things**:



1. New ways to create images

Traditional 3D rendering, NeRF, and Gaussian Splatting.



2. Perception engineering

Understanding what humans can see, hear, and feel, then using that knowledge to design better technology.

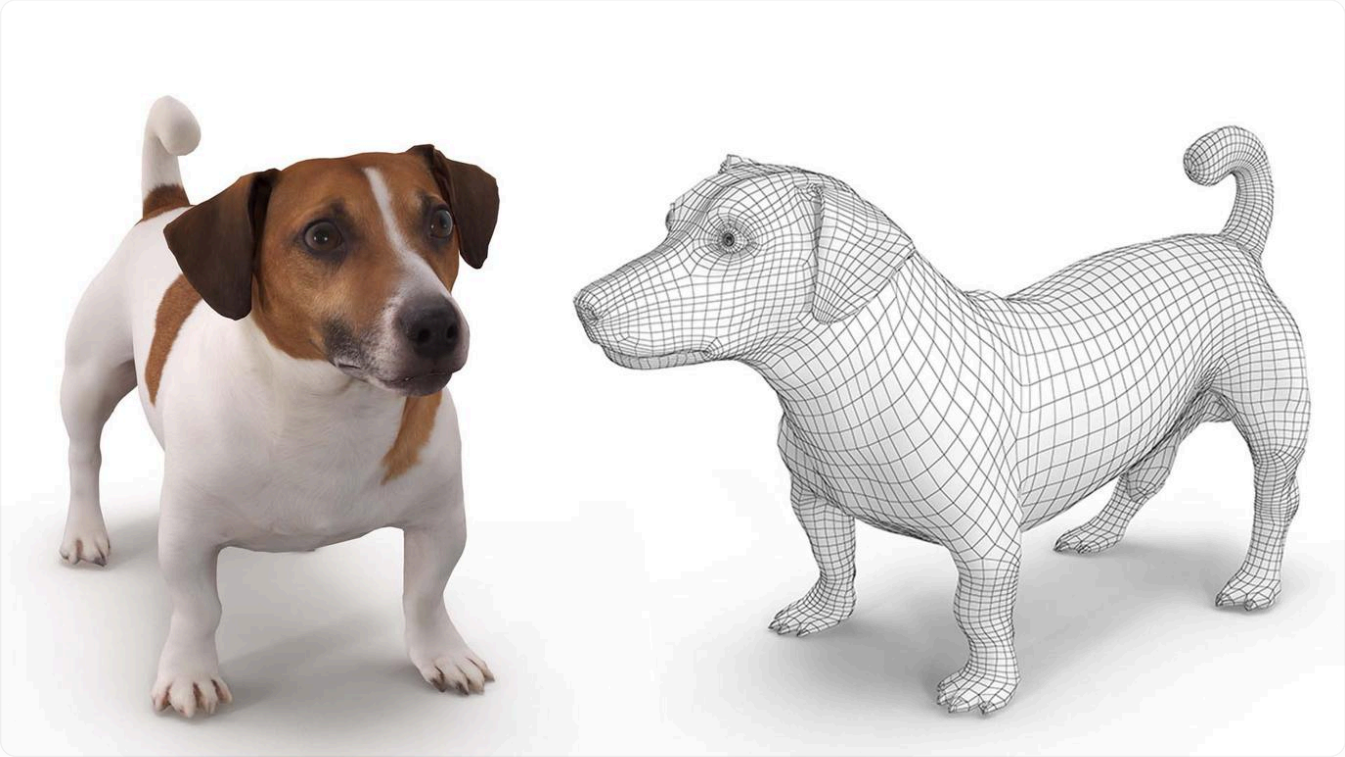
3. Real-life applications

Designing displays, smartphone vibrations, headphones, and virtual reality using human perception.

One-line memory trick: New rendering methods create the scene; perception engineering makes the experience better for humans.

Part 2: Traditional rendering vs NeRF vs Gaussian Splatting

1. Traditional 3D rendering



Imagine building a toy house from small blocks. First, you create the 3D structure. Then, the computer converts it into a 2D image on your screen.

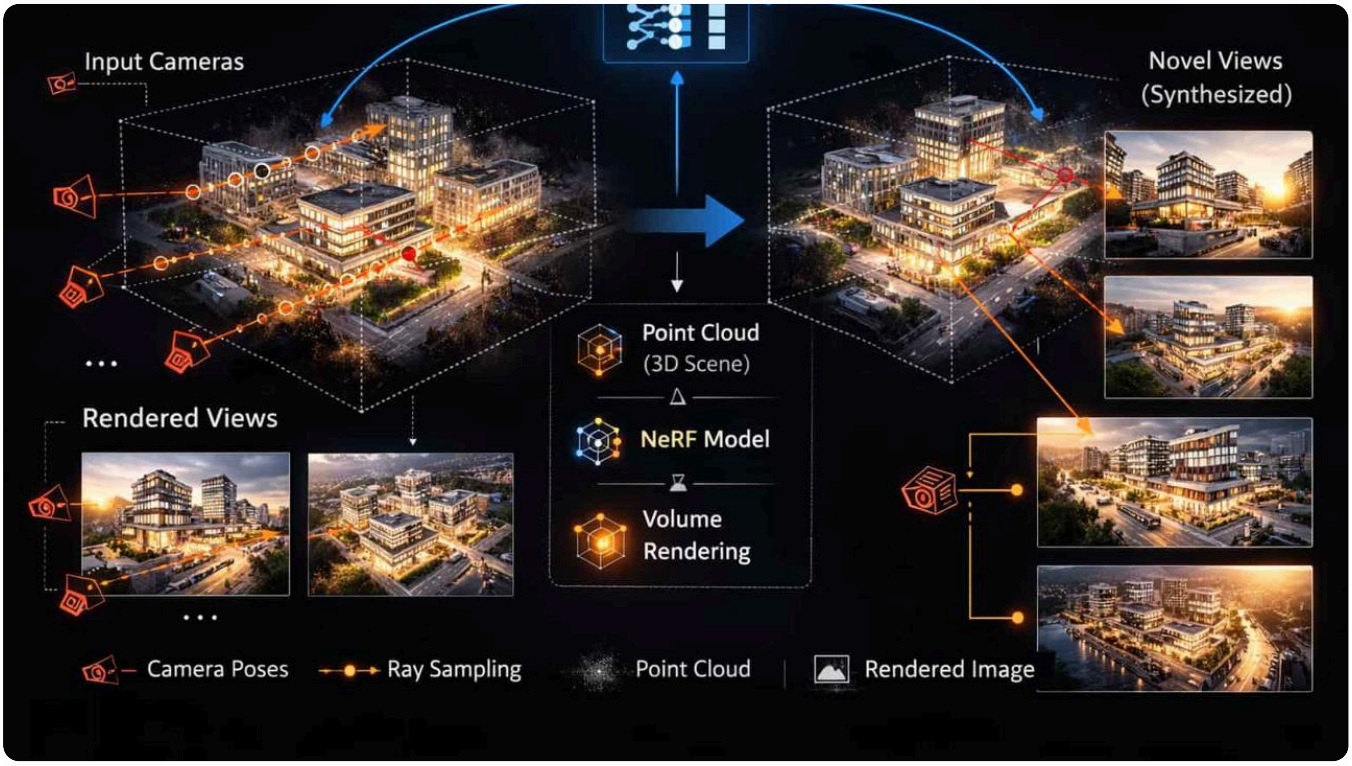
The process is:

3D model → graphics pipeline → 2D image

This is called **forward rendering**.

- Input: A 3D model, usually made using polygons or a mesh.
- Output: An image on the screen.
- Advantage: Well-established and suitable for real-time rendering.
- Limitation: You need to create the 3D geometry.

2. NeRF — Neural Radiance Fields



Imagine taking photographs of your classroom from different corners. A neural network learns information about the scene from those pictures. It can then generate an image from a new camera position.

That is the basic idea of **NeRF**.

Input images → neural network learns a radiance field → new-view image

This is an example of **inverse rendering**: instead of starting with a ready-made 3D model and producing an image, the method learns a scene representation from images.

What does NeRF learn?

The lecture describes a five-dimensional input:

- **3 values for position:** x, y, z
- **2 values for viewing direction:** commonly represented by two directional angles

The model predicts:

- **Color:** what the point looks like from that direction.
- **Density:** how much the point contributes to the volume or how opaque it is.

The color may change when you view the same point from another direction.

Remember NeRF

- **N** = Neural
- **R** = Radiance
- **F** = Field

Advantages:

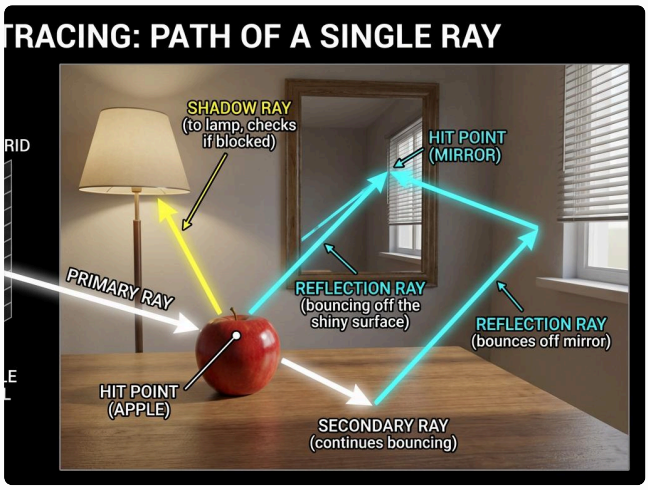
- Can generate realistic images from new viewpoints.
- Can represent complex lighting and appearance.

Disadvantages:

- Training can take a long time.
- It can require considerable computing resources.

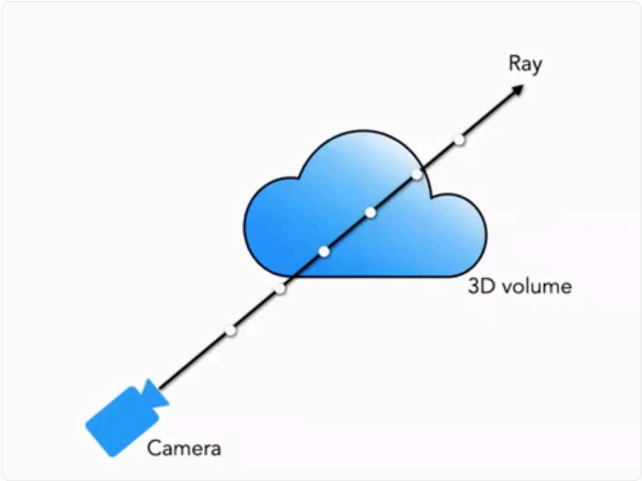
3. Ray tracing vs ray marching

These two names sound similar, but their basic ideas are different.



Ray tracing

Follows rays to find surface intersections and calculate effects such as shadows and reflections.



Ray marching

Samples points along a ray through a field or volume and combines their contributions to produce a pixel.

Memory trick: Ray tracing looks for surfaces; ray marching samples along the journey.

In the lecture's comparison:

Feature	Ray tracing	NeRF ray marching
Main representation	Surfaces and geometry	Radiance field or volume
Main operation	Find intersections and light interactions	Sample points along rays
Used for	Shadows, reflections, refractions, etc.	Rendering images from a learned scene
Neural network required?	Not necessarily	Yes, for the standard NeRF approach

Ray marching is not exclusive to NeRF; it is also used in other rendering techniques.

4. Gaussian Splatting (3D Gaussian Splatting)

Imagine a picture made from thousands of soft, colored dots. Each dot has a position, shape, color, and opacity. Together, these dots represent a 3D scene.

This is the basic idea of **Gaussian Splatting**.

Instead of relying on a conventional polygon mesh, it represents the scene using many 3D Gaussian elements.

The lecture highlights properties such as:

- Position
- Shape and orientation, represented using covariance information
- Opacity
- Color
- Spherical-harmonic coefficients to represent view-dependent appearance

How does it work?

1. Start with images of a scene.
2. Estimate a collection of 3D Gaussian elements.
3. Adjust their parameters to reduce the difference between rendered images and reference images.
4. Render the scene from a new camera position.

The lecture mentions **stochastic gradient descent**, an optimization approach that uses sampled data to help reduce errors.

Advantages and disadvantages

Advantages	Disadvantages
Fast interactive rendering is possible	Can use a lot of memory
Does not require a conventional mesh	Editing the scene can be difficult
Can produce realistic new viewpoints	Meeting strict VR frame-rate requirements remains challenging
Does not require the standard NeRF neural-network training process	Still an active research area

Important: Gaussian Splatting does not mean that no optimization or preparation is required. It means the standard approach does not train a NeRF-style neural network to represent the scene.

5. Photogrammetry vs NeRF vs Gaussian Splatting

Photogrammetry is a method of reconstructing 3D geometry from photographs.

Method	Main idea	Strong point	Main limitation
Photogrammetry	Reconstruct 3D geometry, often a mesh, from images	Detailed surfaces and measurements	Needs image processing and reconstruction
NeRF	Learn a radiance field using a neural network	Realistic novel views	Training and computing costs
Gaussian Splatting	Represent a scene using 3D Gaussian elements	Fast interactive rendering	Memory use and editing challenges
Traditional 3D pipeline	Render an existing 3D model	Established real-time rendering	Requires a model or geometry

Exam memory trick:

- Photogrammetry = photographs to geometry.
- NeRF = photographs to a learned radiance field.
- Gaussian Splatting = scene represented by Gaussian elements.
- Traditional rendering = model to image.

Part 3: What is the future of rendering?

The lecturer explains that newer methods are developing quickly, but there are still challenges in making them suitable for VR.

1. Why is Gaussian Splatting not perfect for VR yet?

VR headsets need images to be generated very quickly and smoothly.

The lecture refers to a target of **90 Hz**.

$$\text{Time per frame} = \frac{1}{90} \text{ seconds} \approx 11.11 \text{ ms}$$

This means the headset needs a new frame approximately every 11.11 milliseconds.

Gaussian Splatting can be fast, but memory requirements, editability, and consistent real-time performance are still challenges.

2. Unity, Unreal Engine, and Godot

These are game engines. They use the underlying graphics pipeline but provide their own tools and rendering options.

Unity

The lecture describes three rendering pipelines:

- **URP:** Universal Render Pipeline; designed to support a broad range of hardware.
- **HDRP:** High Definition Render Pipeline; targets higher-end graphics capabilities.
- **Built-in Render Pipeline:** Unity's older, built-in rendering system. The lecture's three-pipeline explanation focuses on URP and HDRP, but Unity also has this built-in option.

Unreal Engine

Commonly used for high-quality graphics and games. It can also target mobile devices, although performance depends on the project and hardware.

Godot

An open-source game engine with rendering features and community support.

The key idea is that a game engine's rendering pipeline sits on top of the basic graphics pipeline. A pipeline is the series of steps used to turn scene data into an image.

Part 4: Perception Engineering — the most important concept

The lecturer emphasizes this as one of the biggest takeaways from the course.

1. What is perception?

Perception is how our brain interprets what our senses receive.

For example:

- Your eyes receive light, and your brain experiences brightness.
- Your ears receive sound waves, and your brain experiences loudness and pitch.
- Your skin senses vibration, and your brain experiences touch.

2. What is perception engineering?

Imagine designing a smartphone. You could make it vibrate very strongly, but that might waste battery power. Instead, you study how sensitive human skin is to different vibration frequencies and use that information to design a vibration that feels clear while using less energy.

That is perception engineering.

The process is:

1. Measure human perception

↓

2. Find patterns and useful numbers

↓

3. Use the numbers to design technology

↓

4. Improve the human experience

Examples include:

- Designing screen brightness and image encoding.
- Choosing vibration settings for smartphones.
- Improving headphones and audio compression.
- Making VR experiences more comfortable and realistic.

Memory trick: Measure → Understand → Design → Improve.

The lecture also introduces **psychophysics**, which studies the relationship between a physical stimulus and the perception it produces.

For example:

Physical stimulus	Human perception
Light intensity	Brightness
Sound intensity	Loudness
Vibration	Touch sensation
Sound arriving at different times at the ears	Sound direction

Part 5: Threshold — how much can we detect?

A **threshold** is a stimulus level or change used to measure when a person can detect something according to a defined experimental rule.

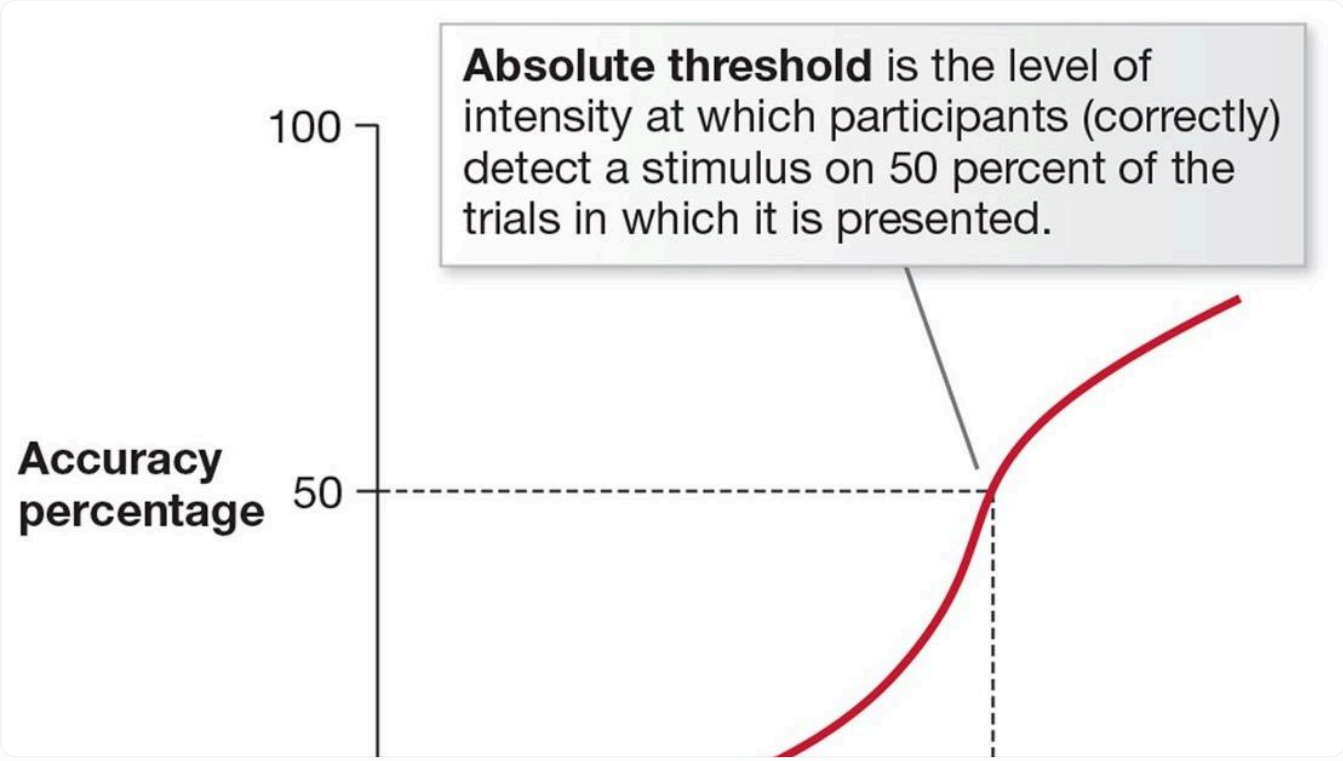
There are two main types in the lecture.

1. Absolute threshold

The absolute threshold measures the smallest stimulus level that can be detected according to a chosen criterion.

Example: Imagine a dark room. A tiny light appears, but you cannot see it. As its brightness increases, you eventually start noticing it.

That detection level is the **absolute threshold**.



The lecturer describes a psychometric curve, which shows how often a person detects a stimulus at different intensities.

- At very low intensity, detection is rare.
- At higher intensity, detection becomes more frequent.
- At sufficiently high intensity, detection may be nearly certain.

The lecturer uses the **50% detection level** as the threshold for this course, while noting that other studies may use 75% or another defined criterion.

Easy example

Suppose a light is shown 100 times at a particular brightness. You report seeing it 50 times.

Under the lecture's 50% rule, that brightness represents the estimated threshold.

2. Difference threshold

The difference threshold measures the smallest change between two stimuli that a person can reliably detect under a chosen experimental rule.

It is also called the **Just Noticeable Difference (JND)**.

Example:

- You hold a bag weighing 1 kg.
- Someone adds a tiny amount of weight.
- You may not notice the change.
- If more weight is added, you eventually notice that the bag is heavier.

The smallest noticeable change is the JND.

Absolute threshold	Difference threshold
Can I detect the stimulus?	Can I detect a change?
Example: Can I see this faint light?	Example: Can I tell which light is brighter?
Measures detection level	Measures detectable difference

Memory trick:

- Absolute = *Is it there?*
- Difference = *Is it different?*

3. Point of Subjective Equality (PSE)

The **PSE** is the comparison-stimulus level at which two stimuli are judged equal according to the experiment's response criterion.

Example: You compare two light spots. One has a fixed brightness; the other changes. The PSE is the comparison brightness at which you judge them equal most often, according to the chosen method.

The lecture also mentions comparison responses such as “weaker” and “stronger.”

Remember:

- **Threshold:** a detection or discrimination boundary.
- **JND:** the smallest detectable difference under the experiment's rule.
- **PSE:** the level that feels equal to the reference.

Part 6: Weber's Law, Fechner's Law, and Stevens' Power Law

These three names are important for exams. Learn their differences carefully.

1. Weber's Law

Weber's Law says that the detectable change in a stimulus is approximately proportional to the original stimulus level, over a range where the law applies.

$$\frac{\Delta I}{I} = k$$

Where:

- I = original stimulus intensity
- ΔI = smallest detectable change
- k = Weber fraction, approximately constant for that stimulus and range

Example

Suppose a person notices a change of 10 units when the original stimulus is 100 units.

$$\frac{\Delta I}{I} = \frac{10}{100} = 0.1$$

If the same fraction applies at an intensity of 1,000 units:

$$\Delta I = 0.1 \times 1000 = 100$$

So the change would be 100 units, not 10 units.

Memory trick: Weber = proportional change.

A fixed percentage of change is more important than a fixed number of units.

This law is approximate; it does not apply perfectly to every stimulus or every intensity.

2. Fechner's Law

Fechner built on ideas about detectable differences and proposed a logarithmic relationship between physical stimulus intensity and perceived sensation.

$$S = k \log \left(\frac{I}{I_0} \right)$$

This is a common form of Fechner's relationship, where I_0 is a reference intensity and k is a constant.

The important idea is that **perceived sensation grows more slowly than physical intensity** in this model.

For example, doubling a stimulus does not necessarily make it feel twice as strong.

Fechner is often called the **father of psychophysics** because of his foundational work connecting physical stimuli with measurable sensation.

3. Stevens' Power Law

Stevens proposed a power-law relationship between physical stimulus intensity and perceived sensation.

$$S = kI^n$$

Where:

- S = perceived sensation magnitude
- I = physical stimulus intensity
- k = proportionality constant
- n = exponent that depends on the stimulus and measurement method

The lecturer gives examples of different exponents for loudness, brightness, and electric shock. These values depend on the experimental conditions and should not be treated as universal constants.

Understand the difference

Law	Main idea	Formula to remember
Weber's Law	Detectable change is proportional to baseline intensity	$\Delta I / I = k$
Fechner's Law	Sensation follows a logarithmic relationship	$S \propto \log I$
Stevens' Power Law	Sensation follows a power relationship	$S = kI^n$

Super-easy memory trick:

- **Weber = Ratio**
- **Fechner = Log**
- **Stevens = Power**

Part 7: How perception engineering is used in real life

1. Gamma correction and display brightness

Human vision does not respond to brightness in a perfectly linear way. Gamma encoding helps represent image brightness in a way that works better with display and image-processing systems.

The lecture connects this to image encoding, decoding, and compression.

Remember: Gamma techniques help encode brightness; they do not magically increase the number of bits available.

2. Why do smartphones vibrate around 250 Hz?

The lecturer explains that the palm is particularly sensitive to vibration around **250 Hz** in the tactile sensitivity data discussed in class.

Imagine two vibration motors:

- Motor A vibrates near the frequency where your skin is highly sensitive.
- Motor B vibrates at a frequency where your skin is less sensitive.

Motor A may need a smaller vibration displacement for you to feel it. That can help reduce the energy required for a noticeable vibration.

This is the idea behind using perception measurements to design haptic feedback.

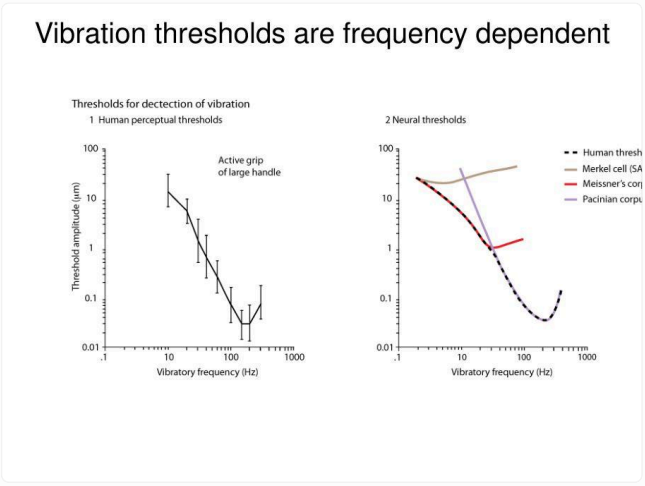
Important: Around 250 Hz is the lecture's example, not a rule that every smartphone must use that exact frequency. Actual phone designs depend on the actuator, desired sensation, and hardware.

3. Tuning curves

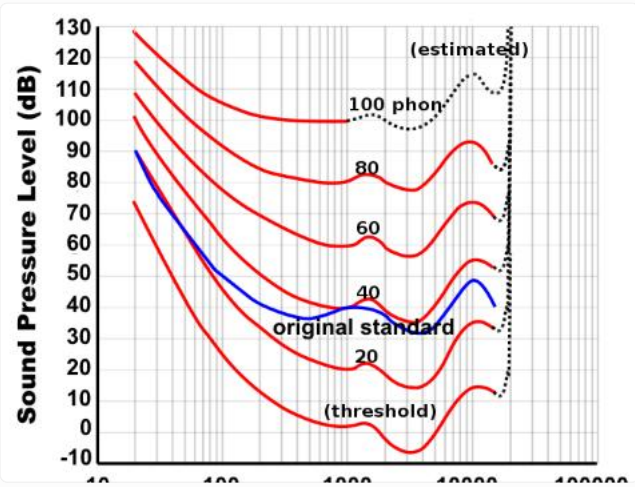
A **tuning curve** shows how sensitive a person is to different stimulus frequencies or levels.

For touch, it can show the vibration displacement required for detection at different frequencies.

For hearing, it can show the sound pressure level required to detect different sound frequencies.



Tactile tuning curve



Auditory tuning curve

Shows sensitivity to vibration frequency.	Shows the sound level needed to hear different frequencies.

In the lecture's auditory example, human hearing is particularly sensitive in parts of the **1–4 kHz** region, with the curve reaching high sensitivity around a few kilohertz.

Why are tuning curves useful?

They help engineers decide:

- Which frequencies need more signal detail.
- Which changes are likely to be noticed.
- How to design vibration motors and audio devices.
- Where compression may reduce information with little perceptual effect.

4. Difference thresholds in audio

Suppose two tones have frequencies of 100 Hz and 101 Hz. Can you hear the difference?

That depends on the listener, sound level, listening conditions, and experimental method.

The lecture's central point is that the smallest detectable frequency change varies with the original frequency.

For example, if a difference threshold is 3% at a particular frequency under a particular condition, a change smaller than 3% may be difficult to detect.

This is useful in **perception-based audio encoding**: engineers can prioritize differences that listeners are likely to notice rather than treating every numerical change as equally important.

The lecturer also discusses the role of frequency differences in distinguishing neighboring piano notes.

5. Spatial audio and sound localization

How does your brain determine whether a sound comes from the left, right, front, or back?

It uses several clues, including:

- **Interaural Time Difference (ITD):** the sound reaches one ear slightly earlier than the other.
- **Interaural Level Difference (ILD):** the sound is louder at one ear than the other.
- **Interaural Phase Difference (IPD):** the phase relationship between signals arriving at the ears.

Another important term is **HRTF**, or Head-Related Transfer Function. It describes how the head, ears, and upper body change sounds arriving from different directions.

HRTFs help create realistic spatial audio in headphones and VR.

Memory trick: Spatial sound = time + level + phase + HRTF.

Part 8: Important terms — one-minute revision

Term	Easy meaning
Forward rendering	3D model → image
Inverse rendering	Images → reconstructed scene representation
NeRF	Neural network learns a radiance field from images
Ray marching	Samples points along a ray
Gaussian Splatting	Represents a scene using 3D Gaussian elements
Photogrammetry	Reconstructs 3D geometry from photographs
Perception engineering	Uses human perception measurements to design technology
Psychophysics	Studies the relationship between stimuli and perception
Absolute threshold	Minimum stimulus level detectable under a chosen criterion

Difference threshold	Minimum detectable change
JND	Just Noticeable Difference
PSE	Point of Subjective Equality
Weber's Law	Detectable change is approximately proportional to the original stimulus
Fechner's Law	Logarithmic sensation relationship
Stevens' Power Law	Power relationship between stimulus and sensation
Tuning curve	Shows sensitivity at different frequencies or levels
ITD	Difference in arrival time between the ears
ILD	Difference in sound level between the ears
HRTF	Describes how the head and ears affect sound from different directions

Part 9: MCQs for exam preparation

Try answering the questions first, then press **Check answers** to see your score.

Practice quiz

0 of 16 answered

1. What is forward rendering?

- A. Image to 3D scene
- B. 3D model to image
- C. Sound to vibration
- D. Image compression only

2. What does NeRF stand for?

- A. Neural Radiance Fields
- B. New Rendering Format
- C. Natural Ray Function
- D. Neural Reflection Filter

3. Which method learns a radiance field using a neural network?

- A. Photogrammetry only
- B. Standard Gaussian Splatting only
- C. NeRF
- D. Gamma correction

4. What is the main idea of Gaussian Splatting?

- A. Represent a scene with 3D Gaussian elements
- B. Use only sound waves
- C. Remove all color
- D. Convert every image into text

5. What does ray marching do?

- A. Only measures sound

- B. Samples points along a ray
- C. Only builds polygon meshes
- D. Measures reaction time

6. What does perception engineering use?

- A. Only computer hardware specifications
- B. Human perception measurements
- C. Only polygon counts
- D. Only network speed

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7. What does psychophysics study?

- A. The relationship between physical stimuli and perception
- B. Only programming languages
- C. Only 3D printers
- D. Only camera lenses

8. What does the absolute threshold measure?

- A. Whether two objects have equal colors
- B. The minimum detectable stimulus level under a criterion
- C. The speed of a GPU
- D. The number of polygons

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9. What is another name for the difference threshold?

- A. HRTF
- B. JND
- C. URP
- D. NeRF

10. Which law uses the ratio ΔI/I?

- A. Stevens' Power Law
- B. Fechner's Law
- C. Weber's Law
- D. Snell's Law

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11. Which law describes a logarithmic relationship between stimulus and sensation?

- A. Weber's Law
- B. Fechner's Law
- C. Stevens' Power Law
- D. Moore's Law

12. Which law uses $S = kI^n$?

A. Stevens' Power Law

B. Weber's Law

C. Fechner's Law

D. Ohm's Law

13. What is a tuning curve used to show?

A. A computer's file names

B. Sensitivity at different frequencies or stimulus levels

C. Only the size of a 3D model

D. A network's IP address

14. What does ITD mean in spatial audio?

A. Interaural Time Difference

B. Internal Texture Design

C. Image Transfer Data

D. Interactive Touch Display

15. What is the purpose of an HRTF?

A. To measure GPU memory

B. To describe how the head and ears affect incoming sound

C. To train every NeRF model

D. To calculate polygon counts

16. Why is around 250 Hz discussed in the lecture's smartphone example?

A. It is the only frequency humans can hear

B. The lecture uses it as an example of high tactile sensitivity

C. Every phone must vibrate at exactly 250 Hz

D. It is the standard frame rate for VR

Check answers

Part 10: Last-minute memory sheet

Memorize these eight lines before the exam:

1. **Traditional rendering:** 3D model → 2D image.
2. **NeRF:** neural network + radiance field + new viewpoints.
3. **Gaussian Splatting:** many 3D Gaussian elements; fast rendering is possible, but memory and editability are challenges.
4. **Perception engineering:** measure human perception and use the measurements to improve technology.
5. **Absolute threshold:** can I detect it?
6. **Difference threshold/JND:** can I notice the change?
7. **Weber = ratio, Fechner = logarithm, Stevens = power.**
8. **Spatial audio:** ITD, ILD, IPD, and HRTF help model how we hear sound direction.

Most important exam advice: If you remember only one big idea from this lecture, remember that good VR is not just about powerful hardware or clever software. It is also about measuring how humans see, hear, and feel, then designing the technology around those measurements.

You said:

[Music] So today you know in the south India it is called the aid puja in the other parts of the India it is maybe sarasi puja. So sarasi puja and aya puja it may be related to our topics that we are going to talk about today. So maybe you know you can see how it is related. We are going to talk about a new topic uh called spatial audio today and in the last class we talked about the visual rendering pipeline but uh in uh in the rest of the another one week or two weeks we will be talking about the nurse audio rendering pipeline. Okay. Um there is one uh announcement. Uh basically we are talking about the next homework that next homework is on the visual audio display. Uh sorry uh the next homework is on uh now visual rendering pipeline. It is related to the now unity shader tutorial. Our TAs are going to talk about uh giving you a tutorial on Unity shader. They will announce the date sometime know coming Tuesday or Wednesday on the know uh one of the earlier tutorial dates whatever the tutorial dates are announced. Please look at the discord they will announce it shortly. So we have been looking at the visual rendering. Now we'll start talking about the audio rendering. Before we talk about the audio rendering, we briefly introduced the uh perception engineering in the last class. We will uh review perception engineering before we get into audio because we are going to use lot of one of perception engineering concepts in the understanding the audio as well. Right. So we started at this perception engineering saying that now uh the basics of fundamentals of the entire VR course the foundation of the entire VR course is the human perception. So it is not about the hardware, it is not about the software. We have to understand the human perception because human perception and is is what the experience is. XR is about experience and we need to do the quantitative objective measurement of the human perception. Right? So we also talked about um uh the um also talk about the the several no uh SI units of uh no existing uh existing uh no physical parameters but we wanted to add what could be the the SI unit or not SI unit what is the unit of measurement of the perception. Yes. So we wanted to add the perception measurement unit is what we wanted to see right. So in the last class I have given you a hint but know I did not know particularly mention it. I hope I wanted you to guess what could be this unit right. So we introduced a concept of the threshold which is measuring the limits of the sensitivity and two kinds of the thresholds we looked at it absolute threshold and difference threshold and we looked at how do we measure it we gave an example of know brightness uh in the screen for finding the absolute measurement absolute threshold and difference threshold and we talked about the um we talked about the um uh Feschner's law right uh because Fesner is the one who who introduced the concept of uh the sensation unit called the J&D J&D is a just noticeable difference which is measuring the uh uh difference threshold right so in this particular graph now we saw this the x-axis is the physical intensity of the no stimula Right? And then y-axis is the sensation or perceived intensity of the uh the the the stimuli. This is the perception measurement. Fesner is the one who introduced J&D just noticeable difference and he said no this first J&D zero is the the basic absolute threshold or one J&D is uh above this uh no absolute threshold. What is the just noticeable difference? There is the first G&D, second J&D, third G&D, fourth G&D, 5G and so on. So your given intensity of the uh stimuli can be said it's about 8 G and D or 5 G and D. Means above the basic absolute threshold now five different uh uh five J&Ds away or five just noticeable differences away each one comes as a simple no junk in between this two things we cannot perceive in between S2 and S1 we cannot perceive in in between S2 and S3 we cannot perceive so therefore this is a unit of perception J and D can be taken as a the like just like the other SI unit this J&D is is one of the you know units right then later on we saw that Stevenson came and he introduced the the power law the power factor can be you know another measure measure of it right so but although threshold is one of the mostly used thresholds there are other measures of psychophysical experiments that can go into the the SI units list accuracy reaction time point of subjective equality different ratio scales and there are other no measures also in the literature which we will know detail I'll we'll have a detailed discussion in the separate course on know psychophysics and we looked at several applications of this perception measures units Right. So in the visual perceptions we looked at um uh the the visual perception we looked at uh um uh the gamma display how the the y-axis is the y-axis is the perceived magnitude of the the actually intensity x ais is the actual intensity luminescence and yaxis is the y-axis is the perceived intensity. You can see that the ice have the uh upward bending of the u uh upward bending of the curve. According to our Steven's law, this has the power factor about 6 or whatever it is. Right? So this is called the gamma. This can be no uh whereas the the camera we saw that it is equal proportional which is not detected not perceived by the camera. It is directed by the camera and this difference is what is now making the now difference the photographs what you see on your cell phone and your real eye whatever is watching right and this is one of the applications of the the psychophysical curve uh that's in the physical uh sorry visual we can I can quote a lot of example but because of the time I'm just showing only one example each this is the the second c tuning c we saw In the tactile perception we saw in the as far as the mobile phone is concerned it is all vibrating at about uh 240 hertz that is because that is the frequency at which we have the highest sensitivity right similarly audio also we saw we have the maximum sensitivity of about 4 kHz and we looked at several other parameters from this human and tuning curve of the audio perception right so uh just to highlight uh just to know uh make sure that you understand what is the unit of the measurement I wanted you to know uh tell me what is the now uh unit of perception right so please u what do you what is the unit of the perception can you type in

please what is the unit of perception yes J and Yeah. So what is J&D? Just noticeable difference. You know what a J&D is. How do we measure J&D? Also we talked about it. It is a very important parameter or I want you to know take this as a an example take-home message from the entire course. When you when you look at the literature's when you look at the VR course know reading material now this parameter you will come across very often if you want to interpret a graph know your perception graph this understanding of this J&D is is very very important. Therefore, I want you to know u make sure that you understand what this is and uh now apply in uh several I'll know several concepts whenever we have come across as um uh this VR concept uh related topics. Okay. Okay. So that is uh that is with the perception engineering. Now let us continue this understanding of the perception to the spatial audio which is our next topic. Right. So as far as the now audio rendering is concerned we are going to talk about the fundamentals of the spatial audio uh we are going to talk about the spatial audio and the rendering techniques. Now interactive design and sound design and some advanced techniques for audio rendering right. So why do we want to study the no audio rendering? So know we are all using this know stereophones or headphones right and why do we need to study audio rendering? confuse whatever existing in the very it looks like it is very well established right audio is the one of the modality stimuli which is very very well studied maybe for the last 200 years or even more than that right then comes the visual and then comes the haptics so audio has been no thoroughly researched for 200 years and there's a lot of literature particularly from the perception point of view, right? So, how that is going to be useful in enhancing the the experience in the virtual reality is what what we are going to see. Before we see I want you to understand and the importance of this topics, right? Um can you tell me uh maybe from your point of view why do we need to know study this? Let me draw a parallel from light and from from light and sound. For example, in the last topics we talked about the ambient light and then diffused light and then specular light, right? And we saw that now the final rendering involves all the three combinations, right? Ambience is the now background noise, background light. Diffuse is now coming from the rough surfaces, right? Whereas specular is coming from the shiny surfaces, correct? And and an object will have all the three combinations. Correct? So do we have a similar things in this? Correct. So uh then and how the way we rendered in the visual rendering can we also now follow the similar things right and sound in ARVR is actually is going to be very very different. How it is going to be different understanding the know how sound in ARVR is different is very important to give you the know much know better experience of uh immersion. Now have you if if you are watching a movie for example if you switch off the sound what will happen will it be interesting the the experience of watching a movie without with and without the sound is a lot right and what we are going to see is that with a normal sound and with a know spatial sound is going to be very very different is what we are going to see how and why we let us see before we you know start discussing I want you to think about now what are the differences between the light and the sound so we know uh from the from our school days also we've been looking at the light and sound but do we know the differences between light and sound you all have had know engineering physics Right? Engineering physics we all discussed in detail about light and sound. But do we know the differences between light and sound? So that now we will review it over here. It will be helpful in understanding the the spatial audio. Right. Uh I would like you to know type in the your understanding of the differences between the light and sound. Yeah. So you are saying light is faster than sound. Okay. mechanical and electromagnetic. Okay. Um, very good. Light has dual nature. Sound is now wave nature. Very good. Sound will make it immersive in VR. Light contains photons but sound is just a no provocation. Very good. Effect of light is visible and effect of sound is audible. Okay. Forms of energy slow versus fast. needs medium versus do not need medium to travel. Very good. Light is electromagnetic whereas sound is mechanical. Very nice. Any other thoughts? We are looking at the the differences between light and sound. Any other thoughts? Mechanical waves affect vestibular system while light waves affect optical and nervous system. Speed is different. Obviously you have know somebody has pointed out earlier. Okay. This is from the neuroscience point of view. Okay. Sound can break objects and light can generate heat. Okay. Light is more important for human than audio. Is that correct? Okay. I'm not very sure. Any other thoughts? Can you Yeah. Some of you have mentioned that now electromagnetic whereas the other one is a mechanical wave. Very nice. But know can you go a little deeper in in that direction? Right is perceived by eyes and the sound is perceived by ears. Very good. Okay. All right. Let us look at so uh let us look at the light and sound from the no wave nature. Light is a uh no light is a longitudinal wave because it is a mechanical wave. It is not electromagnetic wave some of you have rightly pointed out right and there the motion of the matter is happening. So this way uh the movement of the uh the sound particles is the longitudinal right and this is the you know uh the the displacement is very high where this displacement is very low. So we can talk about like this mechanical wave displacement and here now very high this compressed state can be represented by this cenosidal wave. This is a mechanical wave. Whereas light is a transverse wave. Transverse wave means the particle is moving perpendicular to the the direction of the motion. Let us say this is the direction of the motion and this is the direction of the motion. Here the particle is moving up and down. Here now the matter which is having a know definite mass that is moving back and forth. Here the the light it is a photon. It doesn't have the mass that is moving up and down. In fact, no the electromagnetic wave is not only it has two perpendicular waves right in the perpendicular to the direction of the the light. One is the uh one is the electric wave another one is the the magnetic wave. Correct? and and this doesn't travel through the know uh vacuum whereas light travels through the vacuum and one more important and factor is

that because it is a transverse wave it can be polarized. Okay. So polarization is mainly due to the the transverse waves otherwise you know polarization would not happen. Yeah. These are the now uh important differences I would like you to now remember when comparing with the earlier visual rendering uh techniques what we are talking about. Right. Right. Given this differences between the light and sound, what is the difference between know uh no stereo sound and the surround sound? Then we can talk about what is the difference between stereo surround and the spatial sound. Is is there any relation? Is there any differences? Let us see. Right. Stereo stereo sound which we are everyday listening. Correct. We are talking about the the the fixed speaker setup. We have the stereos headphones which is attached to your ears. That's all fixed firmly fixed to your ears. When you turn your head, this head headphones are also turning along with you. Right? Is there any other differences? Can you think about head when turn head sound also turns with the head and in the stereo probably we can and uh we can hear sound coming from within the head. It cannot come from outside the head. the the stereo can make the sound coming somewhere in between you know one ear and the other ear. It cannot come outside the ears. Right? That is one of the major differences of the the stereo. But surround means there are fixed speakers in the room around the room and it is giving you the the the the feeling of of sound coming from somewhere outside the your ears. Correct? Yes. But oh um it can make the surround sound can make the sound coming as if it is coming from the front, left, right or back. Right? Is that good enough for virtual reality? What is missing? What do we want? What what do we want now extra in virtual reality in order to make it much more immersive? Correct. [Music] [Music] The spatial audio has more than the stereo as well as the sound. It is not only coming from the front, left and right and back as in the surround and the stereo but also above and below. This is one of the major differences. It is closer to the real world experience whereas surround is not real world experience. It is completely immersive. That is why it is not completely immersive. And we are going to talk about this is achieved we by now a function called transfer function called head related transfer function and it is achieved by something called bural rendering and it is achieved using the head tracking. Whereas in the sound there is no headtracking. All the all the speakers are fixed and then it is going to render the sound. irrespective of where you are, irrespective of which directions you have turned your head. Whereas surround sorry spatial audio is going to track your head and then depending upon on the direction of the head it is going to render different uh uh different audio uh that is using this head related transfer function. Okay. Now given the differences between the now surround and stereo and then spatial audio let us look at and uh now uh how a personalized spatial audio is possible which is not possible in the surround audio. Personalized spatial audio is that human anatomy is very unique to each individual. Particularly what happens is our ears have know ears play a major role in u in uh in determining how we perceive our sounds spatial audio right sound interaction with our ears is going to be very very important that has to be captured our ears you know each individual has a different ears and then therefore Now the head related functions personalized head related function is going to be different and that is going to make this personalized. Now spatial audio we are going to see in details in the coming slides. Spatial audio can be rendered using the fixed speakers as well. It is not just that we can generate uh spatial audio only with a headset which is moving your head moving through the head. We can put up the fixed speakers as well. But unlike the surround audio, we need to have the head tracking. Unlike the surround audio, we need to have the uh HRTF head related transfer function. Unlike the surround audio, we need to do the cross talk cancellation. What is cross talk cancellations? We will see in the coming slides. Those are the important differences right? Very recently I want you to know experience when you after maybe after the class when you go home Netflix has introduced a spatial audio. Okay. When you go to Netflix know search for know spatial audio supported what do you call movies or whatever it is and then see whether you can experience right in your home. Right. Apple AirPods are there. There are already commercial objects are there. And there are smart speakers are there giving you the spatial audio. And there are headphones with 3D audio are there. And there are sound bars. Sound bars are it's like know all the uh uh uh speakers are over here. But it can generate the spatial audio. It it generates uh so source uh virtual source of uh sound coming from above or below or back depending upon or using this the arrangement of object in your room. It initially scans your the room and then know personalize your know head related transfer functions and then renders this know spatial audio. There are commercial systems are coming up but we wanted to see how it could be improved in a virtual reality. In a virtual reality, I wanted you understand that the the audio is one of the most important things and uh it is going to give a lot of differences in the experience. If you do a good spatial audio, the experience is going to be way different like you know when you watch a movie with and without audio. So the experience is totally different. Correct? Similarly, if you have a very good spatial audio in a VR, the experience is going to be entirely different. Right? So that is why we wanted to see the fundamentals of the spatial audio and then see how we can render it. We can improve the improve in the virtual reality experiences. Correct? The history way goes way back 200 years. It is not very recent. Correct? So very recently 2012 10 years ago or 14 years ago Dolphie Dolby now Atmos and 4D sound bombers have been introduced and there are some uh basically now theaters which uh know uh which uh supports this uh spatial audio right 2006 there is a wave field synthesis has been introduced this is the first And spatial audio rendering has been demonstrated. It is not very new in a virtual reality. No application is going to be made it much more simpler. But 2006 itself first demonstration. Uh 1990s recombinant media has demonstrated the spatial audio. Ambiox is is another demonstration of the spatial audio. Amosics is something like this

particular arrangement of speakers around a person who is sitting in a chair and then know giving a impression of of the sound coming from now somewhere uh in the in the uh anywhere in the space what within this particular limited space right ambient sonics. So then uh the first one was the first two channel audio system that's a stereo uh uh stereo system which was introduced in 1981 but that time itself people have released the the technical difficulty of giving the the spatial audio. Although they started with the stereo systems, two channel audio system um it was now restricted only to within the two years but people have realized it and that there has been steady uh improvement and to know make no special audio right there are other names in the know books you will come across but I want you to be aware of the other names as well. This is called the immersive audio. It's nothing but a spatial audio object based audio ambiox or bural rendering bural audio all this means the same as you know spatial audio this is the now ambiox uh logo which I thought I'll take it from their website and then put it over there right and there are now formats introduced in the recent time uh 2012 Dolby almost at most and there is The MUS format H is a special format for the spatial audio 3D GTSX Sony 360 reality audio and AF. This is a new format open standard which Samsung and Google together they have introduced very recently. Right? All the formats also I would like you to be now familiar with. Before we get into the uh details of it, can we is there any questions on this? Do movie theaters have spatial audio? Uh very highend movie theaters will have the spatial audio otherwise they will have the regular surround system audio will be there. So as I told you there's a difference between the surround system as well as the spatial audio. Spatial audio means it has to take care of your head tracking. It has to take care of of otherwise uh you know it is not called the spatial audio. Uh do let us take all the things later. This I have answered. Do 3D earphones expensive or 3D earphones are expensive? There are some now spatial audio earphones are there. Yes. Uh the technology uh improves mean it is little bit of expensive. Yes, you are right. It is expensive. How did they find how to program sound in movies to make it come from a particular speaker in surround sound speaker set? This is a well addressed problem. What we are in this class we are talking about is the the surround sound and and uh not the surround sound spatial sound and by understanding this uh how it is spatial sound is rendered you will understand how the surround sound is working as well right is it possible we can introduce spatial audio in our final project that is using the mobile phone cardboard phone and then headphones yes your final homework is going to be using the spatial audio So you can introduce your spatial audio in one of the the earlier Unity run projects and that is going to be your the final homeworks in future. Can we get online class with a spatial audio zoom? Yes definitely online classes is it going to be useful? So spatial audio know we can experience it but is it going to add value to the know in the lecture classes unless the teacher is running from one corner to another corner and you need to keep track of keep tracking of it is it going to be useful that is the no uh question you need to answer yes it is possible you can do but know is it adding value in question three and two is sound quality good I feel that some applications don't have sound quality and some have now quality good quality sound. The sound quality is good but you need to program it. The 3D spatial sound you need to program it. How are you going to program it is what we are going to see in the classes right. Surround sound is better in case of zoom live classes. Definitely surround sound is better but what we are talking about is taking your ex experience of zoom live classes to the next level. Say in a Oculus imagine that you are attending the class in Zoom in Oculus and with a no spatial audio the experience is going to be different and will that be different will that add value you need to see suppose your friend and your peer sitting in the classroom he's asking a question from some corner and you need to talk to him you need to identify where the where he is talking from right so maybe there it may be useful Spatial audio can be useful in sleep music and and study music. Yes, there are a lot of applications coming uh uh uh in the internet but using the spatial audio. Yeah, you are absolutely correct. We could use spatial audio for hypnotism definitely that's poss possible right now. Uh coming back to so having know looked at what is the spatial audio what is the differences between stereo and audio uh surround audio let us look at now some of the psychopysics or some of the perception based studies for the spatial audio. Once we understand the perception of the spatial audio perception parameters then we can go ahead and design and and g for spatial audio or the rendering techniques for spatial audio or how do we improve or how do we give a better immersion and for the spatial audio right some of the perception and psychophysical parameters which you already seen in the last class for example that audio tuning curve we saw Correct. The audio tuning code we saw what is that it has the for example frequency versus as uh the threshold uh we saw that now at 4,000 and heads around 4,000 heads we have the maximum sensitivity. This is a threshold curve and from 1,000 to no uh 4,000 is where the maximum uh sensitivity of our ear is right and we have zero dB uh sound pressure level at 3 kohz uh at 3 kHz is about know uh this is where if you look at the 3 0 uh we have about somewhere here 3 kohz we have 40 dB SPL sound pressure level at 100 Hz 40 dB is going to be about over here that is going to be about 100 Hz 40 SPL it must be somewhere now it should be coming somewhere here right and difference limit that is a J&D is 3% at 3 kilog Z this is not in this slide but now we saw that now we can differentiate the know the frequency uh at.3% level uh that is at the 3 kHz the uh difference J&D at 100 Hz is actually 3% at a very low frequency it is 3% J&D is actually increasing J&D is not necessarily constant like knowber said weber said it is a constant but it is not necessarily constant Here in this case it is actually changing you can see we saw that the piano keys are are now differentiated at frequency level at 6% level right and we also saw that this curve also now uh DL for loudness there's a J&D for loudness is about know 12%. We cannot differentiate loudness which is within 12% difference. Right? Steven's power factor for

loudness is about 33. It is now bogging or no bending know very high. Therefore increase amplitude by 10 times loudness increases by only 2% because of this 33. Similarly reducing the amplitude by 99% and the loudness reduces only know 20%. These all are the facts. Now you all may have experienced it because for maybe for your neighbor room if you are staying in a hostel your neighbor room maybe you know blasting you know his favorite music right. So if he increases amplitude by 10 times now the loudness increases only by two that's because of this uh now power factor. Similarly if he reduces the amplitude by 99%. If you close your doors as airtight and you could reduce the amplitude by 99% but the loudness reduces only 20%. Although only 1% of the amplitude but the loudness reduces 20% because as because of this 33 um power factor right the most important psychophysical parameter relevant to spatial audio is the localization. Localization is the now we are very good at know determining where the what is the direction of the sound which is coming where is it coming from right so uh although we may be talking now so all of a sudden somebody is calling we turn our head towards the direction we have a very good turn on our localization right so um but that localization depends upon frequency. Suppose if we plot the error of localization for example error in in terms of the angle and and this error gets maximum around how much 3,00 3 kilohz at lower frequency the error is very very less at very high frequency also error is very less Why is this this error is getting maximum at the know 3 kHz or 4 kHz? We saw in the earlier graph 3 to one to one to now 4 kHz is where the maximum sensitivity right the sensitivity is traded with the localization error which is which is needed now do we need the no sensitivity or localization. So the nature has decided that no sensitivity is much more important than the localization to compensate the localization. Now the nature has some other mechanism. We'll find out what are the other mechanism. Whereas to compensate the the the sensitivity there are no other mechanisms. Right? That is why nature has chosen the sensitivity as more important than the localization. So how are we hearing this mechanism? So uh to understand how we actually hear uh all the spatial sound particularly we need to now understand and this three parameters right so when we have the head over here and there is a source of sound coming from and this the sound is a mechanical wave which is propagating and it's a mechanical wave is a sperical wave and the wave friends which is going hit this left ear or right ear first and then much later the the left ear there is a difference in the time that time is called the interoral time difference ITD we call it as right so you can see that as the wave is now traveling the intensity of the the uh wavefront also reduces you can see That right has this much intensity. Uh the intensity is very high. The left when it reaches over here and it it is now intensity is very less. This is a inter level differences. Inter interoral level differences that the amplitude difference also is a cue for us to know uh to see the spatial hearing. Right? This inter time differences. We saw this is a right and then left. This is the interoral time differences. There also can be third one interphase differences. This is interoral phase differences. The phase of the signal from right to left there may be slight differences that also can be perceived. These are the three parameters which can be used to find out where the uh to help us in the localization of the the the the sound source. Right? All those things are related in a function called transfer function called header related transfer function which we are going to see. [Music] [Music] Before we see the head related function, I wanted to do now uh dwell little bit more into the interoral time differences. Can we calculate what could be the the the the range of interal time differences? Suppose if you have a source like this and the wave front is coming over here, right? It reaches the the right ear first and then finally it reaches the the left ear. So the interoral difference time difference could be this and this. This is the difference. This is the difference. Correct? And suppose if this is a head and this is the uh this is the what do you call uh now the parallel between this one with the wave front and this is causing the theta and if you know the radius is r and we can say this is the r theta and this distance is going to be r sin theta. Therefore, interal time difference it we can say is equal to R theta plus R sin theta divided by C that is the the sound speed. Okay. So r let us take it's about know 20 cm and theta let us calculate the maximum theta let us say no maximum over here in in radians let us say right so we know the maximum it's going to be this source somewhere in the in the in the uh right side it means it is going to be now maximum theta that is going to be now pi Right? So if you substitute for the value of a C let us say 340 m/s the sound speed what we get is the ITD it's about u it's about uh uh 6 milliseconds 65 or 6 milliseconds you can see that this is actually less than 1 millisecond interal time difference is about 65 milliseconds. Right? How does this affect our our uh our uh no spatial hearing? Right? So here is another graph where it shows the face difference is not much perceivable to human being. The ear is sensitive to interal phase differences only at lower frequencies and the sensitivity phase uh begins to deteriorate above about 1 kHz. So here is the two waves where we are talking about 1 kohz wave and 3 kohz sine waves. If you sum these two things, we get uh this time domain signal. And uh uh in this graph, the same 1 kohz and 3 kohz sine waves. But before summed up, this 3 kohz is inverted. Inverted and summed. And because it is inverted and summed, the shape of the wave is much different from the the other wave. Still we don't differentiate between this signal and this signal. What this is showing is that our sensitivity to phase is actually very very less over the now 1 kohz. Okay. But below 1 kohz there may be some phase differences sensitive but not above now 1 kohz. Right. Um I'm going to show you uh there is one more factor which we have to talk about it and um this ITD whatever we talked about the ITD and ID level differences and IPD phase differences right all those things uh is not going to help with within certain areas. For example, here is suppose if there is a there is a object sound object either slightly to the the left side of it uh to the right side and then front that is actually know uh sounds similar to know an object which is uh at the at the right side but behind we cannot

differentiate between the sound source in these two locations the front as well as the back. Similarly, uh the same level up and down as well. The front up down uh or the right up down we cannot differentiate it. The whole area in which we cannot differentiate the sound location is called the cone of confusion. In fact, there can be now lot of applications of this cone of confusion now which we are going to talk about later. Right? So coming back to the now localization uh local uh location dependent attenation lab this uh map. So the atonation what we talked about the the the uh the the the um the level differences right? We talked about the the interoral level differences as it is reducing that reducing depends upon the location dependent it depends upon where exactly it is and better localization with one and hide if I'm going to tell you now how it is. So here is a here is a a map which talks about the frequency versus whether or no it is a front top back or bottom or uh back. So suppose if you have this uh uh head let me see whether I have the the uh graph. So in the front you can see that this is a color map which is talking about the gain and know the higher the gain means uh this is a red color right. So it shows that at very high frequency we have this red region in the front. It means it means we have a the higher gain in the front region. Okay. So it is not atonating. It is not um it's not um atoniating um in the uh here it is going to be u now negative means it is highly atonating. Wherever blue color you see it means it is atonating highly atonating. You can see that it is at the bottom it is highly atonating at the back it is atonating and the back you know here also it is atonating top also it is atonating so it depends upon where we are looking at it's the atonation depends upon the location that is what this graph actually shows and this atonation depends of uh it also depends upon the the shape of the air okay everybody Each one of us have a different shape of the ear and this different shape actually modifies the the audio sound audio signal before it gets into the ear right. So that is also no changing the atonation map. So if you have uh this is called the actually this is called the HRTF. This particular map is called the HRTF. You know where exactly now atonation happening where exactly your maximum gain is there for each individual. This is capturing the the uh the the uh the anatomy due the sound and perception due to the anatomy of the the ear shape and other factors all included. This is one example of a HRF one now one representation of HR HRTF I'm going to show you no other HRTFers. Okay. So this HRDF can be measured for several people and then can be generalized and then generalized HRF is what is going to be useful for you in the the Oculus settings unity setting for example. But if you use your own HRDF uh it is much more effective right. So how are you going to get out your own HRDF is what uh we will see in the coming classes right. So uh again um this year shapes is changing the HRDF is what I wanted to now show you over here location for example what happened is we is a very interesting study people have studied people have conducted somebody called Hoffman in 1998 what he did it is he put a small putty putty you know what is it it's a elastic material or no uh hyper elastic or visco elastic material. He added this elastic material and then changed the shape of a human being and then he conducted the the and he calculated the the location dependent attenuation map basically HRDF and it turned out to be the HRDF is different and it it holded because of the change in the shape of the ears. Okay. So uh this is the original HRTF elevation map and this is the modified because of the the putty added to the years. You can see now lot of differences particularly you can see that now here there's a lot of red whereas the original one only here there is a red you can see that now a lot of changes. There is other ways of know looking at the the HRTF as well. Yeah. Suppose if we know say now theta is a lateral angle. Lateral angle is this one. Okay. And pi is the polar angle. Polar angle is this way. Let us put the no lateral angle in the x-axis and then polar angle in the y- axis and uh now uh this seems to be the French then this will be the top this will be the bottom and this is going to be the left and this is going to be the right and um uh this u now intensity of the the graphs will tell you whether the the the the localization error is actually uh lesser more. Uh if it is a darker it is lesser if it is white it is more more error localization error. You can see that at the top it's a highly no white at the bottom also it's a highly white right and uh um and about um 45° 2 can we say yeah so uh with this one let us say what are the now what are the So the best accuracy is in the front of the listener. You can see that the whole region is actually darker. Right? The best localization accuracy is in the front of it. Optimal performance between minus 45 to + 30. We are talking about minus 45 to now + 30 around this. This is the optimal performance. The worst is above and behind. This is the uh worst and then behind is no here. We are talking about here right. And then in the front of front hemisphere we are talking about front hemisphere responses are towards the loc locations just above horizontal plane. There's also arrow marks. You can see the arrow marks you know points or know gives the direction of of the error. So you can see that in the the front hemisphere front hemisphere is nothing but like now we are talking about the front of the hemisphere. In the front he hemisphere all the arrows are know towards towards the horizontal plane towards just above the horizontal plane. You can see that all the arrows are now above the horizontal plane. In the rear hemisphere the responses are biased towards as uh towards the left and right poles. What are the poles over here? The top is a pole and the bottom is a pole and all the uh all the know errors are actually know pointing towards the the pole and that is the observation. So this observation we may have to know use it in our you know in our uh programming it right. So given this psychopysics of the the the spatial audio can we make use of it? Can we apply for it? Do you know any you know uh application of this as all the psychophysical or psychopysics of the spatial audio to any such such applications. Let us see what is atonation. Atonation is the reduction in the know amplitude of the signal or intensity of the signal we can say right atonation weakening of the signal. Yes, I prefer recorded lectures over live lectures on. Okay, so I I I was asking what is the now application of this all this HRDF one very interesting recent application

of the HRDF is this a robo has been designed in Japan. Now people have uh uh have designed a robo with uh with several shapes of the ears so that robo can behave or can perceive sound just like a human being. So robos can have the HRTF. So change in the uh sorry robo with the artificial ear shape that also now can do the uh can do the localization and we can change the localization depending upon the shape of the air that is one application people are actually looking at it. So all the airports and other gadgets you know sound bars whatever we discussed it's all using this understanding of the psychopysics. So that's all we can say it's a perception engineered objects right apart from this uh know the localization there are many other factors we can actually look at it. For example, here we are talking about know how time and the phase can be used as a cues in in uh in spatial audio. Right? So this graph shows us monoral transfer functions of the left ear for several directions in the horizontal plane relative to sound incident coming from the front. In a unequ chamber 2 m loudspeaker distance impulse technique 25 subjects are used complex averaging a is the level differences here we are talking about the level differences x-axis is a frequency y ais is the level differences and amplitude or atonation yeah right level differences basically del right um and then yaxis is the time differences this is the time senses you can see that now uh this this bar is about 10 dB DV level difference. So if that is the case the first graph is going to talk about 30 at a 30° how this u monoral transfer function is not the header transfer function monoral transfer function is with respect to the frequency how this this graph is changing with know other angles. For example, this is a 60. You can see how it is changing. And this is 90, 120, 150, 180. 180 is until the the extreme know right or left. And then you're talking going behind 210, 240, 270, 300, 330. Given this graphs, can we make anything out of it? Any observations you can make? Similarly here this graph we are going to talk about the the frequency versus the the time delay. This case it is the level differences amplitude differences. Here we are talking about the the the time delay. Again for 30° it is going to look like this. Now this is a 0 dB line plus.25 minus.25 you can see and this is a 60° 90 similar this one is going to talk about 1 millisecond. If that is the case, you can see the the time delay is going to be no near zero near zero and the maximum we can talk about is about uh about uh you know less than milliseconds because we saw in our earlier the maximum ITD can be 65 milliseconds not uh no it is much less than Now 1 millconds right let us look into the some of the details of this. The first graph we can say the sources to the rear gives rise to a reduced high frequency response compared to those at the front going to the the forward facing shape of the penina. Our ears are facing forward and that is shown in this particular graph. Let us see how it is. So from here it's all rear and this is forward because we are talking about 180 here and this is 210. 180 is the straight right. Correct. And you can see that at in the uh in the 210 all these graphs are in the negative direction. Negative direction means no uh reduced high frequency response. The high frequency responses are here you can see very very reduced. Whereas here above in the front side the high frequency responses are actually better. That is essentially because of the forward facing shape of the pina. Pina is the our ears. Uh similar effect we can see over here also in the in the in the know time difference as well. As far as the level is concerned we can observe one more now psychophysical parameter here. Suppose we have a now two waves. One is the first primary wave and the secondary wave and the secondary wave. Suppose if we control the time delay, how much time delay I need or how much know level differences I need in order to know sound make the secondary wave sound similar. All right. So this is called this graph is showing the relative level difference required of a delayed reflection for it to appear equally loud to an primary source. Okay, that is a graph. So we are talking about it is maximum it is about know 12 milliseconds we have the maximum um uh maximum level difference at this place. After that it is actually reducing it. So until then it is increasing so that as the time delay increases the level difference required to make it sound and know same as the primary source is actually increasing maximizes at about 12 milliseconds and then come over there. Can we use that information somewhere? you you can actually design or use it in some particular applications, peculiar applications, right? So we can actually trade time and level differences. So we can exchange between the whether we want to do the level differences or time differences. Right? This is a binaural time intensity trading effect by one of the famous authors. So x-axis is the interal time delay. Y axis is the interal up amplitude difference. You can see that as we increase the interal time delay and with below this as curve it is know we have a tendency towards the earlier sound and over this curve we have the tendency towards the the louder sound. If the level difference is very high then we perceive the louder sound second. Whereas uh if it is the uh if the delay is here for example then we have the first we perceive towards the ear sound. So you can see also see that this number is about 65 as we earlier calculated 65 after this after the 65 it saturates we cannot have more than the 6 that is a maximum delay we could see right and uh within this 65 we can trade the no time and level if you if you have this uh the amplitude more then we tend to perceive the the louder sound. If you have the amplitude lesser then we tend to perceive the earlier sound that's all it matters right. So this is the time amplitude trading possible. Whereas in this level always towards the earlier sound unless delayed sound is more than the 12 dB and after this it comes the echo right. Um so with this psychopysics of the spatial audio let us see how it can be rendered it can be improved. I wanted to uh I wanted to know uh end today's lecture with a with a practical application of of uh now one of the commonly available material in India the whole of know um northeast we have lot of bamboo and the bamboo is very good in absorbing the the sound. Can you find out why? And it absorbs particularly no sound above thousand htz and it is poor in absorbing sound below thousand htz right I'm sitting in a know studio here in madas npel in a room where it is nicely you know soundproof echo proof how this

echoproof materials are working knowing this psychopysics can you think about how this soundproof rooms can be designed. How bamboo can be no uh incorporated in a soundproof room instead of a synthetic material like this? Why bamboo has such a property? All these questions I would like you to know ponder about and uh with this ponder we'll meet the next class. Meanwhile, let me see whether you have any questions. Okay. I couldn't understand the cues time phase graph. Is there something online resources to better understand this subject? Um I will um I will I'll definitely give you some reading material on this and give you the online materials as well. One application may be in military training training is one of the very important and context any training not only military training any training where 3D audio is so crucial definitely military is one thing because he has to be highly alert. He cannot see but if there's a small sound he has to accurately now localize it and then turn and then target the enemy. Yes, you are absolutely correct. How and where the measurements were made to plot this elevation maps and then grass. How measurements are done? We will see in the next class. Like in the visual rendering we put many cameras right to get the visual uh feel. Similarly here we put a lot of mics and uh uh looking at the the mics signal processing we can find out what is the uh what is the the where the sound is coming from or put a small mic in the ear itself the human ear and then see how much uh signal is reaching his ear. Okay. There are several measurement methods and there are several databases available which we'll see in the next class. Can we summarize the Q slide one more time with more example? Okay, that's a good idea. Yeah. So we are talking about now Q's time Q's and phase cues. Correct. In this uh graph we are talking about monal. It is not a bal monal. Here it is a frequency graph and this is the level difference. This is the amplitude difference. Right? As a frequency increases how the um amplitude of a particular signal coming from the front of the face is actually changing uh at a suppose if you keep a mic at a 30° how is it changing? Right? So uh this is the graph and uh uh the the horizontal line is the 0 dB and it is know above horizontal line it means positive. It means you know there is the there is the uh it is not a decay it it is um the the level difference is more positive level difference. So one full one full from here to here is the 10 dB. So we can say that from I know 2.5 yeah from 2.5 to 2.5 is five and we are talking about the 10 dB uh difference. So uh this place it is actually 0 dB. It means it is equal to the the the reference and if it is negative it is actually reducing decaying. it is positive. It is not uh increasing it. The the main observation what I want you to make is that from the front to left this is the sorry from the front to back this is the back this is the front in the front all the high frequencies are not decaying whereas in the left all the high frequencies are know decaying this is what it mentioned over here right so uh Yeah, the graph shows that reduced high frequency response compared to those at the front rowing to the slightly forward- facing shape of the pina. This change from the front to the left rear is because of the forward facing pina. Our ear the shape of the ear is actually know making the difference is what the message I wanted to you know convey. And this is the level differences we are talking about. Now secondary wave, secondary time delay wave. This is another technique we are talking about. How do I create a secondary wave in order to make the uh uh uh secondary level difference to to match with the the primary level. This is the the relative level required of a delayed reflection. That's a secondary wave for it to appear equally loud to an ADR primary source. So it is up to 12 milliseconds it is increasing it and then after this it reduces. So we can actually trade as well time and level trading as also possible. So the maximum uh time difference is about 0 65 which we have seen it and until then we can trade level and uh time after that is not possible and then of know beyond know certain know this is about um uh 30 hertz this is only echoing is possible right so I hope uh now I'll I'll send you some reading materials on this I hope this will clear because of micro structures on the bamboos. You can look on it. Fibrous materials absorb sound dampening. Finally summarize a Q slide one more time with more examples. Sure, I will do small sparrows love to stay within the bamboo plant. So maybe that is why we cannot hear them. Maybe uh I haven't observed it. Level difference versus group difference. Um I don't know what is group difference. What you mean? from the previous topic spatial audio in theaters. Do floors also have sound reflectors or do they absorb the sound? um uh I don't have the idea maybe you can uh actually know look at theaters design you know audio design itself is a big profession you can look into it where can we find PDF of this PPSTS I'm going to share the PPS of all the lectures in the same lovely website you can now look at it along with the reading materials in each of the slides I'll give you the the the link to the reading materials you can now uh dwell deep into it. Okay. Yes. All right. So, thank you for our uh listening. So, to summarize it, psychopysics of the spatial audio is very important. We looked at what is the difference between the spatial audio and the normal stereo audio or the surround audio and understanding of this psychopysics. We are going to design the rendering techniques or the spatial audio uh uh techniques or the compression techniques in the next classes and we'll look at some of the advanced topics uh in the next coming classes. We'll stop here. Stay safe. Until then uh uh goodbye. See you. [Music] lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

Spatial Audio — Easy Notes for a 7th-Class Student 🎧

Let's understand the lecture in very simple words, with everyday examples, memory tricks, important formulas, and MCQs. The most important idea is this:

Spatial audio makes you feel that a sound is coming from a particular place around you — in front, behind, left, right, above, or below.

Imagine playing a VR game. A bird chirps above your head, footsteps approach from behind, and a car passes on your left. Spatial audio helps you identify where each sound is coming from, making the virtual world feel real.

1. First, revise perception engineering

Perception engineering means studying **how humans sense and experience things**, such as light, sound, touch, and brightness.

In VR (Virtual Reality), we want to create an experience that feels real to a human being. So, we need to understand how humans actually perceive the world.

What is the unit of perception?

Your lecturer emphasized **JND — Just Noticeable Difference**.

JND means the *smallest difference between two stimuli that a person can notice*.

Example: Brightness

If you slowly increase your phone's brightness, at some point you notice that it has become brighter. The smallest noticeable change is a JND.

Example: Weight

If two objects feel almost identical in weight, you may not notice the difference. Add enough weight to one, and you can detect it.

Remember these two thresholds:

Threshold	Simple meaning
Absolute threshold	The weakest stimulus you can detect
Difference threshold	The smallest change you can notice; measured using JND

Memory trick: Absolute = *Can I detect it?* Difference = *Can I notice a change?*

Other perception concepts

- **Stevens' power law:** The physical strength of a stimulus and how strong it feels are not always proportional.
- **Frequency:** How many sound-wave vibrations happen per second, measured in hertz (Hz).
- **Loudness:** How loud a sound feels to a person.
- **Sensitivity:** How easily you can detect a stimulus.

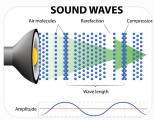
Your lecturer highlighted that human hearing is especially sensitive around the **2–5 kHz range**, with a commonly cited peak near 4 kHz. The exact sensitivity depends on the listening conditions and person.

2. Light vs sound

This comparison helps explain why audio rendering is different from visual rendering.

Light

- An electromagnetic wave.
- Can travel through a vacuum.
- Travels at about 3×10^8 metres per second in a vacuum.
- Human eyes detect it.
- Can be polarized.



Sound

- A mechanical wave.
- Needs a medium, such as air, water, or a solid, to travel.
- Travels at about 340 m/s in air under ordinary conditions.

- Human ears detect it.
- In air, sound waves are longitudinal: air particles vibrate back and forth along the direction the wave travels.

Easy memory trick: Light can travel through empty space; sound needs something to travel through.

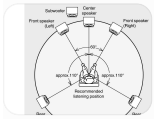
A small correction to the lecture's wording: light is a **transverse electromagnetic wave**, not a longitudinal mechanical wave.

3. Stereo vs surround vs spatial audio

This is one of the most important exam topics.

A. Stereo sound

- Usually uses two audio channels: left and right.
- Sound can seem to move between the ears.
- Often feels as if the sound is inside your head when using headphones.
- Does not, by itself, reliably create a complete 3D world that stays fixed as you turn your head.



B. Surround sound

- Uses multiple speakers arranged around the listener, or processing that simulates such a setup.
- Can create sounds from the front, sides, and rear.
- Traditional fixed-speaker surround systems do not automatically track your head.

C. Spatial audio

- Creates the impression of sound at a particular 3D location.
- Can represent sounds in front, behind, left, right, above, or below.
- Often uses head tracking in VR so the sound appears to remain in the virtual world's location as you turn.
- Can use headphones or suitable loudspeaker systems.

Feature	Stereo	Traditional surround	Head-tracked spatial audio
Left/right sound	Yes	Yes	Yes
Front/rear placement	Limited or simulated	Yes	Yes
Above/below placement	Not inherently	Depends on setup	Can be simulated
Head tracking	Not normally	Not normally	Often used in VR
Personalized ear effects	Not usually	Depends on system	Can use personalized HRTF

Memory trick: Stereo = 2 channels; surround = sound around you; spatial = sound positioned in 3D space.

These categories can overlap: spatial audio can also use stereo headphones or surround-speaker arrangements.

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4. What is HRTF?

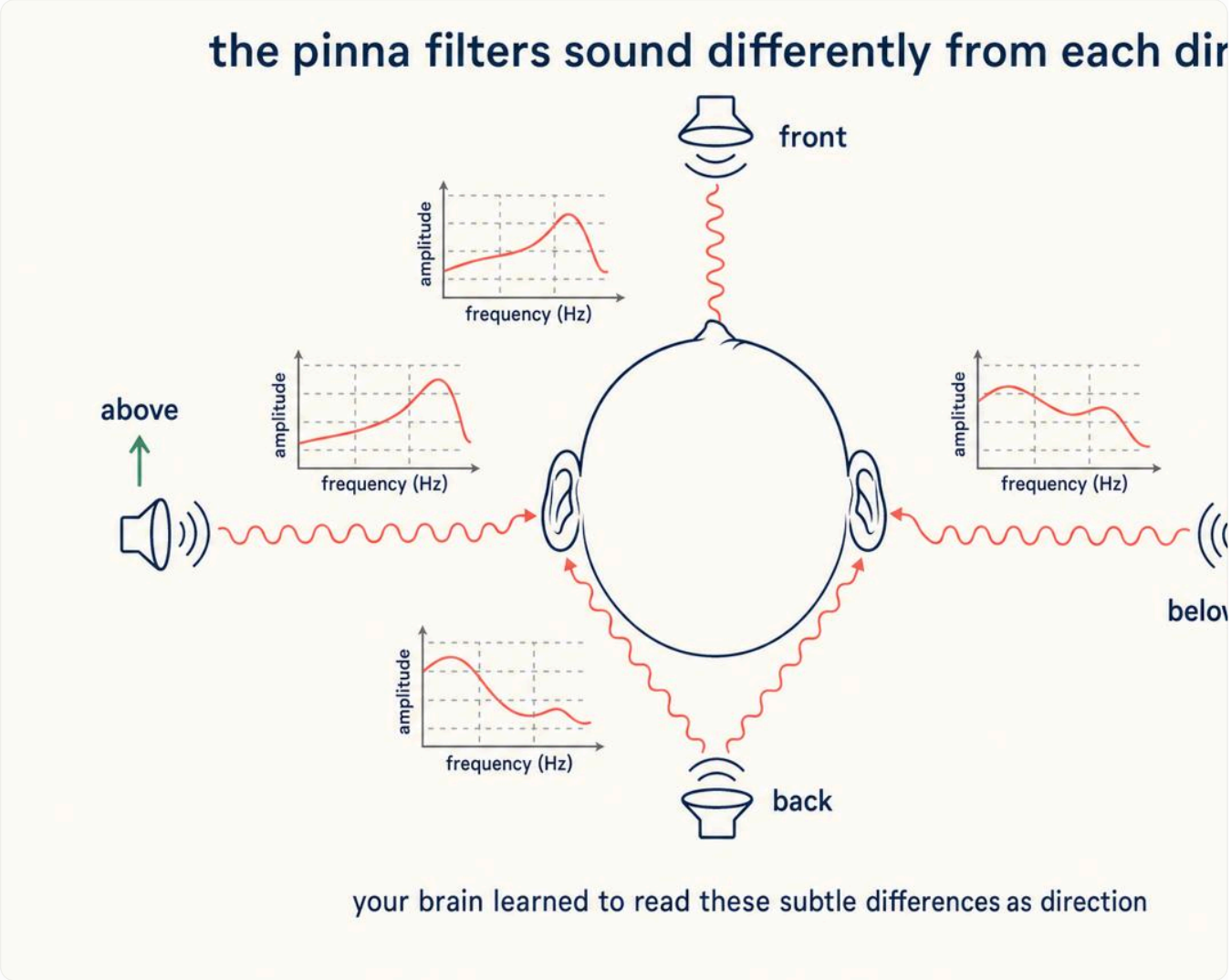
HRTF stands for **Head-Related Transfer Function**.

It describes how your head, ears, and body change a sound before it reaches your eardrums.

Imagine a friend claps:

- In front of you, the sound reaches your ears in one way.
- Behind you, your ears receive a slightly different sound.
- Above you, the shape of your ears changes the sound differently again.

Your brain learns to use these differences to estimate where the sound came from.



Why is HRTF important?

The shape of everyone's ears is a little different. Therefore, different people can have slightly different HRTFs.

- **Generic HRTF:** A general model used for many people.
- **Personalized HRTF:** A model adjusted to an individual's ears and anatomy.

A personalized HRTF can make virtual sound locations feel more accurate for that listener.

Memory trick: HRTF = *How your Head and ears Transform sound before it Reaches you.*

How is HRTF used in VR?

1. The computer knows where a virtual sound source is located.
2. It processes the sound using an appropriate HRTF.
3. It sends the processed sound to the left and right ears.
4. Head tracking detects the direction you turn.
5. The audio is updated so the virtual sound appears to stay in its world location.

For example, suppose a virtual bell is on your right. If you turn your head toward it, the system adjusts the sound reaching both ears so the bell still appears to be in the same place in the virtual world.

5. Binaural rendering and cross-talk cancellation

Binaural rendering means creating the appropriate sound for the left ear and right ear separately, so your brain perceives a 3D sound location.

It is especially useful with headphones.

Cross-talk cancellation matters when using loudspeakers. Sound from the left speaker can reach both ears, and sound from the right speaker can also reach both ears. This mixing is called cross-talk.

Cross-talk cancellation uses signal processing to reduce the unwanted mixing and help produce the intended spatial impression.

Memory trick:

- **Binaural:** Two ears.
- **Cross-talk:** Sound from one speaker leaks to the other ear.
- **Cancellation:** Reduce that unwanted leakage.

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6. The three main sound-localization cues

How does your brain know where a sound is coming from? Your lecturer explained three important cues.

1. ITD — Interaural Time Difference

The difference in the time a sound takes to reach the two ears.

Example: A sound coming from your right reaches your right ear slightly before your left ear.

Memory trick: **T = Time.**

2. ILD — Interaural Level Difference

The difference in sound level or loudness between the two ears.

Example: A sound coming from your right may be louder at your right ear because your head partly blocks the sound reaching the left ear, especially at high frequencies.

Memory trick: **L = Level.**

3. IPD — Interaural Phase Difference

The difference in the phase of a sound wave arriving at the two ears.

Example: At lower frequencies, the wave at one ear may be at a slightly different point in its cycle from the wave at the other ear.

Memory trick: **P = Phase.**

Cue	What is compared?	Simple example
ITD	Arrival time	Which ear hears it first?
ILD	Sound level	Which ear hears it louder?
IPD	Wave phase	How do the wave cycles differ?

Important: ITD and ILD are especially useful for left-right localization. IPD is most useful at lower frequencies; phase becomes ambiguous at higher frequencies.

How small is ITD?

The difference is tiny. For a human head, the maximum interaural time difference is roughly **0.6–0.7 milliseconds**, depending on the model and head size.

The transcript appears to have lost a decimal point when mentioning “65 milliseconds.” That value would be much too large for ordinary left-right ear-arrival differences.

Remember: **the sound reaches one ear first, but the delay is less than one millisecond.**

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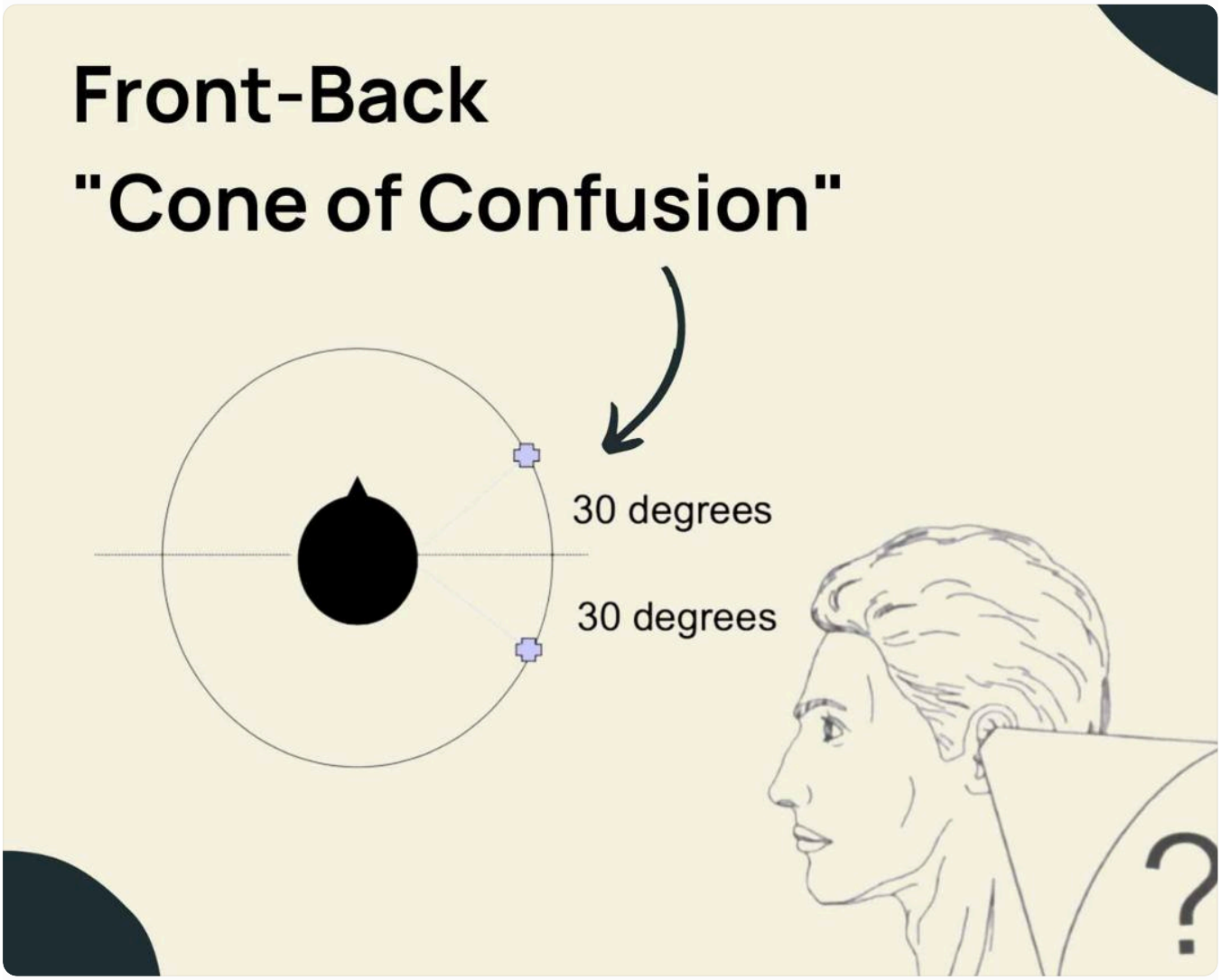
7. What is the cone of confusion?

This is another important concept.

Sometimes, two sound sources in different positions produce similar cues at your ears. Your brain may find it difficult to tell which position is correct.

For example, a sound coming from a position in front of you might initially seem similar to one coming from behind you.

The set of locations that produce similar localization cues is called the **cone of confusion**.



How does the brain solve this problem?

It uses additional information:

- The shape of the outer ears (pinnae).
- Small head movements.
- Changes in the sound as you move your head.
- Familiar sounds and experience.

Memory trick: Cone of confusion = *Your ears have trouble deciding exactly where the sound is.*

8. What is attenuation?

Attenuation means a reduction in the strength or level of a sound.

Think of a friend speaking to you:

- Nearby, their voice sounds strong.
- Far away, their voice often sounds weaker.
- If a wall blocks the sound, it may become weaker still.

Your head and outer ears also affect sound strength differently depending on the direction and frequency of the sound.

Why does ear shape matter?

The outer ear is called the **pinna**. Its shape modifies sound before it enters the ear canal.

Your lecturer's main observation was that sounds arriving from different directions can have different high-frequency responses. These direction-dependent changes help us distinguish sound locations, particularly front from back and above from below.

A change to the ear's shape can change its HRTF. That is one reason a personalized HRTF may work better than a generic one.

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9. Localization accuracy: where do we hear best?

Localization means identifying the direction or position of a sound source.

Your lecturer described experiments in which people were generally more accurate at locating sounds in the front region than in some positions above or behind them.

The main takeaways from the lecture are:

- Localization accuracy depends on direction.
- Some directions are harder to identify than others.
- The outer ear helps resolve ambiguities.
- Sound frequency affects localization.
- HRTF captures important direction-dependent changes.

This is useful in VR games, navigation, military training simulations, robotics, and any situation where a person needs to identify where a sound is coming from.

10. Time and level trading

Imagine two sounds:

- The first sound arrives early.
- The second sound arrives later but is louder.

Which one will you perceive as the main sound?

Your perception depends on both the arrival-time difference and the level difference.

This is called **binaural time-intensity trading** in the lecture.

In simple words, the brain uses timing and loudness together when deciding where a sound seems to come from.

The lecture also discusses delayed reflections: a reflection's level and delay affect whether you hear it as part of the original sound or as a separate echo.

Memory trick: Your brain considers both *when* the sound arrives and *how loud* it is.

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11. Why does bamboo absorb sound?

Your lecturer ended with an interesting question: why can bamboo and other fibrous materials help absorb sound, particularly at some frequencies?

Soundproofing materials work in different ways. Some absorb sound, while others block it or reduce vibrations.

Fibrous or porous materials can allow sound waves to enter tiny spaces. Some of the sound energy is converted into a small amount of heat through friction and other losses.

However, **bamboo does not automatically absorb all sound well**. Its performance depends on its structure, thickness, arrangement, and the frequency of the sound. The lecture specifically suggests that the bamboo example is more effective above about 1 kHz than below it.

How are soundproof rooms designed?

- Absorbing materials reduce reflections and echoes inside a room.
- Heavy, airtight barriers can help block sound from entering or leaving.
- Special room shapes and treatments help control reflections.
- Bass frequencies often require thicker or specially designed absorbers.

Remember: Sound absorption reduces reflected sound; sound blocking reduces sound passing through a barrier. They are related, but they are not the same.

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12. History and applications of spatial audio

The lecture mentions that spatial audio is not a brand-new idea. Researchers have explored it for many decades.

Some names and formats mentioned in your lecture include:

Name	What to remember
Stereo	Two-channel audio
Wave Field Synthesis	Uses multiple sound sources to reproduce a sound field
Ambisonics	A method for representing and reproducing sound around a listener
Dolby Atmos	An immersive audio format that can represent sounds at different positions
DTS:X	An immersive audio format
Sony 360 Reality Audio	A spatial music format
MPEG-H	An audio standard that supports immersive and object-based audio
Binaural audio	Audio designed for the perception of sound at both ears
HRTF	Models how the head and ears affect sound from different directions

The lecture also mentions spatial audio in headphones, smart speakers, soundbars, movie theatres, and VR headsets.

Real-world applications

- **VR games:** Hear footsteps behind you.
- **Military training:** Identify the direction of a sound.

- **Movies:** Make sounds seem to come from different places.
- **Robotics:** Help robots estimate sound-source direction.
- **Virtual classrooms:** Help users locate virtual speakers around a room.
- **Music and entertainment:** Create a more immersive listening experience.

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13. Important formulas and terms to remember

Term	Easy meaning
JND	Smallest noticeable difference
ITD	Difference in sound arrival time between the ears
ILD	Difference in sound level between the ears
IPD	Difference in sound-wave phase between the ears
HRTF	How the head and ears modify sound
Binaural rendering	Rendering sound for the two ears
Attenuation	Reduction in sound strength
Localization	Identifying where a sound comes from
Cone of confusion	Locations that produce ambiguous localization cues
Cross-talk	Sound intended for one ear also reaching the other
Head tracking	Detecting the direction in which the listener turns their head

For the time difference calculation, the basic relationship is:

$$\text{Time difference} = \frac{\text{Path-length difference}}{\text{Speed of sound}}$$

The lecture's simplified head model estimates the path-length difference using the head radius and the direction of the sound. The result is a very small time difference, generally less than one millisecond.

One more caution about the transcript: it appears to have transcription errors in its discussion of Stevens' power law and loudness. Remember the central concept rather than memorizing the suspicious numerical claims: **physical sound intensity and perceived loudness do not increase in a simple one-to-one relationship.**

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14. MCQs for exam preparation

Try answering these questions before looking at the answer key.

Spatial Audio Quiz 0/15 answered

1. What does JND stand for?

- A. Just Normal Difference
- B. Just Noticeable Difference
- C. Joint Noise Detection
- D. Just Natural Direction

2. What is the main goal of spatial audio?

- A. Make every sound louder
- B. Remove all background sounds
- C. Create the impression of sound at a particular 3D location

D. Increase the number of video frames

3. Which type of wave is sound in air?

A. Electromagnetic transverse wave

B. Longitudinal mechanical wave

C. Light wave

D. Radio wave

4. Which type of wave is light?

A. Longitudinal mechanical wave

B. A wave that requires air

C. Transverse electromagnetic wave

D. A sound wave

5. What does HRTF stand for?

A. Head-Related Transfer Function

B. High-Range Time Frequency

C. Head Rotation Tracking Format

D. Human Response Timing Factor

6. Which cue measures the difference in arrival time at the two ears?

A. ILD

B. IPD

C. ITD

D. JND

7. Which cue measures the difference in sound level between the ears?

A. ITD

B. ILD

C. HRTF

D. IPD

8. IPD refers to a difference in what?

A. Sound colour

B. Wave phase

C. Head size

D. Video brightness

9. What is the cone of confusion?

A. A room with echoes

B. A speaker arrangement

C. A set of sound locations that can be difficult to distinguish

D. A soundproof material

10. Why is head tracking useful in VR spatial audio?

- A. It changes the user's hearing ability
- B. It updates audio as the head turns
- C. It blocks all outside sounds
- D. It increases the speed of sound

11. What is attenuation?

- A. An increase in sound frequency
- B. A reduction in sound strength
- C. A type of headphone
- D. A change in head direction

12. What is binaural rendering designed for?

- A. Only the left ear
- B. Only the right ear
- C. The sound received by both ears
- D. Only a room microphone

13. What is cross-talk in loudspeaker-based spatial audio?

- A. A speaker becoming silent
- B. Unwanted sound from one channel reaching the other ear
- C. A change in sound speed
- D. An increase in screen resolution

14. Which part of the ear helps create direction-dependent sound changes?

- A. Pinna (outer ear)
- B. Eyelid
- C. Tongue
- D. Fingertip

15. What is the difference between sound absorption and sound blocking?

- A. They always mean exactly the same thing
- B. Absorption reduces reflected sound; blocking limits sound transmission
- C. Absorption only works for light
- D. Blocking always increases echoes

Check answers

15. Last-minute revision — memorize these 10 points

1. **Perception engineering:** Study how humans experience the world.
2. **JND:** The smallest noticeable difference.
3. **Stereo:** Usually two audio channels.
4. **Surround:** Sound reproduced around the listener.

5. **Spatial audio:** Sound appears to occupy positions in 3D space.
6. **HRTF:** Describes how the head and ears change sound.
7. **ITD:** Difference in arrival time.
8. **ILD:** Difference in sound level.
9. **IPD:** Difference in phase.
10. **Cone of confusion:** Different sound positions can produce similar cues.

The most useful memory line for the entire lecture is:

"Time, Level, Phase, Ear Shape."

- Time → ITD
- Level → ILD
- Phase → IPD
- Ear shape → HRTF

These are the key ideas that connect human hearing to spatial-audio rendering in VR.

You said:

[Music] So we are still talking about the the chapter 11 audio and uh last time we talked about the perception engineering and we looked at the the like the SI units there is a unit for the perception we call it as just noticeable differences, right? And uh the main component of perception engineering is the uh measuring of the thresholds, right? The absolute threshold and uh and the difference threshold. We saw the the sensation unit in the y-axis over here. But we saw we we thought it is the uh it is the now J&D, right? And uh this is what has produced or suggested and this is one of the main reasons he's called as the father of psychopysics. And then Stevenson came in the last 50 years and he proposed the power law and he saw not only threshold but also the other measures as well to measure the performance as well as the the appearances. Correct. As far as the uh psychopysics is concerned and we looked at the perception engineering how this psychopysical experimental numbers, perception measures can be useful for designing now very good objects. Say for example this gma display we looked at it we try to understand the difference between how our camera is viewing and our eyes are viewing it. And we looked at the tactic perception tuning curve and we looked at the nu audio perception n tuning curve and uh that is where now we started. So just to highlight it I wanted to now uh make sure that everybody knows what is the unit of imperception. Last time you all know correctly answered again I wanted you to be now those who have not answered I want them to know that now J&D is a unit of the perception right and so uh so far I hope there is no question and there is one question of it let's see can we can we oops Can we join the summit? Yes, you can join the summit. Everybody can join the summit. What about the final exams and project? Yes, it is included. But just for this XR summit calculation alone and the final exam and projects are not included but eventually for the entire course this will be included. Okay. Can one join the summit online? Uh we will that modality is still being considered. We will tell you. Okay. So we started looking at the spatial audio and uh we looked at the fundamentals why we need to look at the spatial audio and we started looking at the differences between the uh the mech mech sound as well as the light. We uh looked at these differences between the stereo and surround and uh spatial audio has to be know all the directions. It is not just the left and right all the directions and uh now it includes main thing is that it includes the head tracking. Correct. So it is also called the you know personalized spatial audio personalized head related transfer functions. We looked at it head tracking is very important as we mentioned head related transfer functions and uh uh and the uh cross talk uh cancellation has to be used to avoid the uh avoid the the uh speakers or mics uh picking up signal from the other speakers. So it can be you know uh the personalized spatial audio can be using the headphones or the fixed speakers fixed speakers again we need the head tracking afated transformation and then cross related uh cancellation this we will see in the now uh in some details in today's class we looked at some history we looked at the other names now the special there are different formats. We looked at it. We started looking at the now uh the psychopysics of the the spatial audio and we look we saw that now from,000 htz to know 4,000 heads is where we have a very good sensitivity the lowest sensitivity the best correct and we looked at the zero db special at 3 kohz. So if you look at it 0 dB at about 3 kHz it's about 3 kHz right and 40 dB sound pressure level at 100 Hz for 40 dB at 100 and difference that is the J&D is.3% and 3 kHz so the frequency the change in the frequency we can perceive is a.3 percentage at 3 kHz but at the 100% 100 Hz it is 3% as a change our piano is designed at 6 herz we saw and then this J&D for loudness is 12%. The change in the intensity of the sound we can perceive only that 12%. Right? The Steven's power factor is about know 33 and therefore now we saw in last two classes that the increase in the amplitude by 10 times loudness increases only by two times because of this power factor and reducing the amplitude of the sound by 99%. But the loudness reduces only to 20% because of this no the ste power factor right and we saw this uh localization is a important parameter for no uh for understanding the spatial sound right so uh error of localization if you plot it in the y-axis that is the localization is the the perception of of the

angle at which the the sound is coming from. Correct? That is the uh our our error error of locating a sound the angle of sound coming from that is the error of localization correct that depends upon the frequency and at about 3 kHz we have the maximum error of localizations and why we saw in the last class there are now three hearing mechanism. We saw interoral time differences, interoral level differences. Level is the amplitude, right? And the this two things are encoded in a in a function called HRTF. It should be called HRTF. It is normally written H or T. Yeah. Right. Yeah. So this is the ITD interal time difference the time between uh time of arrival of one wave of one wave in one her one year for example here and then another wave comes over here. There is a time difference time difference and this time difference is called the ITD interoral time difference. the level difference the amplitude can also be different. So if the source is over here the uh this is nearer to this year therefore the amplitude of the wave hitting this year is now a little higher than the amplitude of the wave hitting the the left ear. So if this is it. So now let's see we were looking at the ITD and interoral level differences and an IPD interoral phase differences. These are the now main hearing mechanisms we saw in the last class. Um so in the last class we also saw that at for a source is the maximum the interal time difference to be 6 5 milliseconds. We did the derivations in the last class. So or a simple example we saw that and I I would like everybody to know remain that now we also saw that now our perception to the faces of the signal is very poor. The air is very uh uh insensitive to interal phase differences or only at the high frequency. At low frequency it has no good sensitivity but at the high frequency it it is very poor and it start deteriorating about 1 kilob right. There is one area one uh con kind of shapes which in which we have no total confusion of where the sound is coming from. If you have now top or or know bottom we will not be able to differentiate. If it is left or right we will not be able to differentiate within this cone is a sound is is placed the source is placed anywhere we will have a total confusion. This is called no uh cone of confusion. We saw this. So uh uh this particularly the the the ear shape is actually deciding the you know our uh uh uh filtering sound filtering the spatial sound filtering right. So uh this shape is actually now changing the spectral character of the sound what we are perceiving it the spatial sound what we are perceiving it and therefore this is captured in a map called the HRTF that's a head related transfer function this is a frequency versus a front or bottom or back top the uh how this spectral characterist statistics is is amplified or it is not amplified. It is deteriorating. This map is showing here. You can see that red color means uh it is uh very good and blue color means it is deteriorating. Right? So you can see that the front and at the higher frequencies those are all very nice but at the back it is looking bad and at the bottom and the back as well it is looking bad right. So uh this is actually capturing the the spectrum um in the in the entire space of the 3D sound which is coming to the the human being. Right? There is one representation similarly there is you know there are other way of representing the HRTF we are going to see. So uh this HRTF this is the HRTF right it over here this is HRF so uh there's another way of representing the HRTF uh by this HRTF is getting changed if you alter your ear for example here there is some noy or some no uh material is used to change the shape and that changes the hatch. Hard is measured basically by by putting some now mics in the ear right and know locating sources at a particular distance at different angles different elevation and then looking at how the amplitude of the sound and is actually decreasing or increasing it. Okay. So the footy used to change the uh the ear shape and that changes the uh HRDF and therefore localization right. Uh by this way we can actually know personalize the localization. Suppose if you have a sphere is surrounded the human head and if you say no theta is the the is the lateral angle and know is the polar angle and if you plot later angle in the x-axis the polar angle in the y-axis then we can divide Okay, what is front? What is know left or right? Bottom or top or back. Right. So we can divide each of this know different uh areas or domains and then check how this localization error is actually uh in different locations. Right? This is also another way of looking at the HRDF. This numbers mentioned in this graph it is telling you what is the localization error and there's also small arrows even at every point. What is the direction in which it is you know it is the the errors are directed to particular area right that's also is given. So lighter means uh the error is uh very high darker means the error is now zero. Correct? If you look at it now, we saw that uh uh the last class also the best accuracy is in the front of the listener because you can see that at the the front is what the the black area is. You remember the black means it is you know zero near zero error and optimal performance is between the minus 45 to 30°. Correct? If you look at this is this uh you can see this is the uh this is the best performance area right let me put a rectangle over there you can see this is about uh minus 45 to - 45 to this is + 30 right this is area plus west is above and behind so wherever white is there this is a top and bottom also is this this area this also was in the front of the hemisphere responses are biased towards locations just above the horizontal plane. If you look at the now the the arrows right in the front plane and the responses are vas towards the location just above the horizontal plane. So you can see that it is actually pointing uh pointing and upward. So upward in the sense if you look at this particular sphere we are talking about know theta you know phi is is in the positive direction here this is going to be positive directions right you'll have to look at the positive pi so this is a positive pi this is towards pi this is a positive is what it means now in the rear hemisphere responses are bias bias is forward and towards left and right ps. So forward and uh you know this is um left and uh left and right pole. Uh you can see that the arrows are now pointing to uh this way right. This is towards the pole correct. Pole is what? Hold is now top and bottom. It should be around top and bottom. Uh so here also you can see that it is uh uh towards or away. Here it shows the away from the now pole. Here also away from the pole. Okay. So I think this response is the rear hemisphere. This is in the rear hemisphere. The response is biased forward and towards left and right

poles. Okay, that direction should be very right now. I have a question for you. Uh what do you think? Why best accuracy in the front of the listener? As far as the uh as far as the uh the localization is concerned, we have the best security in the front of the why can you answer in the slider? So the flow is all aligned in the frontal region. Any other answers? [Music] [Music] flow is okay sorry uh symmetry no I saw the correction in the subsequent slide is it a HRTF it is HRTF not HDR interated transfer function so Yeah. So, uh the the best secure is in the front of the listener because our our shape of the ear is know facing forward. Yes. Due to the shape of the ears and it is relative position we have seen it. Correct. You're right. So yes and in the last class we saw that how this HRTF can be now uh now engineered use for now making a useful applications. In Japan some of the now robots are given this specific shape of the ear so that robots can do the localization just like a human being right. So robot with artificial ear shape they can do the localization as well. There were you know while locating a sound coming from anywhere in the space we use lot of cues and this cues are changed because of now many anatomy as well. So because of the head size it has a barrier at a high frequency but not at low frequency. We we have the shape not air shape not only ear but also the other thing shoulders and head size also is changing it shape of the pina reflections and resonances of the sound changes the spectrum of the sound at the eard drum depending upon the direction of the head shoulders reflections change in the sound level differences due to the extra distance of the sound that has to travel from one ear to the other ear right so HRDF F is nothing but the sum of all the effects as a spectra for every source location, every direction, every elevation. This has to be found. Right? So, uh it is not just one graph. Maybe there are multiple graphs in order to capture the entirety of the the the the complexity of a 3D sound. Right? Uh this is what we call it as a cues in time as well as this uh phase. Here is one way of capturing this HRTF knows and time and uh phase. Uh so here is one mon transfer function right and the HRTF can be bal or monal. Here is the monal transfer function head related function of the left ear alone. This is monal means it's only one year. Therefore, here is the left for several directions in the horizontal plane. Uh there are there are now the angles are written over here. This is uh with respect to the front, right? The angle is written as 30 and then no uh 60 90 120 150 and so forth. Right. Um related to sound incident from the front. Front is the sound origin source an echo chamber 2 m loudspeak distance uh in front of you. 2 m loud loudspeak is the source impulse technique. It gives a impulse and this is a measure of the uh spectrum at the at the now year left year for 25 subjects and an average right it's a complex complex average A is the level differences there's a amplitude differences and then B this graph is the no is a uh time differences there's a blue delay we are talking about this This is the amplitude difference. This amplitude effects, right? So we can learn a lot of things from this particular graph. What are the things we can learn? There are high frequency peaks. You can see in this one there are high frequency uh peaks over here. Right? Sources to the rear gives reduced high frequencies. Uh rear means what? So if front is the zero and your right is let us say 90° your rear will be north anything from 90° to north uh probably 270 right if you look at the the rear particularly after 180 right from this one if you look at all the graphs the high frequency are all now diminishing So zero means it is a dB right we are talking about the amplitude dB with respect to the the front source right so zero means there is no difference now negative means the amplitude is actually decreasing it positive means the amplitude is increasing it right you can see in this graph all these things are now decreasing the high frequencies are decreasing in the amplitude right it shows that the sources This to the rear gives reduced high frequency. That is because of the the ear shape. Ear shape is playing a major role and that is very evident in this diagram as well. There is a 8 kohz overhead projection. 8 kohz you can see that there is a peak over here and that is because of the overhead projection. 3 kohz to know 6 kohz it is in the know frontal projection and 1.22 to to 12 Hz in the real uh real projection. Okay, this mon HRTF is somehow is used by the prime to provide the direction information and this is where we have this you know localization information coming from. But if you use this monal HRDF with a moving sources this is more useful the error localization actually reduces spectral characteristics independent of the source one spectrum can be now very useful with the moving sources localization or lateralizations there's a small differences differences in the monoral HRDF right so uh uh for left and Right. The HRF are different. Monoral HRDF are different and that differences in the monoral HRF is used for the localizations or lateralizations. And apart from this, ITD is one of the major cues. Interal time difference is one of the major cues for finding the localization. Right? Now if you look at the binaural HRTF right and and binaural there is one particular no interesting phenomena we wanted to look at it and that is the echo right so there is know uh is there threshold for the echo when do you call your secondary wave as an echo just like in the uh in the vision We have a fusion threshold and two sounds separated by a delay which is below the uh echo threshold perceived as a single threshold single sound. This is where the fusion is happening. The two sounds are fused together otherwise otherwise it is considered as an echo. Right? What is this number? What is this echo threshold? 5 milliseconds. In a small click sound whereas in a complex sound it is 50 milliseconds. There is an echo threshold. Above this threshold the second wave is heard as echo. And if uh sec sound source locations are different first reach sound determines the location within this particular echo threshold. If two waves are reaching the air within this particular uh echo threshold the first reached sound location precedence effect or house effect. This is called the precedent effect. We don't consider the second sound at all. So this is called the president and effect. We are going to see the president's effect. Reflections arriving later than 1 milliseconds after the the uh after the the direct sound increases the perceived level and width. Uh so although we are talking about only 5 millconds as the income threshold if the

the second sound coming after 1 millcond but before 5 millisecond then the first sound is perceived now with a higher amplitude and then higher width higher within the sense the more spectrum right reflections within 5 to 35 milliseconds 10 dB lower than the direct sound. So for complex we are talking about the 50 milliseconds 50 milliseconds uh as a threshold. So if it is 5 to 35 milliseconds reflection then the original sound is actually perceived 10 dB louder. This is what's shown in this particular diagram. Suppose if you have the the delay in the x-axis and now the uh the echo perceived echo level in the y-axis about 5 milliseconds to 35 milliseconds we have no about 10 dB the sound is perceived as a 10 dB louder right uh Yeah. Uh this graph is showing the 10 dB louder. The same thing, same graph is shown slightly different way. Uh this is also called the S effect. H effect effect is showing the relative of level required of a delayed reflection secondary reflection for it to appear equally loud to an earlier primary source. the the this echo is actually adding its uh amplitude that is called the pass effect and uh uh that is the no there is a threshold right the secondary level difference x-axis is the delay yaxis is the amplitude secondary level amplitude and it is almost the same as the the loss graph the level differences okay and this the delay as well as the the intensity can be traded with each other for now uh this is what is used in the surround sound. If you want to make the the sound coming from the nearer speaker and to be perceived stronger than the the the the further away speaker then we can use the delay or the the later one as the louder. Correct. So where to use louder sound where to use the delay uh we can decide depending upon the effect what we wanted to uh convey. So this graph if you look at it this is a binaural time intensity trading effect we can uh trade it for example suppose if you have a secondary wave at let us say 0.5 milliseconds then if the amplitude is now lower than this particular point that is about 4 dB or 5 dB Then now the sound is perceived towards the ear sound. If the amplitude is above this particular point then the sound is perceived towards the the louder sound the later sound. Right? But this effect we can trade this trade between the the amplitude as well as the delay only up to certain point only up to this point right this point is the uh 65 milliseconds which we have seen earlier. This is a maximum delay uh we can expect it after this. After this this is actually saturating. We have about uh now 12 dB always towards the ear sound unless delay sound is more than 12. Within this if the sound is lesser than 12 dB we perceive towards the ear sound. Otherwise if it is above uh perceived closing later side right and after this you can see that it is after 30 milliseconds then it becomes a echo. Yeah. So uh with the understanding of this psychoacostistic or psychophysical characteristic of the audio now let us see how this can be useful for know uh rendering 3D sound right how do we use the psychopysics to design 3D spatial rendering right let us look turn off binaural synthesis. We wanted to uh we wanted to design a rendering technique with both ears technique to capture the effect of the outer ear. So we have to first know capture the the effect of the outer air and both ears right by directly recording the signals of the both ears right we can put a dummy head and a microphones in the dummy head and then we can capture the real world experience but the problem is that it doesn't have the now orientation H so we'll have to change the W head to another orientation and do the same again right so measurement of the transfer function from an acoustic source to both as there is another way of doing it we'll have to transfer the HRTF from an acoustic source to both s and then that can be useful for uh that is that is how you know it can be captured right uh that is the direct recording Uh so uh the in measuring the transfer functions from an acoustic source basically it filters the monoronic signals of the virtual source with a pre-captured HRTFS uh monoh HRTF are there and we can and that's how we can actually capture the HRTFS with the head tracking and then database of the monohs we can actually now completely now measure the the transfer functions from an acostic source to both the ears there may be many HRTFs needed mono HRTFs needed for every possible directions right plus we need to do the head tracking okay once we have this technique or capture the the effect of the outer re now we need to render it. So rendering starts with the capturing first then only we can render it. Right? And then rendering techniques whether we can use the headphones or live speakers. Suppose if you use the loudspeakers let us say right loudspeakers or the headphones both they use the delay technique or the amplitude level techniques like what we have seen in the psychopysics or the HRDF techniques. So this is three ways of knowing the the 3D sounds. There are two rendering techniques we are going to discuss. Now uh first is the higher order ambioxics. The second is the wave field synthesis. Let us look at each one of this now very briefly. Okay. So uh what what is ambionics? Let us see and then we will see what is higher order ambionics and then how this ambionics is different from surround sound or uh stereo sound. Let us take a look at it. Okay. So uh ambionics is a way of capturing the entire spatial sound. How do we capture it? How do we synthesize it? Capture it and then render it as well. Right? So first of all it has to be captured and then rendered. Let us say it is captured. Now how do we render is a challenge uh because it is all all around it is called the ambiox ambisonics and we are talking about higher order amics right. So to introduce before I introduce the uh uh higher order amasonics, I wanted to uh I wanted to uh show you uh take you to the the of Yeah. So all of you have had a linear regry on your undergraduate, right? All of you have had studied about the orthogonal basis functions but as a maybe mathematical function. So here I want to give a intuitive idea of what this orthogonal basis function is and how this can be useful for our ambious products. Okay. So suppose that we have a function f over here. This is a function f. This graph this curve over here. And we have a basis function. Basis functions are very let us say this is linear faces functions. Let us say linear oops I have linear I am writing with the mouse. So please excuse me. So we have this linear basis functions. This is one function. This is the second function. This is the third basis function. Right? This the first basis functions. This is all let us say for the simplicity sake it is just

shifted by one unit let us say some arbitrary unit. Now we wanted to find out how this particular basis is contributing to the function. So if you integrate this function with the uh the original function with the basis function let us say we get a constant c1. Similarly we have this the second basis functions we integrate it with the original function and we get the coefficient C2. Similarly the third coefficient and three. Now what this orthogonal basis function says is that we can represent the the original function f with a combination of of the coefficients and the the basis functions. Just by multiplying with the coefficients with the each of the basis functions and summing it up we will get the original functions. Since we are gives a linear one what we get uh the summed up also is going to be linear approximation of the the original function. Right? So uh basically what is happening over that take the C1 and then multiply the basis function first and then C2 multiply the basis function the second basis function take the C3 and multiply the third basis function and then this is the result because basically this the constant is actually either amplifying it here it has reduced it here it is not doing anything here also it is not doing anything now we sum up all the together and this is how the final and f is reconstructed. Right? So now this basis functions is all linear but what is actually needed is a highly nonlinear or nonlinear basis functions. Right? There are some nonlinear basis functions called legendary polymers is what considered for for acoustic waves. Why? Because legendary polymer suits very nicely for a spherical wave. Sound wave is propagating know as a spherical wave front, isn't it? So now legendary polymers are know adapted for uh for representing the original wave with the polymers with the orthogonal basis functions of this polymers. Uh usually this LP there's a legitary polymers or complex numbers whereas since we are talking about only real numbers so we take only the real numbers and it is called the associated legendary poly right. So what you are seeing here this is the kind of polomials we are talking about instead of simple now linear basis functions like this we are going to use several functions of polomials like this and this as polomials are of the order several different order this is the no zero order this is the first order this is the second order and as we go uh increase the the the complexity of the the colon also increase and therefore we can approximate the original function and much better that's what the whole idea is Thanks. [Music] HW9: COLOR PERCEPTION, THE MAGIC
OF CHROMATIC
ADAPTATION
Download
Start with downloading the following VR Cardboard application:
● Chromatic Adaptation.apk
What is Chromatic Adaptation?
Chromatic adaptation is the human visual system’s ability to adjust to changes in light, allowing perceived colors of objects to remain relatively constant under varying lighting conditions. For example, a white piece of paper appears white whether viewed under sunlight, fluorescent light, or LED light. This effect, known as color constancy, helps us maintain stable color perception in different environments. Without chromatic adaptation, these papers would appear slightly different from each other under each type of light.
Our visual system adapts best to light sources with a spectrum similar to natural daylight (https://en.wikipedia.org/wiki/Standard_illuminant). When exposed to such light, our vision fully adapts, and colors appear natural. However, under artificial or monochromatic lighting, like intense red nightclub lights (which have a narrow spectral power distribution), our ability to adapt is reduced, causing distorted color perception.
What Do We Know About Chromatic Adaptation?
Chromatic adaptation has been studied a lot in labs, but research in virtual environments is still developing. Models that mimic how our eyes adjust to different lighting are called Chromatic Adaptation Transforms (CATs). The most famous one, from 1902 by von Kries, helps map color stimuli from one lighting condition to another, ensuring they look the same under different lights.
Why Does Chromatic Adaptation Matter?
Chromatic adaptation is important for digital displays, color design, and photography. It affects how we see colors in everyday objects, art, and fashion. Studying this helps improve color accuracy on screens and makes virtual reality more realistic. It also helps us understand how the brain processes color.
In practical terms, a reliable CAT can improve how colors are displayed on screens and in photos, preventing mistakes that frustrate consumers. For example, CATs are used in cameras to

simulate how our eyes adjust to light, helping photos look more like what we actually see—like when you take a picture of a sunset, and it doesn't quite match real life. CATs help fix that!

VR: A Powerful Tool for Studying Chromatic Adaptation

Virtual reality provides a controlled setting where we can precisely manipulate lighting and color conditions, something difficult to achieve in real-world experiments. For example, we can even simulate different lighting for each eye, which is fascinating.

Before using the app, try a mental experiment: What if your right eye sees the environment under cool white LED light, and your left eye sees it under warm LED light? This is what this homework will let you experience. Spoiler alert: at first, you'll notice a clear difference between the two eyes, but after a few minutes—like magic—the differences will disappear! Although we've spoiled the surprise, we're confident you'll still be amazed when you experience it yourself. This shows just how powerful VR is for studying chromatic adaptation.

Completing This Homework

Once you enter the scene in the APK, you will see 6 fruit canvases, 4 color block canvases, and one central canvas labeled "tutorial" with a color wheel. Start with the tutorial by focusing on the tutorial canvas for one second to enter. Inside, you can practice the "color space interaction" that you will use in all other canvases. In other canvases, the color wheel will be invisible, but your head movements will still interact with the color space.

Here's how it works:

1. Moving your head around the circle will let you choose the hue (color) and saturation (intensity of the color).
2. Rotating your head around a specific color will adjust its value (grey level) .
3. Gazing at a specific color (hue + saturation + value) for more than a second will select that color and exit the experience.

Note: If you're not ready to select, avoid gazing at the color for too long, or it will be selected and you'll have to start over.

You can read more about HSV (Hue, Saturation, Value) here: https://en.wikipedia.org/wiki/HSL_and_HSV.

After the tutorial, practice on the fruit canvases. In these, your left eye will see the fruit, and your right eye will show the color space. Move your head just like in the tutorial to find the color that matches the fruit. Once you select a color, you'll exit the canvas, and your results will appear on a new canvas beside the one you visited.

Once you're comfortable, move on to the color block canvases. This is where we need your answers for the questions below. Try to match the colors using the same method. After each match, report the resulting HSV values and the total time it took to find the best match, which will also be displayed upon exiting each scene.

This Is a Human Subject Experiment

Just like your previous homework on Vection, this homework also serves as an example of how human subject experiments are conducted. We provide you with instructions on how to complete the experiment but will not reveal the hypothesis or share the collected data until all submissions are complete and analyzed.

We may use this data for research and publication purposes, so your participation is an important contribution to the scientific community. All data will be analyzed in aggregate, and we will not disclose which answers came from specific individuals. To ensure privacy, the data will be pseudonymized, meaning it will be associated with a unique identifier rather than your name. Once everyone has completed the homework, we will reveal the results and discuss them with you.

Question 1: Consent to Participate and Use Data

I consent to participate in this color perception study, understanding the nature, risks, and benefits involved. I acknowledge that I have the right to withdraw from the study at any time without penalty. I also consent to the use of my data collected during this homework for further research and analysis, including potential publication.
If I choose not to consent to the use of my data for research, I understand that I am still required to complete the homework assignment. My grades and status in the class will not be affected in any negative way if I decide not to provide my data for research purposes, and I should not feel obligated to consent.
● I consent.
● I do not consent.
MCQ
Answer is: All answers are accepted as correct.
Before we proceed, let's complete two quick tests to understand your color vision and ensure meaningful results for chromatic adaptation.
Question 2: Color Blindness Test
Take the color blindness test using this link: Color Blindness Test. After completing the test, please provide us with your score.
Example test result: 100 % (18/18)
Numeric array: X% (Y,Z)
Answer is: Any answer in the form above, in which the value for X can be any number in the range from 0 to 100. And both Y and Z can be any number from 0 to 30.
Explanation: The values may vary due to individual physiology.
Question 3: What Is Your Dominant Eye?
To find your dominant eye, roll a piece of paper into a tube and look through it at an object on the wall in front of you. Which eye did you instinctively use to look through the tube? That is your dominant eye.
A. Left eye
B. Right eye
MCQ
Answer is: All answers are accepted as correct.
Final Instructions Before Proceeding
Before you begin answering the questions about chromatic adaptation, please review and follow these important steps:
● Turn Off Night Mode: Ensure night mode is disabled on your phone.
● Set Fixed Brightness: This is incredibly important! Please, disable automatic brightness adjustments and set your phone’s brightness to a fixed level.
● Monitor Eye Strain: Did you experience any eye strain during the experiment?
● Watch for Unusual Observations: Did you notice anything strange or unusual in the visual scene?
● Avoid Closing One Eye: Be mindful of whether you closed one eye to assist with the task, and try not to close one eye during the experiment!
Question 4: The Red Square Canvas
Complete the color matching task in the red square canvas. Afterward, what were the resulting H (Hue), S (Saturation), V (Value) values, and the time it took? These values will appear from top to bottom on the canvas below the red square. Please type all of these numbers in the textbox below.
Example: H:64, S:95, V:100, Time:12s
Numeric array: H:XX, S:YY, V:ZZ, Time:ts
Answer is: Any answer in the form above, in which values XX, YY, and ZZ can be any number in the range from 0 to 255. The t can be any number in the range from 0 to 1000.
Explanation: The values may vary due to individual physiology.
Question 5: The Blue Square Canvas
Complete the color matching task in the blue square canvas. Afterward, what were the resulting H (Hue), S (Saturation), V

(Value) values, and the time it took? These values will appear from top to bottom on the canvas below the blue square.
Please type all of these numbers in the textbox below.
Example: H:64, S:95, V:100, Time:12s
Numeric array: H:XX, S:YY, V:ZZ, Time:ts
Answer is: Any answer in the form above, in which values XX, YY, and ZZ can be any number in the range from 0 to 255. The t can be any number in the range from 0 to 1000.
Explanation: The values may vary due to individual physiology.
Question 6: The Green Square Canvas
Complete the color matching task in the green square canvas. Afterward, what were the resulting H (Hue), S (Saturation), V (Value) values, and the time it took? These values will appear from top to bottom on the canvas below the green square.
Please type all of these numbers in the textbox below.
Example: H:64, S:95, V:100, Time:12s
Numeric array: H:XX, S:YY, V:ZZ, Time:ts
Answer is: Any answer in the form above, in which values XX, YY, and ZZ can be any number in the range from 0 to 255. The t can be any number in the range from 0 to 1000.
Explanation: The values may vary due to individual physiology.
Question 7: The Yellow Square Canvas
Complete the color matching task in the yellow square canvas. Afterward, what were the resulting H (Hue), S (Saturation), V (Value) values, and the time it took? These values will appear from top to bottom on the canvas below the yellow square.
Please type all of these numbers in the textbox below.
Example: H:64, S:95, V:100, Time:12s
Numeric array: H:XX, S:YY, V:ZZ, Time:ts
Answer is: Any answer in the form above, in which values XX, YY, and ZZ can be any number in the range from 0 to 255. The t can be any number in the range from 0 to 1000.
Explanation: The values may vary due to individual physiology.
Observing Visual Anomalies
Question 8:
Did you experience any eye strain during the experiment?
● Yes
● No
MCQ
Answer is: All answers are accepted as correct.
Explanation: The values may vary due to individual physiology.
Question 9:
Did you notice anything unusual or strange in the visual scene?
● Yes
● No
MCQ
Answer is: All answers are accepted as correct.
Explanation: The values may vary due to individual physiology.
Question 10:
Did you have to close one eye during any of the color matching tasks to complete it? (Please remember - try NOT TO CLOSE ONE EYE!)
● Yes
● No
MCQ
Answer is: All answers are accepted as correct.
Explanation: The values may vary due to individual physiology. lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

Chapter 11: Spatial Audio + Homework 9: Color Perception

Easy notes for a 7th-class student • Simple examples • Memory tricks • MCQs with answers

Your lecture contains **two main topics**:

1. **Spatial Audio:** How our ears and brain understand where a sound comes from.
2. **Chromatic Adaptation:** How our eyes adjust to different lighting so that colors appear more stable.

Let's learn them one by one in simple language.

Part 1: Spatial Audio — Easy Notes

1. What is perception engineering?

Imagine you hear a bell ringing. You can notice that it is loud, soft, near, far, or coming from your left.

Perception engineering means studying how humans sense and understand things, then using that knowledge to design better technology.

For example:

- Designing displays that look natural to our eyes.
- Designing headphones that make sounds seem to come from different directions.
- Designing VR experiences that feel more realistic.

Two important thresholds

Absolute threshold
The smallest amount of a stimulus you can detect.
Example: The quietest sound you can just hear.

Difference threshold (JND)
The smallest change you can notice between two stimuli.
Example: Noticing that one sound is slightly louder than another.

JND means *Just Noticeable Difference*.

Remember: JND is a measure used in perception studies. It is not an official SI unit like metres or seconds.

Important scientists

- **Ernst Heinrich Weber:** Associated with early work on just-noticeable differences.
- **Gustav Fechner:** Often called the father of psychophysics.
- **S. S. Stevens:** Proposed Stevens' power law, which describes how physical stimulus strength relates to perceived sensation.

Memory trick: Absolute = detect something. Difference = detect a change.

---

2. Light versus sound

Feature	Light	Sound
Type	Electromagnetic wave	Mechanical wave
Needs a medium?	No	Yes
Can travel in a vacuum?	Yes	No
Travels in air	Extremely fast	Much slower
Main human sense	Eyes	Ears

Easy example: Sunlight reaches Earth through space, but sound cannot travel through empty space because it needs a material such as air.

---

3. Mono, stereo, surround and spatial audio

Mono audio — one channel

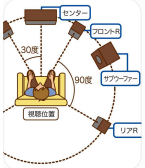
The sound uses one audio channel.

Remember: Mono = one.

Stereo audio — usually two channels

Left and right channels help create a sense of direction between your ears.

Remember: Stereo = left + right.



Surround sound

Several speakers around a room create sounds from the front, sides and rear.

Spatial audio — sound in 3D space

Sound can seem to come from the left, right, front, back, above or below. Head tracking can keep the sound in the correct virtual location as you turn your head.

Most important difference: Stereo mainly gives left-right sound. Spatial audio aims to create a realistic three-dimensional sound environment.

4. Why can we hear the direction of a sound?

Suppose your friend calls you from your right side.

Your right ear receives the sound a little earlier, and the sound may be louder at your right ear. Your outer ears also change the sound depending on its direction.

Your brain combines these clues to estimate where your friend is standing.

The three important hearing cues

1. ITD — Interaural Time Difference

TIME difference between the two ears

Sound reaches the nearer ear slightly earlier.

Example: A clap on your right reaches your right ear first.

2. ILD — Interaural Level Difference

LOUDNESS difference between the two ears

The nearer ear often receives a louder sound, especially at high frequencies, because the head blocks some sound.

Example: A speaker on your right may sound louder in your right ear.

3. IPD — Interaural Phase Difference

PHASE difference between the two ears

The sound waves can be at different stages of their vibration when they reach each ear. This cue is particularly useful at lower frequencies.

Memory trick: T-L-P

- **T** = Time (ITD)
- **L** = Level/loudness (ILD)
- **P** = Phase (IPD)

These cues help the brain locate sounds.

Important correction: The lecture transcript appears to say the maximum ITD is 65 milliseconds. That is almost certainly a transcription error. A typical maximum human ITD is around **0.6–0.7 milliseconds**, or about 600–700 microseconds.

5. What is HRTF?

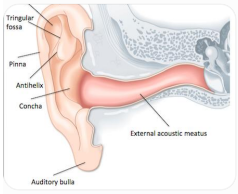
HRTF stands for **Head-Related Transfer Function**.

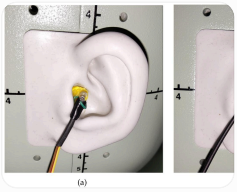
That sounds difficult, but the idea is simple.

Imagine sound travelling from a speaker to your ears. Before reaching your eardrums, the sound is changed by:

- Your head.
- Your shoulders and torso.
- The shape of your outer ears.

HRTF describes these changes for sounds coming from different directions.





Why is HRTF important?

Different people have different head sizes and ear shapes. So, the sound reaching their ears can be slightly different.

A **personalized HRTF** tries to represent the way sound changes for one particular person.

A **generalized HRTF** uses measurements averaged across people or a standard model.

- Personalized HRTF: potentially more accurate for that person.
- Generalized HRTF: easier to use, but may be less accurate for some listeners.

How is HRTF measured?

1. Place microphones near or inside a person's ears, or use a dummy head.
2. Play sounds from different directions and elevations.
3. Record how the sound changes at the ears.
4. Build a collection of measurements for different directions.

These measurements can then help create realistic 3D audio.

Memory trick: HRTF = **Head changes the sound before it reaches the ear.**

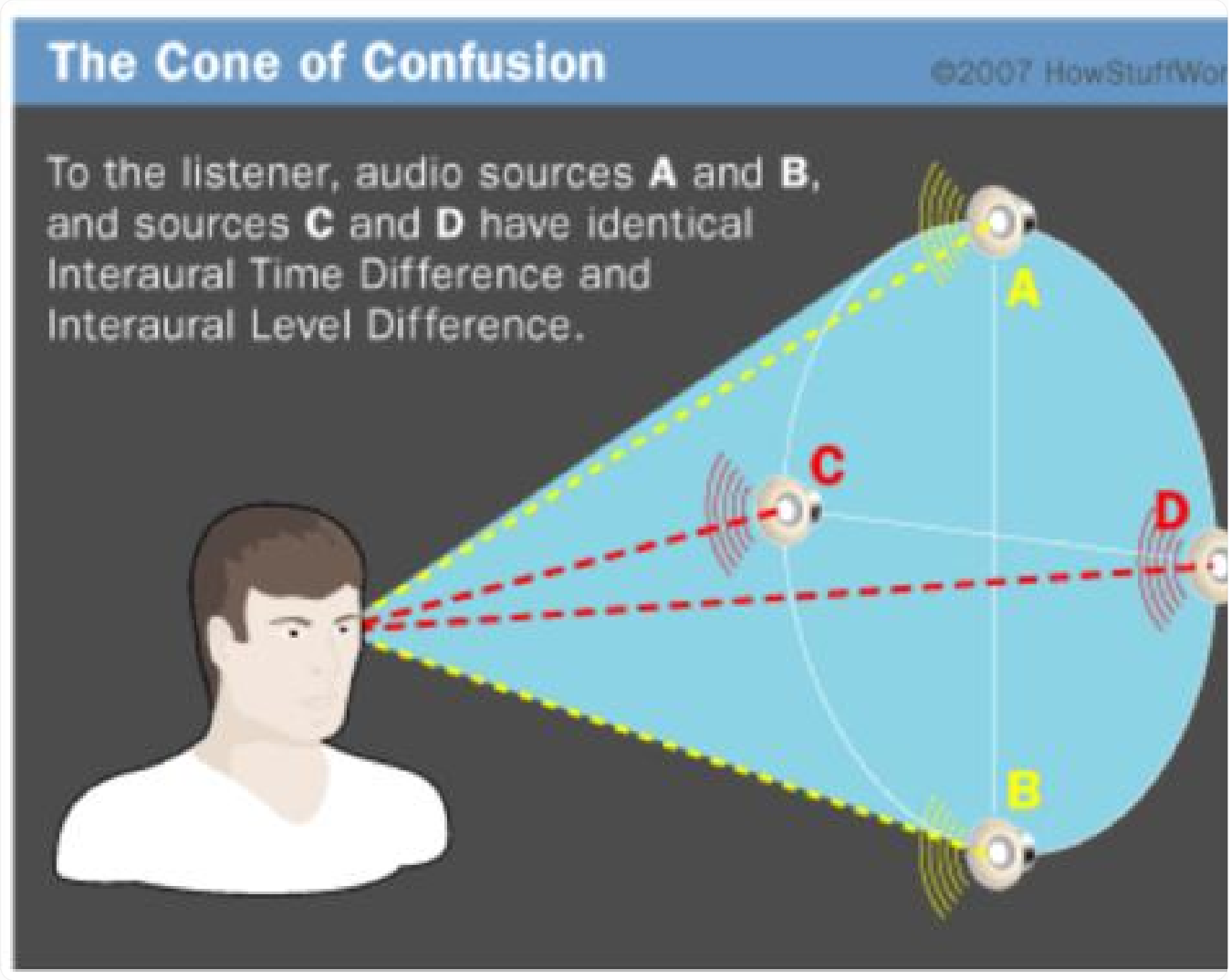
---

6. What is the cone of confusion?

Sometimes, two different sound locations create similar clues at your ears.

For example, a sound in front of you and a sound behind you may confuse your brain.

This problem is called the **cone of confusion**.



How can the brain reduce this confusion?

- It uses the filtering caused by the outer ear.
- It uses additional frequency clues.
- It can use head movement to help determine the correct direction.

The outer ear, called the **pinna**, is especially useful for distinguishing directions such as front versus back and above versus below.

Why is sound localization often better in front?

The lecture explains that the shape and relative position of our ears help us locate sounds in front more accurately.

The lecture gives an approximate best-performance region of **−45° to +30°** in its particular graph. This is a result for that measurement, not a universal rule for every person and experiment.

7. How does sound frequency affect hearing?

Frequency is the number of vibrations per second, measured in hertz (Hz).

- Low frequency: a deep bass sound.
- High frequency: a sharp, high-pitched sound.

Human hearing sensitivity depends on frequency. We are generally especially sensitive to sounds in the few-kilohertz range, roughly around 2–5 kHz.

The lecture also mentions:

- Around 3 kHz, a frequency change of about **0.3%** may be noticeable under the stated experimental conditions.
- Around 100 Hz, the corresponding figure may be about **3%**.

These are approximate values and depend on the experimental conditions.

What is Stevens' power law?

Stevens' power law explains that the strength of a physical stimulus and how strong it *feels* are not always directly proportional.

For sound, multiplying the physical amplitude does not mean the sound feels equally many times louder.

The lecture gives an example where increasing amplitude tenfold results in perceived loudness increasing by about twofold. Treat that as the lecture's simplified example rather than a universal rule for all sounds.

Remember: Physical strength and perceived strength are not the same thing.

---

8. Echo, fusion and the precedence effect

Imagine clapping in a large hall. You hear the direct clap and sound reflected from walls.

Three important terms

- **Fusion:** Two sounds arriving close together may be heard as one sound.
- **Echo:** A delayed reflection heard as a separate sound.
- **Precedence effect (Haas effect):** When similar sounds arrive from different directions close together, the first-arriving sound usually has the strongest influence on the perceived location.

The lecture gives approximate echo thresholds of:

Sound type	Approximate threshold mentioned
Short click	5 milliseconds
Complex sound	50 milliseconds

These are simplified lecture values; actual thresholds depend on the sound and listening conditions.

What is time-intensity trading?

The brain's perception of sound direction depends on both arrival time and loudness.

Changing the delay or loudness of a later sound can change which direction the sound seems to come from.

This relationship is called **time-intensity trading**.

It is useful when designing surround and spatial audio.

Memory trick: First sound = precedence effect; sufficiently delayed reflection = echo.

9. How is spatial audio created?

The lecture explains that we first need to capture or model how sound reaches both ears, then render it for the listener.

Step 1: Capture or measure

Record sounds at the ears or measure HRTFs from different directions.

↓

Step 2: Process the sound

Apply direction-dependent filtering, timing and level differences.

↓

Step 3: Render the sound

Play it through headphones or an appropriate speaker system, optionally using head tracking.

Headphones versus speakers

- **Headphones:** Can use binaural processing to send different signals to each ear.
- **Speakers:** Sound from one speaker can also reach the opposite ear. This is called cross-talk.
- **Cross-talk cancellation:** Processing designed to reduce this unwanted opposite-ear sound in speaker-based systems.

Head tracking measures how the listener turns their head, allowing the virtual sound scene to update accordingly.

Two spatial rendering methods from the lecture

1. Higher-order Ambisonics

Ambisonics represents sound around the listener using mathematical components. Higher orders generally allow a more detailed representation of the sound field.

2. Wave Field Synthesis

This technique uses multiple speakers to reproduce a sound field across a listening area.

The lecture also discusses **basis functions**. Think of them as building blocks: combine several simple mathematical shapes in suitable amounts to approximate a more complicated shape or sound field.

Orthogonal basis functions are designed to have a useful mathematical relationship that helps separate their contributions. The lecture introduces Legendre polynomials as mathematical functions useful for representing sound over spherical directions.

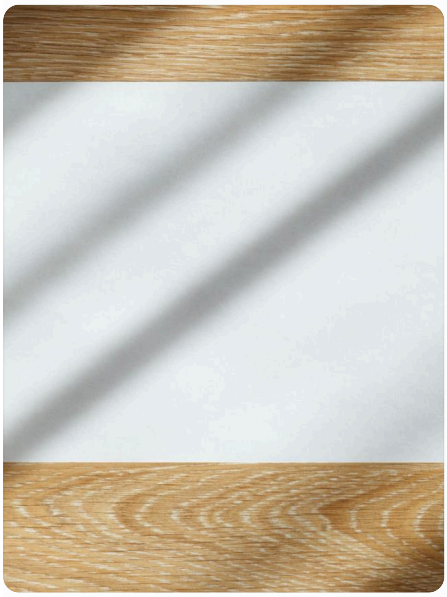
Part 2: Homework 9 — Color Perception and Chromatic Adaptation

This section is particularly important because it explains the homework you must complete.

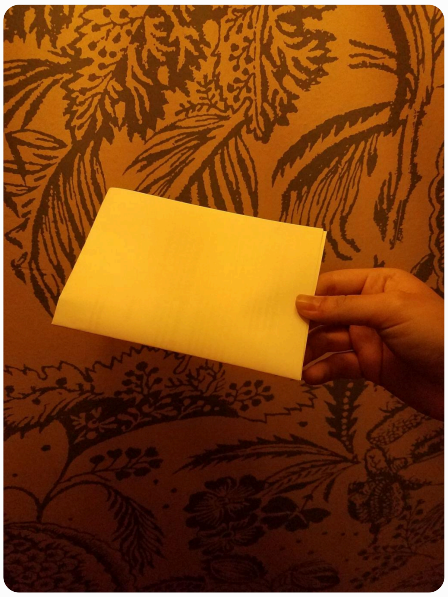
10. What is chromatic adaptation?

Chromatic adaptation is the ability of our visual system to adjust to different lighting conditions so that colors appear relatively stable.

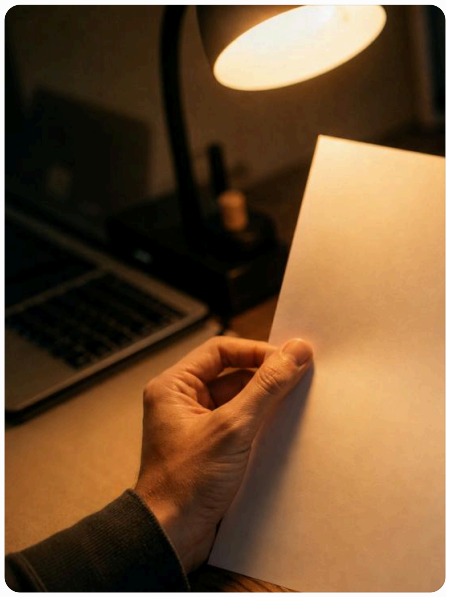
For example, imagine a white sheet of paper.



Daylight



Warm light



Cool light

The light reaching your eyes changes in each environment. However, your visual system adapts, so you can still recognize the paper as white.

This ability is closely related to **color constancy**.

Easy example

Imagine wearing yellow-tinted glasses. At first, everything may look yellow. After some time, your visual system adjusts, and the yellow tint may seem less noticeable.

That is a simple way to think about visual adaptation, although actual chromatic adaptation is more complex.

What happens under colored lights?

Under intense red or other narrow-spectrum lighting, some colors may look very different because the light contains a limited range of wavelengths. Your visual system cannot always compensate completely.

Remember: Chromatic adaptation = eyes adjust to the lighting.

---

11. What is a Chromatic Adaptation Transform (CAT)?

CAT stands for **Chromatic Adaptation Transform**.

It is a mathematical method used to estimate how colors should change between lighting conditions so they appear more consistent.

The lecture mentions the **von Kries adaptation model**, introduced in the early 1900s.

Why are CATs useful?

- Digital displays.
- Photography and cameras.
- Color design and printing.
- Virtual reality.
- Making colors appear more consistent under different lights.

Example: A camera can use color adaptation methods to make a white object look more naturally white under warm indoor lighting.

12. Why use VR to study color perception?

VR lets researchers control the visual environment very precisely.

For example, they can present different lighting conditions to each eye and observe how people respond.

In your homework, the app lets you compare colors while using head movements to select a matching color.

The experiment may help researchers understand how visual adaptation works. The assignment says the hypothesis and combined study results will be discussed after submissions are complete.

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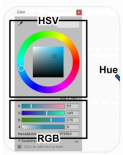
13. How to use the homework app

Follow these steps in order.

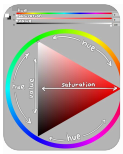
Step 1: Open the VR app

Start the Chromatic Adaptation app and enter the scene.



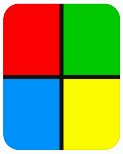
Step 2: Complete the tutorial

Focus on the central tutorial canvas for one second to enter.



Step 3: Learn the head movements

Move your head around the circle to choose hue and saturation. Rotate your head around a chosen color to adjust value.



Step 4: Match the fruit colors

Use the color selector to find a color that matches each fruit. Your left eye sees the fruit, while your right eye sees the color space.

Step 5: Match the four color blocks

Match red, blue, green and yellow. Record the resulting H, S, V values and the time shown after you finish each task.

Important: Looking at a chosen color for more than one second selects it and exits that scene. Avoid staring at a color too long before you are ready to choose.

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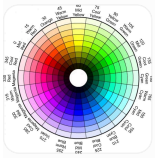
14. HSV — the three parts of a color

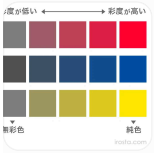
HSV stands for **Hue, Saturation and Value**.



H — Hue

Which color is it? Red, blue, green, yellow, etc.

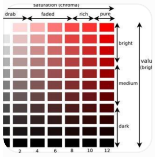
Memory: Hue = type of color.



S — Saturation

How strong or vivid is the color?

Memory: Saturation = strength of color.



V — Value

How light or dark does the color appear in the HSV model?

Memory: Value = brightness.

Quick memory trick

HSV = Color, Strength, Brightness.

- Hue: What color?
- Saturation: How vivid?
- Value: How bright?

The homework allows values from 0 to 255 for H, S and V in the answer format. Use the actual values displayed by your app rather than guessing them.

Part 3: Homework Questions — What should you enter?

The assignment says many answers are accepted because individual results differ. But some answers require you to perform the experiment yourself.

Question	What you should do

Q1: Consent	Choose whether you consent to the use of your data for research. Either answer is accepted.
Q2: Color blindness test	Take the linked test and enter your actual score in the requested format.
Q3: Dominant eye	Use the paper-tube test and select the eye you naturally use.
Q4: Red square	Enter the actual H, S, V and time displayed by the app.
Q5: Blue square	Enter the actual H, S, V and time displayed by the app.
Q6: Green square	Enter the actual H, S, V and time displayed by the app.
Q7: Yellow square	Enter the actual H, S, V and time displayed by the app.
Q8: Eye strain	Answer Yes or No based on your experience.
Q9: Unusual visual effects	Answer Yes or No based on what you noticed.
Q10: Closed one eye?	Answer Yes or No based on what you actually did.

For Q4–Q7, the assignment gives this example format:

```
H:64, S:95, V:100, Time:12s
```

That is only an example, **not the correct result for your experiment**. Your actual numbers may be different.

For Q2, the requested format is `X% (Y,Z)`. For example, if your actual result is 100% (18/18), enter that result.

Before beginning the experiment

- Turn off Night Mode.
- Disable automatic brightness and set a fixed brightness.
- Avoid closing one eye during the matching tasks.
- Notice whether you experience eye strain or anything unusual.
- Record the values and time shown by the app.

If the app causes discomfort, take a break rather than forcing yourself to continue.

Part 4: MCQs for Practice

Try answering these questions before opening the answer key.

20 practice questions 0/20 answered

1. What does JND stand for?

- A. Just Natural Difference
- B. Just Noticeable Difference
- C. Joint Numerical Data
- D. Just Normal Detection

2. What is the absolute threshold?

- A. The largest possible sound
- B. The smallest detectable stimulus
- C. The color of an object
- D. The distance to a speaker

3. Which scientist is associated with Stevens' power law?

- A. S. S. Stevens
- B. Isaac Newton

C. Albert Einstein

D. Charles Darwin

4. Which type of audio usually has one channel?

A. Surround

B. Spatial

C. Stereo

D. Mono

5. Which type of audio aims to place sound all around the listener in 3D?

A. Mono

B. Spatial audio

C. Silent audio

D. Text audio

6. What does ITD measure?

A. Sound color

B. Time difference between ears

C. Head size

D. Screen brightness

7. What does ILD measure?

A. Loudness or level difference between ears

B. Color difference

C. Screen resolution

D. Time of day

8. Which cue means interaural phase difference?

A. HRTF

B. CAT

C. IPD

D. HSV

9. What does HRTF describe?

A. How the head and ears change sound from different directions

B. How a screen produces color

C. How fast a computer runs

D. How to draw a color wheel

10. What is the pinna?

A. The inner part of a computer

B. The visible outer part of the ear

C. A speaker cable

D. A type of light

11. What is the cone of confusion?

A. A color adaptation model

B. A region of sound locations that can create similar localization cues

C. A type of headphone

D. A sound recording file

12. What is the precedence effect?

A. The first-arriving sound usually influences perceived location

B. The loudest sound is always ignored

C. Light travels faster than sound

D. The last sound always wins

13. What is chromatic adaptation?

A. The ear adjusting to loud sounds

B. The visual system adjusting to changes in lighting

C. A method of recording sound

D. A type of speaker

14. What does CAT stand for in color perception?

A. Color Audio Technology

B. Chromatic Adaptation Transform

C. Computer Adaptation Tool

D. Color Angle Test

15. What does H in HSV stand for?

A. Head

B. Height

C. Hue

D. Heat

16. What does S in HSV stand for?

A. Sound

B. Saturation

C. Shadow

D. Speed

17. What does V in HSV stand for?

A. Value

B. Vision

C. Vibration

D. Volume

18. Which setting should be turned off before the color homework?

- A. Night Mode
- B. The keyboard
- C. The camera app permanently
- D. The internet browser

19. Where should the real H, S, V and time values come from?

- A. Guessing
- B. A friend's answers
- C. The app after each matching task
- D. A random website

20. Why is VR useful for chromatic adaptation research?

- A. It lets researchers control visual conditions
- B. It removes all differences between people
- C. It makes every color identical
- D. It does not need a display

Check my answers

Final revision sheet — Memorize these 12 points

1. **JND** = Just Noticeable Difference.
2. **Absolute threshold** = smallest detectable stimulus.
3. **Mono** = one channel; **stereo** = usually two channels.
4. **Spatial audio** = sound positioned in 3D space.
5. **ITD** = difference in arrival time between ears.
6. **ILD** = difference in sound level between ears.
7. **IPD** = difference in sound phase between ears.
8. **HRTF** = how the head and ears change sound from different directions.
9. **Cone of confusion** = different locations can create similar hearing cues.
10. **Precedence effect** = the first-arriving sound usually influences perceived direction.
11. **Chromatic adaptation** = eyes adjust to different lighting.
12. **HSV** = Hue (color), Saturation (vividness), Value (brightness).

One last memory trick: For audio, remember **T-L-P-H**: Time, Level, Phase, HRTF. For color, remember **HSV**: Hue, Saturation, Value. These two sets cover many of the most important terms in your lecture and homework.

You said:

Quiz-11-2026.docx
Document

explain easily

ChatGPT said:

I'll explain your **Quiz 11: Spatial Audio Rendering Pipeline** in very easy words, like you're in 7th class. We'll go through all 10 questions in your document, with simple examples and memory tricks. Quiz-11-2026.docx

Quiz 11 — Easy explanation

Q1. What is spatial audio?

Imagine you are playing a game. A bird chirps on your left, a dog barks behind you, and a plane flies above you.

- **Stereo:** Mainly gives you left and right sound.
- **Spatial audio:** Makes sounds seem to come from different 3D directions, including above and below.
- **Head tracking:** Detects when you turn your head and adjusts the sound accordingly.

Correct answer: C — Spatial audio represents sound in 3D directions and incorporates head tracking.

Memory trick: **Spatial = Sound all around you.**

Q2. What happens when you turn your head?

Imagine a virtual dog standing in front of you. You turn your head 90° to the right.

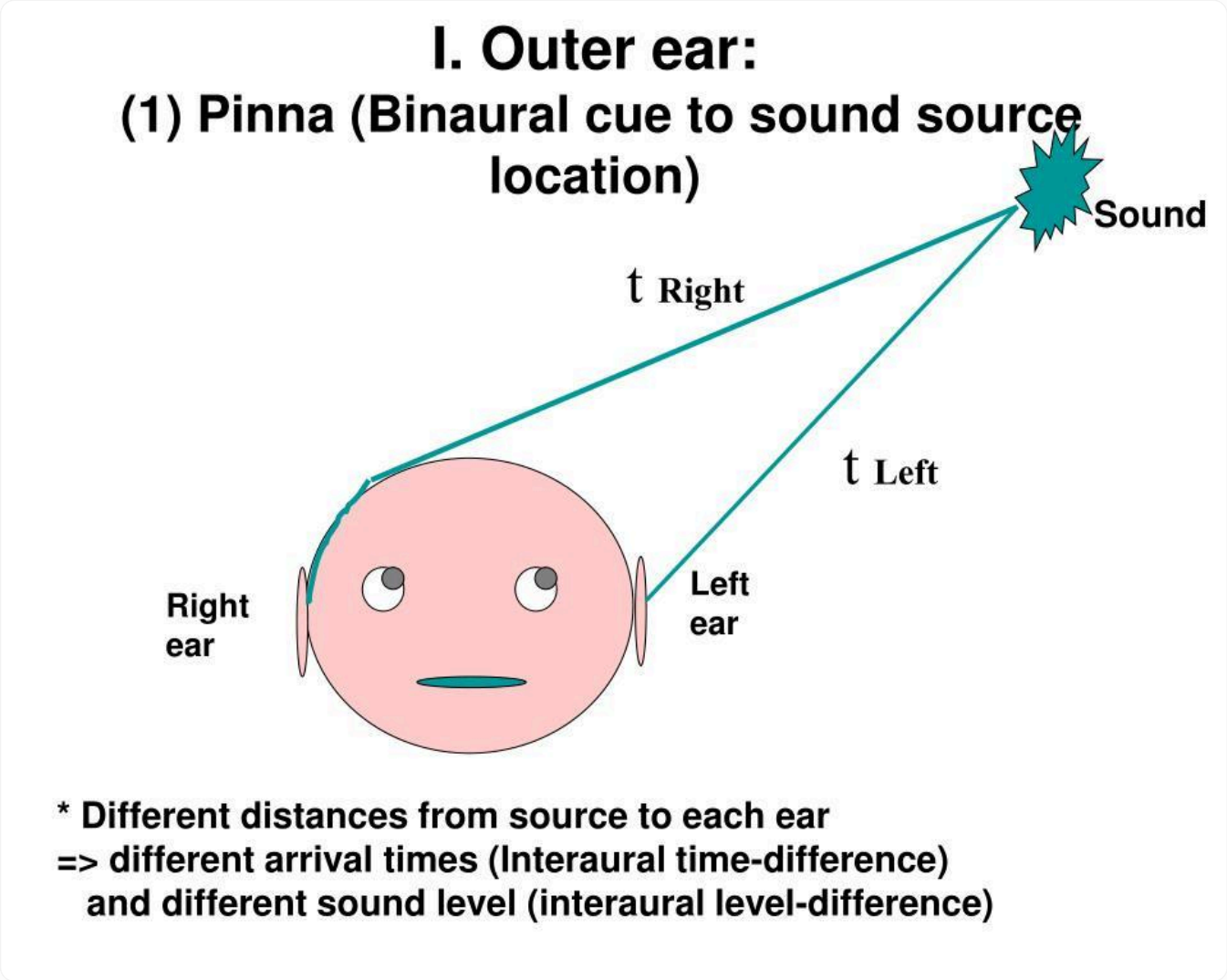
A spatial audio system with head tracking adjusts the sound so the dog still seems to be in its original virtual position.

Correct answer: C — Spatial audio using head tracking and HRTF.

Remember:

- Head tracking = knows how your head moves.
- HRTF = helps make the sound seem to come from the correct direction.

Q3. What is ITD?



Suppose your friend claps on your right side.

The sound reaches your right ear first and your left ear slightly later.

The difference in arrival time is called **ITD — Interaural Time Difference**.

Correct answer: B — Interaural Time Difference.

Memory trick: **ITD = Time Difference.**

Q4. What is ILD?

Your friend is standing close to your right ear and claps.

The sound may be louder at your right ear than at your left ear because your head blocks some sound.

The difference in sound level is called **ILD — Interaural Level Difference**.

Correct answer: D — ILD.

Memory trick: **ILD = Loudness Difference.**

Q5. What are the three major hearing cues?

Our brain uses three important clues to locate sounds.

Short form	Full form	Easy meaning
ITD	Interaural Time Difference	Which ear hears it first?

ILD	Interaural Level Difference	Which ear hears it louder?
IPD	Interaural Phase Difference	How do the sound waves differ in phase?

Correct answer: A — ITD, ILD, IPD.

Memory trick: **T-L-P = Time, Level, Phase.**

Q6. What is the maximum ITD?

Your quiz gives these values:

- Head radius = 20 cm
- Speed of sound = 340 m/s

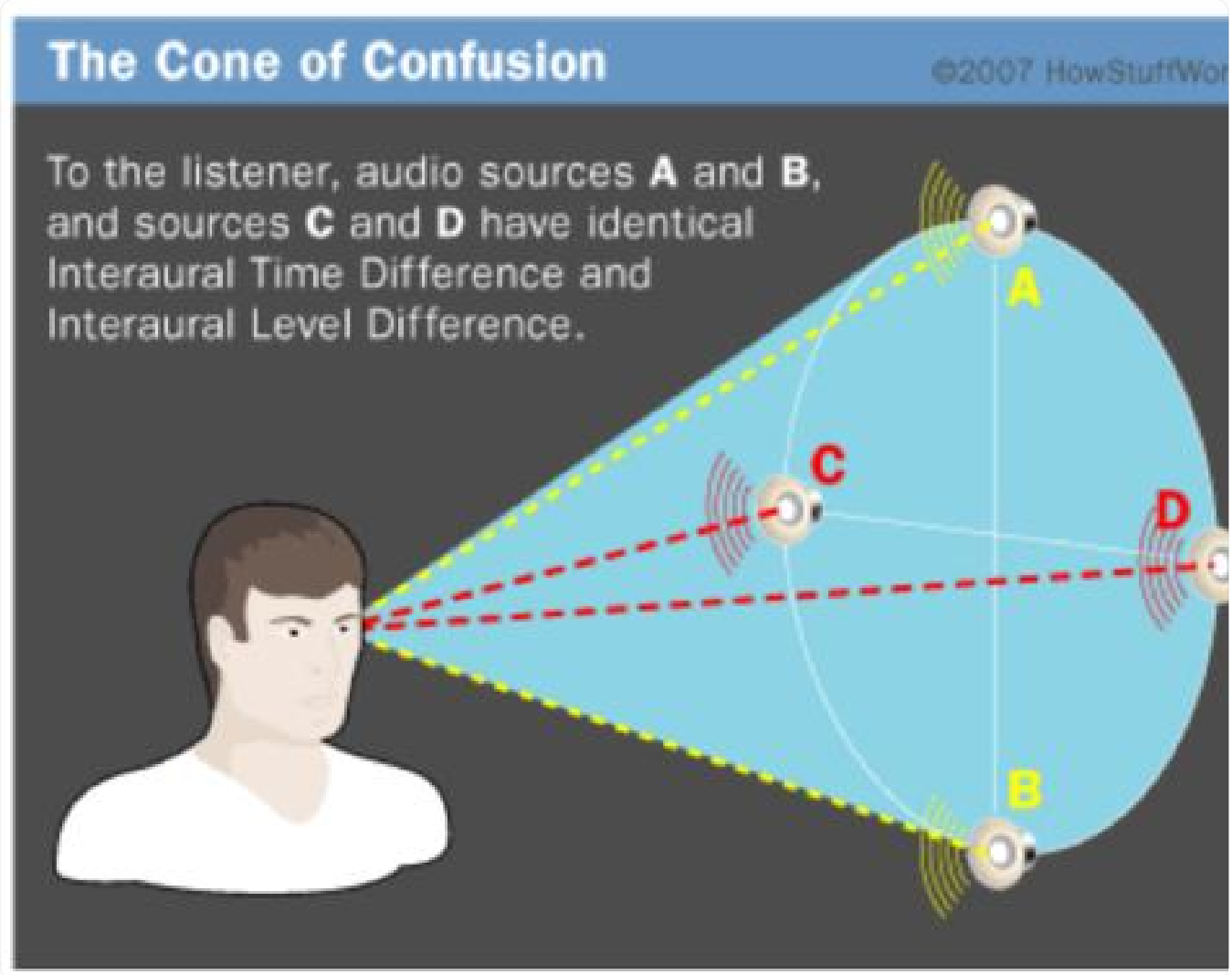
The approximate maximum time difference between the two ears is **0.65 milliseconds**.

Remember that 1 millisecond is one-thousandth of a second.

Correct answer: B — 0.65 ms.

Memory trick: Choose **0.65**, not 6.5 or 65.

Q7. What is the cone of confusion?



Imagine someone makes a sound. Sometimes your brain finds it difficult to decide whether the sound came from the front or the back.

Certain different locations can produce similar hearing clues. This region of ambiguity is called the **cone of confusion**.

Correct answer: B — Cone of confusion.

Memory trick: **Cone of confusion = Confusing sound locations.**

Q8. What does HRTF include?

HRTF does not describe only the outer ear.

Sound is changed by several parts of your body:

- Head
- Outer ear, called the pinna
- Shoulders
- Sound reflections and resonances
- Differences in sound paths to the two ears

All these effects help your brain understand where a sound comes from.

Correct answer: B — HRTF captures the combined effects of the head, pinna, shoulders, reflections, resonances and interaural differences.

Memory trick: **HRTF = How your body changes sound.**

Q9. Why can sounds from behind have weaker high frequencies?

Your outer ears face mostly forward. Their shape changes sound differently depending on where it comes from.

Therefore, sounds coming from behind can have reduced high-frequency content compared with sounds from the front.

This change helps your brain identify sound direction.

Correct answer: A — The forward-facing shape of the pinna.

Memory trick: **Pinna shape helps identify direction.**

Q10. What is the precedence effect?

Imagine clapping in a room.

1. You hear the direct clap first.
2. A reflection arrives shortly afterward.
3. If the sounds arrive close enough together, your brain usually uses the first sound to determine the main direction.

This is called the **precedence effect**, also known as the Haas effect.

Correct answer: C — The first-arriving sound dominates the perceived location.

Memory trick: **Precedence = First sound takes the lead.**

Quick revision: all 10 answers

Q1	**Answer: C** Spatial audio represents 3D directions and uses head tracking.
Q2	**Answer: C** Spatial audio with head tracking and HRTF.
Q3	**Answer: B** ITD = difference in arrival time.
Q4	**Answer: D** ILD = difference in sound level.

Q5 **Answer: A**
ITD, ILD and IPD.

Q6 **Answer: B**
Maximum ITD ≈ 0.65 ms.

Q7 **Answer: B**
Cone of confusion.

Q8 **Answer: B**
HRTF includes combined effects of the head, ears and body.

Q9 **Answer: A**
Forward-facing pinna shape.

Q10 **Answer: C**
The first-arriving sound determines the main perceived location.

Test yourself!

1. Which cue tells us the difference in sound arrival time?

- A. ILD
- B. ITD
- C. HRTF

2. Which system adjusts sound when you turn your head?

- A. Spatial audio with head tracking
- B. Mono audio only
- C. A fixed speaker alone

3. What does ILD measure?

- A. Sound arrival time
- B. Color intensity
- C. Sound level difference

4. What does HRTF help us understand?

- A. How the body changes sound from different directions
- B. How to increase screen brightness
- C. How to measure distance in metres only

5. What is the precedence effect?

- A. The last sound always wins
- B. The first-arriving sound usually determines the perceived location
- C. All sounds disappear

Check answers

The easiest way to remember the entire quiz: Learn **T-L-P** (the three hearing cues), **HRTF** (how the body changes sound), **cone of confusion** (uncertain direction), and **precedence effect** (first sound leads). These ideas cover most of the quiz.

You said:

[Music] Now let us look at the ambiox. Now higher order ambioenics. We are talking about the third order, fourth order, fifth order amiox. First order ambioenics. Here in this graph you can see that uh okay maybe first we'll we'll see this graph. So for you know order uh zero it's only one first order there are three basically there are four basis functions all the four together for the the first order one second order we will have 1 2 3 4 5 + 4 9 will be will be the basis for the second order. So all the night face no basis functions will be used to synthesize the the given uh audio sound right and then third means again now uh seven added this will be all the third level and fourth level. So the higher the order the more basis functions to approximate or to realize the given function. How is this basis functions is actually now understood in a intuitive way. Here is an example. First order let us say uh first order let us look at it 1 2 3 here 1 2 3 let us look at x-axis. This is x axis. This is y axis. This is the z-axis. Right? So uh uh the first order we are looking at the y direction. The second basis function looking at the z-axis and the third basis is looking at the x-axis. This numbers this you know colored indicates what is the direction of this basis functions. These are all basically orthogonal you know basis functions right. So this are all the axis right. So the the orientation is actually given over here. Whereas this is at the origin zero order is the origin. Now in the second order you can look at it. It is the the plane. This is the xy plane. This is now y z plane and this is the know uh zz axis and this is the xz plane and this is the xy plane. So uh you can see the different uh way of combining like this uh this uh this basis functions right. Similarly now third third order the higher order will be even more know further little more complicating it but there are no this amiox are being constructed for synthesizing the 3D sound mostly where is it used it is used in in theaters high-end theaters uh the the latest movies are actually using this higher order aioinx which we are going to see right now I with a uh yeah before I go to the wakeful synthesis the next way of rendering the uh rendering the sound I wanted to see if there is any question on this is there any question on this okay yeah is Is there any question on this so higher order amionic ambioxics I hope I have given you overview of it and there's a lot of mathematics but know if you get a know idea and if you are able to know read the latest papers and And probably now if you want to implement it, you can implement it with this current understanding of how this orthogonal basis function of the uh uh so your question is for ambisonic unable to visualize the third order amonal basis. Okay, let's look at the third order amio sonets. See the third order we are going to talk about know more directions right here is only two two directions x and y in the second order it's only two directions in the first order only one direction in the zero order there's a zero direction okay so we have to talk about three different directions uh in the third order and this three different direction what are different combinations of three different directions. That is what the the bottom row is about. Okay, I hope now you can visualize it. All the combinations permutation in combinations of three different directions, three different combinations of the the the axis is what is giving you the third order you know uh basis functions. Any other questions? Could you provide details for user placement in the graphs? Uh user placement. The question is what do you mean by user placement in the graphs? Where exactly the user is placed? I see the origin represent to the audio source. Oh, I see. Okay. So, we should have put the where the source of the audio is. Yes, I will do that. Okay. Yes, I'm sharing the PPTs. Uh it is in the it is going to be in the list for unable to visualize. I hope now you you are able to visualize it. Could you provide the details of the user placement origin to the uh audio? Okay. Yeah, we will do that. Now let us look at the the next way of song synthesis wave field synthesis. This is another technique. We will uh uh we will find the difference between and how wave field synthesis is different from ambioxic sound. audioics is capturing the the entirety of the 3D sound and then trying to represent and render it. Whereas wavefield synthesis is synthesizing the wave in a way using kind of some you know uh some uh set of speakers. Okay. So um this is coming from actually one particular principle called the heens principle. Higgin's personal principle to be more accurate. Hyen's print principle says every point on a wave for example if this is a wave we are talking about and this is a surface we are hitting this is sound wave frame when it hits every point on a wave is a source of wavelets that spread out in the forward direction at the same speed as the wave itself. Suppose now if you take one point here over here let us say let let me take as a different color. Let us say this is a point and this point becomes a source of secondary wave. This is going to generate spherical waves from here and it is it is in the same speed as the wave itself. The new wave is a line tangent to all the wavelengths. Right? Suppose if we have of this uh sources these are the points taken right and each one has a no spherical wave and then tangent to all this wave is a new wave frank okay that is what this is about so this was proposed some 400 years ago the high principle has been proposed some 400 years ago and we are applying this principle now in the wave synthesis. How would this apply? Let's see. Uh so suppose if this is a small small speakers, these are the small small speakers, right? And we want to know uh generates sound in the speakers in such a way that when user is over here he hears a sound as if coming from a virtual car virtual acoustic source either this car or the bird he should be able to you know feel it just by manipulating the wave because this wave when it When it is reaching a another wave, the now the intersection is itself is causing another new wave, secondary wave and the secondary wave is is now actually now giving a illusion of the virtual sources just by manipulating the delay and the gain. we can

steal the the perception of where the the the 3D sound is there. Right? This is the loudspeaker array and these are the elementary uh wave and the blue colors are the synthesized wave packs. So uh this uh uh hyen principle is used to design wave fields and there are actually no uh theaters built using the wfield synthesis. I'm going to show you some theaters. Here is a hall where now linear array of now speakers are over here. And they all steer the wave to give an impression of know where the the sound is coming from. Not only at the level of where the people are sitting. Here is the the overhead and uh array of speakers to give you the the immersive not only the sound coming from the at the horizontal plane but also the now upper plane as well as bottom plane the sound can be steered to give you the the the feeling of of sound source. There is a sound bar in the market recently Sony or there are other companies which looks like a very small u uh linear array of speakers right and that it uses the the the configuration the geometry of the room to create the the 3D sound actually that is using the wavefield synthesis. So wave field synthesis for example we have the HRTF of 180°. So let us say we have two uh speakers this speakers and this speaker and here is the user over here and uh we can create the virtual surround over here as we have to make it appear coming from the behind also but whereas the speakers are only the front. How do we make it coming from behind? There's a whole challenge, right? So, we will use the HRTF at 110 HR of zero and then HR of plus 110us 110. We will use this three HR and then create the wave synthesis field synthesis. While doing it, we need to be careful to remove the cross talk controller. Otherwise, no the the the the speaker uh sound from one speaker is going to be no is going to be is going to be uh picked up by the other speaker. It is going to interfere with all of the another speaker. Therefore the protop cancellation is very important and uh cross talk cancelceled version of the virtual speakers added to the main left and right channels right this is very important. So u this wave field synthesis is actually it is in it is in the commercial commercialized now. Okay. Um you you all have heard about know uh all versions of the stereo and sur. So 2.0 is the two stereo sound. 3.0 is uh 3.1. What is 3.1? 3.1 is the No, it's three in the front and one is a subwoofer. 5.1 is the three in the front and then two in the back and one is the subwoofer. 7.1 is no three in the front, two in the back and then other two at the side. 7.1 9.1 maybe up and down as well right along with the subovers. So we can keep adding it but this is a surround sound. This is different from the wave field synthesis. Maybe this speaker also can and can now implement this surround wave field synthesis. Right? But S but the wavefield synthesis doesn't need this uh speakers arrangement at all. That is one of the uh major differences between this uh no surround sound and the wave field synthesis. Let me see if there are any questions over here. Can we simulate a higher order using the software? You are talking about the higher order of the uh the uh uh ambisonics. Yes, definitely you can use the higher order. Okay. Yes. So somebody wants uh now wavefield synthesis. Quickly summarize one more before I go to this Derby atmas. Yes, let me go again. Wavefield synthesis uses the highensal principle. Higgin's personal principle is is where the uh every point on the wave can be used as a source of wavelengths that spreads out in the forward direction at the same speed as the wave itself. Basically, wherever there is an intersection, your new wave is generated. that new wave know where it should be generated what is the direction it should be generated that can be determined by position of the speakers and therefore we can steer the the wave the steer the sound wave right that's the whole idea of this high fal principle and the wavefield synthesis and uh this is the speaker array you can see the speaker array and this speaker array. Now uh by controlling the delay and the intensity this wavefront is new wavefront is created and the new wave is actually steered and that new wave is giving illusion of the the the source of the virtual source of the uh the sound. I hope uh now uh it is clear. Now let me go through quickly the the the last point last topic on this particular uh 3D audio sound. There is a new idea of Dolby sound coming all the nowadays. Now uh it is adapting this Dolby Atmos. Dolby Atmos also is a no spatial sound but it is object based. What do we mean by object based? So the 3D sound is considered as an object in the sense while recording while encoding this uh while encoding the source of the audio is considered as the object. So irrespective of while rendering irrespective of where the the speakers are it can be two speakers or three speakers or five speakers of the position of the speakers. Now the source of the uh sound is actually fixed because this is object based. So the Di atmos encodes the the source of the sound as an object. Therefore it doesn't change it is hugely independent of the speaker arrangement. That's what the the whole idea of this address. So the metadata the location of each sound in 3D is the metadata. That's the object. It ensues a similar sound effect regardless of the number of speakers now where they are actually located. Okay. All this is you know Netflix, Disney plays, Apple TV, Amazon Prime, HBO Max the are all using this the whole list is all using this now Dolby Atmos this way. Now uh uh the you would have trueity in the home theaters for example you may have only 5.1 now I may have only 3.1 so how do we now generalize or reduce the the the change in the hardware effects is what the whole idea of object based right uh this you know format is becoming popular. I want you to be aware of it. And the final topic today is the how do we evaluate the 3D sound. We saw the psychopysics of the 3D sound. We saw how do we render this sound using the higher order ambisonic uh atmos right? So now how do we know whether it is correct or not? Wavefield synthesis we might need many number of of speakers. Each speaker spaced no very littleness because the spacing of the the speakers that determines the the the frequency of the sound as well isn't it? So we all know the the uh the uh sampling theorem correct. So uh uh how much degree of externalization how far outside we can hear the 3D sound that is one parameter of evaluation. What room characters can presets how small room or big room or what idea of the room um can be simulated. That's again another way of sim uh evaluation. As far as the head tracking is concerned, can we have the quality of the latency?

How much is the latency? How much is the degree of freedom? What are the artifacts and anomalities are there? And how this artifacts and anomalies are are affecting the quality of the sound that also will be part of the evaluations. Right? There are much more topics of the evaluations and I'm going to share some document on this reading materials and I hope that will be useful for the time being. Okay. So I have quickly summarized there are ultrasonic direction speakers available. How about those technology? Ultrasound you cannot hear that is for now imaging only. So let us not talk but the same technique actually that's also steering you're also steering is this the same hyens who gave the uncertity principle yes you're right you're talking about 400 years right are the specific packages tools available in the game engine like unity QT etc for developing audio experience will this be covered yes so there are are specific packages in Unity we are going to talk about a very simple tutorial how to use the 3D sound in the unity. We are going to give a tutorial in the next class. All right. So, right now we have covered only the theory. Now we'll go to the practical in the next class. All right. If there are no more questions, we'll stop here and uh see you uh on Friday. Till then stay safe. Thank you. Bye. Take care. [Music] [Music] Welcome back. So in the last uh few lectures we have been looking at the spatial audio. The previous topic what we discussed about before there was spatial audio we also looked at the the visual rendering pipelines in all those things. Now we talked about the the psychopysics perception engineering which is a now very important topic uh of uh the entire course to continue the now the topics of visual rendering and then audio rendering today we will talk about the haptic rendering pipeline as well. Okay. So particularly we are going to talk about what is haptics and uh now how haptics is rendered is what we are going to see by the end of this class. Before I start talk talking about uh haptics there are few announcements I wanted to make. Now last week also I made this announcement that we are we are conducting this XR summit at IIT medas November 16th to 17th here. Top 10% of this course are invited to attend and there are no more quiz and no more homework. Your exams are scheduled on next Friday and Saturday. Okay. So uh next class that is the Tuesday's class is going to be a industry lecture coming from next class is that is on Tuesday is an industry lecture by QT one of the leading opensource now GUI 2D GUI and they wanted to extend that to now virtual reality. So it will be very interesting to hear from them. We'll also see whether we can discuss about uh uh all the challenges in VR in the next class. So that would be the the end of this lectures and next Friday we'll have this final exam and Saturday you are given two options two attempts to know do the final exams. You can choose either of them. Right? So this top 10% of this course we will release the eligibility today. We have the list already. We are going to release it today and registration website will open by this week. So you will come here for one week of immersion in IT madas and uh this XR summit for two days is going to be a part of this total immersion in the course. If there are any questions we will discuss it later. So uh we will start with a simple question. The cell phone vibration is at what frequency? You all answered in the know previous lectures. Now the question is the new question is that do you call the cell phone as a haptic device? We discussed this as topic of tactic engine in the iPhones which we have a no contribution. Our know scholar from IIT MAS who was trained in haptics has designed this tactic engine for Apple. Right? But can we call the Apple phones with a tactic engine or any other phones with a vibration as a haptic device? That is a question. What when do you call something a haptic device? If we cannot call Apple phone as a haptic device, then what is the name for this haptic Apple phones? The haptic feedback in the optic uh Apple phones. When do you call something a haptic device? That is a question we are going to answer in this particular course. Right? So virtual reality entire course we are almost at the end of the course. Right? So we are looking at the know the bigger picture of now. Most of the time we are talking about the immersion that is one of the pillars of the course. How do we improve the immersion? We looked at a lot of quality parameters. How do we improve the immersion? There is equally important parameter. Another pillar of this virtual reality is the interaction. These two things are no two pillars. I would call as two eyes of the virtual reality. Without the interaction, however immersion is there, it's not going to be useful. Interaction without immersion is not going to be useful together. Immersion and interaction has to go hand in hand. Only then we can have a know quality virtual reality experience. Right? So most of the virtual reality courses around the globe they focus on immersion but in our course we also talk about the interaction and how do we improve the interaction. If you want to improve the interaction, first of all, we need to understand what is interaction. What are the factors that is affecting the interaction? Therefore, how do we control it? How do we improve on it? How do we measure it? Right? Only then we can improve on it. So interaction in virtual reality, I'm going to show you by the end of this class, you will understand that it is nothing but haptics. There are many ways of interacting with the virtual reality, right? You're all aware of it. We have been doing a lot of homeworks and then assignments in virtual reality. Communication to virtual reality is the interaction, right? We use gaze and speech and gestures, writing, they're all motor acts. They're all part of the interactions. We use our body parts also to interact with virtual reality. What is that we use? We use the heads, hands and legs. the whole body even the eyes also we use to interact with the virtual reality to make future movements we need no sensory memory and cognitive no belief processes in the sense only when we understand that know the current communication or interaction methods well we can design the featured interactions Well, designing very good interactions in virtual reality is still a challenge which we know a lot of research is still going on. For this we need to understand the know sensory memory cognitive belief processes in the sense psychopysics perception engineering has to be understood very well only then we make so haptics plays a major major important in the future of virtual reality. So far all this communications and know uh the

interactions design it's all you know uh it's all uh I would call it as a uh not a systematic design with the haptics we can design systematic interaction with a virtual reality right so uh is this very clear you know why we need to study hortics heartach is not only no a very small part where is in the phones or in your cell phones the entire interaction is what heart is what I'm conveying why it is so I'm going to know take you to the next slides and then present it but I wanted to make sure that no the point which I'm driving it's very clear okay the more the reason why we have to understand uh haptics is very very uh know simple because the entire interaction the two pillars of virtual reality we are talking about 50% of the now the the the pillars is that interaction that we need to understand haptics is a foundation of it therefore it is very very important once you understand haptics you can build better quality virtual reality systems in the future So uh how are we going to understand this uh haptics? We are going to talk about the the the human factors of interactions. We call it as a now human haptics. We are going to talk about how devices can be designed for interaction in the virtual reality. that is the machine haptics and then now software for interaction we are going to call about now computer haptics these are the three topics we are going to see today so uh human haptics machine haptics and computer haptics is the broad topics we'll talk about it in human haptics we are going to talk about know psychopysics of the human haptics neuroscience of the human haptics and biomechanics of the human haptics also we can talk about it meatronics in machine haptics we'll talk about the sensors and actuators that's a meatronics and also the computer systems and computer haptics like computer graphics it's a no yeah software system we are looking at it I offer a one semester full course on haptics where we discuss all those points in uh in uh in very much detail so I'm going to give you only summary and then I'll walk you through uh the important points in this particular has okay so uh as I mentioned perception engineering plays a major role in understanding the haptics I'm just going to quickly review uh just to remind you uh the the important concepts of the perceptual engineering we need to understand the human perception very well important highly highly important right we know what is the unit of the perception J&D right so the J&D is nothing but the difference threshold threshold is one way to measure the perception that's a measuring the limit of the sensitivity and fnel is the one who introduced this now sensation unit there is a J&D first time and he is called the father of psychopysics and here is the physical intensity or stimulus intensity this is the perceived intensity uh in the y-axis and each one is now given a unit that's a J and D and this can be now pointing upward or downward or linear or in any way possible all the stimulus will have a different different curve and that is what now Stevenson has later generalized to power law it's a power factor that is important for every uh stimulus possible he has found out power factors and threshold is only one of the measure there are recently a lot of measures are proposed accuracies and reaction times and point of subject equality and then ratio scales and other measures also has been proposed but among them threshold is the most important things and widely used right and we saw when we talk about the tactile perception tuning curve the reason why our cell phones are now vibrating at 250 htz is that at this particular frequency we have the highest sensitivity So if we take xaxis as a frequency, y-axis is a displacement and at 250 Hz we have the highest uh highest sensitivity it's about.1 and.1 micron right uh micron that much amplitude is good enough to give us the sensation of the sensation of the vibration correct now what is haptics if to search in Google or in the you know literature as well. There are no hundreds of definitions which may be confusing you. So uh um every haptics researchers use their own haptics definitions. What we are going to talk about today is that now our own definition of what haptics is and why we have defined the haptics in a way we did it right. We'll give you the reason also why right. So uh can you just uh no uh give me answer type in your answers. What what do you think is haptics is? Let me let me start now you can you can do the Google also but know without googling also you can type in what do you think is haptics. Let us start with that at uh your feedback on what is what do you mean by haptics now any answers? I wanted to see some responses on what is hard. Can you type in your reactions responses? The question is what do you mean by updates? Sense of touch. Touch and feedback. Tactile simulation. Okay. Anymore. What do you think haptics is? Physical feedback. Okay. Anything else? Let me close this. Close. Physical touch and feedback. Tactile simulation. um touching sensitive any other any sensory interaction. Okay, machine human interaction. Okay, we are asking what haptics means? What do you understand by the word haptics? Sense of pressure on body parts. Okay, any type of interaction Is that correct? Okay. All right. Sense of uh body part. So um interaction where feedback is available. Interaction means no uh without the feedback we can do the interaction as well. Okay. Sense of touch. All right. 3D touch. Okay, I can see a variety of answers on know the simple question of what a haptics is. Right? So let us look at what a haptics is. For light we are are uh no for vision we are we are seeing and using the eyes optics is the know the physics of it and light is a basic element of it. Similarly for auditory we are hearing using the ears. Acoustics is the basic science and then sound is the the basic stimuli. Similarly for haptics it is for touching. Skin is the the organ and the the the biomechanics is a basic science and then force is the the basic stimuli. Still this is not uh not defining very clearly what haptics is right. Haptics it is much more complex than sense of touch. Let us look at what what sense of touch is. Some of you have mentioned that it is a sense of touch. Sense of touch is fine but you know haptics is much more than that. We will see. Sense of touch is also called a sumatic sense. I don't know how many of you realize that that the the biggest sensory system in your body is the the haptics right that is called it is called the somatic sense somatic senses means the entire body the bodily senses right I'm going to write it out to emphasize this bodily sense is called the somatic sense and that is the tactile sense or that's the touch sense some of you mentioned that there is a

tactile sense. Tactile sense is only one of the four major modalities of the Domatic sense. Okay, tactile sense means it is a fine touch and then fine pressure. For example, your shirt is applying a fine touch and fine pressure on your skin. Right? So that is the tactile sense. And there is a no temperature sense as well. your if you bring a know some hot object on your skin you feel warm and cold or cold right so there is a temperature sense also there is also pain also is part of the touch maybe surprising for many of you but now pain sense also is part of it and then the fourth important most important of all is the kinesthesia sense kinesthesia sense is the sense of our position or motion right when I move my hands from one position to another position. All the muscles in the hand actually convey the position to the the brain and that sense is called the kinesthesia sense and that sense also is a important part of the somatic senses. This four modalities together is what the somatic sense and that is the sense of touch. So when we say touch immediately you should remember that tactile sense, temperature sense, pain sen and chesthesia sense together is what touch means right? And can you imagine why I said this is the somatic sense. This tactile sense is on the skin temperature skin. Temperature sense can imagine pain skin sorry pain sense can imagine both are throughout the body even inside the every organ every tissue every cell has this uh the sense isn't it and that is why this sense of touch is called the somatic sense or the bodily sense. In fact, the entire body is the touch sense and in the head all the other senses are located and it is actually using the touch sense to know to do its function. Well, can you imagine how you can hear without the touch vestibular sense or the you know coccia sensor it's all using it's all it's a no highly specialized touch sensor it is you converting the force into electrical signal right then know taste you know know smell we know although we think there is the molecular aspect of it but inside it is the contact that is happening and touch is the the most uh fundamental sense of the entire human body. Can you imagine how your eyes function and uh without the touch sense if you don't have the touch sense you cannot move your eyes well you cannot fixate on an object we cannot do the no all the movement of the eyes so um eye movement is important therefore the touch is important right anywhere wherever movement is there now haptics is there sense opposition kinesesthesia comes into picture if you don't have the kinesthesia you cannot control the moment therefore kinesthesia is very important therefore that is the reason at the beginning of the the lecture I said interaction means it is haptics interaction means it is touch is only the sensory part of it I'm going to show you that there's also control part of it right I hope I have uh emphasis enough on this but for our technology purposes when we talk about the our machine haptics there are no temperature simulator stimulator or the pain stimulator. Therefore now when we say for our course point of view somatic sense means there's a tactile sense and the canesic sense. We are going to simulate how tactile sense and cania sense can be implemented in the technology. [Music] [Music] And this touch can be both active as well as the passive. Okay. So how touch can be active as well as passive? I have this pen with me and this pen I'm going to keep it on my skin. This is a passive touching of the the skin. Whereas my hands it is going to actively touch this pen and then rotate it and then know find out what is the texture on it and it looks very nice rubbery feeling right that is the active part of it. The touch can be both active as well as passive. Correct? So the hand is the hand or hand or finger contacting or receiving sending physical sensation. The whole skin can only be touched. Your shirt is touching the skin but it is only the passively touching it. It is not actively touching it. Right? The sensory and motor systems of touch right. So there is the now sensory system is passive motor system is active. When there is a motor systems involved then it becomes active. Hand is a true organ of touch. It can be active. It can reach out some object and then manipulate the object and then find out more information about the object. Right? Directionality. So um uh passive is only unidirectional whereas active is birectional information is is going and coming. Unidirectional means from this object to skin to brain that is the unid directional. Birectional means no object to skin to brain and then brain ask this hands to move away from this hot object. Maybe that is the birectional. Correct. Now we can even define each of the senses. We saw that tactile sense is a part of the somato sense. Right? We can even go and design or define what tactile sense means to give you much more meaning of what does it mean. Right? So um the tactile sense actually we can talk about no can be mechanical stimuli or thermal stimula chemical stimuli and electromagnetic stimuli it's all passively can be touched right tactile fine pressure is what we saw in the earlier slide and this can be you know any of this stimuli or combination of it that results in this perception of each article pain and pleasure, right? So, uh leading the stimuli can lead to this perception, right? That is what the tactile senses. So our definition of the tactile senses that all the tactile person perceptions are triggered by spatial temporal variations of the stimuli above all the sport stimuli or combination of it resulting in the higher brain functions experience or emotions. These are the experience or emotions we are talking about. Right? By the way it is still not very well known why we feel tickled only by somebody else's tickling. If you try to tickle yourself, you will not be able to feel the tickling, right? It is still a active active uh no uh point of research. Maybe now some of you in the future can come and know unravel the the the mystery behind this tickling. Finally, we'll come to define what is happen. So, we saw what is the sense of touch. We saw what is a somata sense and we saw what is a passive touch and active touch and then we saw what is hapt tactile. We we can even define what is kesthesia. Right? Putting all this together we'll define what is haptics. Haptics is pertaining to sensing. There is a sensing involved and there's also object manipulation involved. It is active through the touch. Right? And this has to be by the human or machine. Even machine can do the a robot can be manipulated. It can touch an object, manipulate an object and it can feel the touch of it, right? And it can be in real or virtual in a virtual reality, right? And in a teleoperated environment. Yeah. Doctor doing the telly

surgery. That's also possible. This is now as generic as possible. So this is very important. It has to have a sense. It has to have a manipulation and it has to be through uh touch. Sensing and manipulation can be through the eyes also. Maybe you can think about it. Right? But through touch is what the the definition of haptics is right. So it is for no with the human to machine human to machine or machine to human or human to human human to machine to human that is also is possible right. So uh in a human interaction in a virtual environment it we can think about what haptics is actually doing. Suppose if we touch a real object, this is a real object and we have our real fingers and we are touching it. There are contact forces coming up and this contact forces actually converted to the convey to the brain and the brain ask the hands to move our you know hands or finger to to acquire more information. Motion results in motion and we get more details of the object. This is how the entire human interaction is happening right when contacting our skin has lot of receptors and then receptors is conveying the forces into electrical impulses and through which the brain is actually reorganizing all the information right we are going to see that this is what we are going to replicating in the machine and uh instead of no uh receptors we are going to put the the sensors actuators, right? And uh uh yeah uh this is a computer which is doing the processing and this is equivalent to the muscular skeletal system. We are talk going to talk about it. We will see in detail how we can build a system that can give you a touch feedback. In fact, if if there is an external environment, when there is a stimulus force, we do the sensation of it and know where we convert the stimulus into electrical impulses. That branch of study is called the neurohysiology and then sensation to perceptions we do and that is the theory of psychopysics and perception to cognition. We do lot of learning and all those things. then we do the motor planning and motor planning is asking the the motor system there's a muscularkeeletal system um to do the action. So this part of motor planning to motor system is called the theory of motor action. So these are the different branches of the cortex that talks in detail about each of this uh the process. So what we are going to discuss is only the high level. We will not go into know all this internal details. So let us take a now look at the our uh small cross-section of our finger table. Where are this receptors? How is this information or forces is converted into electrical impulses? Right? So uh we think you know we have a uh your skin. Right? If I ask you to identify the skin, right? How many how many of you will touch your skin and then show me that is your skin? Everybody will show, right? But do you know that we cannot call this whatever we see is not the real skin. It is actually a dead skin. I'm going to show you that now the top portion of it about half a mm thickness. It is called the epidermis. Epidermis means a layer which is over the skin. Right? So what you are seeing is only the epidermis. It's a dead skin. It has to be washed away every day. Under this epidermis there is a layer called know dermis. That is a real layer. And that layer is where there are lot of no uh interesting things are happening. There are receptors are there. Sweat glands are there. Right? There are a lot of sensors are there. There's a blood vessel, there's a neurons, nerve cells. It's all there. And we are going to quickly go into the details of each of this. And now get a feeling of what is actually happening just like what we did it in the uh visual rendering. Right? We we understood the the anatomy of the eyes before we went into the visual rendering, right? Similarly, we'll do the anatomy of our skin and before we go to the heartic rendering, right? So, basically all the sensors in the skin are are four types. Okay. So, uh there is there that is called some name SA1, SA2, RA1, RA2. Those are the four types, right? Uh each one is doing its own now no sweet functions. I'm going to show you what are those functions, right? Um so uh and it has a name also for example this uh uh this sensor is called the mner carpus school it's all named after the scientist who has invented it maybe 200 years ago or 300 years ago and then another sensor called Merkel disc and then there is a third sensor is called Pinian and then fourth sensor is called the no rufini. This is called the no SA1 and this is called the no SA 2, SA 2 and this is called the RA1. Why we call it as RA um one or RA 2? We will now show you now shortly. Right? So this uh Merkel and Dufini they are all no uh they all all um they are all in the no slow adaptive slow adaptive in the sense and they are are called the SA SA means slow adaptive RA means no rapidly adaptive okay both of them are uh slow adaptive and and This one Merkel and Mason they are in the superficial skin. They are not in the deep skin. Deep means they are no range of field is very high. This is RA2 and SA2. The no their receptive field we call it as in neuroscience it is very high. Whereas now if it is very superficial near to the skin then its receptive field is you know uh smaller among this receptors can you imagine now which receptor is responsible for feeding your cell phone vibration I'm going to show you that um uh uh you will understand which receptor is responsible once I explain this to you. So once we give a now stimulator let us say this is a now ram sig stimuli we give and this this is going to be a um uh or or it can be a cenosidal input as well when you give this two stimuli how the receptors are going to behave right so here is masoner masoner is going to now uh give responses and then stop because it is a slow slow uh no this is masonary so this is a rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rap rapidly adaptive uh once there is no stimulate change then there is not going to be no responses there is a neuronal firing is going to be stopped right whereas slow adaptive it is continuing to fire even if if the stimuli is uh is not changing correct and here also it's a rapidly adapt adapted two and you can see that for the same input this rapidly adapted two is responding differently. See how RA1 has responded, how RA2 has responded, right? This is for now measuring high frequency vibration. Each sensor is acting like a filter, right? So your stimuli will have a range of frequency, right? Each filter is know tuned for particular frequency range, right? and RA2 is for a very high frequency range because it is rapidly adapting very very high whereas this one is rapidly adapting lesser. So SA1 as for the lower frequency

SA2 also you can see now there's a uh some differences between the SA1 and SA2 right so among this sensor now can you tell me what is the uh which receptor would be responsible for your cell phone vibration sensation if you are feeling your sensation in your hands among these four receptors which receptor is responsible So SA1, SA2, RA1, RA2 or you can use the names Masoner, Merkel, Rufini and Pini. Okay. SA1, SA 2, SA 1 and two you are saying. Okay. RA1. All right. Any other answers? RA1 and RA2. Okay. Good attempt. Any other attempts? Which receptor is responsible for your cell phone vibration sensation? Ra2. Okay. All right. Any other? Okay. Right. Um. Yes. So, RA2 is what responsible? RA2 is also called the pini carp carpus. So this is a now microscope now view of the pinian receptors. You can see that now this pinian receptors has looks like a nice grapes right and you can see that the it's also like a grape wine right these are all no receptors. Do you know that our gut has a lot of these receptors and it is feeling the vibration of our no movement and then it is conveying to the brain. Correct. In fact, this Pinian has controversial history of discovery. Okay. So, in fact, know 1741 somebody has invented then 1840 Pini is the one who invented. But we called this receptor in the name of the Pini. But there are some people saying that now 100 years previous to Pini this has been invented as well. Right? There are a lot of controversies which now we will not get into it but among all the four receptors fine is the one which is uh which is responsible for high frequency vibration. Okay. In fact now the tuning curve what we have seen when we are talking about the cell phone vibration that is actually tuning curve for the RA2 particularly for the person. Okay, that way no person is the you know most important for haptic as of know all these four receptors is concerned. Okay, so how does it work? We I'm not going to go to the details of each of these ones in a haptics course we go into the details but just to give an idea when the stimuli is applied what is going to happen suppose if you have a pini the subtask in right if you apply very slowly there is like onion like structures inside like a sheet of no layer by layer right if you apply the stimula slowly the layers slip over each other and finally the information doesn't core to the the core of it. Whereas if you apply very fast that's a high frequency then it reaches the core of it and the core will start sending the information. It's a nice way of know this receptor is know designed and uh know you can uh uh you can uh no make know biomimetic sensors using the know this person receptors. Okay I have a question now me receptors necessary for the tactile perception. So so far we have seen four different type of the receptors right SA1 SA2 RA1 RA2 Pini Merkel Miner and Roffini right you may have impression that this mechs are necessary for the tactile perception is this true can we assume very safely that mechano receptor are absolutely necessary for the tactile perception can I have your feedback back. Yes. True. All right. Without me receptors, we cannot perceive touch. Is that true? Yes. Yes. Yes. Any other answers? Okay. Okay. Oh, yeah. There is somebody is saying no. Uh, can you tell why? If you saying everybody is saying yes, why are you saying no? Can somebody tell me why? If you are saying no, can you reason it out? Yes. We have seen it. If you are saying no, why is there any reason? Okay. All right. Let us look at it. The answer is yes. Uh the answer is actually you know uh we don't need it. The answer is no. Right? So we don't need the mechano receptors as for the tactile perception. I'm going to show you maybe some of you have missed the the the slide which I have shown you when we talk about the when we talk about the the four different receptors. There was a fifth receptor called the nerve endings. Right? Our ear loops don't have any of those receptors. Can you check whether your earloes have any tactile sensation? Does it have a tactile sensation? I'm using my pen to check whether earloss has any tactile sensation. My earlo seems to have it. Can you check your ears also? Does it have the tactile sensation? Yes. Can everybody experience? Is there anybody who cannot feel their ear loops? Yes. Okay. Can you feel your earlobes? But it doesn't have the macro receptors. How is it possible is a question, right? So the there is we have the free nerve endings. Free nerve endings are no uh again is a tactic receptors but it is not specialized into any of the four types. Okay. So the free nerve endings are there throughout the body. Even the blood vessels have it uh any you know wherever nerve is there it it can sense the it can act like a nono receptors that is one of the main reason we call touch sense as the n somatic sense throughout the body interior out on the skin as well that is the touch sense right in fact uh in the Indian philosophy Indian science and Indian philosophy no touch sense This is a major major important and uh in the temples we worship sense as a god and but you know it may not be very apparent for all of you but you know we will we'll get into the details when time permits. Okay. Oops. Right. So now we have seen how this uh touch sensation is working. We also feel the object. How do we it is not just the now the kinesthesia and the basic uh perception of the touch but we also perceive objects through materials and the geometry. When we uh get information about the geometry we get the shape and the size. about the material we find out the texture and thermal right and there is you know these are the six different basic types of ways in which we manipulate an object when we design for virtual reality we may have to if you design it for the six basic types then mostly we'll be able to um we'll be able to you know address most of the interactions for example here we do the lateral motion we have the hands and know we it we'll find the texture right and here is the pressure if there's a ball we press it with the two fingers right and this is a static contact the touching the know temperature of we touch it feel the uh you know temperature this is unsupported weight holding how do you feel the weight you move up and down right there is the weight perception and then enclosure shape volume we enclose our hands to get the volume contour following and we follow an object Right to find out the contour. Right? These are the basic six ways of know perception. [Music] [Music] And the structure when we how do we do it again? It's a multi-layer haptic perception. How do we this texture like a color is the perceptual quantity. It is not a real one. Texture also is a perceptual quantity. It is not a real one. These are the six

different uh here five dimensions. Five dimensions of no texture like RGB two psychophysical dimensions we have of color we have seen right. Similarly now here there are five different dimensions of the texture. Hardness, hard or soft, warmness, warm or cold, you know. Roughness, micro roughness, uneven or relief, fine roughness, rough or smooth, and then friction, moist or dry or it is called sticky or slippery. Right? These are the diff fingers. If you look at it, not the entire hands, no fingers. If you look at it, there are no different ways in which we know uh do the grasping. And uh now we do the contact grasp, we do the precision grasp and we were the we do the power grasp. When you design your ARVR, you need to look at now uh this classification of the finger position in order to know hold the objects in a virtual reality. Right? So the haptics perception know we talk about the tactile and canesthetic and motor together is what the haptics we saw and they involve no spatial factors. the psychophysical factors and then temporal factors and then spatial temporal factors that we discuss very detail in our course in a haptics one semester course but here we will just highlight with one of the our very uh you know uh recent papers that is about know five years ago u out of the body tactile experience okay what do I mean by out of the body no don't think that no the out of the body means It is like you know soul going out and you love experience of the out of the body. No this is we can feel the that tactile even without the skin is what we mean right. So what we did is a very interesting paper a very interesting study. Now this is highly cited in the websites you can go and see. Suppose if you have a know small actuator. This is a small vibrator. We give it in the fingertip. And here is another in the other hand. Right. Here is another finger. Vibrator. Here is another vibrator. And we put the third vibrator in the in the no another end of it. Right. So we can actually say another finger. Let us say this finger. This is third one. And we can feel the vibration in the empty space as well. So to feel the vibration you don't need the skin is what we want to prove it right and it is based on the power law. The Stevenson power law works very well in this and it is is for the the mobile haptics we call it as a mobile haptics. Okay this is a very interesting paper. Some of you are interested please you know go through this paper. So I want to quickly recap what we have done for the human haptics. Haptics is somoto sensory plus motor system manipulation. Sensory system sensory system in general sensory system we have a exterior seption that is now perceiving the exterior interior perceiving the internal things. So all this pain and temperature they are all part of the interior seption and then exterior seption is tactile and canesthetic they're all exterior seption. Exterior reception is a tactile uh pain and temperature are all interior. Proprioception is the canesthetic sense. Tactile sense know SA1 SA2 RA1 RA2. Interiorception is position as well as the the motion. Manipulation is using the motor systems. I hope uh now this page can be taken as a summary of it. If there is any question before I go to the machine haptics, I'll be happy to know take any questions. Yes. Uh unfortunately now uh we'll do a separate entics now later not now. All right. Now let us talk about now machine hearts with our little understanding of in the last 10 minutes or 15 minutes. Let us see how we can convert that into know machines. Now before we converted it into machine I wanted to now give you a complexity of of our hands right we wanted to understand how important and or how significant our hands are right do you know what is the number of degrees of freedom for your hand is it's about 23 degrees of freedom can you imagine what is the most complex now robo degree of freedom in the world is much lesser than 23 degree of freedom so your hand has 23° of freedom and it has so much of know flexibility and it has extreme sensitivities of cutaneous receptors. It the highest sensitivity at your fingertip but here it is much lesser but think about the number of receptors in the entire hand. Yeah, the joints are frictionless. The sensors and actuators are actually frictionless. Right. So how is it happening? How is the friction is reduced and you know in the the external environment we put the lubricants but here how is the lubrication is happening correct and we don't give a damn about it right so it is actually happening without our knowledge consciously unconsciously is happening it right and how is it happening complex adaptive control systems do you feel the weight of your hands do you know what is the weight of your hands can you put a Uh uh have you ever weighed find found out what is your weight of your hands? How do you find out the weight of hand? We can now cut my finger sorry hand entire hand and put it in the weighing balance. That is one way. But how do you actually find the weight of the hand? It's about know two two three kgs. And do you feel the weight of your hands when you move your hands know or manipulate an object? No, we don't feel it. It's a highly complex adaptive know control closed loop control system is there but we we are not at any uh conscious of it right any mechanical device in our real world will have our own intrinsic properties have own inertia friction but here are our hands without any of the things how is it it's a miraculous right the friction mass complaint and viscosity Um all those things are for any object but none of this is there in in our hands at least we are not perceiving it. How is it happening? Right. Visual interfaces exploit the limitations of the visual apparatus. Similarly meaning like what what is the meaning of this particular sentence? Because our eyes cannot continuously see. We cannot you know differentiate between two frames lesser than let us say uh 30 milliseconds and that is what we exploit in designing the displays right the fusion uh threshold right similarly can we exploit some of the limitations of the human hands to design at our haptic devices there is a point I'm driving over here the human haptic system has limitations as well and that has to be exploited to design the haptic devices. Correct. And what is the limitation of the now human haptic systems? Again, we saw the psychophysical parameters. The psychophysical parameters one of the earlier classes I show the the numbers right this numbers tells you what is the limitation of it and those limitations can be exploited to design this. And these are the some of the uh available haptic devices in the market. This is force force based joystick is there in the market and

unfortunately no nowadays it is not popular but a few years ago 10 years ago or 20 years ago it was all very popular right and have you seen now forcebased m or haptic mice? Yeah, you will be able to know remove or move your haptic mouse or your computer mouse but it gives you haptic feedback, force feedback, right? And uh and there are skeletons which gives you a no hand movements or resistance to your movements. There are no haptic glasses which gives you the know resistance your finger movements right and tele manipulation is possible right and this is one particular company is a startup company from MIT which has done the which has made the first haptic device in the world that's about how many years 25 plus know 10 it's about know 35 years 30 years ago right and still it is a market leader and it is bought by some Chinese now companies and that is still ruling the entire haptic know market and they have a different versions of the haptic devices and it is called the you know 3D systems now 3D touch or whatever it is you can look at it if you search for sensible phantom you will get it the cheapest haptic device was made available some know 10 15 years ago it is not there in uh market again and this is again designed developed by one of the MIT you know scholars. Uh it cost only $199. The range of space and the sensitivity resolution it's all very less but it is not available in the market now. So um the haptic devices, desktop devices there are different actuation principles we can classify whatever is available in the market using the kinematic structure whether there is a serial structure like you know your haptic devices and phantom is using or parallel structures. Those of you know coming from the robotics background you will understand whether you are using the parallel structure or serial structures serial manipulators or par manipulators or the hybrid of it both serial as well as the hybrid of it. Actuation principle they use different they use a pneumatic actuator or they hydraulic actuator or different actuators and electric motor also can be simply used. Control principles how do we control it? I'm going to show you that now impedance control is what haptics is using it sometimes admittance controls also different haptic devices use different control systems depending upon the use cases which we talk about this in detail in the other classes in the uh semester long courses. So the device in the earlier slide which is shown is the kinesthetic device or desktop devices. Tactile device alone can be again classified in a many different way and there are some devices as available in the market. Tactile devices give you give you the now vibration feeling. Not only vibration now touch pressure right. So mechanical vibration is one thing. Surface shape changing is another thing and then friction modulation is another thing. So in one of the classes I mentioned about the ITAD which we have designed in IIT Madras that is basically no friction modulation. So now if you search for ITAD this is a ITA interactive touch active display that is the uh display we have designed in I metas that is actually no friction modulation device. Mechanical vibration. There are a lot of mechanical vibrations. Erm, eccentric, no rotation mass, Linear resonant actuators, P electric, voice coil. These are all no usual mechanical vibration actuators. Shape changing. We have the pin array. There are 100 by 100 array are there. Right? There is a electrode array. There are pneumatic array. Now we can add another thing called the ultrasound. ultrasound based array which is giving a very subtle touch over here that's also can give you the know shape surface shape as well you you you can uh you can search for ultrahatics and that is essentially based on the know ultrasound that is a product available in the market maybe some of you have already bought it this commercial devices just to give a overall view of the entire set of devices available In the commercial sites we uh we can draw this graph in the x-axis we put the force and in the y- axis we put the workspaces. So we wanted the lot of workspaces and we want lot of torque or force generations right unfortunately the the sensible devices whatever first MIT has designed is on the lower range and it's also affordable. There are devices designed in uh for example no yeah human size right human size has no lot of forces it can generate right but a workspace can be no very very little uh so all is there any any particular one I wanted to highlight it yeah so all those exoskeletons can be no uh can generate a lot of force It can also have a very large work spaces. Exoskeleton in the sense the skeletons which can be added to your body and then know it helps you to move your hands and it can also resist your resist your motion. This is one device which is very unusual device from CMU Carne University. Each it acts on magnetic principles. Okay. It uses magnets to give you a force feedback but it is highly nonlinear and it's very difficult to develop a control system for this. That's one of the reasons it is not a very popular one and there are devices which can be worn. Variable haptics is another term which you can search the you know Google and there are lot of devices available. But one of the difficulties with the variable haptics is that can you imagine what is the difficulty for according to Newton's law there has to be a reaction for every action force right but if it is generating a force where is the reaction force is going to be delivered it has to be delivered at the back of the hands if you're generating a force at the fingertip where is the action force has to be developed it has to be put it on the back of the finger Right? That is going to destroy the feeling of the haptics. This is one of the major major challenges as far as the design of the haptic interfaces is concerned. Right? There are again control systems when you look at it. It's all highly digital systems. Highly digital system. It's all highly know digital systems and we have know we have developed a system which can give you a very very you know uh very very hard know perception of know objects because it is a sample system because there is a time delay involved the the the haptics feeling is usually soft. It is most difficult to develop a give you a haptic feeling of very very hard surfaces. Right? It's still a challenge know we can achieve only to certain very high stiffness you can develop but hard is know almost near impossible right there are bilateral teley operators are knowings are there using position and that modeling is used to design lot of uh devices as well network is used in understanding the the operator environment haptic uh teleoperation models And one of our colleagues Black Hanford from

Washington University is a leader in this and it's all impedance controller operator and impedance transmitted to the environment. So you can find out what is the minimum impedance we need to now control. This is a implementation summaries and every haptic device we should have a metrics what are the different metrics to look at the quality of it. Now static and dynamic accuracy should be there. Latency should be as minimum as possible. Update rate has to be 1,000 Hz. For devices uh for the uh uh for the visual display it's good enough if you have 30 Hz or 60 Hz. For virtual reality 90 Hz minimum, right? But for haptic devices minimum you to have no 1,000 Hz. thousand Hz we have only one milliseconds for one frame within one one millisecond we need to device we need to decide how much force I need to now give and within that time we have to know generate that force the actuators and then deliver it to the the fingertip right that is a challenge signal to noise ratio and jitter has to be extremely very high because our fingers are very very sharp We we have seen it at 250 Hz we can pick up.1 micron display right distortion has to be no controlled as well these are the metrics quality metrics for haptic devices I met haptics lab is one of the very few labs in the world which can design and develop haptics for a wide applications okay here is a laparoscopic surgical simulator with The haptics feedback has been developed as a part of a PhD work and this is a IVF invitro fertilization training simulator and and it has been marketed and through our startup company it's a it's a product available in the market now you can search for the fertil and then you will get this now product this is uh one of the examples of our now acting device of course we have seen the now The Apple phone haptic engine also is one now fine example and here is one haptic generic haptic device is first haptic device designed in in India even today there's nobody else has designed the haptic device for the past 20 years we have we have been designing it and empowering other labs to use this labs and defy or design know similar haptic devices for example one of our students is a faculty in Indian of science and he started this laboratory haptics lab and he has been designing the haptic devices there. Right? It is a a six degree of freedom position sensing but three degree of freedom force feedback. It can give you 10 degree of 10 mill torque. That's a that's a that's how it is possible. We build it entirely within the campus except the motor which we have to buy from abroad which is a very very specialized motor. Okay. Now coming back to the the main important topic of uh today's class is how do we render the haptics? We have understanding of what is the know the human haptics is what do we uh what do we mean by now machine haptics and how these haptic devices are working. Now how do we render it right? So in a virtual reality scene suppose we have the virtual reality environment in that our hand is represented by a by by a you know what do you call uh what do you a position our hand has to be first of all sensed and that position has to be found out whether it is actually colliding or not with any of the object. Once there is a collision then we have to find out whether how do we respond to the collision. If it is a rigid wall we need to resist the movement of the hands. And so once the collision response is decided we need to compute the forces. Once we know what is the forces then we send it to the graphic display as well as the now force display that is the our actuators. Once we give the uh force and torque values to the actuators then it it renders the force to the hands and visual also it is rendering to the user and the user actually displays their finger and therefore that the position and sensing is changing and the entire uh cycle goes on at 1 kilohertz that is 1,000 htz this has to happen you know for every 1 millisecond. So the challenge is within 1 millisecond I need to find out the collision detection I need to find out the collision response. I need to compute the forces. If the VR is very very complex then it is going to take lot of time right and at the same time we have to you know graphic is happening only at let us say 90 Hz for virtual reality right whereas you know the haptics has to happen this is 1 kohz position sensing also has to happen 1 kohz right that is a challenge there are several rendering types are possible. There is a point based rendering. There is a ray based rendering. The entire hand can be considered as a single point. So we saw that the entire hand has no 23 degrees of freedom. But for haptics we reduce the 23 degree of freedom to three degree of freedom. Right? That is a point based rendering. That is a three degree of freedom. So that is what happened 30 years ago when they first designed the haptic device and still most of the haptic devices 95% of or 99% of the haptic devices available in the market they use three degree of freedom assuming the hand as a three degree of freedom point based rendering. So it is very very difficult to consider the entire hand as a 23 degree of freedom and then find out the interaction. Right? The next uh improvement can be done is a ray based instead of point we'll consider as a line that is a ray based right just like in the visual rendering what we saw as the ray based rendering right path tracing here also in the haptics also it is a ray based rendering which considers the entire hand as one single line and therefore uh where is this this is yeah this is a line this is a point based render This is a line based rendering. And then the next complexity can be it can be a object based. This is your object object based rendering. These are the different type of the rendering is possible. So this is our paper in uh 2007 where we have developed a rendering method to give you the the very very hard uh objects in a virtual reality. Right? So uh due to the interest of time I will go to the now computer haptics. Computer haptics is where we are talking about the all the software design just like what we have talked about the uh the textures and force shading in the visual rendering. Can we talk about the textures and force shading ray based rendering and point based rendering for the for the haptic rendering as well. Can we also do the tissue modeling, static and deformation modeling? This is what we are talking about in the computer. So I'm going to now quickly go through it for shading just like a shading. Shaders we discussed in the visual rendering for haptics also we can find out the shading. Shading is basically now uh filling the forces as in the uh middle. It is basically interpolating between the the two vertices right so smoothly for example if there's a edges

of an object we know that we the force has to be for each edge this has to be determined and then we can smoothly interpolate all the force forces to give a smooth feeling of it right smooth edges or discontinuity avoiding the discontinuity right Averaging the surface normals of neighboring polygons. That is one way. Contacted triangles into three triangles and then a different methods which we have developed it about 20 years ago. Now how do we develop friction and textures? Your ITAD the device what we have designed recently ITAD it uses the friction and textures right. So displaying the detailed geometry of know textures and friction is computationally expensive. So there are some tricks are used. The whole computer graphics is about the tricks. How do we simplify it? There are challenges. Instead of doing the physically realistic, how do we make it know appear realistic? Right? That is a simulation. Simulation by perturbing the reaction forces friction tangential forces opposite to the uh direction of the movements. Right? Texture both tential and normal. We saw in the earlier lecture or earlier slides that what are the different dimensions of the textures perceptual dimensions all those dimensions are to be simulated physically simulated to get the nerve textures deformable objects you know uh virtual reality we some of the times for example in medical simulation we have to know uh render deformable objects there are geometry based deformable objects there are physics based deformable objects it's all highly computationally expensive But there are examples algorithm. These are the two architectures for the deformable object. Basically we have the geometry model basically and uh it can develop the force and displacement separately or or the geometric model you have physics based model determines the displacement separately or the force uh separately. Okay, these are the two different architectures available. Now we have seen human haptics, machine haptics and computer haptics. Now let us quickly look at what are the feature challenges of the haptics. So listening to in the last now 30 minutes or so or is 60 minutes or so what do you think is the most important challenges in the haptics? Can you now give feedback? What is the most interesting challenge in the haptics? We know what is the most interesting challenge in the visual rendering. We saw what is the most interesting challenge in the audio rendering. Now I want you to think about it. What could be the most interesting challenge in the haptics? I'm going to close it. It is not. Yeah. Yeah. So what is the what do you think is the most important challenge as far as the haptics is concerned? Any thought? We saw several topics. Okay. Processing required for optimal experience. Okay. To simulate real life sense. Okay. Degree of freedom. Okay. Realistic experience is difficult. Okay. Sensing feedback. Feedback and action. Very good. To get same degree of freedom. Yes. Less latency. Very good. Less friction. Very good. Okay. You all nailed it very nicely. Um I'm going to Yeah, I'm going to uh I want to take you uh uh take forward this particular message of what is one most important challenge in the haptics as of now today is the actuator. There are several challenges but the most most important and challenge is the actuator. I need to design a very very small actuator but it will give me very big you know forces and tors right I want this is a very lightweight actuator but it should give me wide range of forces right that is the design requirement that is not going to be possible but know we are slowly improving in it the same challenge is in the EV sector as well electric vehicle as well right we want a know very very small actuated but you know very very powerful and very fast actuators right the same problem in the haptics as well. So actuator design is the very very most important problem as of now. In the future I hope some of you will come and then you know invent this actuator right handheld haptics with the different haptics patterns and a multimodel haptic devices. It is not just electromagnetic alone or electric motor or pneumatic or actuator but it can be combination of all those things. Soft haptics. How do we make soft robotics into a into a haptics? Can I use a drone for haptics? Right? These are the topics we wanted to discuss as in detail. High fidelity haptic rendering. So highly realistic is very very difficult as of now. what we could do achieve is only this faking or it is actually approximation I would say understand the human haptics for multimodel haptics and the haptic illusions are there which is still there are lot of psychophysical research needs to be done okay social haptics is coming up now so since you're all familiar with the no uh cell phones we wanted to use your cell phones for social interactions okay now Coming back to the initial question which I asked can you call the knowh the vibration in your cell phone as a haptic feedback and that is what Apple is actually marketing it as right it's a small vibration feedback in your Apple phone cannot be called as a haptic feedback it can only be a tactile feedback but unfortunately Apple is marketing and everybody is following it that's okay if IF you want to use this mobile phone for haptic interaction, it has to give a force feedback, touch feedback, kesthetic feedback. As of now, there is no canistic feedback in the phone. But how do I give you the sense of kesthetic feedback? It need not be real there. It can be no uh mimicked or no faked or approximated. when I give you a high-fi in this through the cell phone camera, I should get a feeling of giving you a high- five. How do we generate the forces? That is what the the mobile companies, top mobile companies are working on it. And uh pretty soon you will feel you will get some knowhes in your mobile phone to give you a hi-fi through your mobile phone. Okay, that's a social haptics I call it as I want to end this lecture with uh with a topic of uh now uh ultimate haptics. What could be ultimate haptics? We saw the human haptics, machine haptics, computer haptics, social haptics. What could be ultimate haptics? Haptics in the sense without which we cannot there is not nothing better than this haptics is what they alternate haptics. Okay. So uh for the interest of time um the ultimate VR the ultimate haptics is one without the haptic devices can I give you a sense of touch the virtual reality I don't have the haptic device but I want to give you the now haptic feedback right without any gadgets how do I program your mind to give you the touch feedback that is going to be the ultimate how do I program your mind directly without any of the visual rendering pipeline without the headset, without the spatial

audio. I want to give you the feeling of virtual reality. How do I do that? That is the ultimate VR ultimate haptics. Here we did a very small experiment. We called some 10 and I medas volunteers to the stage and uh we have inserted needle into their hands. All the 10 people hands but all of them were also hypnotized. Not all are hypnotized. Some are not hypnotized at all. Whoever has not hypnotized they felt the the needle insertion pain. Whoever is hypnotized they have not felt the the pain at all. So it means the pain is only the mental perception is not necessarily of the physical may be there but no it can be withdrawn and using this principle there are lot of virtual reality application coming for no pain relief even now delivery is happening using this virtual reality headset to reduce the pain the cancer patients are suffering with the pain they can be relieved of now pain using the virtual reality systems. Pain is a haptic feeling and that can be reduced using multimodal systems. It has a tremendous applications but we need to understand the mind much better. How do we understand through the psychophysics? So psychophysics is very important. Human perception understanding human perception measuring the human perception is important. Perception engineering is important. I'll stop here. Okay. So, I'll take one or two questions if there is any any questions. Uh I know it is you know 10 minutes past 7:15. So, I will take one or two questions. I hope this topic on haptics is is know interesting to all of you. Where can we find the list of XR summit? I'm going to send you the the the list including the registration fees and Indian temples relevant to this topic. Yes, very much relevant relevant but know this is not the scope of this course. When you come to the campus, we'll discuss in detail. Tesla suit of UK makes full body haptic suit. It's all climb. Have you ever felt it? I want you to feel it and then check whether it is really haptics or not. Yes. So please is know just from the the name haptics. Now don't be carried away. It's all marketing terms. Apple has used this to market its iPhones, right? Similarly, everybody is using haptics to market their suits. Useful for astronauts training in zero gravity. Yes, it is a very very uh extremely difficult but we're all trying with this room. Thank you. Okay. Uh we will stop here. Thank you so much and we will meet uh next uh Tuesday and meanwhile we will send you the the details of uh details of the XR summit. Yes, sure. We will initiate it. Bye. Take care. [Music] lecturer explanation attached so explain easily as i am a 7th class student i should remember all easily and give mcqs too

ChatGPT said:

Spatial Audio + Haptics — Easy Notes for a 7th-Class Student

Your lecture covers two main topics:

1. **Spatial audio rendering** — how technology makes sound seem to come from different places in 3D space.
2. **Haptic rendering** — how technology creates the feeling of touching, holding, or pushing something in Virtual Reality (VR).

I'll explain them with simple examples, memory tricks, important exam points, and MCQs with answers.

Part 1: Higher-Order Ambisonics

1. What is Ambisonics?

Imagine you are standing in the middle of a room. A bird chirps on your left, a car moves behind you, and a plane flies overhead.

Ambisonics is a technique for representing and reproducing sound in 3D space.

It uses mathematical building blocks called **basis functions** to represent the directions and patterns of sound.

Think of basis functions like LEGO blocks: combining more blocks lets us build a more detailed sound representation.

2. What are the orders of Ambisonics?

As the order increases, the number of basis functions increases, allowing a more detailed representation of the sound field.

Order	Number of basis functions	Easy way to remember
Zero order (0)	1	One basic block
First order (1)	4	1 + 3
Second order (2)	9	1 + 3 + 5
Third order (3)	16	1 + 3 + 5 + 7
Fourth order (4)	25	Add 9
Fifth order (5)	36	Add 11

The formula for the total number of basis functions in 3D Ambisonics is:

$$N = (L+1)^2$$

Here, L means the order.

For example, third order:

$$N = (3+1)^2 = 16$$

Memory trick: Add consecutive odd numbers: 1, 3, 5, 7, 9, 11...

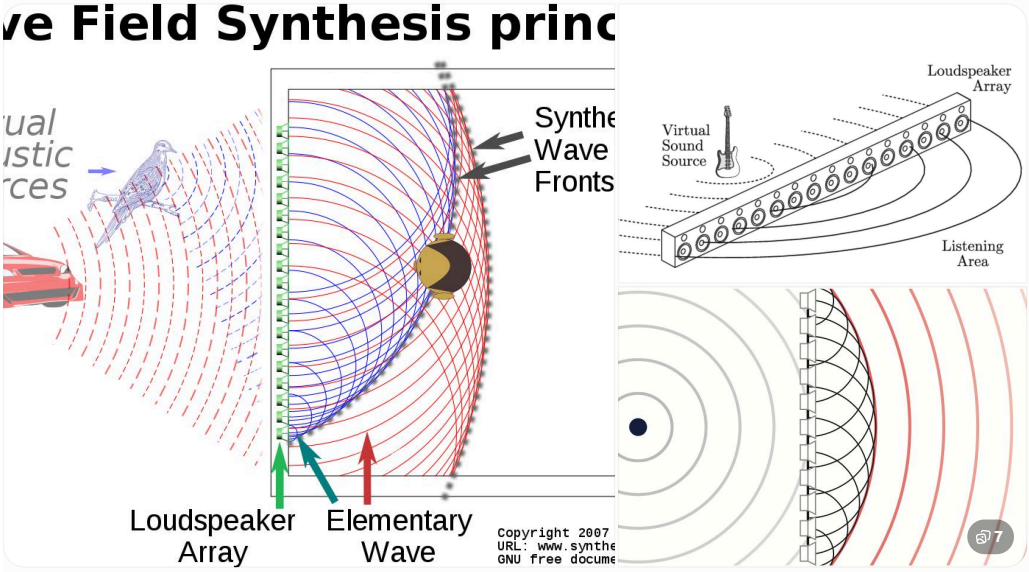
- Zero order represents the basic sound component.
- First order adds directional components along the three axes: X, Y, and Z.
- Higher orders add more complex directional patterns.

The lecture describes these basis functions as **orthogonal**, meaning they are mathematically independent in a particular sense.

Exam point: Higher order = more basis functions = more detailed representation of the 3D sound field.

Part 2: Wave Field Synthesis (WFS)

3. What is Wave Field Synthesis?



Imagine a row of many small speakers arranged along a wall. By carefully controlling when each speaker plays sound and how loud it plays, the system creates a sound wave that makes you feel as if a sound is coming from a virtual location.

This is called **Wave Field Synthesis (WFS)**.

For example, the speakers may be in front of you, but the system tries to make the sound appear to come from a virtual car farther to your left.

4. Huygens' principle

Your lecturer refers to this as the principle behind WFS.

In simple words:

Every point on a wavefront can be treated as a source of tiny secondary waves.

These tiny waves combine to form a new wavefront.

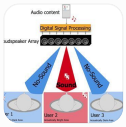
Imagine dropping several small stones into a pond. Each stone creates circular ripples. When the ripples combine, they form a more complex wave pattern.

WFS uses this idea to create and control sound waves using speaker arrays.

5. How does WFS work?

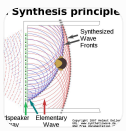
Step 1: Speaker array

Many speakers are placed in a line or arrangement.



Step 2: Control the sound

The system adjusts the timing (delay) and strength (gain) of the speakers.



Step 3: Create the wavefront

The sound waves combine to approximate the desired sound field.

Step 4: Perceived virtual sound

The listener perceives sound as coming from a virtual location.

6. Ambisonics vs WFS

Feature	Ambisonics	Wave Field Synthesis
Main idea	Represents a 3D sound field using basis functions	Reproduces a sound field using controlled sound waves
Main tool	Mathematical basis functions	Arrays of loudspeakers
Important concept	Order and number of basis functions	Huygens' principle, wavefronts
Main controls	Sound-field representation and decoding	Speaker position, delay and gain
Easy example	Building a 3D sound representation from blocks	Many speakers working together to shape sound waves

Memory trick: Ambisonics = *represent the sound field*. WFS = *reproduce the wave field*.

7. What is crosstalk cancellation?

Suppose the left speaker's sound reaches both ears when the system intends it mainly for the left ear. The unwanted sound reaching the other ear can interfere with the intended effect.

This is called **crosstalk**.

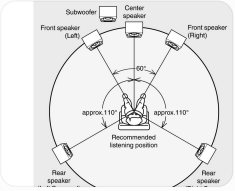
Crosstalk cancellation uses signal processing to reduce this unwanted overlap.

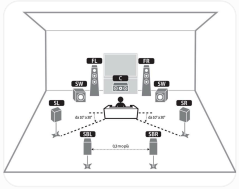
Remember: Crosstalk cancellation = reduce unwanted sound from one speaker reaching the other ear.

Part 3: Surround Sound and Dolby Atmos

8. What is surround sound?

Surround sound uses multiple speakers placed around the listener to create an immersive audio experience.





System	Meaning
2.0	Two main speakers: left and right
3.1	Three main speakers and one subwoofer
5.1	Five main speakers and one subwoofer
7.1	Seven main speakers and one subwoofer

A **subwoofer** produces low-frequency sounds, such as deep bass and explosions.

9. What is Dolby Atmos?

Dolby Atmos is an **object-based audio system**.

Imagine a movie scene where a helicopter flies from the front of the room to the back. Instead of simply assigning its sound to one fixed speaker, object-based audio can describe the helicopter sound's position and movement using audio metadata.

The playback system uses that information to render the sound for the available speaker arrangement.

Metadata means extra information describing the sound, such as its location in 3D space.

10. Compare the three techniques

Technique	Main idea
Ambisonics	Represents a sound field with mathematical basis functions
WFS	Uses speaker arrays to reproduce sound waves
Dolby Atmos	Uses audio objects and positional metadata to render sound

Memory trick:

- **A**mbisonics = **A**rrangement of mathematical components.
- **W**FS = **W**avefront reproduction.
- **D**olby Atmos = **D**escribes sound objects in space.

These are simplified memory aids, not their official expansions.

11. How do we evaluate 3D audio?

How do we know whether a spatial audio system works well? We measure its performance.

Evaluation factor	Simple meaning
Externalization	Does the sound seem to come from outside your head?
Room simulation	Can the system create the feeling of different rooms?
Latency	Is there a delay between an action and the audio response?
Degrees of freedom	Can sound respond appropriately to head movement and direction?
Artifacts and anomalies	Are there unwanted distortions or strange sound effects?
Speaker spacing	Are speakers positioned suitably to reproduce the intended waves?

Important: Lower latency is generally better for responsive head-tracked audio.

Part 4: What is Haptics?

12. The easiest definition

Haptics is the science and technology of touch, sensing, and manipulation.

Imagine playing a VR game:

- You see a virtual ball.
- You reach out and touch it.
- You feel resistance when you push it.
- You move your fingers to explore its shape.

The system that creates or simulates the touch-related experience is using haptics.

13. The two pillars of Virtual Reality

Your lecturer says that good VR depends on two major pillars:

1. Immersion

Feeling as though you are inside the virtual world.

2. Interaction

Being able to act on and manipulate things in that world.

For example, seeing a virtual door is immersion. Reaching out, turning its handle, and feeling resistance is interaction.

Memory trick: Immersion = *being there*. Interaction = *doing something there*.

14. The three branches of haptics

Branch	What it studies	Easy example

Human haptics	How humans sense touch and move	Your fingers feeling a rough surface
Machine haptics	Devices, sensors, actuators and mechanisms	A force-feedback controller
Computer haptics	Software that calculates and renders touch	Code that makes a virtual wall feel solid

Remember: Human → Machine → Computer.

- Human: How do we feel?
- Machine: How can a device create the feeling?
- Computer: How does software calculate the feeling?

Part 5: The Four Main Components of the Somatic Sense

15. What is the somatic sense?

The **somatic sense** refers to bodily sensations, including touch, temperature, pain, and the sense of body position and movement.

Your lecturer emphasizes four important modalities.

1. Tactile sense

Feeling fine touch, pressure, and surface details.

2. Temperature sense

Feeling hot or cold.

3. Pain sense

Detecting potentially harmful stimuli.

4. Kinesthesia

Sensing the position and movement of body parts.

Memory trick: T-T-P-K

- **T**ouch
- **T**emperature
- **P**ain
- **K**inesthesia

Kinesthesia is especially important in VR because you need to know how your hand or fingers move, not just what they touch.

16. Active touch vs passive touch

Active touch	Passive touch
You move to explore something	Something touches you, or you receive a touch sensation
You rotate a pen to feel its texture	Your shirt rests against your skin
Involves sensing and movement	Primarily involves receiving a sensation

Memory trick: Active = *I explore*. Passive = *I receive*.

Part 6: Skin Receptors

17. What are skin receptors?

Skin contains sensory nerve endings and specialized receptors that detect different types of stimuli. They convert physical stimulation into signals that the nervous system can process.

Your lecture focuses on four named mechanoreceptors.

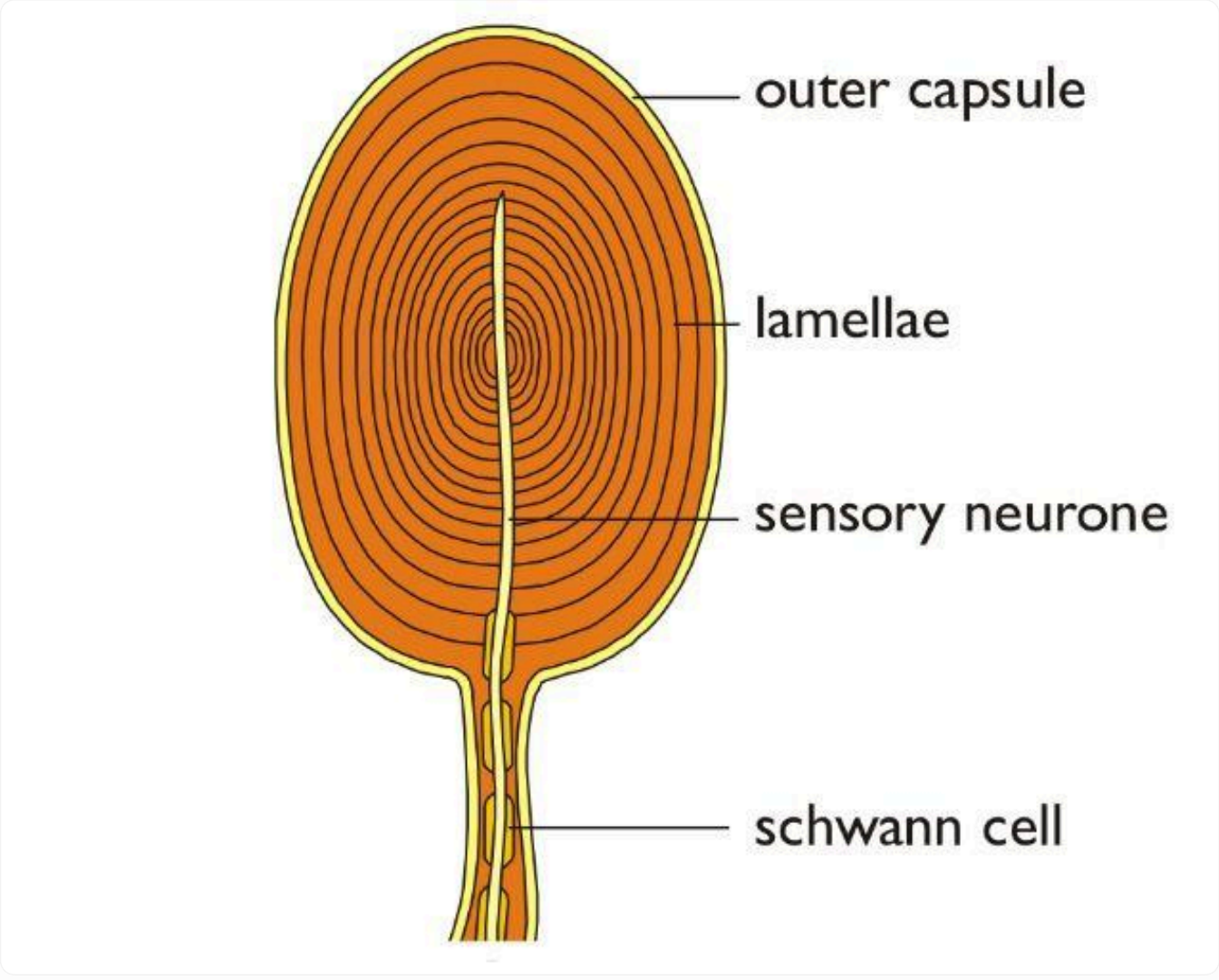
Receptor	Other name	Main role to remember
SA1	Merkel-associated	Sustained pressure and fine spatial detail
SA2	Ruffini-associated	Sustained skin stretch
RA1	Meissner-associated	Changing touch and lower-frequency vibration
RA2	Pacinian-associated	High-frequency vibration

SA means **slowly adapting**. RA means **rapidly adapting**.

- Slowly adapting receptors continue responding during a sustained stimulus.
- Rapidly adapting receptors respond especially strongly when the stimulus changes.

18. Which receptor detects phone vibration?

The lecture's answer is **RA2, also called the Pacinian corpuscle**.



Imagine an onion with many layers. The Pacinian corpuscle has a layered structure. It is highly sensitive to vibration, particularly at higher frequencies.

Your lecturer connects this receptor with the strong sensitivity of the hand to vibration around 250 Hz.

Exam memory: Phone vibration → high-frequency vibration sensitivity → RA2 → Pacinian corpuscle.

One important detail: touch is not dependent only on these four specialized receptors. Free nerve endings also contribute to bodily sensation.

Part 7: How Do We Recognize Objects by Touch?

19. Six basic exploratory procedures

When you touch an object, your brain collects information about its shape, size, texture, weight, and material.

Procedure	What you do	What you discover
1. Lateral motion	Slide your finger across a surface	Texture and roughness
2. Pressure	Press an object between your fingers	Hardness or softness
3. Static contact	Touch without moving much	Temperature and surface properties
4. Unsupported holding	Lift and hold an object	Weight
5. Enclosure	Wrap your hand around it	Shape and volume
6. Contour following	Trace around its edges	Exact shape

Easy memory: Slide, Press, Touch, Lift, Hold, Trace.

For example, to identify a ball:

- Enclosure helps you understand its round shape.
- Pressure helps you determine whether it is soft or hard.
- Lifting helps you estimate its weight.
- Sliding your fingers helps you feel its surface.

20. Five perceptual dimensions of texture

Texture is not just “rough” or “smooth.” The lecture identifies five dimensions.

Dimension	Meaning
Hardness	Hard or soft
Warmness	Warm or cold
Roughness	Micro-level surface roughness
Fine roughness	Rough or smooth surface feel
Friction	Sticky, slippery, moist or dry feeling

These are perceptual descriptions: they describe what a person feels rather than simply giving a physical measurement.

21. Three types of grasp

- **Contact grasp:** Touching an object.
- **Precision grasp:** Holding a small object using the fingertips, such as picking up a pin.
- **Power grasp:** Holding an object firmly with the hand, such as gripping a hammer.

These are useful when designing VR interactions because virtual objects should respond appropriately to different hand positions.

Part 8: Machine Haptics — How Devices Create Touch

22. What is an actuator?

An **actuator** is a device component that produces movement or force.

Think of a phone vibrating because a small motor is moving inside it. That motor is an actuator.

Haptic devices may use motors, pneumatic systems, hydraulic systems, and other mechanisms to create touch or force feedback.

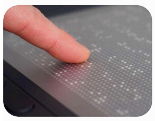
Types of haptic feedback

1. Vibration-based

Creates buzzing or vibration sensations. Example: a game controller rumbling.

2. Force-based

Creates resistance or pushes against your movement. Example: a force-feedback joystick.



3. Shape-changing

Small pins or surfaces move to create patterns that can be felt.

4. Ultrasound-based

Focused ultrasound can create tactile sensations without requiring direct contact with a solid surface.

23. Why is designing a haptic device difficult?

The human hand is very flexible and sensitive. The lecture notes that it has around **23 degrees of freedom**.

A degree of freedom is an independent way something can move.

For example, a simple object sliding on a table may move left-right and forward-backward. A human hand has many more possible movements because of its fingers, joints, and wrist.

Haptic-device designers must balance:

- Large workspace — how far the device allows you to move.
- Force capability — how much force it can generate.
- Accuracy — how correctly it produces the desired sensation.
- Size and weight — whether it is comfortable to use.
- Safety and responsiveness.

Main challenge highlighted by your lecturer: designing small, lightweight actuators that can produce large, fast, controlled forces.

24. Why is low latency important?

Latency is the time delay between an action and its response.

Imagine pushing a virtual wall. If the force comes too late, the wall feels unrealistic.

Your lecture gives these target update rates:

System	Rate mentioned in the lecture
Conventional visual display example	30–60 Hz
VR visual display	At least 90 Hz, as stated in the lecture
Haptic rendering	1,000 Hz (1 kHz)

At 1,000 Hz, the system updates every:

$$\frac{1}{1000}\ \text{second} = 1\ \text{millisecond}$$

Memory trick: Visual = tens of updates per second; haptics = one thousand updates per second in the lecture's example.

The exact required rate depends on the device and application, but haptic feedback generally needs very fast updates to feel stable and responsive.

Part 9: The Haptic Rendering Pipeline

25. How does a virtual object feel solid?

This is one of the most important parts of the lecture.

Imagine moving your hand toward a virtual wall. The system needs to detect your hand, determine whether it touches the wall, calculate a force, and send that force to a haptic device.

1. Position sensing
Find the position of the hand or device.

↓

2. Collision detection
Check whether the hand touches or enters an object.

↓

3. Collision response
Decide what should happen when contact occurs.

↓

4. Force computation
Calculate the force or resistance to apply.

↓

5. Force output
The actuator applies force to the user's hand.

Meanwhile, the visual display shows the virtual contact. The cycle repeats as the hand moves.

Remember the five steps: Sense → Detect → Respond → Calculate → Apply.

Example: touching a rigid wall.

1. The device senses your hand position.
2. The software detects contact with the wall.
3. The software prevents your hand from passing through it.
4. It calculates the resistance force.
5. The actuator pushes back against your hand.

The visual and haptic systems work together, but they can run at different update rates.

26. Three haptic rendering methods

Method	How it represents the hand or contact	Simple idea
Point-based	A single point	Treat the interaction point as a dot
Ray-based	A line or ray	Represent interaction along a line
Object-based	A larger object or geometry	Consider a more detailed object or contact shape

A point-based approach is simpler and less computationally expensive. Detailed object-based approaches can represent more complex contact, but they are generally harder to calculate.

Exam point: More detailed interaction can improve realism but also increases computational complexity.

Part 10: Computer Haptics

27. What does computer haptics do?

Computer haptics is the software side of touch simulation. It calculates what a user should feel when interacting with virtual objects.

Your lecturer discusses three important areas.

A. Force shading

Just as visual shading makes a surface look smooth, force shading helps make contact forces feel smooth when moving across a virtual surface.

For example, imagine sliding your finger along a virtual table. You should not feel sudden, unwanted jumps in force as you cross the boundary between two surface pieces.

B. Texture and friction rendering

Software can simulate the feeling of different surfaces by changing the force feedback.

- A rough surface may produce varying sensations as your finger moves.
- A slippery surface may provide less resistance.
- A sticky surface may produce greater resistance to movement.

C. Deformable-object rendering

Some objects change shape when touched.

Examples include a soft ball, human tissue in a surgical simulator, or a sponge.

The software must calculate how the object deforms and what force the user should feel.

Two approaches mentioned in the lecture are:

- Geometry-based models.
- Physics-based models.

Physics-based approaches model physical behavior more directly, but can be computationally expensive.

Part 11: Advanced Applications and Challenges

28. What is social haptics?

Social haptics explores using touch-related feedback to support communication between people.

For example, imagine a future phone that lets you feel a simulated high-five from a friend who is far away.

The challenge is that a small phone must somehow produce a convincing touch or force sensation.

Your lecturer explains that ordinary phone vibration is better described as **tactile feedback**, rather than a complete haptic interaction involving force and kinesthetic feedback.

29. What is ultimate haptics?

The lecture presents the idea of creating a touch experience without conventional haptic devices.

It explores whether the brain's perception can be influenced so that a person experiences a sensation even when the physical stimulus is absent or reduced.

The lecture's example involves hypnosis and pain perception. The broader lesson is that **perception is not always a simple, direct copy of physical stimulation**.

This is a research idea, not a claim that all touch sensations can currently be created without physical devices.

30. What are the biggest challenges in haptics?

Challenge	Why it matters
Actuator design	Small devices must produce sufficient force quickly
Realistic rendering	Virtual objects should feel believable
Low latency	Delayed forces feel unnatural

High degrees of freedom	Hands and fingers have complex movements
Accurate force feedback	The force should match the virtual interaction
Multimodal feedback	Sight, sound, and touch should work together

Most important according to the lecture: actuator design — making small, lightweight devices that generate large and fast forces.

Part 12: Final Revision Sheet

Use this table for a quick review before your quiz or exam.

Topic	One-line answer to memorize
Ambisonics	Represents a 3D sound field using basis functions
Number of 3D Ambisonics components	$(L+1)^2$, where L is the order
Higher-order Ambisonics	More basis functions for a more detailed representation
WFS	Uses controlled speaker arrays to reproduce sound fields
Huygens' principle	Each point on a wavefront can act as a source of secondary waves
Crosstalk cancellation	Reduces unwanted sound reaching the other ear
Dolby Atmos	Object-based audio with spatial metadata
Immersion	Feeling present in the virtual environment
Interaction	Acting on objects in the virtual environment
Haptics	Touch-related sensing and manipulation
Human haptics	Studies human touch and movement
Machine haptics	Studies devices that produce touch or force
Computer haptics	Software that calculates and renders touch
Somatic sense	Bodily sensations, including touch, temperature, pain and kinesthesia
Kinesthesia	Sensing body-part position and movement
RA2 / Pacinian corpuscle	Highly sensitive to high-frequency vibration
Actuator	Produces movement or force
Haptic rendering	Calculates and applies forces during virtual contact
Point-based rendering	Represents interaction using a point
Ray-based rendering	Represents interaction using a ray
Force shading	Helps create smoother force transitions
Haptic update rate	The lecture uses 1,000 Hz as its example
Major haptics challenge	Designing small, fast, powerful actuators

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Part 13: MCQs for Practice

Try answering the questions before opening the answers.

15-question practice quiz

0/15 answered

Choose one option for each question, then check your score.

1. How many basis functions are there in third-order 3D Ambisonics?

- A. 4

B. 9

C. 16

D. 25

2. What happens when the order of Ambisonics increases?

A. The basis functions decrease

B. More basis functions become available

C. Sound becomes mono

D. Speakers are no longer needed

3. Which principle is associated with Wave Field Synthesis?

A. Ohm's law

B. Huygens' principle

C. Newton's cooling law

D. Snell's law

4. What is the main idea of WFS?

A. Only using headphones

B. Controlling speaker arrays to reproduce sound fields

C. Removing all sound waves

D. Using only one speaker

5. What does Dolby Atmos use to describe the position of audio objects?

A. Only speaker volume

B. Spatial metadata

C. Skin receptors

D. Collision detection

6. What are the two major pillars of VR highlighted in the lecture?

A. Storage and memory

B. Immersion and interaction

C. Color and brightness

D. CPU and keyboard

7. Which branch of haptics studies devices that generate force feedback?

A. Human haptics

B. Machine haptics

C. Computer haptics

D. Visual haptics

8. Which receptor is especially sensitive to high-frequency vibration?

A. SA1

B. SA2

C. RA1

D. RA2

9. Which exploratory procedure is useful for feeling an object's weight?

A. Unsupported holding

B. Lateral motion

C. Static contact only

D. Contour following

10. What does an actuator do?

A. Stores images

B. Produces movement or force

C. Measures only color

D. Removes latency completely

11. What is the first step in the haptic rendering pipeline described in the lecture?

A. Force output

B. Texture shading

C. Position sensing

D. Audio encoding

12. What is collision detection?

A. Finding whether the hand contacts an object

B. Increasing the screen brightness

C. Changing audio metadata

D. Measuring skin temperature

13. What is the example haptic update rate in the lecture?

A. 30 Hz

B. 60 Hz

C. 90 Hz

D. 1,000 Hz

14. Which rendering method represents interaction using a single point?

A. Object-based

B. Point-based

C. Wave-based

D. Physics-only

15. What challenge does the lecturer highlight as especially important in haptics?

- A. Making screens larger
- B. Designing small but powerful and fast actuators
- C. Removing all visual feedback
- D. Increasing speaker count

Check my answers

The five-minute memory challenge

Before your exam, try to recall these without looking:

1. Ambisonics: $(L+1)^2$.
2. WFS: speaker arrays + Huygens' principle.
3. Dolby Atmos: objects + spatial metadata.
4. VR: immersion + interaction.
5. Haptics: sensing + manipulation through touch.
6. Somatic sense: tactile, temperature, pain, kinesthesia.
7. High-frequency vibration: RA2 / Pacinian corpuscle.
8. Haptic pipeline: position → collision → response → force → output.
9. Haptic update example: 1,000 Hz = 1 ms.
10. Main design challenge: small, powerful, fast actuators.

If you can explain these ten points in your own words, you will have a strong foundation for this lecture.

You said:

explain easily

ChatGPT said:

Week 12 – Assignment 12: Haptics Explained Easily

Let's understand all 10 questions in very simple English, like a 7th-class student, with examples and memory tricks so you can remember the answers for your exam.

Q1. What are the three broad areas of haptics?

Correct answer

Human haptics, machine haptics, and computer haptics

Haptics means the study of touch and physical interaction.

1. Human haptics

How humans feel and explore objects using touch.

2. Machine haptics

How machines or robots sense and interact physically with objects.

3. Computer haptics

How computers create touch sensations, such as vibration in a game controller.

Memory trick: H-M-C = Human, Machine, Computer.

Q2. Why is haptics broader than the sense of touch?

Correct answer

Haptics includes sensing as well as manipulation through touch.

Imagine picking up a pen.

- **Sensing:** You feel whether the pen is smooth, rough, heavy, or light.
- **Manipulation:** You move your fingers to hold, turn, or use the pen.

Therefore, haptics includes both feeling objects and doing things with them.

Memory trick: Haptics = Feel + Act.

Q3. What are the four major modalities of the somatic sense?

Correct answer

Tactile, temperature, pain, and kinesthesia.

The somatic sense helps us understand sensations from our body.

Modality	Simple meaning	Example
Tactile	Feeling touch	Feeling a pencil on your skin
Temperature	Feeling hot or cold	Touching a cold bottle
Pain	Feeling something hurts	Touching a sharp pin
Kinesthesia	Sensing body movement and position	Knowing your finger is bent even with your eyes closed

Memory trick: T-T-P-K = Touch, Temperature, Pain, Kinesthesia.

Q4. A pen touches your skin, but you do not move your hand to explore it. What is this?

Correct answer: Passive touch

Passive touch: Something touches you, but you are not actively exploring it.

Example: Someone places a pen on your hand while your hand remains still.

Compare this with **active touch**: you move your fingers over the pen to discover its shape and texture.

Memory trick:

- Passive = Person stays still.
- Active = You actively explore.

Q5. What is the main difference between passive and active touch?

Correct answer

Passive touch is unidirectional, whereas active touch involves bidirectional interaction.

Let's simplify these words:

- **Unidirectional = one-way interaction.**
- **Bidirectional = two-way interaction.**

Passive touch — one-way

An object or person touches your skin, and you receive the sensation without actively exploring it.

Active touch — two-way

You move your hand and receive feedback from the object while exploring it.

Memory trick: Passive = Receive. Active = Move + Receive feedback.

Q6. Which receptor senses high-frequency vibration around 250 Hz?

Correct answer: Pacinian receptor / RA2

Our skin contains different sensory receptors. Think of them as tiny sensors.

Pacinian receptor (RA2)

Detects rapid, high-frequency vibrations and is located deeper in the skin.

Example: Feeling a phone vibrate strongly against your hand.

Here are the four receptors mentioned in the assignment:

Receptor	Main role to remember
Merkel (SA1)	Fine details, edges, sustained pressure
Ruffini (SA2)	Skin stretch
Meissner (RA1)	Light touch and lower-frequency vibration
Pacinian (RA2)	High-frequency vibration

Memory trick: Pacinian = Powerful, fast vibration.

Q7. What is the difference between SA and RA receptors?

Correct answer

SA receptors continue responding during sustained stimulation, whereas RA receptors primarily respond to changes.

SA and RA describe how receptors respond over time.

SA = Slowly adapting

Imagine pressing your finger against a table and keeping it there. An SA receptor continues sending information while the pressure remains.

SA = Sustained Awareness.

RA = Rapidly adapting

Imagine a phone starting to vibrate. An RA receptor responds strongly when the stimulus changes, such as when vibration starts or stops.

RA = Responds to change.

Easy way to remember: SA keeps responding; RA focuses on changes.

Q8. How do we know the earlobe can feel touch without the four specialized mechanoreceptors?

Correct answer

The earlobe can exhibit tactile sensation despite lacking the four specialized mechanoreceptors.

The four receptors discussed in the lecture are Merkel, Ruffini, Meissner, and Pacinian.

The question challenges the idea that these four receptors are absolutely necessary for *every* tactile sensation.

The key point is that the earlobe can still exhibit touch sensation even though it lacks those four specialized mechanoreceptors.

Exam tip: Remember the exception — *the earlobe can exhibit tactile sensation.*

Q9. What information can we obtain by touching an object?

Correct answer

Geometry → shape and size; material → texture and thermal information.

When we touch an object, we can learn different things about it.

Geometry = Shape and size

Is the object round, square, long, or small?

Material = Texture and thermal information

Is it rough or smooth? Does it feel cold or warm?

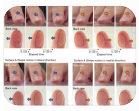
Memory trick: Geometry tells you the shape; material tells you what it feels like.

Q10. What are the six basic ways of interacting with or perceiving an object?

Correct answer

Lateral motion, pressure, static contact, unsupported weight, enclosure, contour following.

These are six ways we use touch to understand an object.



1. Lateral motion

Sliding your fingers across an object helps you identify its texture.

2. Pressure

Pressing an object helps you judge its hardness or softness.

3. Static contact

Keeping your fingers in contact with an object without sliding them.

4. Unsupported weight

Holding an object to judge its weight.

5. Enclosure

Wrapping your hand around an object to understand its overall size and shape.

6. Contour following

Moving your finger along an object's boundary to discover its shape.

Memory trick: L-P-S-U-E-C.

- **L** — Lateral motion
- **P** — Pressure
- **S** — Static contact
- **U** — Unsupported weight
- **E** — Enclosure
- **C** — Contour following

Final revision table — all 10 answers

Q.	Correct answer	Key idea

1	Human haptics, machine haptics, computer haptics	Three areas
2	Sensing + manipulation through touch	Feel and act
3	Tactile, temperature, pain, kinesthesia	Four body senses
4	Passive touch	No active exploration
5	Passive is unidirectional; active is bidirectional	One-way vs two-way
6	Pacinian receptor / RA2	High-frequency vibration
7	SA continues; RA responds mainly to changes	Sustained vs changing stimulus
8	Earlobe can exhibit tactile sensation without the four specialized mechanoreceptors	Exception
9	Geometry → shape/size; material → texture/thermal information	What we learn by touch
10	Lateral motion, pressure, static contact, unsupported weight, enclosure, contour following	Six exploratory procedures

Test yourself

Choose one answer for each question, then check your score.

1. Which receptor is best associated with high-frequency vibration?

- Merkel / SA1
- Ruffini / SA2
- Meissner / RA1
- Pacinian / RA2

2. You keep your hand still while someone touches it. What type of touch is this?

- Active touch
- Passive touch
- Contour following
- Enclosure

---

3. Which receptors continue responding during sustained stimulation?

- RA receptors
- SA receptors
- Only Pacinian receptors
- None of them

4. Which exploratory procedure helps you judge an object's weight?

- Static contact
- Contour following
- Unsupported weight
- Lateral motion

---

5. What does geometry tell you about an object?

- Only its temperature

- Shape and size
- Only its vibration
- Only its weight

Check my answers

You said:

now all the syllabus is completed for vr course nptel now based on the total course that we have discussed give some 50 mcqs covering the whole course so that it will be like practice that in exam based on this course mcqs will be given so give total summary and give mcqs to prepare well and better attempt in exam give full summary as i can remember very easily as i am a 7th class student

Worked for 50s

ChatGPT said:

NPTEL Virtual Reality (VR) — Complete Revision + 50 Practice MCQs

Let's prepare as if you're getting ready for the NPTEL exam. I'll use **simple English, easy examples, memory tricks, and exam-style multiple-choice questions** so you can revise without memorizing difficult paragraphs.

The revision below covers the VR topics we've discussed together, especially **perception engineering, visual perception and chromatic adaptation, spatial audio rendering, and haptics**. I can't verify from the conversation context that these represent every lecture in the full NPTEL course, so I'll keep the revision grounded in the topics we have covered.

Part 1: Full syllabus summary in simple language

1. Perception engineering

Perception engineering means designing technology based on how humans see, hear, and feel things.

For example, a VR headset should make images look natural, sounds seem to come from the correct direction, and controllers feel realistic.

Important terms

Term	Simple meaning	Example
Sensation	Detecting something using your senses	Your ears detect a sound
Perception	Understanding what you sense	You recognize that the sound is a bell
Absolute threshold	Smallest stimulus you can detect under specified conditions	The quietest sound you can hear
Difference threshold / JND	Smallest change you can notice	Detecting that one light is slightly brighter
Weber's law	A noticeable change often depends on the original stimulus	A small weight increase is easier to notice on a light object than a heavy one
Fechner's law	Perceived intensity grows approximately with the logarithm of stimulus intensity	Doubling physical intensity doesn't necessarily feel twice as strong
Stevens' power law	Perceived intensity follows a power relationship with physical intensity	Different senses respond differently as stimulus strength increases

Remember:

- Absolute threshold = *Can I detect it?*
- Difference threshold = *Can I notice the change?*
- Sensation = *Detect.*
- Perception = *Understand.*

2. Vision, colour, and chromatic adaptation

VR uses visual information to make a virtual world look convincing.

The HSV colour model

H — Hue

The basic colour: red, green, blue, and so on.

S — Saturation

How strong or pure the colour looks. Lower saturation makes a colour look more washed out.

V — Value

How light or dark the colour looks.

Memory trick: HSV = Colour name, Colour strength, Brightness.

Chromatic adaptation

Chromatic adaptation is how our visual system adjusts to the colour of the lighting around us.

Example: A white paper may look slightly yellow under a warm bulb, but your brain gradually adjusts and it appears more nearly white.

- **Chromatic adaptation:** The overall adjustment to illumination colour.
- **Colour constancy:** The ability to perceive an object's colour as relatively stable under different lighting conditions.
- **Chromatic Adaptation Transform (CAT):** A mathematical transformation used in colour science to model adaptation between viewing conditions.
- **von Kries model:** A simple model that adapts the responses of colour-sensitive channels by scaling them.

Why use VR in visual experiments?

VR can let researchers control lighting, backgrounds, and virtual objects while keeping other conditions similar. This helps them study how people perceive colour under different conditions.

Remember: Chromatic adaptation = *eyes and brain adjust to the lighting.*

3. Spatial audio rendering

Spatial audio makes sounds appear to come from particular positions around you, not just from the left and right speakers.

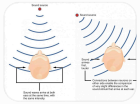
Example: In VR, a person speaks from behind you. When you turn your head, the voice should still seem to come from that person's location.

Types of audio

Type	Easy meaning
Mono	One audio channel
Stereo	Left and right audio channels
Surround sound	Sound comes from multiple directions around the listener
Spatial audio	Sound is positioned in 3D space, potentially including height and head-tracked direction

The three important binaural cues

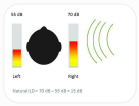
Binaural means **using both ears**.



ITD — Interaural Time Difference

Sound reaches one ear slightly earlier than the other.

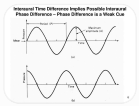
Memory: T = Time.



ILD — Interaural Level Difference

Sound is louder at one ear than the other. The head can block some sound, especially at higher frequencies.

Memory: L = Level.



IPD — Interaural Phase Difference

The sound wave reaches the two ears at different phases.

Memory: P = Phase.

One-line memory trick: ITD, ILD, IPD = Time, Level, Phase.

HRTF — Head-Related Transfer Function

HRTF describes how sound is changed by the listener's body before reaching the eardrum.

It accounts for effects involving the:

- Head
- Outer ears (pinnae)
- Shoulders
- Sound reflections and resonances

HRTF helps a VR system create the impression that a sound is coming from a particular direction.

Example: A sound from behind can have different spectral characteristics from a sound in front.

Other spatial-audio concepts

Concept	Simple explanation
Head tracking	Detecting how the listener turns or moves their head
Cone of confusion	A set of possible sound locations that can create similar binaural cues, making direction ambiguous
Precedence effect	When similar sounds arrive close together, the first-arriving sound usually dominates perceived location
Haas effect	A commonly used term related to how a delayed reflection can fuse with the first sound and affect perception
Ambisonics	A method for representing and reproducing a sound field around the listener
Wave Field Synthesis (WFS)	Uses an array of loudspeakers to recreate a desired sound field over a listening area
Time-intensity trading	Time and level differences can influence perceived sound direction in related ways

Important numerical formula

For a simple spherical-head approximation, maximum ITD can be estimated as:

$$\text{Maximum ITD} \approx \frac{2r}{c}$$

Here, r is head radius and c is the speed of sound.

For a head radius of 0.11 m and sound speed of 340 m/s:

$$\frac{2(0.11)}{340} \approx 0.000647 \text{ s}$$

That is approximately **0.65 milliseconds**.

Exam trap: Milliseconds are thousandths of a second.

4. Haptics — touch and physical interaction

Haptics is the study and technology of touch. In VR, it helps you feel that you are interacting with objects.

Example: A controller vibrates when a virtual object is hit.

Three broad areas of haptics

1. **Human haptics:** How humans sense and explore objects.
2. **Machine haptics:** How machines and robots physically sense or interact with objects.
3. **Computer haptics:** How computers create touch-related interactions and feedback.

Memory trick: H-M-C = Human, Machine, Computer.

Haptics is more than touch

Haptics includes:

- **Sensing:** Feeling an object's shape, texture, pressure, or temperature.
- **Manipulation:** Moving, grasping, pressing, or turning an object.

Four modalities of the somatic sense

Modality	What it means
Tactile	Touch and surface sensations
Temperature	Hot and cold
Pain	Harmful or painful sensations
Kinesthesia	Sensing body movement and position

Passive touch versus active touch

Passive touch

Something touches you, but you do not actively explore it.

Active touch

You move your hand or fingers to explore an object.

For this assignment, passive touch is described as unidirectional and active touch as involving bidirectional interaction.

Four mechanoreceptors

Mechanoreceptors are sensory receptors that respond to mechanical stimulation, such as pressure, skin stretch, or vibration.

Receptor	Type	Main idea to remember
Merkel	SA1	Fine details and sustained pressure
Ruffini	SA2	Skin stretch
Meissner	RA1	Light touch and lower-frequency vibration
Pacinian	RA2	High-frequency vibration

SA means slowly adapting: continues responding during sustained stimulation.

RA means rapidly adapting: responds particularly to changes in stimulation.

Memory trick: Pacinian = fast vibration; Merkel = details.

Six exploratory procedures

These are ways of using touch to learn about objects.

Procedure	Example	What it helps reveal

Lateral motion	Slide fingers across a surface	Texture
Pressure	Press a sponge	Hardness or softness
Static contact	Keep a finger still against an object	Surface properties
Unsupported weight	Hold an object	Weight
Enclosure	Wrap your hand around an object	Overall size and shape
Contour following	Trace an object's edge	Exact shape

Memory trick: L-P-S-U-E-C — Lateral, Pressure, Static, Unsupported, Enclosure, Contour.

What can touch tell us about an object?

- **Geometry:** Shape and size.
- **Material:** Texture and thermal information.

One detail from your assignment: it identifies the earlobe as an example of tactile sensation that can occur despite lacking the four specialized mechanoreceptors discussed in the lecture.

Part 2: 50 exam-style practice MCQs

Try answering each question before checking the answer key. These are practice questions based on the topics we discussed, not official NPTEL exam questions.

Section A — Perception engineering (Q1–Q8)

Q1. What is perception engineering mainly concerned with?

A. Making computers physically heavier
B. Designing technology based on human perception
C. Increasing internet speed
D. Storing files

Q2. What is sensation?

A. Detecting information through the senses
B. Remembering a story
C. Solving a mathematical problem
D. Creating a computer program

Q3. What is perception?

A. Only detecting light
B. Only detecting sound
C. Understanding or interpreting sensory information
D. Measuring computer memory

Q4. The absolute threshold refers to:

A. The loudest possible sound
B. The smallest stimulus detectable under specified conditions
C. The difference between two colours
D. The speed of light

Q5. The difference threshold is also called:

A. HRTF
B. JND
C. HSV
D. ITD

Q6. A student notices that a second bag is slightly heavier than the first. Which concept explains the smallest noticeable change in weight?

A. Absolute threshold
B. Difference threshold
C. Chromatic adaptation
D. Head tracking

Q7. Weber's law relates the noticeable change in a stimulus to:

A. The original stimulus intensity
B. The number of computer users
C. The size of a screen only
D. The colour of the background only

Q8. Which law describes perceived intensity approximately as a logarithmic function of physical intensity?

A. von Kries model
B. Fechner's law
C. Weber's law
D. Precedence effect

Section B — Vision and colour (Q9–Q16)

Q9. In the HSV colour model, H stands for:

A. Height
B. Hue
C. Hearing
D. Haptics

Q10. In HSV, saturation describes:

A. How strong or pure a colour appears
B. The object's physical weight
C. Sound direction
D. The size of an image

Q11. In HSV, value mainly represents:

A. Vibration frequency
B. Colour brightness or lightness
C. Sound intensity
D. Object movement

Q12. What is chromatic adaptation?

A. The eyes and visual system adjusting to the colour of illumination
B. The ears adapting to sound
C. The hand adjusting to pressure
D. A computer increasing its storage

Q13. Which statement best describes colour constancy?

A. Every object always emits the same amount of light
B. An object's perceived colour remains relatively stable under different lighting
C. All colours have the same brightness
D. Colours cannot be seen in dim light

Q14. What is the purpose of a Chromatic Adaptation Transform (CAT)?

A. To estimate a person's weight
B. To model colour adaptation between viewing conditions
C. To calculate sound delay

D. To detect hand movement

Q15. The von Kries model is associated with:

A. Scaling colour-sensitive channel responses during adaptation
B. Creating vibration in a controller
C. Detecting sound echoes
D. Tracking a robot's weight

Q16. Why can VR be useful for studying colour perception?

A. It removes the need for human participants
B. It allows controlled changes to lighting and virtual scenes
C. It makes all colours identical
D. It prevents any change in visual perception

Section C — Spatial audio (Q17–Q33)

Q17. What is the main purpose of spatial audio?

A. To make every sound equally loud
B. To make sounds appear to come from positions in 3D space
C. To remove all background sounds
D. To change sound into colour

Q18. Which audio type typically uses left and right channels?

A. Mono
B. Stereo
C. Single-channel text
D. Silent audio

Q19. Which statement best describes mono audio?

A. It uses one audio channel
B. It always requires eight speakers
C. It always includes head tracking
D. It is the same as full 3D audio

Q20. What does binaural mean?

A. Using both eyes
B. Using both ears
C. Using both hands
D. Using two screens

Q21. Sound reaches the right ear slightly before the left ear. Which cue is involved?

A. ILD
B. ITD
C. IPD
D. HSV

Q22. Sound is louder at the nearer ear than at the farther ear. Which cue is involved?

A. ITD
B. IPD
C. ILD
D. JND

Q23. IPD stands for:

A. Interaural Phase Difference
B. Internal Perception Display
C. Interactive Position Detection

D. Interpersonal Pressure Difference

Q24. Which set contains the three major binaural cues discussed in the lecture?

A. HSV, CAT, JND
B. ITD, ILD, IPD
C. SA, RA, HMC
D. Mono, colour, pressure

Q25. What does HRTF describe?

A. How a computer stores sound files
B. How the head, ears, and body affect sound before it reaches the ears
C. How the eyes adjust to light
D. How a hand detects weight

Q26. Why is head tracking important in VR spatial audio?

A. It changes the colour of objects
B. It adjusts the sound presentation as the listener turns their head
C. It makes every sound mono
D. It measures skin pressure

Q27. What is the cone of confusion?

A. A colour-blindness condition
B. A region of possible sound locations that can produce similar binaural cues
C. A type of vibration receptor
D. A method of displaying images

Q28. What usually happens in the precedence effect when similar sounds arrive very close together in time?

A. The later sound always determines location
B. The first-arriving sound tends to dominate perceived location
C. Both sounds disappear
D. The listener hears only higher frequencies

Q29. A delayed reflection follows the original sound. Under suitable conditions, what may the listener perceive?

A. Two completely unrelated colours
B. One fused sound whose location is dominated by the first arrival
C. No sound at all
D. Only a tactile sensation

Q30. A simple spherical-head approximation estimates maximum ITD using:

A. $r \times c$
B. $2r/c$
C. c/r
D. $r + c$

Q31. If the head radius is 0.11 m and sound speed is 340 m/s, the approximate maximum ITD is:

A. 0.65 milliseconds
B. 65 milliseconds
C. 6.5 seconds
D. 340 milliseconds

Q32. What is Ambisonics used for?

A. Representing and reproducing a sound field around the listener
B. Measuring the temperature of skin
C. Changing image saturation
D. Measuring the weight of a virtual object

Q33. Wave Field Synthesis (WFS) aims to:

A. Recreate a desired sound field using an array of loudspeakers
B. Create a colour wheel
C. Detect pressure through the fingertips
D. Calculate an absolute visual threshold

Section D — Haptics (Q34–Q50)

Q34. Which combination gives the three broad areas of haptics from your assignment?

A. Visual, auditory, thermal
B. Human, machine, computer
C. Mechanical, optical, neural
D. Passive, active, virtual

Q35. Why is haptics broader than simply the sense of touch?

A. It includes sensing and manipulation through touch
B. It includes only temperature
C. It excludes movement
D. It includes only sound

Q36. Which option lists the four major somatic-sense modalities from the assignment?

A. Vision, hearing, taste, smell
B. Tactile, temperature, pain, kinesthesia
C. Pressure, vision, hearing, balance
D. Hue, saturation, value, brightness

Q37. Someone places a pen on your skin while you keep your hand still. This is:

A. Active touch
B. Passive touch
C. Contour following
D. Enclosure

Q38. Which statement matches the lecture's distinction between passive and active touch?

A. Passive is unidirectional; active involves bidirectional interaction
B. Passive is always bidirectional; active is always unidirectional
C. Both exclude sensation
D. Both require the hand to remain still

Q39. Which receptor is most strongly associated with high-frequency vibration?

A. Merkel / SA1
B. Ruffini / SA2
C. Meissner / RA1
D. Pacinian / RA2

Q40. Merkel receptors are associated especially with:

A. Fine details and sustained pressure
B. High-frequency vibration only
C. Sound direction
D. Colour adaptation

Q41. Ruffini receptors are associated especially with:

A. Skin stretch
B. Hearing
C. Colour brightness
D. Sound phase

Q42. Meissner receptors are associated especially with:

A. Light touch and lower-frequency vibration
B. High-frequency vibration only
C. Visual tracking
D. Colour constancy

Q43. What does SA mean in SA1 and SA2?

A. Sound adaptation
B. Slowly adapting
C. Spatial audio
D. Sensory absence

Q44. What does RA mean in RA1 and RA2?

A. Rapidly adapting
B. Repeated audio
C. Relative adaptation of colour
D. Receptor absence

Q45. Which statement about SA and RA receptors is correct?

A. SA receptors continue responding during sustained stimulation; RA receptors primarily respond to changes
B. SA receptors detect only sound
C. RA receptors continue responding identically during every unchanged stimulus
D. Both have exactly the same response characteristics

Q46. Which observation challenges the claim that the four specialized mechanoreceptors are essential for every tactile sensation?

A. The palm contains no receptors
B. The earlobe can exhibit tactile sensation despite lacking the four specialized mechanoreceptors discussed
C. Pacinian receptors detect only temperature
D. Kinesthesia eliminates touch

Q47. Which pairing is correct?

A. Geometry → shape and size; material → texture and thermal information
B. Geometry → temperature; material → sound direction
C. Geometry → pain; material → colour only
D. Geometry → vibration frequency only; material → joint angle only

Q48. Sliding your fingers across a surface to determine whether it is rough or smooth is:

A. Enclosure
B. Lateral motion
C. Unsupported weight
D. Static contact

Q49. Holding an object to judge how heavy it is corresponds to:

A. Contour following
B. Static contact
C. Unsupported weight
D. Lateral motion

Q50. Tracing the edge of an object with your finger to discover its shape is:

A. Pressure
B. Enclosure
C. Contour following
D. Passive touch

Part 3: Answer key with easy explanations

Try to answer all 50 before reading this section.

Q	Answer	Easy explanation
1	B	Perception engineering uses knowledge of human senses to design technology.
2	A	Sensation means detecting information through the senses.
3	C	Perception means understanding what the senses detect.
4	B	Absolute threshold is the smallest detectable stimulus under specified conditions.
5	B	JND means Just Noticeable Difference.
6	B	Difference threshold means noticing a change.
7	A	Weber's law relates the noticeable change to the original stimulus.
8	B	Fechner's law describes a logarithmic relationship.
9	B	H = Hue.
10	A	Saturation describes colour strength or purity.
11	B	Value represents how light or dark a colour appears.
12	A	Chromatic adaptation is adjustment to the illumination colour.
13	B	Colour constancy helps an object appear to retain its colour under changing illumination.
14	B	CAT models colour adaptation between viewing conditions.
15	A	The von Kries model scales colour-sensitive channel responses.
16	B	VR allows researchers to control lighting and virtual scenes.
17	B	Spatial audio places sounds in 3D space.
18	B	Stereo normally uses left and right audio channels.
19	A	Mono uses one audio channel.
20	B	Binaural means using both ears.
21	B	ITD = difference in arrival time.
22	C	ILD = difference in sound level.
23	A	IPD = Interaural Phase Difference.
24	B	The three cues are ITD, ILD, and IPD.
25	B	HRTF describes how the body changes sound reaching the ears.
26	B	Head tracking updates sound as the listener turns their head.
27	B	Similar binaural cues can leave several possible sound locations.
28	B	The first sound usually dominates perceived location.
29	B	A close reflection may fuse with the original sound.
30	B	The simple estimate is $2r/c$.
31	A	$2(0.11)/340 \approx 0.000647$ seconds = 0.65 ms.
32	A	Ambisonics represents a sound field around the listener.
33	A	WFS uses loudspeaker arrays to recreate a sound field.
34	B	The three areas are human, machine, and computer haptics.
35	A	Haptics involves sensing and manipulation.
36	B	The four modalities are tactile, temperature, pain, and kinesthesia.
37	B	Passive touch happens without active exploration.

38	A	This is the distinction given in your assignment.
39	D	Pacinian / RA2 detects high-frequency vibration.
40	A	Merkel / SA1 is associated with fine details and sustained pressure.
41	A	Ruffini / SA2 is associated with skin stretch.
42	A	Meissner / RA1 is associated with light touch and lower-frequency vibration.
43	B	SA means slowly adapting.
44	A	RA means rapidly adapting.
45	A	SA continues responding; RA responds particularly to changes.
46	B	This is the exception identified in your assignment.
47	A	Geometry tells shape and size; material tells texture and thermal information.
48	B	Sliding the fingers is lateral motion.
49	C	Holding an object to judge its weight is unsupported weight.
50	C	Tracing an object's edge is contour following.

Part 4: The most important memory sheet

Read this twice before your exam.

Vision

- HSV = Hue, Saturation, Value.
- Hue = which colour.
- Saturation = colour strength.
- Value = lightness or darkness.
- Chromatic adaptation = adjustment to illumination colour.
- CAT = Chromatic Adaptation Transform.
- von Kries = scaling colour-sensitive channel responses.

Spatial audio

- Mono = one channel.
- Stereo = left and right channels.
- ITD = Time difference.
- ILD = Level difference.
- IPD = Phase difference.
- HRTF = effects of the head, ears, and body on sound.
- Head tracking = sound presentation changes as the head turns.
- Cone of confusion = ambiguous possible sound locations.
- Precedence effect = first arrival tends to dominate location.
- Ambisonics = sound-field representation.
- WFS = sound-field recreation with loudspeaker arrays.
- Maximum ITD approximation = $2r/c$.

Haptics

- HMC = Human, Machine, Computer.
- Haptics = sensing + manipulation.
- Somatic modalities = Tactile, Temperature, Pain, Kinesthesia.
- Passive touch = no active exploration.
- Active touch = actively explore.
- Merkel / SA1 = details and sustained pressure.
- Ruffini / SA2 = skin stretch.

- Meissner / RA1 = light touch and lower-frequency vibration.
- Pacinian / RA2 = high-frequency vibration.
- SA = Slowly adapting.
- RA = Rapidly adapting.
- LPSUEC = Lateral motion, Pressure, Static contact, Unsupported weight, Enclosure, Contour following.

Part 5: How to prepare for a better exam attempt

1. **First round:** Read the summary and memory sheet.
2. **Second round:** Attempt all 50 questions without looking at the answers.
3. **Third round:** Revise only the questions you got wrong.
4. **Final revision:** Practise the differences that are easy to confuse — ITD vs ILD vs IPD, SA vs RA, sensation vs perception, and hue vs saturation vs value.

Use this score guide for your practice test:

Score	What it means
45–50	Excellent revision
38–44	Good preparation; revise mistakes
30–37	Review the memory sheet and retry
Below 30	Study one topic at a time, then repeat the test

These score ranges are just practice guidelines, not predictions of your official NPTEL result.

My most important advice: Don't memorize only the letters A, B, C, and D. Understand *why* the answer is correct, because an exam may change the example or rearrange the options while testing the same concept.